

MASTERING THE MARKET CYCLE

GETTING THE ODDS
ON YOUR SIDE

HOWARD
MARKS

MASTERING THE MARKET CYCLE

GETTING THE ODDS
ON YOUR SIDE

Howard Marks

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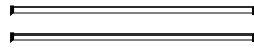
*With All My Love
to Nancy
Jane, Justin, Rosie and Sam
Andrew and Rachel*

“When I see memos from Howard Marks in my mail, they’re the first thing I open and read. I always learn something.”

—Warren Buffett

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INTRODUCTION



Seven years ago I wrote a book called *The Most Important Thing: Uncommon Sense for the Thoughtful Investor*, regarding where investors should direct their greatest attention. In it I said “the most important thing” to nineteen other things as well. There is no most important thing in investing. Everyone of the twenty elements discussed in *The Most Important Thing* is absolutely essential for anyone who wishes to be a successful investor.

Vince Lombardi, the legendary coach of the Green Bay Packers, is famous for having said, “winning isn’t everything, it’s the only thing.” I’ve been able to figure out what Lombardi actually meant by that statement, there’s no doubt he considered winning the most important thing. Like I can’t say an understanding of cycles is everything in investing, but for me it’s certainly right near the top of the list.

Most of the great investors I’ve known over the years have had an exceptional sense for how cycles work in general and where we stand in the current one. That sense permits them to do a superior job of selecting portfolios for what lies ahead. Good cycle timing—combined with an effective investment approach and the involvement of exceptional people—has accounted for the vast bulk of the success of my firm, Oaktree Management.

It’s for that reason—and because I find something particularly interesting about the fluctuations of cycles—and because where we stand in the current one of the things my clients ask about most—and finally because I’ve been written about the essential nature of cycles—that I decided to write *The Most Important Thing* with a book devoted entirely to an exploration of cycles. I hope you’ll find it of use.



Some patterns and events recur regularly in our environment, influencing our behavior and our lives. The winter is colder and snowier than the summer, and the daytime is lighter than the night. Thus we plan ski trips

and sailing trips for the summer, and our work and recreation for and our sleeping at night. We turn on the lights as evening draws turn them off when we go to bed. We unpack our warm coats as approaches and our bathing suits for the summer. While some people the ocean in winter for exhilaration and some elect to work the night free up their days, the vast majority of us follow the normal patterns, making everyday life easier.

We humans use our ability to recognize and understand patterns our decisions easier, increase benefits and avoid pain. Importantly, on our knowledge of recurring patterns so we won't have to reconsider decision from scratch. We know hurricanes are more likely in September, we avoid the Caribbean at that time of year. We New Yorkers schedule visits to Miami and Phoenix for the winter months, when the differential is a positive, not a negative. And we don't have to in January and decide anew whether to dress for warmth or

Economies, companies and markets also operate pursuant to patterns. of these patterns are commonly called cycles. They arise from naturally occurring phenomena but, importantly, also from the ups and downs of human psychology and from the resultant human behavior. Because psychology and behavior play such a big part in creating them, these aren't as regular as the cycles of clock and calendar, but they better and worse times for certain actions. And they can profoundly affect investors. If we pay attention to cycles, we can come out ahead of past cycles, understand their origins and import, and keep alert for one, we don't have to reinvent the wheel in order to understand every investment environment anew. And we have less of a chance of being blindsided by events. We can master these recurring patterns for betterment.



It's my primary message that we should pay attention to cycles should say "listen to them." [Dictionary.com](https://www.dictionary.com) supplies two closely distinct definitions for the word "listen." The first is "to attend closely purpose of hearing." The second is "to heed." Both definitions are what I'm writing about.

In order to properly position a portfolio for what's going on in the environment—and for what that implies regarding the future of the the investor has to maintain a high level of attention. Events matter to everyone who is operating in a given environment. But not to them equally in the sense of paying attention, being aware of and thus potentially figuring out their import.

And certainly not everyone heeds equally. By “heed” I mean “obey, listen to, mind, be guided by or take to heart.” Or, in other words, “listen to and follow its dictates.” Perhaps I can better convey this “heed” by listing its antonyms: ignore, disregard, discount, reject, neglect, shun, flout, disobey, tune out, turn a deaf ear to, or be unheeding. Invariably, investors who disregard where they stand in cycles are bound to suffer serious consequences.

In order to get the most out of this book—and do the best job with cycles—an investor has to learn to recognize cycles, assess them for the instructions they imply, and do what they tell him to do. (See the author’s note below regarding my use of male pronouns.) If an investor follows in this sense, he will be able to convert cycles from a wild, unpredictable force that wreaks havoc, into a phenomenon that can be understood to his advantage of: a vein that can be mined for significant outperformance.



A winning investment philosophy can be created only through the combination of a number of essential elements:

- A technical education in accounting, finance and economics provides a foundation: necessary but far from sufficient.
- A view on how markets work is important—you should have your own. As you set out to invest, but it must be added to, questioned, reshaped as you proceed.
- Some of your initial views will come from what you’ve read, so reading is an essential building block. Continuing to read will enable you to increase the efficacy of your approach—both embracing those ideas you find appealing and discarding those you don’t. Importantly, it’s great to read outside the strict boundaries of investing. Legendary investor Charlie Munger often points to the benefits of reading broadly. Insights and processes in other fields can add greatly to effective investment approaches and decisions.
- Exchanging ideas with fellow investors can be an invaluable source of growth. Given the non-scientific nature of investing, there’s nothing as being finished with your learning, and no individual has a monopoly on insight. Investing can be solitary, but I think practicing it in solitude are missing a lot, both intellectually and interpersonally.
- Finally, there really is no substitute for experience. Every year I come to view investing differently, and every cycle I’ve lived through has taught me something about how to cope with the next one. I recommend a long career and see no reason to stop any time.

Writing my book has given me a wonderful vehicle for acknowledging people who have contributed to my investment insight and the texture of my working life.

- I've gained a great deal from reading the work of Peter Berns, Kenneth Galbraith, Nassim Nicholas Taleb and Charlie Ellis.
- I've continued to pick up pointers from the people I cited in *Important Thing* and others, including Seth Klarman, Charlie Munger, Warren Buffett, Bruce Newberg, Michael Milken, Jacob Rothschild, Todd Combs, Roger Altman, Joel Greenblatt, Peter Kaufman and Doug Kass. And since Nancy and I moved to New York in 2013 with the kids, I've been fortunate to add Oscar Schafer, Jim Tisch and Ajit Menon to this circle. Each of these people's way of looking at things has helped mine.
- Finally I want to return to the most important collaborators, co-founders: Bruce Karsh, Sheldon Stone, Richard Masson and John Keele. They honored me by adopting my philosophy as the basis for Oaktree's investment approach; applied it skillfully (and thus earned recognition for it); and helped me add to it over the thirty-plus years we've been associated. As indicated in what follows, Bruce and I exchanged ideas and backed each other up almost daily over that period, and my give-and-take with him—especially in the most difficult moments—has played a particularly indispensable part in the development of our approach to cycles on which this book is based.

I also want to thank the people who played important parts in the book's creation: my talented editor at HMH, Rick Wolff; my resourceful assistant, Levine, who brought me to Rick; my great friend Karen Mack Goldsmith, who pushed me at every turn to make the book more appealing; and my highly supportive long-time assistant, Caroline Heald. I particularly want to cite Prof. Randy Kroszner of the University of Chicago's Booth School of Business, who helped out by reviewing the chapters on the economic cycle and monetary intervention with it.



Since knowledge is cumulative but we never know it all, I look forward to learning more in the years ahead. In investing, there is nothing static about the world, since the environment is always changing, and investors' strategies must respond to the environment or it will change further. Thus, I look forward to things in the future that I don't know now, and I look forward to the insights that will come in my memos and books yet to come.

Author's notes:

1. As I did in *The Most Important Thing*, I want to issue an apology for my consistent use of male pronouns. It can be a habit for someone who started to write more than sixty years ago. It's much easier and more attractive to write "he" than "he/she." Alternating between "he" and "she" seems forced. And I dislike using "they" when the subject is a single person. The exception: I've been privileged to work with over the course of my career with women who I absolutely think every bit as much of them as professionals and investors as I do their male counterparts.
2. Also as in *The Most Important Thing*, in order to make my book more readable, I will borrow from time to time from the client memos I've written over the years starting in 1990. I will also borrow from my first book. I could go to the trouble of reinventing the wheel and writing new subjects anew, but I won't. Instead, I'll lift key passages and memos that I think make their point clearly. I hope my readers won't make those who buy this book feel they've received less than their money's worth.
In order to advance the purposes of this book, I will occasionally delete a few words or delete a few from the passages I cite, or rearrange paragraphs in an order different from that in which they appeared in the original. Since they're my passages, I think it's okay to do this without noting it in every case. But I do it only to increase clarity or to alter their meaning or make them more correct with the benefit of hindsight.
3. And finally as in *The Most Important Thing*, I'll be dealing with a topic that—like investing in general—is complex and involves many elements that overlap and can't be neatly segregated into discrete chapters. Since some of those elements are touched on in multiple places, you'll likewise find some instances of repetition where I repeat noteworthy quotations from others or citations from my book and memos that I can't resist using more than once.
4. Please note that when I talk about "investing," I'll assume the investor is buying, holding or, as we say, "being long" in the expectation that certain assets will appreciate. This is as opposed to selling securities that one doesn't own in the hope they'll decline. Most investors aren't always "long" rather than "short," but most of the time they are. The number of people who sell stocks short or ever get into short positions that is, whose short positions have a total value exceeding that of the stocks they own—is tiny relative to those who don't. Thus, in

- I'm going to speak exclusively about investing in things because they're expected to rise, not selling assets short in the hope
5. Lastly, whereas I first conceived of this book as being just cycles, as I wrote I came up with ideas on lots of other asset selection and "catching falling knives." Rather than disc I've included them, too. I hope you'll be glad they're here: prov bonus rather than straying from the mission.

I

WHY STUDY CYCLES?



The odds change as our position in the cycles changes. If we change our investment stance as these things change, we're being passive regarding cycles; in other words, we're ignoring the cycles and tilt the odds in our favor. But if we apply some insight regarding cycles, we can increase our bets and place them on more aggressive investments when the odds are in our favor, and we can take money off the table and increase our defensiveness when the odds are against us.

Investing is a matter of preparing for the financial future. To define the task: we assemble portfolios today that we hope will benefit from the events that unfold in the years ahead.

For professional investors, success consists of doing this better than the average investor, or outperforming an assigned market benchmark. The performance of which is determined by the actions of all the investors. But achieving that kind of success is no small challenge: although it's easy to generate average investment performance, it's quite hard to generate above-average.

One of the most important foundational elements of my investment philosophy is my conviction that we can't know what the "macro future" has in store for us in terms of things like economies, markets or geopolitics. Put it more precisely, few people are able on balance to know the macro future than others. And it's only if we know more than others that we can achieve that kind of success. That kind of success consists of having better data; doing a superior job of interpreting the data we have; knowing what actions to take on the basis of our interpretation; and having the emotional fortitude required to take those actions) that our forecasts will lead to outperformance.

In short, if we have the same information as others, analyze it the same way, reach the same conclusions and implement them the same way, we shouldn't expect that process to result in outperformance. And it's difficult to be consistently superior in those regards as it relates to

So, in my view, trying to predict what the macro future holds is not likely to help investors achieve superior investment performance. Very few investors are known for having outperformed through macro forecasting.

Warren Buffett once told me about his two criteria for a decision based on macro information: it has to be important, and it has to be knowable. “Everyone knows” that macro developments play a dominant role in determining the performance of markets these days, “macro investors” who have shown rather unimpressive results. It’s not that the macro is an unsolvable matter, but rather that very few people can master it. For most things that are knowable (or not knowable well enough and consistently enough for long enough to outperform).

Thus I dismiss macro prediction as something that will bring investment success for the vast majority of investors, and I certainly include myself in that group. If that’s so, what’s left? While there are lots of details and nuances, I think we can most gainfully spend our time in three general areas:

- trying to know more than others about what I call “the knowable”: the fundamentals of industries, companies and securities,
- being disciplined as to the appropriate price to pay for a particular asset based on those fundamentals, and
- understanding the investment environment we’re in and deciding our strategic position for our portfolios for it.

A great deal has been written on the first two topics. Together, they constitute the key ingredients in “security analysis” and “value investing.” They involve judgments regarding what an asset can produce in the future—usually in terms of earnings or cash flow—and what those prospects make the asset worth today.

What do value investors do? They strive to take advantage of discrepancies between “price” and “value.” In order to do that successfully, they (a) quantify an asset’s intrinsic value and how it’s likely to change over time, (b) assess how the current market price compares with the asset’s value, past prices for the asset, the prices of other assets, and “theoretical fair” prices for assets in general.

Then they use that information to assemble portfolios. Most of the time, their immediate goal is to hold investments offering the best available value propositions: the assets with the greatest upside potential and/or the least exposure to downside risk. You might argue that assembling a portfolio should consist of nothing more than identifying the assets with the highest value and the ones whose prices most understate their value. That might be true in general and in the long term, but I think another

profitably enter into the process: properly positioning a portfolio for likely to happen in the market in the years immediately ahead.

In my view, the greatest way to optimize the positioning of a given point in time is through deciding what balance it should aggressiveness and defensiveness. And I believe the aggressiveness/defensiveness balance should be adjusted over time in response to changes in the state of the investment environment number of elements stand in their cycles.

The key word is “calibrate.” The amount you have invested, your allocation of capital among the various possibilities, and the of the things you own all should be calibrated along a continuum runs from aggressive to defensive. . . . When we’re getting value cheap, we should be aggressive; when we’re getting value expensive, we should pull back. (“Yet Again?,” September 2017)

Calibrating one’s portfolio position is what this book is mostly about.



One of the key words required if one is to understand the reason cycles is “tendencies.”

If the factors that influence investing were regular and predictable, for example, if macro forecasting worked—we would be able to say “it will happen.” Yet the fact that that’s not the case doesn’t mean we’re helpless in contemplating the future. Rather, we can talk about what might happen or should happen, and how likely they are to happen. Things are what I call “tendencies.”

In the investment world, we talk about risk all the time, but there is no universal agreement about what risk is or what it should imply for investment behavior. Some people think risk is the likelihood of losing money, (including many finance academics) think risk is the volatility of asset prices or returns. And there are many other kinds of risk—too many to cover here.

I lean heavily toward the first definition: in my view, risk is the likelihood of permanent capital loss. But there’s also such a thing as opportunity risk: the likelihood of missing out on potential gains. Put them together and we see that risk is the possibility of things not going the way we want.

What is the origin of risk? One of my favorite investment philosophers, late Peter Bernstein, said in an issue of his *Economics and Portfolio* newsletter titled “Can We Measure Risk with a Number?” (June 2007):

Essentially risk says we don't know what's going to happen. . . . We walk every moment into the unknown. There's a range of outcomes and we don't know where [the actual outcome is] going to fall within the range. Often we don't know what the range is.

You'll find below a few ideas (summarized very briefly from the treatment provided in my memo "Risk Revisited Again" of June 1994). I think they follow directly from the starting point provided by Bernstein. They might help you understand and cope with risk.

As retired London Business School professor Elroy Dimson says, "The future means more things can happen than will happen." For each event in economics, business and markets (among other things), if only one outcome could happen—if there could be only one outcome—and if it was predictable, there would, of course, be no uncertainty or risk. And with no uncertainty regarding what was going to happen, in theory we could know exactly the position of our portfolios to avoid loss and garner maximum gains. But in practice, and in investing, since there can be many different outcomes, uncertainty and risk are inescapable.

As a consequence of the above, the future should be viewed as a fixed outcome that's destined to happen and capable of being predicted as a range of possibilities and—hopefully on the basis of insight into their respective likelihoods—as a probability distribution. Probability distributions reflect one's view of tendencies.

Investors—or anyone hoping to deal successfully with the future—form probability distributions, either explicitly or informally. If it's done properly, those probabilities will be helpful in determining one's proper course of action. But it's still essential to bear in mind that even if we know the probabilities, that doesn't mean we know what's going to happen.

Outcomes regarding a given matter may be governed by a probability distribution in the long run, but with regard to the outcome of a particular event there can be great uncertainty. Any of the outcomes included in a probability distribution can occur, albeit with varying probabilities, since the process through which the outcome is chosen will be affected not only by the merits of the outcomes but also by randomness. To invert Dimson's statement, even though many things can happen, only one will. We may know what to expect "on average," but that may have no connection with what actually will happen.

In my way of thinking about it, investment success is like the lottery. Both are determined by one ticket (the outcome) being chosen from a bowlful (the full range of possible outcomes). In each case, the outcome is chosen from among the many possibilities.

Superior investors are people who have a better sense for what ticket to buy in the bowl, and thus for whether it's worth participating in the

other words, while superior investors—like everyone else—don't know exactly what the future holds, they do have an above-average understanding of future tendencies.

As an aside, I want to add a thought here. Most people think they deal with the future by formulating an opinion as to what's going to happen, perhaps via a probability distribution. I think there are actual requirements, not one. In addition to an opinion regarding what's going to happen, people should have a view on the likelihood that their opinion will prove correct. Some events can be predicted with substantial confidence (will a given investment grade bond pay the interest it promises?), some are uncertain (will Amazon still be the leader in online retailing in a year?), and some are entirely unpredictable (will the stock market go up or down next month?). It's my point here that not all predictions should be treated as equally likely to be correct, and thus they shouldn't be relied on equally. Most people are as aware of this as they should be.



A good way to think about the superior investor described above is someone whose insight into tendencies permits him to tilt the odds in his favor.

Let's say there are 100 balls in a jar, some black and some white. If you know the color, should you bet? Will it come up?

- If you don't know anything about the contents of the jar, betting is just a matter of guessing: uninformed speculation. The situation is similar if you know there are 50 black and 50 white balls. You can bet wisely on black as white, but doing either wouldn't give you a better than a 50:50 chance of being right. Thus betting would be unprofitable unless you're offered odds that are at least even—and unless you're offered odds that avoid paying an admission charge (in investing, a commission or bid-ask spread) to play. Betting on black or white at even odds is not very profitable other than if you got lucky, and luck isn't something you can count on. Betting in the absence of a knowledge edge about the contents of the jar wouldn't be dependably profitable.
- But what if you do have special insight regarding the contents of the jar? Let's say you know there are 70 black balls and 30 white. That allows you to win more often than you lose. If you can bet \$10 against someone who gives you even odds, you'll win \$10 70% of the time and lose \$10 only 30% of the time, for an expected profit of \$4 per 10 picks. (Note: these will be the outcomes on average over a large number of trials, but they are subject to significant variation in the short run due to randomness.)

- Of course, your betting partner will only give you even odds on black (a) if he doesn't know the balls are 70% black and 30% white, (b) if he doesn't know that you do know. If he knew about the contents of the jar, he would give you only 30:70 on black, and the bet would be back to being profitless.
- In other words, in order to win at this game more often than you have to have a knowledge advantage. That's what the superior investor has: he knows more than others about the future tendencies.
- Yet it's important to remember what I said earlier: even if probabilities—that is, even if you do have superior insight into tendencies—you still don't know what's going to happen. Even if the ratio of balls in the jar is 70 black to 30 white, you don't know the color the next one picked will be. Yes, it's more likely to be white, but it'll still be white 30% of the time. When there are as well as black in the jar, and especially when random forces are at work when the next ball is chosen, there can be uncertainty regarding the outcome.
- But all this being said, there doesn't have to be certainty in the game to be worth playing. A knowledge advantage regarding the future tendencies is enough to create success in the long run.

And that brings us to the payoff from understanding cycles. The superior investor doesn't know much about it:

- He doesn't fully understand the nature and importance of cycles.
- He hasn't been around long enough to have lived through many cycles.
- He hasn't read financial history and thus learned the lessons of past cycles.
- He sees the environment primarily in terms of isolated events rather than taking note of recurring patterns and the reasons behind them.
- Most important, he doesn't understand the significance of cycles and what they can tell him about how to act.

The superior investor is attentive to cycles. He takes note of repeating patterns, gains a sense for where we stand in the cycles that matter, and knows those things have implications for the future. This allows him to make helpful judgments about cycles and when to act in them. Specifically:

- Are we close to the beginning of an upswing, or in the middle of one?
- If a particular cycle has been rising for a while, has it reached its peak, or are we now in dangerous territory?

- Does investors' behavior suggest they're being driven by greed or fear?
- Do they seem appropriately risk-averse or foolishly risk-tolerant?
- Is the market overheated (and overpriced), or is it frigid (and underpriced) because of what's been going on cyclically?
- Taken together, does our current position in the cycle imply we should emphasize defensiveness or aggressiveness?

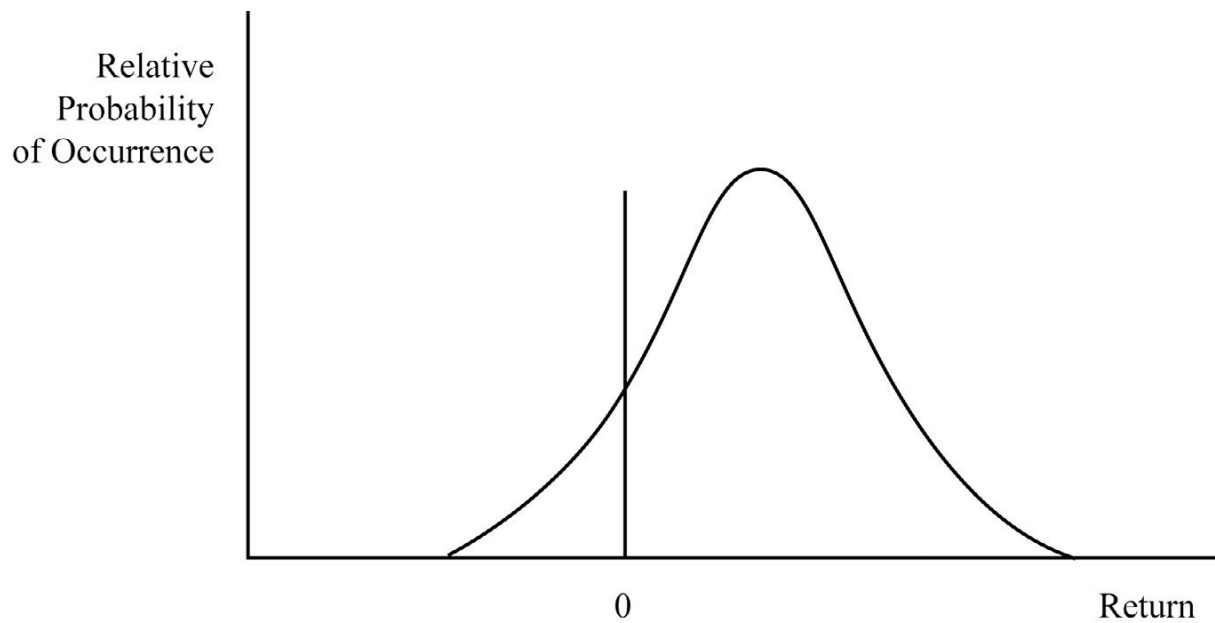
Attention to these elements gives the superior investor an edge that enables him to win more often than he loses. He understands the tendencies or tendencies, thus he knows something that others don't about the color of the balls in the jar. He has a sense for whether the chances of winning exceed the chances of losing; thus he is able to invest more when they are favorable and less when they aren't. Importantly, all these things can be assessed on the basis of observations regarding current conditions. As we'll see in later chapters, we can tell us how to prepare for the future without requiring that we predict the future.

Remember, where we stand in the various cycles has a strong influence on the odds. For example, as we'll see in later chapters, opportunities for investment gains improve when:

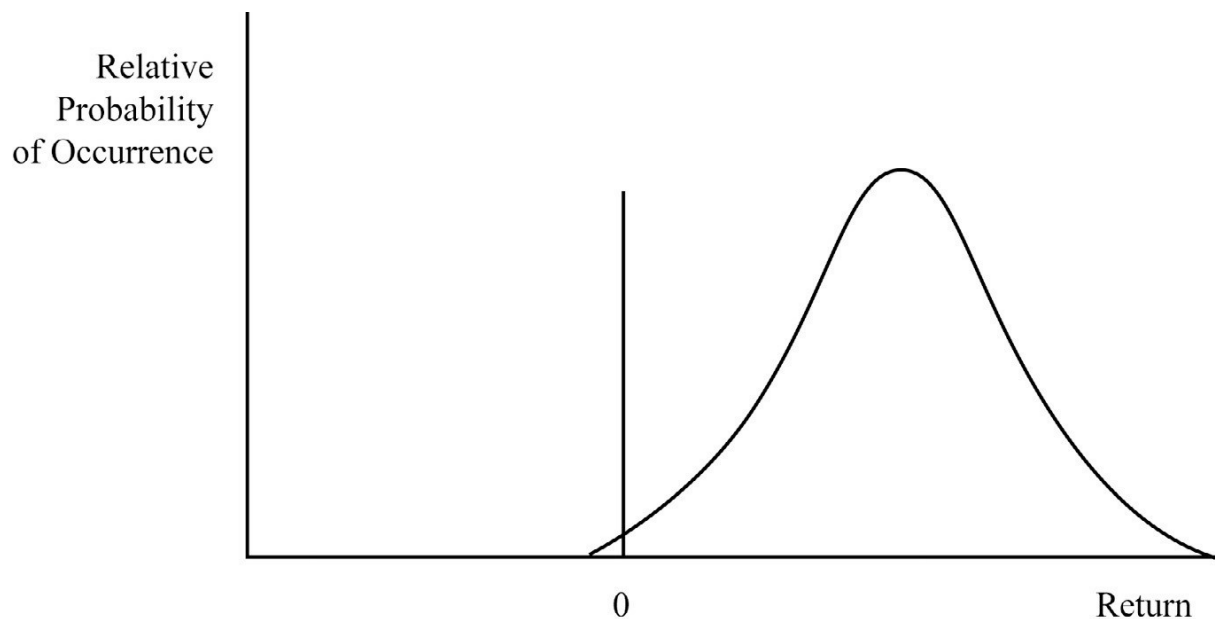
- the economy and company profits are more likely to swing upward than downward,
- investor psychology is sober rather than buoyant,
- investors are conscious of risk or—even better—overly concerned about risk, and
- market prices haven't moved too high.

There are cycles in all these things (and more), and knowing where we stand within them can help us tilt the odds in our favor. In short, understanding movement through the cycle repositions the probability distribution governing future events. Perhaps I should illustrate with regard to investment.

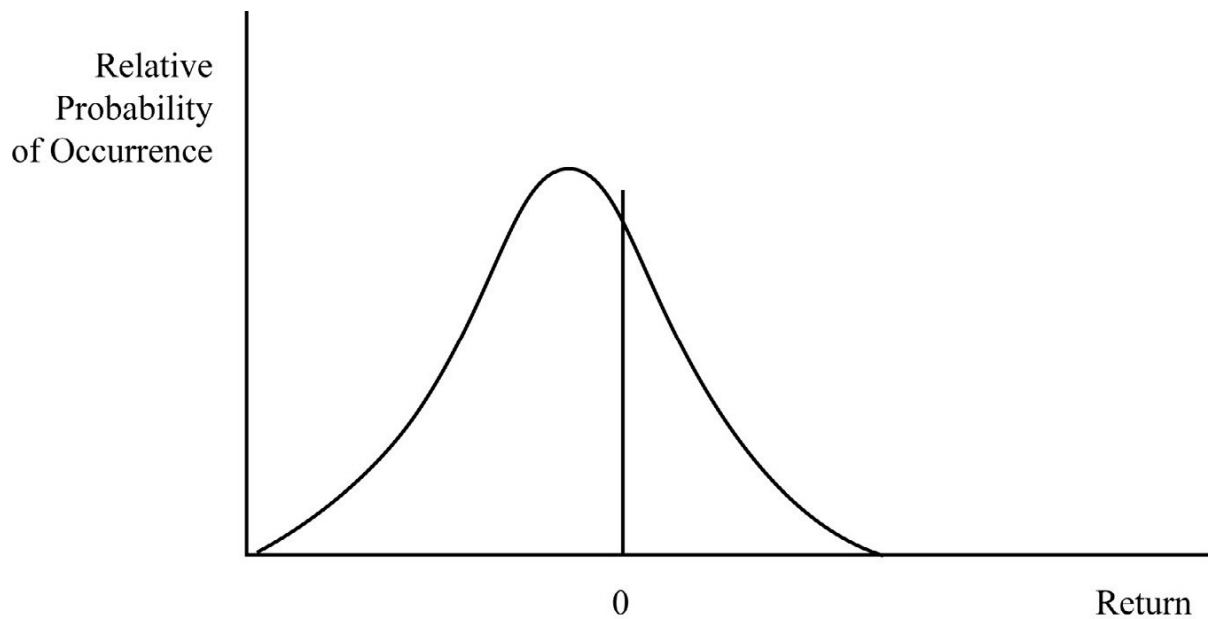
When our position in the various cycles is neutral, the odds are "normal."



When the cycles are positioned propitiously, the probability distribution shifts to the right, such that the outlook for returns is now tilted. Our favorable position in the cycles makes gains more likely and so.



But when the cycles are at dangerous extremes, the odds are against meaning the likelihoods are less good. There's less chance of gain and more chance of loss.



The same is true when our position changes in only a single cycle. For example, regardless of what's going on with regard to the economy, company profits (that is, as the academics say, *ceteris paribus* "things being equal"), the outlook for returns will be better when investors are depressed and fearful (and thus allow asset prices to fall) and worse when they're euphoric and greedy (and drive prices upward).

The odds change as our position in the cycles changes. If we follow our investment stance as these things change, we're being passive regarding cycles; in other words, we're ignoring the chance to tilt the odds in our favor. But if we apply some insight regarding cycles, we can increase our odds. We can place them on more aggressive investments when the odds are in our favor, and we can take money off the table and increase our defensive investments when the odds are against us.

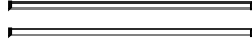
The student of cycles doesn't know for a fact what's going to happen—any more than someone with insight regarding the balls in the jar knows what color ball will come out next. But both have a knowledge advantage regarding what's likely. The student's knowledge of cycles and where we stand at a point in time can make a big contribution to the odds that must be present in order for an investor to achieve superior results. A ball-chooser who knows the ratio is 70:30 has an advantage. So an investor who knows better than others where we stand in the cycle. It's the purpose of this book to help you become that person.

In that interest, I'll describe a number of cyclical processes that take place in real time. The oscillations might seem extreme, and in fact they may be, since they're chosen from the experience of a half-century or so. And they may give the impression that the events under discussion were compressed in time, whereas in truth they took months and years.

develop. But these examples are real, and I hope they'll make my clear.

II

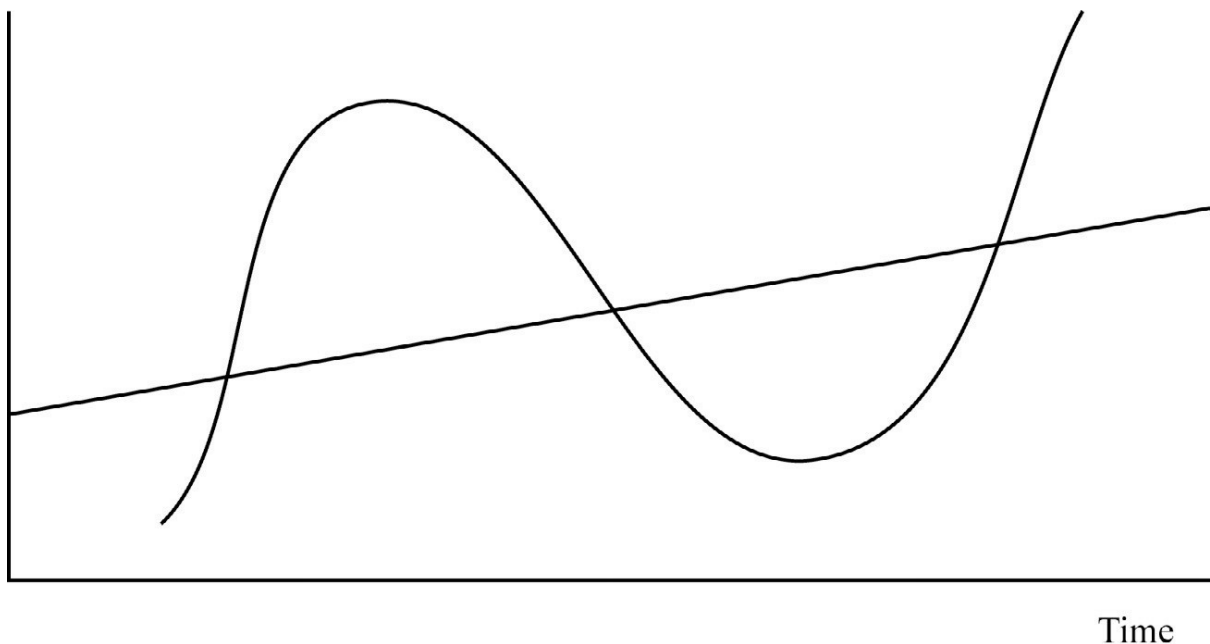
THE NATURE OF CYCLES



Most people think of cycles in terms of a series of events. People understand that these events regularly follow each other in a usual sequence: upswings are followed by downswings, and then eventually by new upswings. But to have a full understanding of cycles, that's not enough. The events in the life of a company can be viewed merely as each being followed by the next, but—more importantly—as each *causing* the next.

When I meet with Oaktree clients, they almost always ask them to make sense of what's going on in the world or in the market. They usually want to know about one particular cycle or another and where they stand in it. I invariably pull out a sheet of paper and make a discussion.

There's usually a line that stretches from lower left to upper right. The line fluctuates up and down around it. Together they look like this.



When I started to organize for the task of writing this book, I found my Oaktree bag and found a large number of such drawings. I put them in the course of describing several different phenomena, and annotated differently. But each one related to a cycle worthy of The chapters in this book will generally be devoted to those cyclical phenomena.

Before moving ahead with my discussion of cycles, I want to say something I mentioned in *The Most Important Thing*. I confess between discussing the ups and downs of cycles and the side-oscillations of pendulums, applying the cycle label to some phenomena and (as seen in chapter VII) the pendulum label to others (usually those to psychology). Sometimes I'll talk about a given phenomenon as sometimes as a pendulum. But when pressed, I find it hard to choose between the two or to say why one gets one label and not the other.

I tend to think about things visually, so perhaps I can use an analogy to describe the connection between cycles and pendulums. As I will explain in length later, cycles oscillate around a midpoint (or a secular trend). Pendulums hang over a midpoint (or norm) and swing back and forth from there. But if you take the hang-point of the pendulum, turn it on its side, and drag it from left to right as it oscillates, what do you get? A cycle.

There really is no fundamental difference. I'll even admit that a pendulum is little more than a special case of a cycle, or perhaps just a cycle in disguise. My reasons for referring to cycles and pendulums as cycles and others as pendulums are clear to me. I hope they are clear to you as well. Or, at minimum, I hope my use of the terms does not detract from what you take from this book.

The bottom line is that, in the world investors inhabit, cycles and pendulums swing back and forth. Cycles and pendulum swings come in many forms and relate to a wide variety of phenomena, but the underlying reasons for them—and the patterns they produce—have a lot in common. They tend to be somewhat consistent over time. Or as Mark Twain once said (although there's no evidence he actually said it), "It doesn't repeat itself, but it does rhyme."

Whether Twain said it or not, that sentence sums up a lot of what this book is about. Cycles vary in terms of reasons and details, and the timing of the ups and downs (and the reasons for them) will occur forever, producing changes in the investment environment—and thus in the results that's called for.

The central line in my drawings constitutes a midpoint around which the cycle oscillates. It sometimes has an underlying direction or secular trend ("secular" as in "of or relating to a long term of indefinite duration," Webster's *New Collegiate Dictionary*), and that's usually upward. So, over

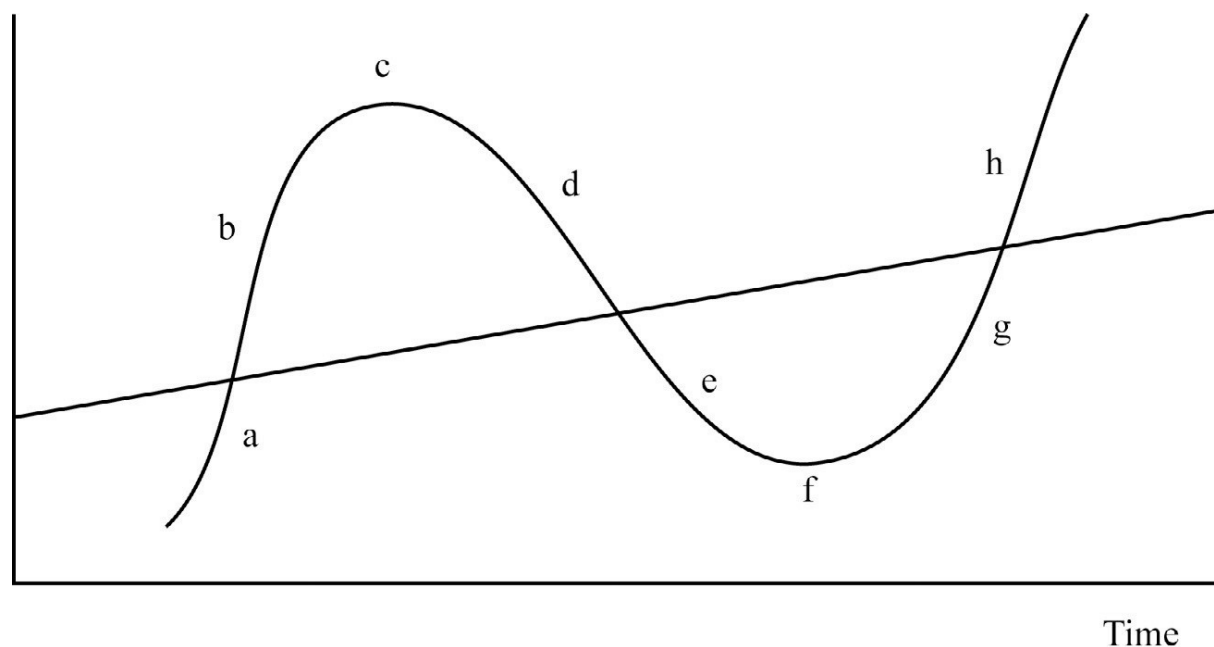
time and in the long run, economies tend to grow, companies' profits increase and (largely because of those things) markets tend to these developments were scientific or wholly natural, physical processes. Economies, companies and markets might progress in a straight line at a constant rate (at least for a while). But of course, they're

The fact is that the performance of these things is heavily influenced in the short run by, among other things, the involvement of people, which is far from steady. Rather they fluctuate from time to time, often because of things we can lump under the broad heading of "psychology." Thus people's behavior varies. . . . certainly as the environment varies, and in the absence of changes in the environment, too.

It's the oscillation of things around the midpoint or secular trend that the book is largely about. The oscillation bedevils people who don't understand it, are surprised by it or, even worse, partake in and contribute to it. I've said before, it often presents profit opportunities for those who understand, recognize and take advantage of cyclical phenomena.

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It's clear from looking at my drawings for a few seconds that the oscillation of cyclical phenomena can be understood as taking place in a few identifiable phases:



- a) recovery from an excessively depressed lower extreme or "low" toward the midpoint,
- b) the continued swing past the midpoint toward an upper extreme "high,"
- c) the attainment of a high,

- d) the downward correction from the high back toward the midpoint mean,
- e) the continuation of the downward movement past the midpoint toward a new low,
- f) the reaching of a low,
- g) once again, recovery from the low back toward the midpoint,
- h) and then, again, the continuation of the upward swing past midpoint, toward another high.

It's important to note from the above that there cannot be said to be a "starting point" or "ending point" for a cycle. Any of the stages can be described as representing the beginning of a cycle, or any stage in between.

The simplistic narrator may find it easy to talk about the beginning of a cycle, but someone a bit more sophisticated can find that extremely difficult. Here's what I wrote on this subject in "Now It's All Bad?":

Henry Kissinger was a member of TCW's board when I was on it, and a few times each year I was privileged to hear him hold forth on world affairs. Someone would ask, "Henry, can you explain yesterday's events in Bosnia?" and he'd say, "Well, in 1722. The starting point is that chain reaction-type events can only be understood in the context of that which went before."

If someone asks, "How did we get to this point?" or "What caused us to reach that high (or low)?" invariably the explanation has to be based on events that went before. But that being said, it may not be easy to know just how far back to reach for the starting point for your narrative.

People often ask me "what caused the cycle to begin?" or "at what point did the end of the cycle?" I consider these improper questions, since cycles neither begin nor end. Better questions might be: "what caused the current cycle's up-leg to begin?" or "how far have we gone since the beginning of the current cycle?" or "are we close to the end of the down-leg?" You might wonder whether we're close to the end of a cycle, as long as you define the cycle from one peak to the next, or from trough to trough. But—in such a definition—cycles do not have a defined beginning, and they will never end.



The cycle oscillates, as I mentioned, around the midpoint. The midpoint of the cycle is generally thought of as the secular trend, norm, mean, average, or "happy medium," and generally as being in some sense as "right and

The extremes of the cycle, on the other hand, are thought of as excesses to be returned from, and generally they are. While the thing cycling tends to spend much of the time above or below it, eventual movement back in the direction of the mean is usually the rule. Movement from either a high or a low extreme back toward the often described as “regression toward the mean,” a powerful and reasonable tendency in most walks of life. But, thinking back to the stages listed above, it can also be seen that the cyclical pattern consists as much of movement from the reasonable midpoint toward potentially imprudent extreme (phases b, e and h in the preceding diagram) as it does going from an extreme back toward the midpoint (a, d and g).

The rational midpoint generally exerts a kind of magnetic pull, bringing things that are cycling back from an extreme in the direction of “normal.” It usually doesn’t stay at normal for long, as the influences responsible for the swing toward the midpoint invariably continue in force and thus cause a swing back from an extreme to proceed through the midpoint and beyond, further, toward the opposite extreme.

It’s important to recognize and accept the dependability of these details vary—the timing, duration, speed and power of the swing, and, importantly, the reasons for them—and that’s likely what’s behind Taiwan’s remark regarding history not repeating. But the underlying dynamics are usually similar. In particular, that means the swingback from a high almost never halts at the midpoint . . . regardless of how “appropriate” the midpoint may be. The continuation of the movement from the midpoint and toward the opposite extreme is highly dependent on the example, markets rarely go from “underpriced” to “fairly priced” and stay there. Usually the fundamental improvement and rising optimism that causes markets to recover from depressed levels remain in force, causing the swing to continue right through “fairly priced” and on to “overpriced.” It doesn’t always happen, but usually it does.



Cycles have more potential to wreak havoc the further they progress from the midpoint—i.e., the greater the aberrations or excesses. If the swing goes from one extreme goes further, the swingback is likely to be more violent, and more damage is likely to be done, as actions encouraged by the operation at an extreme prove unsuitable for life elsewhere in the system.

In other words, the potential for havoc increases as the movement from the midpoint increases: as economies and companies do “too well,” stock prices go “too high.” Advances are followed by mere corrections, bull markets by bear markets. But booms and bubbles are followed by more harmful busts, crashes and panics.



What is this midpoint around which things cycle? As I said, it's astride a secular trend. For example, an economy's gross domestic may exhibit a secular annual growth rate of, say, 2% for growth will be faster in some years and slower—even negative—in The performance in individual years is usually part of a cycle underlying secular trend.

Importantly, the secular growth rate may be subject to a longer-term, more gradual one. You have to step back further to example, societies tend to follow long-term patterns of rise and the Roman Empire, for example—and the short term we talk about ups and downs around the long-term trend (see [pages 48–51](#)).

The same can be true for industries. But since long-term cycles occur decades and centuries, rather than quarters and years—and thus they can entail time frames that exceed any observer's lifetime—they can be detected in real time and hard to factor into a decision process.

Here's what I wrote on the subject in my memo "The Long Cycle" (2009):

There's an old story about a group of blind men walking down the road in India who come upon an elephant. Each one touches a different part of the elephant—the trunk, the leg, the tail or the tusk—and comes up with a different explanation of what he'd encountered based on the small part to which he was exposed. We are those blind men. Even if we have a good understanding of the events we encounter, we don't easily gain the overall view needed to put them together. To the time we see the whole in action, our knowledge is limited to the parts we've touched.

One of the most important lessons from the study and remember the events of the past and (b) be conscious of the cyclical nature of things. Up close, the blind man may mistake the elephant's leg for a tree—and the shortsighted investor may think an uptrend (or a downtrend) will go on forever. But if we step back and view the long sweep of history, we should be able to begin to understand the long-term cycle also repeats and understand where we stand in it.



This is a good time for me to make one of the most important points about the nature of cycles. Most people think of cycles in terms of individual events above and recognize them as a series of events. And most people

that these events regularly follow each other in a usual sequence: are followed by downswings, and then eventually by new upswings.

But to have a full understanding of cycles, that's not enough. the life of a cycle shouldn't be viewed merely as each being next, but—much more importantly—as each *causing* the next. For example:

- As the phenomenon swings toward an extreme, this movement builds up energy, which it stores. Eventually its increased weight makes it difficult for the swing to continue further from the midpoint, and it reaches a maximum beyond which it can no longer proceed.
- Eventually it stops moving in that direction. And once it does, then pulls it back in the direction of the central tendency with the energy it has amassed powering the swingback.
- And as the phenomenon in question moves from the extreme toward the midpoint, the swing imparts momentum to it that causes it to overshoot the midpoint and keep moving toward the opposite extreme.

In this way, a cycle in the economic or investment world consists of a series of events that give rise to their successors. The process described by the three bullet points above sounds like a physical one, governed by forces such as gravity and momentum. But as I mentioned above and we'll see in the next chapter, the most important deviations from the general trend—and the variation in the timing, speed and extent of those deviations—are largely produced by fluctuations in psychology.

If you consider the human psyche—rather than physical attributes—as the source of much of the energy or momentum, these three points do a good job of also explaining the swings and oscillations that investors have been challenged to deal with. In the chapters that follow, some of the important content will consist of descriptions of ways in which the different types of cycle generate their successors.

Owing to this view of cycles as progressions of causative events, this book contains several step-by-step accounts of progressions that took place in the past. The goal with each progression will be to illustrate what caused each event in the progression, what it meant in the progression, and how it contributed to the events that followed. The recounting of progressions may feel repetitive, and some of the ones covered actually will be touched upon more than once (although with reference to different aspects). But hopefully real-world examples will help readers achieve the goal of understanding cycles and how to position for them.

It's extremely important to note this causal relationship: that the talking about consist of series of events that cause the ones that it's equally significant to note that while cycles occur in a variety to these serial events, cyclical developments in one area also influence in others. Thus the economic cycle influences the profit cycle. Corporate announcements determined by the profit cycle influence investor attitudes. Investor attitudes influence markets. And developments in markets influence the cycle in the availability of credit. . . . which influences economic companies and markets.

Cyclical events are influenced by both endogenous developments (including the cyclical events that precede them) as well as exogenous developments (events occurring in other areas). Many of the latter from all—are parts of other cycles. Understanding these causative interactions isn't easy, but it holds much of the key to understanding and coping with the investment environment.

It must be understood that while I will describe cycles as sequential and discrete, this is not entirely realistic. I will provide a smooth description of each type of cycle in isolation. I will give the impression that each cycle has an independent life of its own. I may give the impression that the swing in a given direction of one type of cycle occurs before the start of a corresponding or resultant swing in another type—i.e., that they operate sequentially and independently. In other words, I will attempt to discuss each type of cycle in isolation In reality, they don't operate in isolation.

My description might suggest that the different cycles are independent of each other and self-contained. It may seem that something happens in cycle A, which affects cycle B, which affects cycle C, which might feed all back to influence cycle A. This may give the impression that cycle A holds after it has influenced cycle B, and while cycle B's influence on cycle C takes place. But that's not the way it is.

The interrelationships among the various cycles are nowhere near what my descriptions will be. The various cycles operate on their own, but they also continuously affect each other. I try to tease out the various cycles from the mind and treat them separately, and that is the way this book will be organized. But the well-behaved, isolated cycles I'll describe are analytical concepts. In life they're really a jumble of interrelated cycles that can't be separated entirely. A affects B (and C), and B affects C, and they all influence D, which influences all of them. They're all connected with each other, but we must think about them in an orderly fashion to understand cycles and their effects.

Finally, perhaps under the heading of “miscellany,” I want to say more things about the nature of cycles that are essential for a better understanding (starting here with a few observations from my November 2001 memo, “You Can’t Predict. You Can Prepare.”):

- *Cycles are inevitable.* Every once in a while, an up- or down- cycle occurs for a long time and/or to a great extreme, and people don’t know when it’s different.” They cite the changes in geopolitics, institutions, technology or behavior that have rendered the “old rules” obsolete. They make investment decisions that extrapolate the recent trend. But then it usually turns out that the old rules do still apply, and the cycle continues. In the end, trees don’t grow to the sky, and few things go to the moon. Most phenomena turn out to be cyclical.
- *Cycles’ clout is heightened by the inability of investors to learn from the past.* As John Kenneth Galbraith says, “extreme brevity of the market’s memory” keeps market participants from recognizing the recurring nature of these patterns, and thus their inevitability:
 When the same or closely similar circumstances occur again, sometimes in only a few years, they are hailed by a new, often youthful, and always supremely self-confident generation as a brilliantly innovative discovery in the financial and larger economic world. There can be few fields of human endeavor in which history counts for so little as in the world of finance. The experience, to the extent that it is part of memory at all, is dismissed as the primitive refuge of those who do not have the insight to appreciate the incredible wonders of the present.
Short History of Financial Euphoria, 1990)
- *Cycles are self-correcting,* and their reversal is not necessarily caused by exogenous events. The reason they reverse (rather than continue forever) is that trends create the reasons for their own reversal. Like to say success carries within itself the seeds of failure, or the seeds of success.
- *Seen through the lens of human perception, cycles are often viewed as less symmetrical than they are.* Negative price fluctuations are called “volatility,” while positive price fluctuations are called “profit.” Collapsing markets are called “selling panics,” while surges receive more benign descriptions (but I think they may best be seen as “panics”; see tech stocks in 1999, for example). Commentators call “investor capitulation” at the bottom of market cycles, “capitulation at the top,” when previously prudent investors throw in the towel and buy.

Although this may be underestimated and overlooked, in my experience financial cycles generally are largely symmetrical. Every cycle moves in an “other side,” meaning every upswing is invariably followed by—perhaps better said, leads to—a downswing, and vice versa.

“Boom/bust”—that’s a phenomenon that’s widely talked about and generally understood, and it’s a good illustration of cycle symmetry. Most people understand that busts follow booms. Somewhat fewer grasp the point that the busts are caused by the booms. From the latter, it makes sense that booms usually won’t be followed by modest, gradual and painful adjustments and (b) on the other hand, we’re unlikely to have a bust if we haven’t had a boom.

It must be noted, however, that this symmetry only applies in one direction, not necessarily to the extent, timing or pace of movement (the point that Nick Train makes—you’ll meet him in the next chapter.) An upward movement may be followed by a downward movement of greater or lesser magnitude. The downward turn may commence just after the apex is reached, or things may stay at a high for a long time before they come to correct. And, perhaps most importantly, it can take years for a boom to grow to its full extent. But the bust that follows may seem like a freight train; as my long-time partner Sheldon Stone says, “The air coming out of the balloon goes much faster than it went in.”

Let’s return to what Mark Twain is supposed to have said: “History repeats itself, but it does rhyme.” Grasping this concept is absolutely essential to an understanding of cycles. What Twain must have meant by this is that if he indeed was responsible for it—is that where the details of one event to another in a given category of history (say, the ascension and fall of demagogues), the underlying themes and mechanisms are consistent.

This is true of cycles in finance, and absolutely true of financial history. As you’ll see later, the Global Financial Crisis of 2007–08 occurred largely because of the issuance of a huge number of unsound sub-prime mortgages, and that took place in turn because of an excess of optimism, a lack of risk aversion, and an overly generous capital market, which led to the behavior surrounding sub-prime mortgages. Thus the narrow-minded literalist might say, “I’ll definitely turn cautious the next time mortgage financing is made readily available to unqualified home buyers.” But that aspect of the Crisis need never recur for the lessons of the Global Financial Crisis are valuable. Rather, the themes that provide warning signals in financial history are the general ones: that excessive optimism is a dangerous thing; that risk aversion is an essential ingredient for the market to be safe; that overly generous capital markets ultimately lead to unwise financing, and that there is danger for participants.

In short, the details are unimportant and can be irrelevant. But they are essential, and they absolutely do tend to recur. Understanding that tendency—and being able to spot the recurrences—is one of the most important elements in dealing with cycles.

Finally, I want to bring in the definition of insanity that Alvin Karpis is credited with: “doing the same thing over and over and expecting different results.” When people invest in things after they’ve been carried aloft by “everyone knows” they’re both flawless and underpriced—thinking they can get high returns without risk of loss—that’s insanity. Such beliefs have been defrocked in the aftermath of every bubble. But many people—either because they think that bubbles tend to be followed by crashes, or blinded by their eagerness to get rich quickly—buy into the next one nevertheless.

Securities and markets that have benefitted from fabulous appreciation are much more likely to succumb to a cyclical correction than they are to continue to appreciate *ad infinitum*. Try telling that to the eager investor who says “it’s different this time.”



The length of this chapter and the large variety of topics covered are indicative of the multi-faceted and challenging nature of cycles. For this reason, cycles have to be understood both analytically and intuitively. In addition to the many other aspects of investing, those who possess the latter ability will go the furthest. Can an intuitive understanding be taught? Yes, to some degree, but most fully to those who start off with a natural sense of insight. In short, some people just tend to “get it” (whatever that means) and some don’t.

Courses in accounting, finance and security analysis will equip the student with the technical knowledge that is necessary for success but, in themselves, those courses are far from sufficient. The main element missing from them is an understanding of cyclical phenomena and how they develop as discussed in this book. Some cues will be found in the newly established fields of behavioral economics and behavioral finance, and I commend them to your attention. Psychology is an essential component in understanding the human factors that matter so much to investors.

The greatest lessons regarding cycles are learned through experience, as in the adage “experience is what you get when you didn’t get what you wanted.” I know so much more about this today than I did when I was a young security analyst at First National City Bank 48 years ago.

However, since we usually see only one major cycle per decade, an investor who’s going to rely solely on the amassing of experience for his progress must better have a lot of patience. I hope what you read here will add to your understanding and speed your education.

The ancient Greek historian Thucydides stated in *History of the Peloponnesian War* that he would be satisfied “if these words of judged useful by those who want to understand clearly the events happened in the past and which (human nature being what it is) time or other and in much the same ways, be repeated in the future.” A good description of my goal here as well.

III

THE REGULARITY OF CYCLES

This effort to explain life through the recognition of patterns—and thus to come up with winning formulas—is complicated, in part because we live in a world that is beset by randomness and in part because people don't behave the same from one instance to the next, even when they intend to. The realization that past events were largely affected by these things—and thus that future events aren't fully predictable—is unpleasant, as it makes life less subject to anticipation, rule-making and rendering safe.

In the fall of 2013, in response to something I'd written in *Important Thing*, I received an email from Nick Train of Lindsell London-based money management firm. Nick took issue with my use of word “cycle” to describe phenomena like those I'm discussing here. Nick and I had a healthy email colloquy regarding the issue and met for a spirited lunch.

By the time the main course arrived, it had become clear that what motivated Nick to write was his conviction that for something to be as cyclical, its timing and extent have to be regular and thus like a radio cycle or sine wave, for example, rises and falls in a regular, repeating pattern, with the same amplitude, frequency and ending point every time.

Dictionary.com defines a cycle in physics as “a complete alternation in which a phenomenon attains a maximum and minimum value, returning to its final value equal to the original one,” and a cycle in mathematics as “a permutation of a set of elements that leaves the original cycle unchanged.” In other words, these scientific and mathematical cycles follow patterns so regular that they end up back where they started. That happens because the timing and path of the fluctuations is the same. Score one for Nick.

But economies, companies and markets—and certainly investors' psyches and behavior—are not regular in this way. I asserted at our

Nick eventually agreed, that things can be cyclical without exhibiting degree of consistency. It's all a matter of your definition of Here's part of what I wrote to Nick to follow up:

What I claim is that, usually, things rise and fall. Most natural things have a birth/death cycle, and investor psychology has a very pronounced cycle of rising optimism (and price appreciation) followed by rising pessimism (and price declines). You may think that's simplistic and unhelpful. But one of the main points is that when something rises, investors have a tendency to think it'll never fall (and vice versa). Betting against those tendencies can be very profitable.

Little in the world—and certainly not in the investment world—is regular enough in time to profit from applying a mechanistic pattern. But that doesn't mean you can't take advantage of up-and-down cycles.

I don't think fluctuations have to end up back where they started to be called a cycle. Many cycles end up higher than they started—i.e., they are cycles around an underlying secular uptrend—but that doesn't mean they're not cyclical, or that it's not desirable to ride the up-cycle and avoid the down-cycle, as opposed to staying on the sidelines throughout.

The *Cambridge Dictionary's* definition of a cycle—for use in the non-technical world—is “a group of events that happen in a regular, one following the other, and are often repeated.” I'm happy with that one; it reflects the sense in which I think about cycles and oscillations in



While I don't agree with Nick Train's objection that the irregularity of the phenomena I'm discussing disqualifies them from being described as cyclical, there is a great deal to be understood about their irregularity and what we can learn from it.

The most important thing to note here is that, as I said in the previous post, the things I call cycles do not stem completely—or sometimes at all—from the operation of mechanical, scientific or physical processes. They would be much more dependable and predictable if they did—but much less potentially profitable. (This is because the greatest profits come from seeing things before others do, and if cycles were totally dependable and predictable, there would be no such thing as superiority in seeing them.) Some cycles are based on an underlying principle (and sometimes not), but much variation is attributable to the role of humans in creating cycles. The involvement of

process enables their emotion- and psychology-induced tendencies to influence cyclical phenomena. Chance or randomness also plays a big role in some cycles, and human behavior contributes to their existence, too. They are a big part of the reason these cycles exist, but also—along with randomness—for their inconsistency and thus their undependability.



We humans have to live in the real world. As described earlier, we look for patterns and rules that will permit us to live more easily and profitably. Perhaps it started with early man's experience with daily and annual cycles. He may have learned the hard way that it was unsafe to visit a certain place at the time of day when mother lions went there with their cubs. And he learned through experimentation that certain crops did better when planted in the spring than in the fall. The more absolute the rules, the better they are to be. It now seems to be ingrained in the human brain to look for patterns.

But this effort to explain life through the recognition of patterns—to come up with winning formulas—is complicated, in large part, because we live in a world that is beset by randomness and in which people experience the same from one instance to the next, even when they intend to. The realization that past events were largely affected by these things—and that future events aren't fully predictable—is unpleasant, as it makes life more subject to anticipation, rule-making and rendering safe. Thus people look for explanations that would make events understandable . . . beyond that which is appropriate. This is as true in investing as in other aspects of life.

I found some interesting statements on this subject in *The Drunkard's Walk*, a 2008 book about randomness by Leonard Mlodinow, a faculty member at Caltech. Here's the first, from his book's prologue:

To swim against the current of human intuition is a challenge. The human mind is built to identify for each event a definite cause, and can therefore have a hard time accepting the influence of unrelated or random factors. And so the first step is to recognize success or failure sometimes arises neither from great skill nor from great incompetence but from, as the economist Armen Alchian would say, "fortuitous circumstances." Random processes are fundamental in nature and are ubiquitous in our everyday lives, yet most people do not understand them or think much about them.

In a chapter about the unpredictability and capriciousness of the movie industry, Mlodinow describes producer William Goldman's view

the subject:

Goldman didn't deny that there are reasons for a film's box office performance. But he did say that those reasons are so complex that the path from green light to opening weekend is so vulnerable to unforeseeable and uncontrollable influences that educated guesses about an unmade film's potential aren't much better than the flips of a coin.

Mlodinow goes on to discuss how random elements apply to a baseball:

The result of any particular at bat (that is, an opportunity for a hit) depends primarily on the player's ability, of course. But it also depends on the interplay of many other factors: his health; the weather; the sun, or the stadium lights; the quality of the pitches; the game situation; whether he correctly guesses how the pitcher will throw; whether his hand-eye coordination works just perfectly; whether he takes his swing; whether that's the first time he's met at the bar; whether he's too late or the chili-cheese dog with garlic fries he had for breakfast has soured his stomach. If not for all the unpredictable factors, he would either hit a home run on every at bat or fail to do so on every at bat].

We know that a variety of factors influence outcomes in all areas of life, and many of them are random or unpredictable. That certainly includes developments in economics and investing. Even if one's income is determined by one's individual's propensity to consume, which may be affected by the weather, which may be affected by which country wins the World Cup (and that, in turn, by how a particular team performs off a defender's shin). A company may issue a favorable earnings report, but whether its stock rises or falls as a result will be influenced by what competitors do, whether the central bank chooses that week for an interest rate increase, and whether the earnings announcement comes in that week in the market. Given this degree of variability, the cycles of boom and bust with certainly aren't regular and can't be reduced to reliable decision rules.

I can give you an example from the world of high yield bonds: I've found it quite annoying. At one point in my experience, a view arose that bonds tend to default around the second anniversary of their issuance. That would be a very helpful bit of knowledge: to avoid default, one would have to do is sell all bonds as they approach that anniversary and buy back the ones that have survived it. (Of course, this rule ignores

of what price sellers would receive for bonds that are nearing date—since everyone knows it poses risk—and how much they’d have to buy back the ones that have cleared it.)

Perhaps a cluster of second anniversary defaults occurred and that notion became popular. But coincidence is very different from causation. Is that phenomenon dependable? What were the reasons for it? Will it repeat? Should you bet on it? In particular, the history of defaults probably covered only twenty years or so at that time, making it difficult to know whether the experience and sample size were sufficient to justify that observation. Rather than intellectual rigor, I prefer to think the two-thirds rule was based more on people’s thirst for simple, helpful rules, and an excessive tendency to extrapolate without any real foundation.

I think it would be better to recognize that bonds default in a wide variety of influences—like those that contribute to the success of a hitter in baseball—and that most defaults have absolutely nothing to do with the number of years that have elapsed since the bonds were issued. Invert Mark Twain’s purported remark, history may rhyme, but it rarely repeats exactly.



I’m firmly convinced that markets will continue to rise and fall, and I know (a) why and (b) what makes these movements more or less immune to prediction. But I’m sure I’ll never know when they’re going to turn up or down, how long they’ll go after they do, how fast they’ll move, when they’ll reach the midpoint, or how far they’ll continue on the opposite side. It’s a great deal to admit uncertainty about.

I have found, however, that the little I do know about cycle timing gives me a great advantage relative to the majority of investors, who know even less about cycles and pay less heed to them and their implications for appropriate action. The advantage I’m talking about is probably all that I can achieve, but it’s enough for me. It’s been the source of satisfaction that my Oaktree colleagues and I have enjoyed over the last 22 years. A lot of what I want to pass on in this book.

IV

THE ECONOMIC CYCLE

The output of an economy is the product of hours worked per hour; thus the long-term growth of an economy is determined primarily by fundamental factors like birth rate and the rate of productivity (but also by other changes in society and environment). These factors usually change relatively little from year to year, and only gradually from decade to decade. Thus the average rate of growth is rather steady over long periods of time.

Given the relative stability of underlying secular growth, one might be tempted to expect that the performance of economies would be consistent from year to year. However, a number of factors account for the variability, causing economic growth—even as it follows the underlying trendline on average—to also exhibit annual variability.

The economic cycle (also known—mostly in the past—as “the business cycle”) provides much of the foundation for cyclical events in the world and the markets. The more the economy rises, the more likely it is that companies will expand their profits and stock markets will rise. I’ll briefly review here on the factors that influence economic cycles. But I want to make the confession I volunteer whenever I discuss it as a proud proclamation?): I’m no economist.

I took courses in economics as both an undergraduate and a graduate student. I think about economics. I deal with economics as a professional investor. And I consider myself to be largely an “economic realist.” Most of my most decisions for logical reasons based on the relationship between value, risk and potential return. But my thinking about economics is based largely on common sense and experience, and I’m sure I’ll write things that with which many economists will disagree. (Of course, they also disagree with each other. The workings of economics are quite unclear and impractical, and thus it’s for good reason that it’s called “the dismal science.”)

The main measure of an economy’s output is GDP, or gross domestic product, the total value of all goods and services produced for final

economy. It can roughly be viewed as the result of multiplying hours people spend working by the value of the output produced (Earlier in my career it was called gross national product, but that's gone out of style. The distinction between the two lies in the treatment of foreign manufacturers operating in a given country: GDP includes that country's output, while GNP does not.)

The main questions most people (and certainly most investors) care about with regard to the economy are whether we'll have growth or not in a given year, and what the rate of change will be. Both of these are part of what I call the short-term economic cycle. (I'll introduce other considerations shortly.)

When we think about U.S. GDP growth in a given year, we usually start with an assumption in the range of 2% to 3% or so and then adjust for the specific circumstances. But the starting point for each year's growth is invariably positive. For example, as last year began there was a lot of discussion about the rate at which GDP would grow. The optimists thought it'd be nearly 3%, and the pessimists thought it might not reach that. But about everyone thought it would be positive. The official definition of a recession is two consecutive quarters of negative growth, and very few people thought GDP growth would fall into negative territory—last year's growth was positive thereafter.

Long-Term Economic Trends

Many investors are concerned about year-to-year economic growth: high or low, positive or negative. The developments they're asking about are short-term considerations. They're important, but they're not everything. In the long run their importance fades and long-term considerations become more relevant.

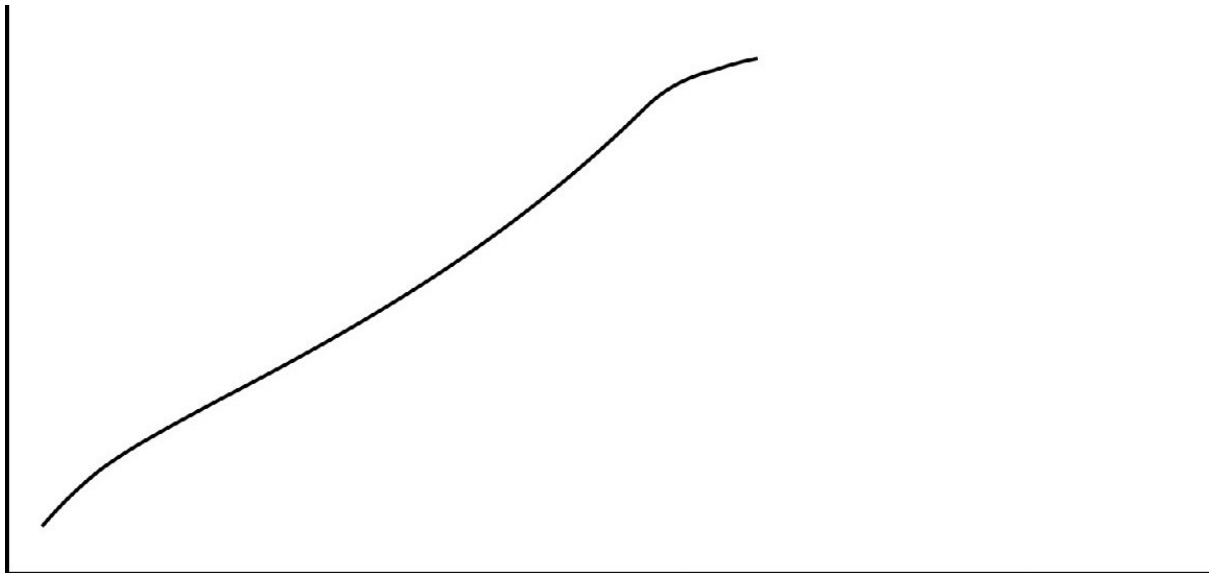
As I mentioned early on, most of the cycles that attract investors consist of oscillations around a secular trend or central tendency. The oscillations matter a great deal to companies and markets in the short run, but changes in regard to the underlying trendline itself will prove to have greater overall significance. The *oscillations around the trend* will be important in the long run (admittedly after causing much elation or distress in the short years), but *changes in the underlying trend* will make the biggest difference in our long-term experience.

In January 2009, I wrote a memo entitled "The Long View" on this subject. I'm going to quote extensively from it here.

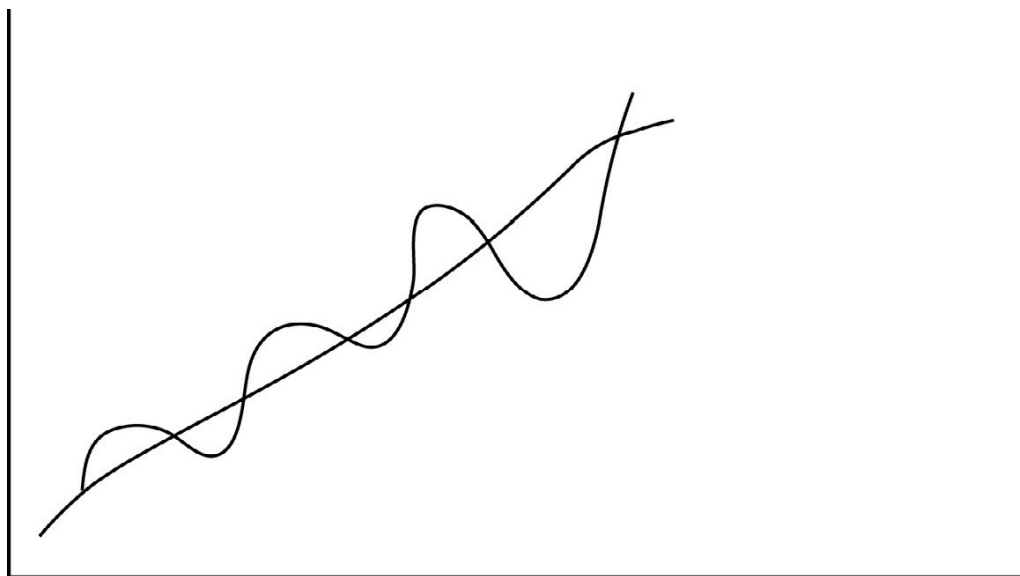
First, I described a number of “salutary secular trends” that the markets had been riding over the preceding decades. I’ll list them below, omit the descriptions that accompanied them in the memo:

- macro environment
- corporate growth
- the borrowing mentality
- popularization of investing
- investor psychology

The developments enumerated above constituted a strong tailwind behind the economy and the markets over the last several decades, and they produced a long-term secular uptrend.



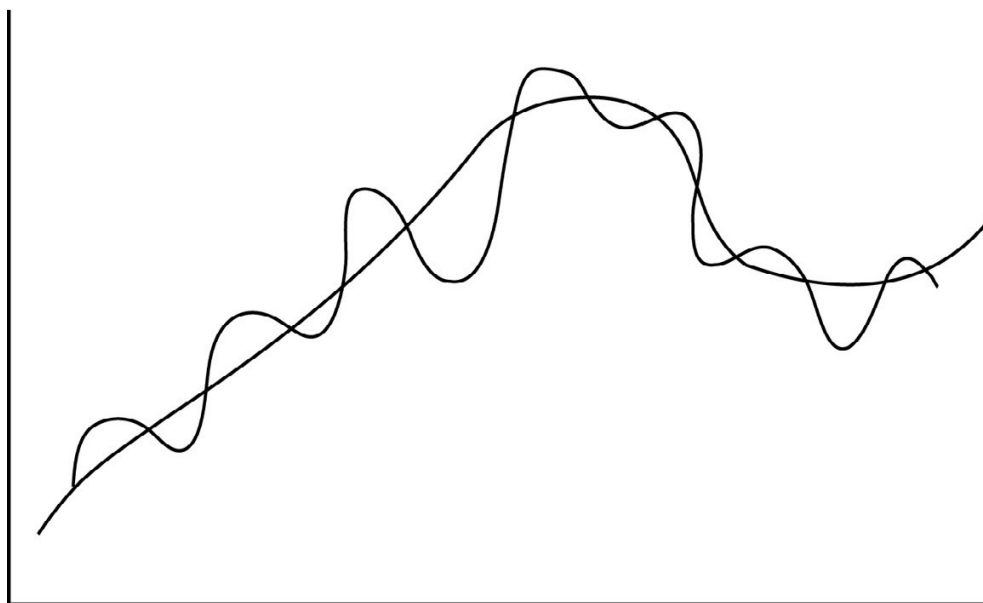
Yet, despite the underlying uptrend, there’s been no straight line. The economy and markets were punctuated every few years by cyclical bouts of short-term fluctuation. Cycles around the trendline made for frequent ups and downs. Most were relatively small and brief, but in the 1970s, economic stagnation set in, inflation reached 16%, the average stock lost almost half its value in two years, and *Business Week* magazine ran a cover story trumpeting “The Death of Equities” (August 13, 1979). No, my forty years in the market haven’t been all wine and roses.



From time to time we saw better economies and worse—slowdown and prosperity, recession and recovery. Markets, too, rose and fell. These fluctuations were attributable to normal economic cycles and to exogenous developments (such as the oil embargo in 1973 and the emerging market crisis in 1998). The Standard & Poor's 500 had a few down years in the period from 1975 to 1999, but none in which it lost more than 7.5%. On the upside, however, 16 of those 25 years showed returns above 15%, and seven times the annual gain exceeded 30%.

Despite the ups and downs, investors profited overall, and investing became a national pursuit, and Warren Buffett, one of America's richest men, got that way by buying common stocks and whole companies. A serious general uptrend was underway, reaching its zenith in 2007.

Until mid-2007, my 39 years of experience as a money manager had been limited to part of the long-term story. Perhaps what looked like an underlying long-term uptrend should have been viewed instead as the positive part of a long-term cycle incorporating downturns as ups. Only when you step back and look at the big picture can you gauge its proportions.



The main thing I want to discuss here is my realization that the cycles in the long-term trend, not just short-term cycles around it, we've been living through the positive phase of a big one.

Before I progress to a discussion of the short-term economic cycle, most investors concern themselves with, I'm going to spend some more time on the long term: the factors that shape it and the current outlook. That I'll turn to the matter of the short-term economic cycle.

I have mentioned that one of the main determinants of each year's economic output is the number of hours worked. In turn, the fundamental factor underlying increases in hours worked is population growth. Growth in the population means there are more people each year to make and sell products (and also more people to buy and use them, encouraging production). More production equals more GDP. If the population is growing, hours worked tend to grow, and so does GDP. Births are one of the main reasons for the usual presumption that economic growth will be positive. On the other hand, if the population is shrinking, positive GDP growth faces a significant headwind.

Population growth doesn't vary much from year to year. The number of people of child-bearing age doesn't change much in the short term, nor does their tendency to have children. These things do change over decades or longer, however, so they cause changes in long-term population.

What are the kinds of things that can alter a nation's birth rate (the average number of children each couple has)?

- rules like China's longstanding but recently revised one-child

- wars (like World War II, which depressed the birth rate but gave the Baby Boom when it ended),
- economic conditions, which, among other things, alter people's about whether they can afford to have children, and
- social mores, like the recent tendency of young Americans to family formation

Changes in birth rates generally take place over long periods of time when they do, they require years to affect GDP growth. Take China's child policy, for example. You might say the shift was sudden: one 2015 the policy was in full force, and the next day its phase-out announced. True as far as it goes. But while people who already certainly might have gotten busy producing another on that new day, take roughly twenty years for that second child to become a able to contribute to China's economic output. Thus the bottom year-to-year changes in GDP are unlikely to be attributed significant changes in the birth rate.

The other principal element in the GDP equation—the value of the produced in each hour of labor—is determined by “productivity.” Changes in productivity are fundamental determinants of changes in the long-run growth of GDP. Whatever the rate of population growth might be, GDP will grow faster if productivity is rising or slower if it's falling. And looking at the derivatives, the rate of growth in GDP will accelerate if the rate of productivity is rising and decelerate if it's falling. It's all just math.

Changes in productivity, like changes in birth rate, take place in degrees and gradually, and they require long periods to take effect, primarily from advances in the productive process. The first big gains occurred during the Industrial Revolution of roughly 1760 to 1830, when human labor was replaced by machines driven by steam and water, when large factories replaced the work that was done less efficiently in shops and at home. The second major gains occurred in the late 19th and early 20th centuries, when electricity and automobiles replaced older and less-efficient forms of power and transportation. The third major change in the latter half of the 20th century, when computers and other forms of automated control began to take the place of humans in guiding machinery. And, of course, the fourth wave is underway now, during the Information Age, as massive advances in information acquisition, storage, and application—and such activities as metadata and artificial intelligence—are permitting tasks to be accomplished that weren't dreamed of in the past.

Remember, each of these changes took place gradually, over decades. Each made a massive difference in GDP, but even with these, there weren't sharp accelerations and decelerations from year to year. The rate of gain

productivity tends to remain relatively steady for years, and certainly short-term cycles of economic recession and recovery generally aren't attributed to changes in it.

It's clear that trends in hours worked and in output per hour combine to determine long-term trends in national output. But what factors produce changes in those two? Here's a partial, indicative list:

- **Demographic movements**—The migration of millions of Chinese farmers to cities is an example of what I'm talking about here. By increasing the availability of workers, this migration fueled China's expansion as a site of low-cost manufacturing, and it is contributing to the expansion of China's consumer class. Another example is immigration from Latin America to the United States. America, like other nations, is experiencing a declining birth rate. But ongoing immigration from south of the border—some of it illegal—takes the place of expanding the U.S.'s supply of productive labor and rate of consumption.
- **Determinants of inputs**—The number of hours worked can be determined from the number of people working, and certainly from the factors that are interested in working.
 - “Workforce participation” reflects the percentage of people in the working age who are either employed or looking for work.
 - The unemployment rate (the percentage of people participating in the workforce who don't have a job) rises and falls in response to changes in consumer and business spending (and thus to changes in demand for goods, and in the need for workers to produce those goods).
 - The number of hours worked by each person with a job varies with economic conditions—businesses shorten work weeks when demand for goods is low, and they authorize overtime work when demand is high (until demand is strong enough to justify hiring or another shift).
- **Aspiration**—The profit motive and the desire to live better are among the forces that drive workers (and thus societies) to work harder and produce more. It might be tempting to think of these things as purely economic, but they aren't. For example, the profit motive was pretty much absent from the economic system under the Soviets, and the willingness to work more is constrained in other economies (to wit, I've watched workers clock out at European banks—not to prove that they had worked 5:00 as in the U.S., but rather that they had left by 5:00 and not exceeded the 35-hour work week).

- **Education** —The deterioration of public education in the to have a negative effect on workers' ability to contribute economy in the future, as well as their ability to generate incomes with which to consume. These negative trends are likely to work counter to the positive effects from the influx of immigrants.
- **Technology** —Innovation causes new businesses to come into but brings about the demise of old ones. It both creates and eliminates them. In short, it provides a hyper-example of the nature of economic evolution: it creates winners *and* losers. New technologies overtake human effort as well as old technologies are not in any way "safe," as they, too, can be displaced today's terminology—disrupted. Technology epitomizes the pattern of rise and fall, life and death . . . and rebirth.
- **Automation** —The ability to replace human labor with a particularly interesting factor. On the one hand, automation can be viewed as additive to the economic cycle, since it increases or the amount of output that is generated per hour of mechanization of agriculture, for instance, allowed many fewer farmers to produce much more food at much lower cost than ever before. On the other hand, automation decreases the hours of labor applied to production. Today we see factories run by just a few workers that years ago might have had a hundred. Thus the net effect of automation on GDP might be neutral or positive but, since it has the effect of eliminating jobs, automation might have the effect of reducing employment, and thus incomes, and thus consumption.
- **Globalization** —The integration of nations into a world economy adds to total world economic output, in part because of benefits from specialization, or it may not, leaving it a zero-sum (or even negative-sum) exercise. But clearly, globalization can have differential effects on individual nations' economies (and create winners and losers within a nation). The massive increase in the number of factory workers described above certainly accelerated China's economic growth over the last thirty years by permitting it to become a leading exporter of the world. However, that same trend caused developed nations to export a lot of goods from China that they otherwise might have produced themselves, thus curtailing their own GDP. The few million manufacturing jobs estimated to have been lost to China since 2000 certainly made U.S. economic growth lower than it otherwise would have been, although one would need to take into account the effect of importing low-priced goods from China to estimate the total impact on the U.S. economy.

The U.S. was blessed with an intact infrastructure coming out of II, and it benefitted greatly from the Baby Boom in post-war birth created a massive upsurge in economic growth. American produced the best in the world, and American corporations were wildly competitive in the yet-to-globalize world, American workers could remain the best-paid from competition from goods produced more cheaply elsewhere. Improving management techniques and rapid increases in productivity were further contributors. Thus secular economic growth in the U.S. was rapid, contributing to demand for consumption and thus creating a virtuous cycle from which many benefitted. . . . but it was not one that could continue unchanged.

More recently, economic growth appears to have slowed in the U.S. (as well as elsewhere). Is this a short-term cyclical change relative to the underlying long-term trend, or a change in the long-term trend itself? It will take many years before we know definitively. But there has arisen a school of thought blaming it on “secular stagnation”—that is, a fundamental change in the long-term trend.

Gains in population and productivity have declined in the U.S., as they have in other developed nations. Taken together, these two things suggest that GDP will grow more slowly in the U.S. in the coming years than in the years following World War II. It is postulated that the major advance in productivity of the recent past will not be replicated in the future. The availability of much-cheaper labor elsewhere makes it unlikely that the U.S. will be able to compete on price to produce the manufacturing goods it requires; this has obviously negative implications in the U.S. for the well-being of among less-skilled and less-educated Americans, income inequality, and standards of living relative to people in other countries. These issues, of course, played an obvious part in the 2016 presidential election.

Changes in population growth and productivity growth can require time to take effect, but clearly they can affect countries’ economic growth. In the 20th century, the U.S. surpassed Europe as an economic power. Japan seemed to sprint forward in the 1970s and 1980s, threatening to overtake the world, until the late ’80s, when it fell back into negligible growth. The emerging markets—and especially China—were the site of rapid growth. Over the last few decades, and while their growth is slower at present, they may well outgrow the developed world in the next few decades. India has human resources that can make it a rapid-growth economy if it increases its efficiency and reduces corruption. And frontier nations like Nigeria and Bangladesh stand behind the emerging nations, waiting for their rapid growth.

Societies rise and fall, and they speed up and slow down in terms of economic growth relative to each other. This underlying trend in growth clearly follows a long-term cycle, although the short-run ups and downs around it are more discernible and thus more readily discussed.

Short-Term Economic Cycles

As I mentioned earlier, economic forecasters and the consumers of product are usually preoccupied by the rate of GDP growth in a year or two. In other words, they're concerned about the growth in the upward swing of the short-term economic cycle and its duration, as well as whether it'll go negative for two quarters in the downward swing, thus be termed a recession. These things represent short-term fluctuations around the long-term growth trend as illustrated a few pages back. The factors that produce the long-term trend change little from quarter to quarter, and year to year, why should short-term changes be of great concern? Why should they even occur? Why isn't there just growth at the rate—say 2%—every year?

These questions provide a good opportunity to introduce some of the themes of the book's protagonists: psychology, emotion and decision-making processes. Births and productivity often tend to be viewed as independent and almost mechanical variables. Births result from procreation, and the reasons for it and the rate at which it takes place are usually quite steady over time. Likewise, the rate of change in the level of productivity—in output per hour of labor—is viewed as being dictated largely by technological gains and their dissemination. In other words, although economies are made up of many parts, the level of economic growth isn't thought to be highly reflective of the ups and downs of people's lives.

But, in fact, it is. While the long-term trend sets the potential economic growth rate, the actual level of each year's GDP will vary relative to the trend, which the trend dictates. . . . largely because of the involvement of people.

Birth rate may determine the long-term trend in the number of hours worked, but other factors can introduce variation in the short run. The willingness to work is not constant. There may be times when conditions discourage people from seeking a place in the workforce, as previously mentioned, and also times when world events alter the level of consumption.

The most obvious example is the ability of world events to discourage economic activity. The sub-prime mortgage crisis and financial institution meltdown that reached its apex with the bankruptcy of Lehman Brothers in September 2008 discouraged consumers from buying, investors

from providing capital, and companies from building factories and expanding workforces. These cutbacks occurred even among people who hadn't lost jobs, suffered home foreclosure or seen declines in the value of their portfolios. These developments quickly affected the overall economy, the result was a serious recession lasting from December 2007 to June 2009.

If the number of workers working and the amount they earn were constant, we might expect the amount they spend on consumption to be similarly constant. But it isn't. Spending fluctuates more than employment and earnings because of variation in something called "the marginal propensity to consume": of every additional dollar earned, it determines the percentage that will go to consumption. Because this propensity is constant in the short run, consumption can vary independently of income.

Earners may choose to spend a higher percentage of their earnings on consumption because:

- the daily headlines are favorable;
- they believe election results presage a stronger economy, or lower taxes;
- consumer credit has become more readily available;
- asset appreciation has made them feel richer; or
- their team won the World Series.

The fourth of these factors—the so-called "wealth effect"—is particularly noteworthy. Asset owners (a) are probably unlikely to fund consumption by selling their stocks or homes and (b) should recognize that asset values can prove ephemeral and thus aren't a good reason to alter spending. Yet asset appreciation does tend to lead them to spend more. This phenomenon demonstrates the contribution of psychology to economic behavior, and its role in short-term economic variation.

It's particularly important in this vein to note the extent to which economic expectations can be self-fulfilling. If people (and companies) believe the future will be good, they'll spend more and invest more. If they believe the future will be good, and vice versa. It's my belief that most companies concluded that the Crisis of 2008 wouldn't be followed by a V-shaped recovery. It had been the rule in the last few recessions. Thus they declined to expand factories or workforces, and the resulting recovery was modest and slow in the U.S. (and even more anemic elsewhere).

Another reason for short-term variation concerns inventories. Businesses may overestimate the demand for their products in a given period and increase production such that it exceeds the amount they can sell. They may hold production constant but encounter surprisingly soft demand. In either case, more goods will be produced than sold. The excess will be

to inventories. That, in turn, is likely to cause production in subsequent periods to be adjusted downward until inventories are restored to levels. In this way, additions to and reductions in inventories often cause short-term ups and downs in economic output.

These are only a few examples of the factors that can cause the economy to vary in a given quarter or year from the growth rate and productivity gains that might suggest. They are the types of factors that are not “mechanical” or reliable in nature. They arise from human behavior, and thus they are uncertain and unpredictable.



This leads me to add a few words about economic forecasts. Many investors predicate their actions on forecasts that they make themselves or obtain from economists, banks or the media. And yet I doubt many such forecasts contain information that’s likely to add value and lead to investment success. (For a more extensive discussion of “what we don’t know,” see chapter 14, *Most Important Thing*.)

Here’s how I view the foundation for considering this issue:

- In investing, it’s easy to achieve performance that is equal to the average investor or a market benchmark.
- Since it’s easy to be average, real investment success must consist of outperforming other investors and the averages. Investment success is largely a relative concept, measured on the basis of relative performance.
- Simply being right about a coming event isn’t enough to ensure relative performance if everyone holds the same view and as everyone is equally right. Thus success doesn’t lie in being right rather in being more right than others.
- Similarly, one doesn’t have to be right in order to be successful; one can be wrong rather than others.
- Success doesn’t come from having a *correct* forecast, but from having a *superior* forecast. Can such forecasts be obtained?

Most economic forecasts consist of extrapolations of current levels and long-term trends. And since the economy usually doesn’t depart from those levels and trends, most extrapolation forecasts turn out to be accurate. But those extrapolation forecasts are likely to be commonly shared, reflected in the market prices for assets, and thus not generators of superior performance—even when they come true. Here’s how Nobel Prize–winning economist Milton Friedman put it:

All these people see the same data, read the same material, and spend their time trying to guess what each other is going to say. [Their forecasts] will always be moderately right—and almost never of much use.

The forecasts that are potentially valuable are those that correctly forecast a significant deviation from long-term trends and recent levels. If a forecaster makes a non-conforming, non-extrapolation prediction that turns out to be correct, the outcome is likely to come as a surprise to the other market participants; they scramble to adjust their holdings to reflect it, the result is a net gain for the few who correctly foresaw it. There's only one catch: major deviations from trend (a) occur infrequently and (b) are hard to correctly predict, most unconventional, non-extrapolation forecasts turn out to be incorrect, and anyone who invests on their basis is usually below average.

So these are the possibilities I see with regard to economic forecasting.

- Most economic forecasts are just extrapolations. Extrapolations are usually correct but not valuable.
- Unconventional forecasts of significant deviation from trend would be very valuable if they were correct, but usually they aren't. Forecasts of deviation from trend are incorrect and also not valuable.
- A few forecasts of significant deviation turn out to be correct and are valuable—leading their authors to be lionized for their acumen—but it's hard to know in advance which will be the few right ones. Since the overall batting average with regard to them is low, unconventional forecasts can't be valuable on balance. There are forecasters who are famous for a single dramatic correct call, but the majority of forecasts weren't worth following.

Taken together, these three conclusions on economic forecasting aren't very encouraging. Thus it's not for nothing that John Kenneth Galbraith said, "We have two classes of forecasters: those who don't know—and those who don't know they don't know."

Secular changes in long-term economic cycles are hard to predict. The correctness of forecasts of such changes is hard to assess. The forecasts of short-term economic cycles, too, are hard for any one person to consistently predict better than others. It's tempting to act on economic predictions, especially since the payoff for correct ones is theoretically high. But the difficulty of being able to do so correctly and consistently mustn't be underestimated.

Here is what I believe to be the bottom line on economic

- The output of an economy is the product of hours worked per hour; thus the long-term growth of an economy is determined by fundamental factors like birth rate and the rate of gain (but also by other changes in society and environment). These usually change relatively little from year to year, and only gradually from decade to decade. Thus the average rate of growth over long periods of time. Only in the longest of time frames does the secular growth rate of an economy significantly speed up. But it does.
- Given the relative stability of underlying secular growth, one is tempted to expect that the performance of economies would be consistent from year to year. However, a number of factors cause annual variability, causing economic growth—even as it follows the trendline on average—to also exhibit annual variability. These factors can perhaps be viewed as follows:
 - **Endogenous**—Annual economic performance can be influenced by annual variation in decisions made by economic units: for consumers to spend or save, for example, or for businesses to expand or contract, or to add to inventories (calling for increased production) or subtract from inventories (reducing production relative to what it might otherwise have been). Often these decisions are influenced by the state of mind of economic actors, such as consumers or the managers of businesses.
 - **Exogenous**—Annual performance can also be influenced by (a) man-made events that are not strictly economic, such as the occurrence of war; government decisions to change tax rates or to adjust trade barriers; or changes caused by cartels in the production of commodities, or (b) natural events that occur without the involvement of people, such as droughts, hurricanes and earthquakes.
- Long-term economic growth is steady for long periods of time, but subject to change pursuant to long-term cycles.
- Short-term economic growth follows the long-term trend on average, but it oscillates around that trendline from year to year.
- People try hard to predict annual variation as a source of economic investing profit. And on average they're close to the truth over a long time. But few people do it right consistently; few do it

than everyone else; and few correctly predict the major deviations trend.



I often find—miraculously—that just as I’m just about to conclude I’m writing, the perfect example pops up in real life or in Thus, as I was completing the first draft of this chapter on June was reported that a majority of voters in the United Kingdom leave the European Union.

This decision was generally unexpected: the British pound and London stock market had strengthened in the days leading up to the London bookmakers were giving odds that Brexit would be voted much for forecasting.

This decision may have significant ramifications—economic, social and political—for the U.K. and Europe, but also for the rest of the negative impact on the psyches of consumers, investors or business may cause near-term economic growth to slow. So may increases in barriers and reduced global efficiency.

Further, there’s a chance that this event (and follow-ons such as possible departure of Scotland and Northern Ireland from the United Kingdom) will alter long-term growth for the nations directly involved possibly for others. This event may be cited fifty years from now as changed the growth trajectory of major parts of the world economy, the whole, and as having contributed to a redirection of the

Certainly there’s a good chance the economic environment will in the years ahead from what it would have been if the Brexit the other way—that is, that Brexit will cause a turn in Britain’s economic cycle. We just can’t be sure how, to what extent, or what on effects on other economies will be.

V

GOVERNMENT INVOLVEMENT WITH THE ECONOMIC CYCLE



Extreme economic cyclicalities are considered undesirable. Too much strength can kindle inflation and take the economy so high that a recession becomes inevitable. Too much weakness, on the other hand, can cause companies' profits to fall and can cost people their jobs. Thus it is part of the job of central bankers and Treasury to manage cycles.

Since cycles produce ups and downs that can be excessive, the tools for dealing with them are counter-cyclical and applied with a counterforce of their own—ideally inverse to the economic cycle itself. However, like everything else involving cycles, managing them is far from easy. If it weren't, we wouldn't see the extremes we do.

In most of the world, capitalism and free markets are accepted as constituting the best system for allocating economic resources and encouraging economic output. Nations have tried other systems, such as socialism and communism, but in many cases they have either switched wholesale to or adopted aspects of free markets.

Despite the widespread acceptance of the free-market system, markets are rarely left entirely free. Government involvement takes many forms, from the enactment and enforcement of laws and regulations to direct participation in the economy through entities like the U.S.'s government agencies. Perhaps the most important form of government involvement, however, comes in the attempts of central banks and national governments to control and affect the ups and downs of economic cycles.

Central Banks

Over the centuries, significant power and responsibility has been vested in central banks such as the U.S. Federal Reserve Bank. Whereas in their primary roles may have been to issue currency and exchange it for gold or silver, today central banks are concerned primarily with managing economic cycles.

Early on, many central banks issued currency. As time passed and banks took responsibility for economic cycles, their primary concern usually shifted to inflation. In particular, the world went through periods of hyperinflation with inflation running in the hundreds of percent per year, as Germany during the post-World War I Weimar Republic. Thus central banks turned to managing inflation. The goal wasn't to eliminate inflation, but to control it. It is accepted as (a) having a variety of salutary aspects and (b) being unavoidable—but to control it.

The reasons for inflation are somewhat mysterious and, like many other economic processes described here, unreliable and sporadic. Sometimes a given set of circumstances will give rise to inflation, and at other times the same circumstances will lead to more or less inflation, or to none at all. In general, inflation is viewed as a result of a strong upward-sloping economic cycle.

- When the demand for goods increases relative to the supply, “demand-pull” inflation.
- When input costs to production such as labor and raw materials increase, there can be “cost-push” inflation.
- Finally, when the value of an importing country's currency declines relative to that of an exporting country, the cost of the imports can rise in the importing country.

The cost of goods can escalate for any of these reasons. That's why central banks are concerned with inflation. But, as I just said, sometimes these events can occur without an accompanying acceleration of inflation. And sometimes inflation can increase without these things being present. There is a large psychological component to inflation that influences all of this.

Since inflation results from economic strength, the efforts of central bankers to control it amount to trying to take some of the life out of the economy. They can include reducing the money supply, raising interest rates, and selling securities. When the private sector purchases securities from the central bank, money is taken out of circulation; this tends to reduce the demand for goods and thus discourages inflation. Central bankers who are strongly dedicated to keeping inflation under control are called “hawkish.” They tend to do the things listed above sooner and to a greater extent.

The problem, of course, is that actions of this kind are an either-or proposition. They can accomplish the goal of keeping inflation under control, but they can also restrain the growth of the economy, with effects that can be beneficial.

The issue is complicated by the fact that in the last few decades, central banks have been given a second responsibility. In addition to controlling inflation, they are expected to support employment, and, of course, employment does better when the economy is stronger. So central banks encourage this through stimulative actions such as increasing the money supply, decreasing interest rates, and injecting liquidity into the economy by buying securities—as in the recent program of “quantitative easing.” Central bankers who focus strongly on encouraging employment lean toward these actions and are called “doves.”

The bottom line is that most central bankers have two jobs: to control inflation, which requires restraining the growth of the economy, and to support employment, which calls for stimulating economic growth. In other words, their dual responsibilities are in opposition to each other, and their job requires a delicate balancing act.

We have discussed the fact that the economy is cyclical, growing at some times and weakly (or contracting) at others. An economic boom tends to encourage employment but can cause inflation to accelerate. Stagnation or contraction, on the other hand, discourages inflation but can lead to unemployment. So the job of the central banker is to counter-cyclically: that is, to limit the extent of cycles, slowing them down in times of prosperity in order to keep inflation under control, and speeding them up during slowdowns to support employment.

But just as the investor’s insight into cycles is limited, so is the central banker’s. His two tasks—stimulating the economy and controlling inflation—obviously can’t both be done at once. Is it time for stimulus now? Which ever one is chosen, how much of it? If interest rates are raised, as they have been since the Global Financial Crisis, in order to provide a brake on economic growth, how weak (also as it has been), can rates be raised without an increase in inflation without choking off the economy’s tepid recovery? These cycles are challenging for investors to understand and predict, they are even more so for central bankers to manage.

Governments

Governments have a greater variety of responsibilities than central banks, but only a small portion of which are related to the economic matters discussed here.

central banks, they also are charged with stimulating the economy appropriate, albeit not directly with controlling inflation. In their work with the economy, treasuries, too, are concerned with regulating the cycle: not fast and not too slow.

Governments' main tools for managing the economic cycle are fiscal, defined as being concerned primarily with taxing and spending. Thus, if governments want to stimulate their countries' economies, they can cut taxes or increase government spending and even distribute stimulus checks, making more money available for spending and investment. On the other hand, if they think economies are growing so fast as to be at risk of overheating, they can set the scene for a resulting slowdown—governments can increase taxes or cut spending, reducing demand in their economies and thereby slowing economic activity.

The ultimate topic under this heading concerns national deficits. In the distant past, most governments ran balanced budgets. In short, they were not able to spend more money than they brought in through taxes (or borrowing). But then the concept of national debt arose, and the ability to borrow was introduced, creating the potential for deficits: that is, for governments to spend more than they take in.

I seem to remember from my youth that there was active debate regarding the propriety of countries having national debt, but we no longer have much resistance on that subject. It is generally accepted that countries can borrow money, although questions do arise from time to time about how much is prudent. The answer generally seems to be “not too much more than now.”

The economic theory propounded by John Maynard Keynes in the 1930s dwelled heavily on the role of governments vis-à-vis cycles. Modern macroeconomics focuses on the role of aggregate demand in determining the level of GDP, in contrast with earlier approaches that emphasized the role of the supply of goods. Keynes said governments should manage the economic cycle by influencing demand. This, in turn, could be accomplished through the use of fiscal tools, including deficits.

Keynes urged governments to aid a weak economy by stimulating it by running deficits. When a government's outgo—its spending—exceeds its income—primarily from taxes—on balance it puts funds into the economy. This encourages buying and investing. Deficits are stimulative, and thus Keynes considered them helpful in dealing with a weak economy.

On the other hand, when economies are strong, Keynes said governments should run surpluses, spending less than they take in. This removes money from the economy, discouraging spending and investment. Surpluses are contractionary and thus an appropriate response to booms. However, the use of surpluses to cool a thriving economy is little seen these days.

to be a wet blanket when the party is going strong. And spending you bring in attracts fewer votes than do generous spending programs. surpluses have become as rare as buggy whips.



Extreme economic cyclicalities are considered undesirable. Too much strength can kindle inflation and take the economy so high that a recession is inevitable. Too much weakness, on the other hand, can cause companies' profits to fall and cost people their jobs.

Thus it is part of the job of central bankers and Treasury to manage cycles through the techniques described above. Since cyclical ups and downs can be excessive, the tools for dealing with them are counter-cyclical and applied with a cycle of their own—ideally inverting the economic cycle itself.

However, like everything else involving cycles—such as knowing when we are and what to do about them—managing them is far from easy. If we weren't, we wouldn't see the extremes we do.

VI

THE CYCLE IN PROFITS

The process that determines a company's profits is complex and multivariate. The economic cycle has a profound effect on some companies' sales but less on others. Largely because of different operating and financial leverage, a given percentage change in sales has a much greater impact on profits for some companies than others.

These days, as I said earlier, the normal growth rate for the economy is about 2–3% per year. Growth might come in at 1% or so in a recession or hit 4% or 5% in boom times (or during a recovery from a recession). The annualized growth rate might even turn negative by a couple of percentage points in tough times, and if it stays negative for two successive quarters, we are in a recession. So there are fluctuations, but they are moderate: annual U.S. GDP almost always falls between 5% and minus 2%, and even the extreme extremes are seen only once every decade at most.

Does that mean companies' profits also are stable from year to year? Far from it. Profits can gain much more than 5% in good times and decline more than 2% in bad. They, too, follow a cycle—one that is influenced by the economic cycle, but that rises and falls much more than the economy as a whole. So profits are more volatile than GDP. The question is what factors cause the profit cycle to perform differently from the economic cycle.

First, the ups and downs of the economy absolutely are very important in determining the rise and fall of corporate profits. More GDP means—more than anything else—more consumption, and thus stronger demand for goods and services. That, in turn, means greater volumes sold and higher selling prices. More work and higher wages, and thus still more consumption. All those things together mean increased revenues for businesses.

By definition, the collective sales of all businesses are one and the same as GDP, and they reflect the same rate of change. But that doesn't mean that all companies follow the same pattern.

Sales are responsive to the economic cycle in some industries, and in they aren't. And some respond a lot, while others respond just a

- Sales of industrial raw materials and components are directly responsive to the economic cycle. When business collectively increases its output—that is, when GDP expands—it takes more chemicals, metals, plastic, energy, wire and semiconductors to do so, and vice versa.
- On the other hand, everyday necessities like food, beverages and medicine aren't highly responsive to the economic cycle. People generally consume them regardless of what's going on in the (But demand isn't absolutely constant: people trade down in recessions—buying cheaper food and eating at home rather than in restaurants—and they trade up in times of prosperity. And, sadly, people who struggling financially may cut down even on their consumption of “necessities” when forced to choose between food, medications payments.)
- Demand for low-cost consumer items (like everyday clothing, newspapers and digital downloads) isn't very volatile, while demand for luxury goods and vacation trips may be.
- Purchases of big-ticket “durable goods”—things like cars and homes, individuals and trucks and factory equipment for businesses—are highly responsive to the economic cycle. First, the fact that they're durable means they last a long time, so replacement can be deferred in economic weakness. Second, because they cost a lot, they're affordable in bad times and easier to afford in good times. And third, businesses generally need more of them when business is good and less when it isn't. These things make the demand for durables highly responsive to the economic cycle.
- Demand for everyday services generally isn't volatile. If they're necessary (like transportation to work) and low-priced (like haircuts), demand won't be highly sensitive to changes in the economy. Services like these have a limited shelf life and can't be stored; they have to be purchased continually. But demand still can vary based on economic conditions: for example, a haircut can be made to last two weeks rather than three.

In addition, sales of some products respond to cycles other than the economic cycle. Because durables are expensive and can be paid for over their long life span, demand for them can go up and down (everything else being equal) in response to fluctuations in the credit cycle that make financing more and less readily available. And some things are influenced by non-cyclical developments:

example, the demand for new cell phones and laptops responds to technological reductions, new product introductions and improvements in technology.

For the most part, however, economic growth dominates the process through which sales are determined. Sales generally rise strongly when GDP growth is strong and less so (or they decline) when it isn't.



But the linkage between economic growth and profit growth is imperfect. This is because the movements of the economic cycle aren't the only thing that influences sales (as just shown), and also because a change in sales doesn't necessarily result in an equivalent change in profits. One of the main reasons for this latter phenomenon is the fact that most businesses are characterized by leverage of two types. These are elements that magnify the response of profits to a change in sales. The meaning of leverage is more immediately obvious from the British word for it: gearing.

First, businesses are subject to operating leverage. Profits equal revenues minus costs (or expenses). Revenues are the result of sales, and we know that sales fluctuate for a large number of reasons. So do costs, and revenues and costs fluctuate in different ways, particularly in response to changes in sales.

Most businesses have some costs that are fixed, some that are semi-fixed, and some that are variable. For example, take a taxicab company:

- It has its headquarters in an office building, but when ridership increases a bit, it doesn't need to add more office space. This is a fixed cost.
- It has a fleet of taxis. The current fleet can accommodate an increase in ridership, but if trips increase enough, it may need to add additional cabs. Thus the expense for cabs is semi-fixed.
- Its cabs are powered by gasoline. If an increase in business requires more cabs to go $x\%$ more miles, their consumption of gasoline is also increased by that same $x\%$. For the taxi company, then, the cost of gasoline is variable.

The sum of the above means that if the company's ridership (and thus its revenues) increases by 20%, its spending on office facilities won't increase, its spending on cabs probably won't increase initially, but may later, and its spending on gasoline will increase immediately and proportionately. Thus the taxicab company's total costs will increase with an increase in ridership, but usually less than its revenues. This will cause its profit margin to increase, meaning the increase in operating profits will be considerably greater than the increase in sales: that's operating leverage. In general, it's higher

companies for whom a larger percentage of costs are fixed and lower ones whose costs are more variable.

Operating leverage is great for companies when the economy does well and sales rise. But when the opposite happens, it's less good: profits can decline more than sales, and if conditions are bad enough, profits can turn into losses. Companies can, however, take actions to limit the effect on profits when sales decline. These can include laying off employees and closing stores. Such actions in an economy measures usually need time to take effect; (b) they sometimes involve increased expense in the short run, as in the case of severance pay; (c) they generally can limit the negative effect but not eliminate it; and (d) they don't always work as well as projected.

The second form of leverage affecting most companies is financial leverage. Let's say a given company's operating profits decline by 33%, from \$3,000 to \$2,000. If the company's \$30,000 capital has been sourced entirely through equity, meaning it hasn't borrowed, its capital and doesn't have to make any interest payments, that decline will flow through to the company's net income—the “bottom line”—and net income would fall by 33%.

But most companies are financed with a combination of equity and debt. Debtholders occupy a senior position relative to the equity investors. They are said to be in the “first-loss” position; this means the equity investors bear the loss if profits decline. If profits decline, and then all losses, until the equity is exhausted, fall to the debtholders. As long as there's enough profit to cover the interest, the outcome for the debtholders remains unchanged—they merely receive the interest payments they were promised. (That's why bondholders' notes are called “fixed income securities”: the outcome is fixed.)

Let's assume the capital structure of this company consists of \$15,000 of debt (requiring annual interest payments of \$1,500) and \$15,000 of equity. That means the \$1,000 decline in operating profits reduces net income from \$1,500 (\$3,000 of operating profit before interest payments, minus \$1,500 of interest) to \$500 (\$2,000 minus \$1,500). In other words, the decline in operating profit (from \$3,000 to \$2,000) causes this decline in net income to decline by 67% (from \$1,500 to \$500). The magnitude of the decline in operating profit on the net income illustrates financial leverage.



The process that determines a company's profits is complex and varies. The economic cycle has a profound effect on some companies' sales but not on others. Largely because of differences in operating and financial leverage, a given percentage change in sales has a much greater impact on some companies than for others.

And, of course, idiosyncratic developments can have a very significant impact on profits. These can include things like management's decisions regarding inventories, production levels, and capital investment; technological advancements (on the part of a company, its industry competitor, or companies in competing industries—see below); changes in regulation or taxation; and even developments exogenous to the industry, or even the business world, such as weather, war and fads. The economic cycle is the backdrop for changes in companies' sales and profits, but the deviation from the cycle-based expectation is vast. Idiosyncratic developments are the main reason.

I'll take a moment here to deal with the subject of technology (I'll devote a separate chapter to it). "Disrupt" is the word of the day. The ability of new technologies to disrupt traditional industries can shake up competition and dismantle the incumbents' profit margins. Take, for example, the newspaper industry. As recently as the 1990s:

- Newspapers were considered an indispensable source of information.
- Most people bought a newspaper daily—or maybe one on the way to work and another on the way home—and the cost was small.
- Even if you bought the newspaper on Monday, you still had another on Tuesday; there was no "shelf life" or protracted usefulness.
- Newspapers were one of the very few ways local businesses like movie theaters and used-car dealers could reach their customers, and a newspaper from one city generally couldn't compete for local advertising from another.
- Competition came primarily from other newspapers, television and radio. Once a newspaper was strongly established in a city, however, it would be hard to displace—thus newspapers were viewed as "moats."

Because this combination of factors was seen as making a company's position largely impregnable, newspaper company stocks were thought of as "defensive," benefiting from highly stable revenues and profits.

Who would have thought that the Internet and other forms of mass communication would significantly impact the fortunes of newspapers in less than twenty years? Today many companies compete to bring information directly to consumers. Newspapers are struggling to maintain their market share and profitability, as "free" has come to characterize many aspects of the digital world, knocking newspapers' business models off stride.

Newspapers provide an excellent example of the ability of an external factor to influence a company's sales and profits, completely apart from the economic and traditional profit cycle. But isn't technology cyclical itself?

Technologies are born, they prosper, and then they are replaced by ones. The innovation of a few years ago can be supplanted more rapidly than ever these days, and the list of industries that are perceived as being vulnerable to disruption seems to shrink every day.

Thirty or forty years ago, it seemed as if the world was a place that provided a relatively unchanging backdrop for life, and economic and social developments—including cycles—played out against that unchanging backdrop. Today, largely because of technological developments (both technological, social and cultural), nothing seems unchanging. In fact, much appears to be changing too fast for most of us to keep up with it.

VII

THE PENDULUM OF INVESTOR PSYCHOLOGY

In business, financial and market cycles, most excesses on the upside—and the inevitable reactions to the downside, which also tend to overshoot—are the result of exaggerated swings of the pendulum of investor psychology. Thus understanding and being alert to excessive swings is an entry-level requirement for avoiding harm from cyclical extremes, and hopefully for profiting from them.

So far we have discussed the economic cycle, governments' efforts to influence the economic cycle, and the profit cycle. To a great extent these things provide the backdrop or environment for investing. And they tend to be exogenous to investing—independent processes that operate on their own. But anyone who thinks these things are “mechanical” in their operation, and in full control of investment results underestimates the role of emotion, or, I tend to say interchangeably, emotion. (Psychology and emotion are certainly different elements, but I see no meaningful way to distinguish between the two in terms of their effect on the investment environment.)

First, swings in emotion/psychology strongly influence the economic cycle and corporate profit cycles, as indicated. And second, they play a very large part in causing ups and downs in the investment world—especially in the short run.

As I noted in chapter I, there is no fundamental distinction between the economic cycle and pendulum swings. In fact, I could have made life easier for all by titling this chapter “The Cycle in Psychology” and fitting this phenomenon within a consistent nomenclature. But, for some unspecified reason, I introduced the notion of the “pendulum” in emotion/psychology in my second memo to clients, “First Quarter Performance” (April 1991). I haven't found a reason to distance myself from it in my subsequent work, so I'm going to continue to refer to it here.

To introduce the pendulum, I'm going to borrow from what I wrote in 1991:

The mood swings of the securities markets resemble the movement of a pendulum. Although the midpoint of its arc best describes its location, the location of the pendulum “on average,” it actually spends more of its time there. Instead, it is almost always swinging toward one of the extremes of its arc. But whenever the pendulum is near either extreme, it is inevitable that it will move back toward the midpoint sooner or later. In fact, it is the movement toward the extreme itself that supplies the energy for the swingback.

Investment markets make the same pendulum-like swing:

- between euphoria and depression,
- between celebrating positive developments and obsessing over negatives, and thus
- between being overpriced and underpriced.

This oscillation is one of the most dependable features of the investment world, and investor psychology seems to spend much more time at the extremes than it does at a “happy medium.”

I returned to the topic in “It’s All Good” (July 2007). Before I make a new observation, I listed a half-dozen additional elements in which pendulum swings are seen:

- between greed and fear,
- between optimism and pessimism,
- between risk tolerance and risk aversion,
- between credence and skepticism,
- between faith in value in the future and insistence of concrete value in the present, and
- between urgency to buy and panic to sell.

I find particularly interesting the degree to which the polarities listed above are interrelated. When a market has been rising strongly for a while, we invariably see all nine of the elements listed first. As the market’s been declining, we see all of the elements listed second. Rarely do we see a blend of the two sets, given that the elements of each are causally related, with one giving rise to the next.

A lot of what I wrote then about the pendulum corresponds directly to what I wrote about cycles back in chapter I. There’s a swing from one extreme to the other; then the attainment of an extreme that can’t be exceeded; and then a swingback toward the midpoint, fueled by

of momentum. It might be said that the pendulum also tends to regress to the mean or midpoint, but—like most cycles—it usually overshoots and continues toward the extreme opposite from whence it came.



Why is the pendulum important? In essence, the too-strong upward and downward swings of the cycles I'm covering in this book largely—and represent—psychological excesses in action.

- The trendline rate of growth in economic output and corporate earnings is moderate, and when participants' pro-cyclical decisions cause growth to be abnormally rapid (other than in times of recovery from recession), this usually represents a too-optimistic expansion from which there will be a retreat.
- Likewise, it seems rational that, in the long run, stocks overall should provide returns in line with the sum of their dividends plus growth in corporate profits, or something in the mid-to-high single digits. When they return much more than that for a while, that is likely to prove to have been excessive—borrowing from the future and thus rendering stocks risky—meaning a downward correction is in order.

In business, financial and market cycles, most excesses on the upside are followed by the inevitable reactions to the downside, which also tend to overshoot—the result of exaggerated swings of the pendulum of psychology. Thus, understanding and being alert to excessive swings is an entry-level requirement for avoiding harm from cyclical extremes, and hopefully for profiting from them.

The norms in terms of growth and appreciation are in so many ways "healthy." And if the participants built their behavior around those norms—instead of occasionally building up hopes for more and thus setting the stage for eventual moves toward less—the world would be a more temperate, and less-error-prone place. But that's not the nature of the beast.

I touched on the inapplicability of norms in "The Happy Medium" (July 2004, but now with data brought up to date through 2016):

Putting it all together, fluctuations in attitudes and behavior cause the stock market to be the ultimate pendulum. In my 47 years of the investment business, starting with 1970, the annual returns on the S&P 500 have swung from plus 37% to minus 37%. Averaging out good years and bad years, the long-run return is

usually stated as 10% or so. Everyone's been happy with that typical performance and would love more of the same.

But remember, a swinging pendulum may be at its midpoint "on average," but it actually spends very little time there. The same is true of financial market performance. Here's a fun question (with an illustration): for how many of the 47 years from 1970 through 2016 was the annual return on the S&P 500 within 2% of "normal," that is, between 8% and 12%?

I expected the answer to be "not that often," but I learned that it had happened *only three times!* It also surprised me to learn that the return had been more than 20 percentage points away from "normal"—either up more than 30% or down more than 10%—more than one-quarter of the time: 13 out of the last 47 years. So the one thing that can be said with total conviction about stock market performance is that the average certainly isn't the norm. Market fluctuations of this magnitude aren't nearly fully explained by the changing fortunes of companies, industries or economies. They're largely attributable to the mood swings of investors.

Lastly, the times when returns are at the extremes aren't randomly distributed over the years. Rather they're clustered, due to the fact that investors' psychological swings tend to persist for a while—paraphrase Herb Stein, they tend to continue until they stop. Most of those 13 extreme up or down years were within a year or two of another year of similarly extreme performance in the same direction.

How about an example of the pendulum in action?

One of the most time-honored market adages says that "markets fluctuate between greed and fear." There's a fundamental reason for this: *people* fluctuate between greed and fear. In other words, sometimes people feel positive and expect good things, and when that's the case, they become greedy and fixate on making money. Their greed causes them to make investments, and their bidding causes markets to rise and appreciate.

But at other times, they feel less good and their expectations turn negative. In that case, fear takes over. Rather than enthuse about making money, they worry about losing it. This causes them to shrink from buying—eliminating the upward impetus beneath asset prices—and perhaps to sell, pushing prices down. When they're in "fear mode," people's emotions bring negative forces to bear on the markets.

Here's part of the discussion of the swing between greed and fear from "The Happy Medium" (July 2004):

When I was a rookie analyst, we heard all the time that “the market is driven by greed and fear.” When the market environment is in healthy balance, a tug-of-war takes place between optimists interested in making money and pessimists seeking to avoid losses. The former want to buy stocks, even if they have to pay a price above yesterday’s close, and the latter want to sell them, even if it’s on a downtick.

When the market doesn’t go anywhere, it’s because the sentiment behind this tug-of-war is evenly divided, and the people—or forces—on the two ends of the rope carry roughly equal weight. The optimists may prevail for a while, but as securities are bought and become more highly priced, and then the pessimists gain sway and sell them down.

It didn’t take long in my early days, however, for me to realize that often the market is driven by greed or fear. At the times that I can count, large numbers of people leave one end of the rope free. Either the greedy or the fearful predominate, and they move the market dramatically. When there’s only greed and no fear, for example, everyone wants to buy, no one wants to sell, and few people can think of reasons why prices shouldn’t rise. And so they do—often in leaps and bounds and with no apparent governor.

Clearly that’s what happened to tech stocks in 1999. Greed was the dominant characteristic of that market. Those who weren’t participating were forced to watch everyone else get rich. “Prudent investors” were rewarded with a feeling of stupidity. The buyers moving that market felt no fear. “There’s a new paradigm,” was the battle cry, “get on board before you miss the boat. And by the way, the price I’m buying at can’t be excessive, because the market is always efficient.” Everyone perceived a virtuous cycle in favor of tech stocks to which there could be no end.

But eventually something changes. Either a stumbling block materializes, or a prominent company reports a problem, or an exogenous factor intrudes. Prices can also fall under their own weight or based on a downturn in psychology with no obvious cause. Certainly no one I know can say exactly what it was that burst the tech-stock bubble in 2000. But somehow the greed evaporated and fear took over. “Buy before you miss out” was replaced by “Sell before it goes to zero.”

And thus fear comes into the ascendancy. People don’t worry about missing opportunities; they worry about losing money. Irrational exuberance is replaced by excessive caution. Whereas in 1999 the sky-high forecasts for a decade out were embraced warmly, in

investors chastened by the corporate scandals said, “I’ll never trust management again” and “How can I be sure any financial statements are accurate?” Thus almost no one wanted to buy the bonds of scandal-plagued companies, for example, and they sunk to giveaway prices. It’s from the extremes of the cycle of fear and greed that the greatest investment profits, as distressed debt demonstrated in 2003.

“Greed/fear” is the most obvious psychological or emotional coin along which investors swing, and in many ways the most illustrative. What are some of the other key emotional or psychological swings? Most move in a manner similar to the greed/fear pendulum, and usually this is a coincidence. The various parameters are interrelated. Here are a few examples.

Underlying the swing between greed and fear is the swing between euphoria and depression. As previously described, for example, it may be that positive events give rise to greed. Rather, positive events encourage euphoria, which abets greed (and vice versa for negative events, depression and fear). Euphoria and depression are the foundation emotions that give rise to the swings that follow.

Euphoric investors may be excited about current developments and what may arise in the future, and this may accentuate their fixation with expectation of—profit. On the other hand, depressed investors are unlikely to feel positive enough to be greedy. If you think about it, euphoria is inconsistent with fear, and depression is inconsistent with greed.

In a similar vein, investors also swing between optimism and pessimism. Positive events generally give rise to expectations of further gains and positive outcomes . . . a state better known as optimism. Optimism underlies greed; it makes no sense to think people can be greedy if they are forward to invest when their expectations are negative. Clearly, optimism and pessimism encourage other emotions and influence behavior.



The next phenomenon I want to touch on is the tendency of investors to fluctuate between credulousness and skepticism, and to fluctuate between being entranced by the possibility of profit in the future and insistent on value in the here and now.

Sometimes—usually when things are going well in the world and asset prices are rising—investors become willing to swallow favorable stories regarding future developments, buy into appreciated assets, and shoulder elevated risk. But when things take a turn for the worse, they become likely to reject even reasonable projections and decline to buy, largely

because prices are down (even though that increases the likelihood that they are cheap).

Some investors spend their time working hard to quantify this year's earnings and the growth thereafter. Others strive to value real and intellectual property and business advantages (and predict what others will pay for them). Still others try to deduce the value implications of mergers and acquisitions, balance sheet restructurings and private/public transactions. In all of these ways and many more, it's the job of those in the investment business to predict the future and put a value on it.

Let me give you an example. In 2000–01, our distressed debt fund invested a few hundred million dollars in bankrupt telecom companies. In each case, the purchase price implied a value for the company that was a small fraction of the amounts that had been invested in hard assets such as switching gear or fiber-optic cable. If we could resell the equipment for a higher percentage of its cost than we had paid, the investment would be profitable.

The first sale went well, and we made a quick 50%. But soon thereafter, people stopped showing up to bid on these assets. The party to whom we had sold the first company thought he had gotten a bargain, but in later instances the possible buyers shied away from assets that were turning out to be in heavy oversupply. And that brings me to my point. In 1999, investors accepted at face value their telecom companies' rosy predictions of the future, and they were willing to pay handily for that potential. But in 2001, they saw the potential was largely empty and wouldn't pay a dime for it, given that the excess capacity vastly exceeded its current needs and no one could imagine the excess being absorbed in their lifetime. This cycle in investor willingness to value the future is one of the most powerful cycles that exists.

A simple metaphor relating to real estate helped me to explain this phenomenon: What's an empty building worth? An empty building (a) has a replacement value, of course, but it (b) has no revenues and (c) costs money to own, in the form of taxes, insurance, minimum maintenance, interest payments and opportunity costs. In other words, it's a cash drain. When investors are in a pessimistic mood and can't see more than a few years out, they can only think about the negative cash flows and are unable to imagine a time when the building will be rented and profitable. But when the mood turns upward and interest in future potential runs high, investors

envision it full of tenants, throwing off vast amounts of cash salable at a fancy price.

Fluctuation in investors' willingness to ascribe value to possible future developments represents a variation on the full-or-empty conundrum. Its swings are enormously powerful and mustn't be underestimated. ("The Happy Medium," July 2004)



The superior investor is mature, rational, analytical, objective and unemotional. Thus he performs a thorough analysis of investment fundamentals and the investment environment. He calculates the intrinsic value of each potential investment asset. And he buys when any discount the price from the current intrinsic value, plus any potential increases in intrinsic value in the future, together suggest that buying at the current price is a good idea.

In order to be able to do all the above, the superior investor maintains an appropriate balance between fear (which is shorthand for risk aversion, fear of loss, and respect for uncertainty and randomness) and greed (a.k.a. ambition, aspiration, aggressiveness and acquisitiveness). All people feel emotions, but the superior investor keeps these conflicting elements in balance. The presence of the two offsetting forces leads to responsible, wise and even-keeled behavior.

But the important points are these:

- Few people are always even-keeled and unemotional.
- For this reason, few investors are capable of staking out a position that balances greed and fear—and staying there—as more-often-than-not, less-positive developments arise.
- To the contrary, most investors swing between being greedy when they're optimistic and fearful when they're pessimistic.
- Most swing to those positions at the wrong time—becoming greedy after the emergence of positive developments has caused prices to become elevated, and becoming more fearful after negative events have caused prices to become depressed.

Here's some of what I wrote about the swing of psychology in "Couch" (January 2016):

There are many more ways in which non-objective, non-rational quirks commonly affect behavior. As Carol Tavris pointed out in her May 15, 2015 *Wall Street Journal* review of Prof. Richard Thaler's book *Misbehaving: The Making of Behavioral Economics* (2015):

As a social psychologist, I have long been amused by economists and their curiously delusional notion of the “rational man.” Rational? Where do these folks live? Even 50 years ago, experimental studies were demonstrating that people stay with clearly wrong decisions rather than change them, throw good money after bad, justify failed predictions rather than admit they were wrong, and resist, distort or actively reject information that disputes their beliefs.

The difficulty of understanding events, their significance and their potential ramifications comes in good part from the kinks in investor psyches, and it contributes to—and feeds back to exacerbate—investors’ responses. Thus investors tend to emphasize just the positives or the negatives much more often than they take a balanced, objective approach. And they tend to become optimistic and eager to buy when good news, positively interpreted, has forced prices up . . . and vice versa. All of this is obvious (especially in retrospect). The equally obviously, understanding and dealing with it presents a potential way to improve results.

The basic point is that psychology does swing, and most people’s behavior swings with it. The fluctuation between greed and fear is typical of the psychological pendulum. In fact, it explains not only the behavior of most investors, but also—taking investors collectively—the behavior of markets. Markets move upward when events are positive and psychology turns up, and they fall when events are negative and psychology turns down.

The pendulum spends only a little of the time at the midpoint. Rather, the pendulum is usually swinging toward one extreme or the other, first recovering from one psychological extreme—either too high or too low—and then continuing toward the other.

The superior investor resists psychological excesses and thus rarely participates in these swings. The vast majority of the highly successful investors I know are unemotional by nature. In fact, I believe that unemotionality is one of the great contributors to their success.

This is one of my most persistent observations and—in a related context—of the questions I’m most often asked is whether people can learn to be unemotional. My answer is “yes and no.” I think it’s possible to train people on the lookout for potential emotional influences and to try to control their effect. But I also think people who are inherently unemotional are much easier. A lack of emotionality is a gift (in investing, but not in other areas, like marriage). It’s not my point that emotional people

can't be good investors, but it will require a great deal of self-restraint.



In addition to the interrelatedness of the various emotional swings on the last few pages, it's also important to note the causal phenomena. Just as positive events give rise to euphoria, and euphoria rise to optimism, and optimism abets an increase in greed, swings of all these elements cause the way investors perceive things to fluctuate between rosy and dark. Investors' perception of events is colored along the various emotional or psychological arcs. And this perception feeds back, creating more euphoria, optimism and greed.

Here's how I put it in "On the Couch" (January 2016):

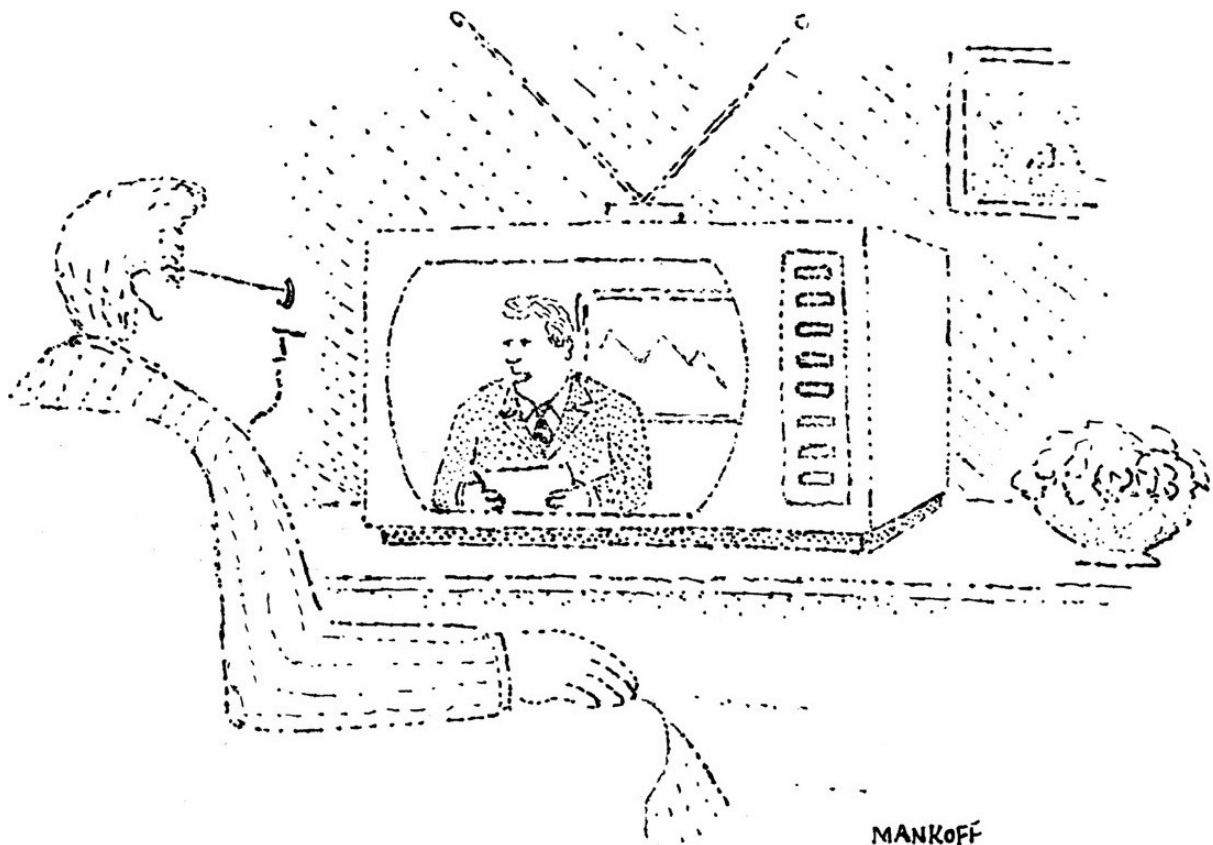
One of the most significant factors keeping investors from reaching appropriate conclusions is their tendency to assess the world with emotionalism rather than objectivity. Their failings take two primary forms: selective perception and skewed interpretation. In other words, sometimes they take note of only positive events and ignore the negative ones, and sometimes the opposite is true. And sometimes they view events in a positive light, and sometimes it's negative. Rarely are their perceptions and interpretations balanced and neutral.

Ever since the events of August 2015 in China, I've repeatedly found myself harking back to one of the oldest cartoons in my collection and still one of the very best:



"Everything that was good for the market yesterday is no good for today."
 © BERNARD SCHOENBAUM

The bottom line is that investor psychology rarely gives equal weight to both favorable and unfavorable developments. Likewise, investors' interpretation of events is usually biased by their emotions. Whatever is going on at the moment, most developments have both helpful and harmful aspects. But investors generally obsess about one or the other rather than consider both. And that recalls another cartoon:



MANKOFF

"On Wall Street today, news of lower interest rates sent the stock market up, but that these rates would be inflationary sent the market down, until the realization stimulate the sluggish economy pushed the market up, before it ultimately overheated economy would lead to a reimposition of higher interest rates." E

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It all seems so obvious: investors rarely maintain objective, rational, neutral and stable positions. First they exhibit high levels of optimism, greed, risk tolerance and credulousness, and their resulting behavior causes asset prices to rise, potential returns to fall, and risk to increase. But then, for some reason—perhaps the arrival of a tipping point—they switch to pessimism, fear, risk aversion and skepticism, and this causes asset prices to fall, prospective returns to rise, and risk to decrease. Notably, each group of phenomena tends to happen in unison, and the swing from one to the other often goes beyond what reason might call for.

That's one of the crazy things: in the real world, things fluctuate between "pretty good" and "not so hot." But in the world of investing, perception often swings from "flawless" to "hopeless." The pendulum careens from one extreme to the other, spending almost all its time at "the happy medium" and rather little in the range of reasonableness. First there's denial, and then there's capitulation.

The world is full of positive and negative events, and on most of each. And some of the events that occur are ambiguous, having elements of both good and bad, making them subject to either positive or negative interpretation.

Take the example of the second cartoon. Low interest rates are good because they stimulate business activity and increase the discounted value of future cash flows. But they're also bad, since the stronger activity they abet can give rise to inflation and thus signal the center that rates should be raised, withdrawing stimulus from the economy. Interpretation may not fluctuate between the extremes several times over the cartoon suggests. But it does fluctuate excessively, and in fact on a dime.

A few years ago my friend Jon Brooks supplied this great illustration of a skewed interpretation at work. Here's how investors react to even good news when they're feeling good about life (which usually means the market is rising):

- Strong data: economy strengthening—stocks rally
- Weak data: Fed likely to ease—stocks rally
- Data as expected: low volatility—stocks rally
- Banks make \$4 billion: business conditions favorable—stocks rally
- Banks lose \$4 billion: bad news out of the way—stocks rally
- Oil spikes: growing global economy contributing to demand—stocks rally
- Oil drops: more purchasing power for the consumer—stocks rally
- Dollar plunges: great for exporters—stocks rally
- Dollar strengthens: great for companies that buy from abroad—stocks rally
- Inflation spikes: will cause assets to appreciate—stocks rally
- Inflation drops: improves quality of earnings—stocks rally

Of course, the same behavior also applies in the opposite direction. When market psychology is negative and markets have been falling for a while, good news is capable of being interpreted negatively. Strong economic data is likely to make the Fed withdraw stimulus by raising interest rates. Weak data is taken to mean companies will have trouble meeting earnings targets. In other words, it's not the data or events; it's the interpretation that fluctuates with swings in psychology.

At the greatest extremes of the pendulum's swing, a process

appearance of a virtuous circle or a vicious circle. When even predominantly positive psychology is rosy, negative developments to be overlooked, everything is interpreted favorably, and things are often thought to be incapable of taking a turn for the worse. Expectation of further advances appears irresistible; past constraints and norms are ignored or rationalized away; and anyone imagining limits to the positive future is dismissed as an old fogey lacking imagination. The potential for gains comes to be viewed as infinite. Asset prices encouraging further optimism.

But on the other hand, when things have been going badly for months or years and psychology is highly negative, it's the potential for improvement that can be forgotten. Unpleasant events are emphasized and positive events ignored. The case for further deterioration seems rock-solid, its errors are imagined, and now it's the downside that seems to be unlimited. Prices falling, resulting in further pessimism.

The virtuous circle and the vicious circle are both unrealistic exaggerations. While they've been dreamed up many times in the past, they've never come out. But that fact doesn't enable most people to resist them while they are rampant.

Again, the superior investor—who resists external influences, remains emotionally balanced and acts rationally—perceives both positive and negative events, weighs events objectively and analyzes them dispassionately. But the truth is that sometimes euphoria and optimism cause most investors to view things more positively than is warranted, and sometimes depression and pessimism make them see only bad and interpret events with a negative bias. Refusing to do so is one of the keys to successful investing.



For a bullish phase to hold sway, the environment has to be characterized by greed, optimism, exuberance, confidence, credulity, daring, risk tolerance and aggressiveness. But these traits will not govern a market forever. Eventually they will give way to fear, pessimism, prudence, uncertainty, skepticism, caution, risk aversion and reticence. . . . Busts are the product of booms, and it's usually more correct to attribute a bust to the excesses of the preceding boom than to the specific event that sets off the correction. (“Now What?,” January 2008)

Usually, when either set of polar extremes is in the ascendancy, the fact is readily observable, and thus the implications for investors should be obvious to objective observers. But of course, the market pendulum to one extreme or the other occurs frequently.

reason that the psyches of most market participants are moving in the same direction in a herd-like fashion.

Few of the people involved actually are objective. To continue the thread from my memo "Everyone Knows" (April 2007), expecting a widespread clinical observation during a market mania makes as much sense as saying "everyone knows the market has gone too far." If many people recognized that it had gone too far, it wouldn't be there. ("It's All Good," July 2007)

VIII

THE CYCLE IN ATTITUDES TOWARD RISK

The rational investor is diligent, skeptical and appropriately risk-averse at all times, but also on the lookout for opportunities with a potential return that more than compensates for risk. That's the ideal. But in good times, we hear most people say, "Risk? What risk? I don't see much that could go wrong: look how well things have been going. And anyway, risk is my friend—the more risk I take, the more money I'm likely to make."

Then, in bad times, they switch to something simpler: "I don't want to risk if I ever make another penny in the market; I just don't want any more. Get me out!"

Now that we've moved on from considering cycles in the abstract to discussing their operation in the investment world, I'm going to provide a brief aside regarding the fundamental nature of investing, in order to lay a foundation for the discussion that will follow. Some of this material may be familiar from earlier chapters.

What is investing? One way to think of it is as bearing risk for profit. Investors try to position portfolios so as to profit from market developments rather than be penalized by them. The superior investor is simply someone who does this better than others.

Do we know what will happen in the future? Some investors think we do—or think they have to act as if they do, because if they don't, they lose their jobs and their clients—or they have been pursuing profit through market forecasts for so long that they've brainwashed themselves into believing it's possible to be right about the future (and have become conditioned to their low past success rates). Other investors—the smarter, more self-aware ones, I think—understand that the future isn't knowable with certainty. They may form opinions regarding future events, but they don't bet heavily on those opinions will prove correct.

Since (a) investing consists of dealing with the future but (b) the future isn't knowable, that's where the risk in investing comes from. If future

were predictable, investing would be easy and profit would be sure. general level of returns might be low in that case because so involved; that's a topic for another day.) But the fact that events predictable introduces risk. Because the events that actually occur may different than those that were predicted, or the market's reaction to may differ from what was expected, a portfolio may be positioned incore for the future that unfolds.

Since risk (that is, uncertainty with regard to future developments, and possibility of bad outcomes) is the primary source of the challenge of investing, the ability to understand, assess and deal with risk is the the superior investor and an essential—I'm tempted to say *the* requirement for investment success.

Finally as to foundation, it's important to recognize that while the investment environment varies over time, at any particular point in time given. What I mean is that we can accept the environment or we can reject it and stay on the sidelines, but we don't have of saying, "I don't like the environment as it is today; one." Or rather we can demand another, but of course that won't materialize.

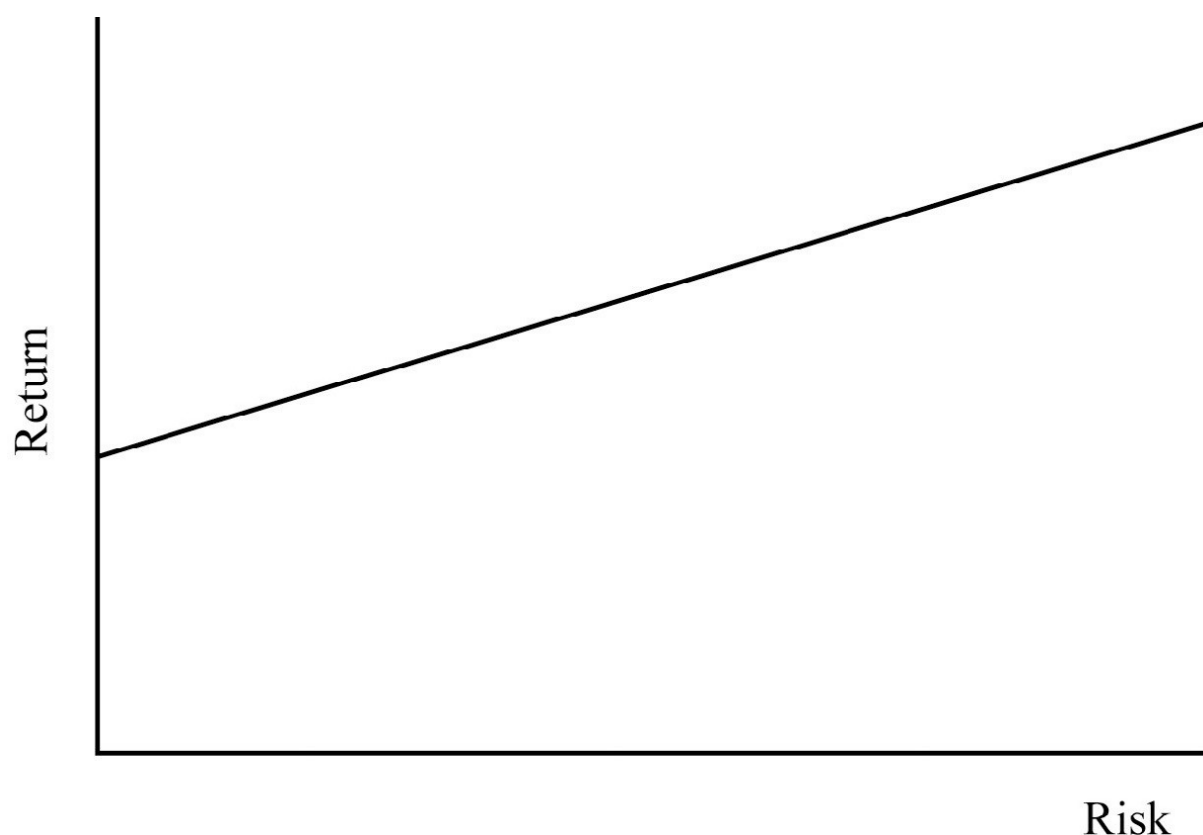
My view that risk is the main moving piece in investing makes conclude that at any given point in time, the way investors collectively viewing risk and behaving with regard to it is of overwhelming shaping the investment environment in which we find ourselves. And of the environment is key in determining how we should behave to risk at that point. Assessing where attitudes toward risk stand in is what this chapter is about—perhaps the most important one in



One of the luckiest breaks I received in my lifetime came opportunity to attend the University of Chicago's Graduate School of Business (since renamed the Booth School) in 1967–69. I, like many in those days, went straight from college to graduate school, perhaps the most efficient route to success, but with the added incentive provided Vietnam War and accompanying draft.

Over the four prior years, I had received a nuts-and-bolts education in finance at Wharton: practical, non-theoretical and qualitative. My choice of graduate school was most fortuitous, as Chicago had just begun to new theory of finance and investment that had been developed, largely in the early 1960s. Thus my Wharton training was paired with juxtaposed against—further study in finance that was almost entirely academic, theoretical and quantitative.

Soon after arriving at Chicago, I was exposed to a graphic much of the foundation for the new theory of investment, and that as the starting point for much of my thinking and writing ever



This graphic has become ubiquitous in the investment world in the since my first exposure to it fifty years ago. Its essence lies in the line slopes up and to the right, indicating a positive relationship between risk and return. This is usually misinterpreted, in my view, as meaning “riskier assets produce higher returns,” and thus “if you want to make more return, the answer is to take more risk.” This formulation cannot be correct. If riskier assets could be counted on to produce higher returns, the definition wouldn’t be riskier.

The linearity with which the relationship between risk and return is presented in the graphic above ignores the fact that at every level of risk there is a range of potential outcomes regarding return, and thus it ignores the unreliability of the relationship. That’s what makes people say “riskier investments produce higher returns.” Instead, I think the graphic should be interpreted as saying, “Investments that seem riskier have to appear to promise higher returns, or else no one will make them.” Words like “appear” and “appear” are the right ones, since they indicate that risk and potential return can only be estimated, and that the investment world doesn’t work like a machine. That makes those words highly appropriate—actually, completely

—for use when discussing investing. (For a more thorough discussion, see chapter 5 of *The Most Important Thing*.)

People who immediately “get” concepts like risk and risk/return usually have an intuitive sense that prepares them to be good investors. I reasons behind my interpretation of the graphic will become clear once I have prompted you to think about it.

Let’s suppose a logical investor is offered two investments with the same expected return, but in one case the return is virtually assured and in the other it is highly uncertain. We would expect him to choose the certain one. People prefer certainty over uncertainty. If a Treasury bond and a startup both seemed likely to return 7%, for example, the vast majority of people would go with the Treasury. Why take the extra risk associated with the startup if no potential increase in return is offered to justify the incremental risk?

Well, that’s the point: most people would prefer a sure 7% over a 7% that is uncertain. In other words, most people are risk-averse. That’s the essential assumption that underlies the “Chicago school” of finance.

To describe risk aversion, I say most people prefer safety and dislike risk—even though I’ve never seen the word “disprefer” in a dictionary. There is a big difference of opinion regarding the propriety of that word, with the linguistic establishment railing against it, but I think it’s a useful word (if it doesn’t exist, it should.)

The widespread dislike for risk and the resultant insistence on incremental return if incremental risk is to be borne are the reasons why long-term Treasuries carry higher yields than short-term Treasuries; why high yield bonds promise higher returns than investment grade bonds; why stock returns are generally expected to return more than bonds; and why venture capital investing is expected to provide higher returns than public equity. I’m only talking here about “expectations” and “promises”—or what “may happen” or “should happen”—rather than things that “will happen” or “are sure to happen.” But that expectation of incremental return generally has been present in order for most people to voluntarily take on added risk.

Because of the natural dispreference for risk, it’s entirely reasonable to expect that if they’re going to bear it, investors have to be offered the possibility of an incremental reward. Non-daredevils will do risky things only if they expect a generous reward for doing so. Nothing else will motivate them.

Risk aversion is an essential element in investing. People’s aversion to risk causes them to police the markets. Because most people are averse to risk,

- they approach investing with caution,
- they perform careful analysis when considering investments, and especially risky ones,

- they incorporate conservative assumptions and appropriate skepticism into their analysis,
- they demand greater margins of safety on risky investments to against analytical errors and unpleasant surprises,
- they insist on healthy risk premiums—the expectation of incremental returns—if they’re going to undertake risky investments, and
- they refuse to invest in deals that make no sense.

These are all essential parts of the investment process. Because risk-averse investors perform them, investing is a rational field in which reasonable propositions are offered. In short, risk aversion is the main element that makes markets safe and sane.

Please note, however, that the above is a normative description, or a description of how things *should* be. These are things the superior investor does, and that all investors *should* do. But the bottom line is that the superior investor does them, and certainly not all investors do them equally at all.

It’s one of the absolute truths that attitudes toward risk change, and that they alter the investment environment. That’s what the rest of this chapter will be about.



Here’s another aside: How is the investment environment formed? It is the result of discussions that take place in the marketplace—either within each investor’s consciousness or among investors, spoken or signaled by their actions. Here’s how I described the creation of the investment environment in “Risk and Return Today” in October 2004:

I’ll use a “typical” market of a few years back to illustrate how it works in real life: The interest rate on the 30-day T-bill might have been 4%. So an investor says, “If I’m going to go out five years, I want 5%. And to buy the 10-year note I have to get 6%.” He wants a higher rate to extend maturity because he’s concerned about inflation and loss of purchasing power, a risk that is assumed to increase with maturity. That’s why the yield curve, which in reality is a reflection of the capital market line, normally slopes upward with the maturity of the asset life.

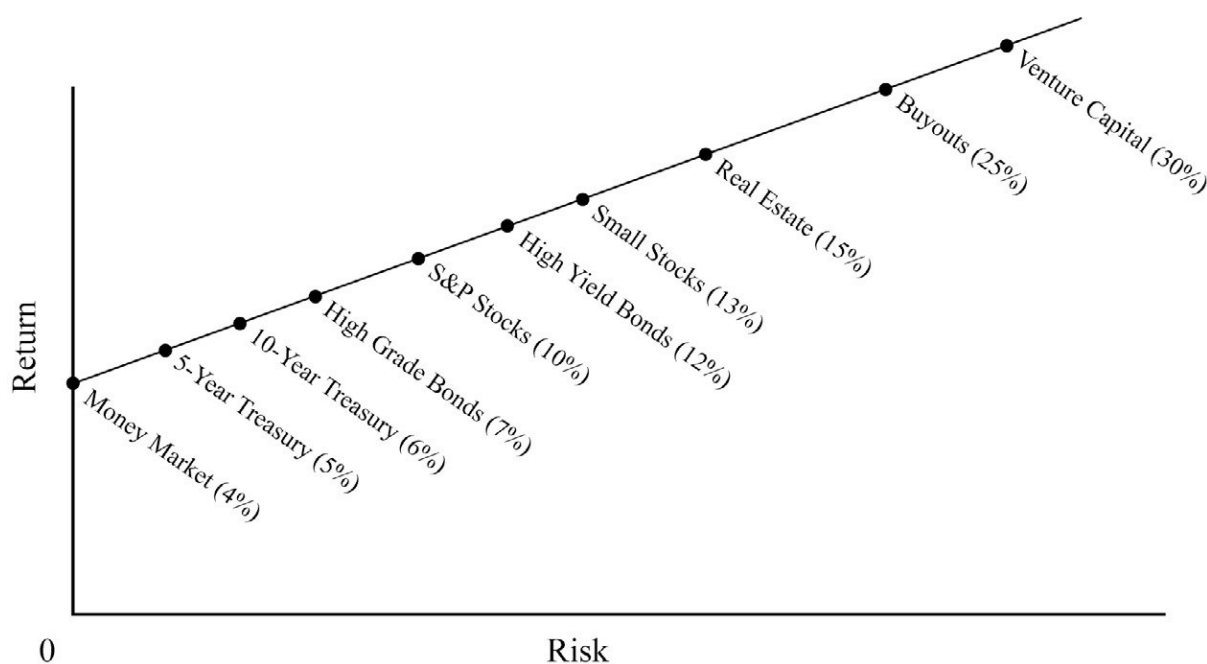
Now let’s factor in credit risk. “If the 10-year Treasury pays 6%, I’m not going to buy a 10-year single-A corporate unless I can get 7%.” This introduces the concept of credit spreads. Our hypothetical investor wants 100 basis points to go from a “government” to a “corporate.” If the consensus of investors feels the same, that’s what the spread will be.

What if we depart from investment grade bonds? “I’m not going to touch a high yield bond unless I get 600 over a Treasury note of comparable maturity.” So high yield bonds are required to yield 12%, for a spread of 6% over the Treasury note, if they’re good buyers.

Now let’s leave fixed income altogether. Things get tougher, because you can’t look anywhere to find the prospective return on investments like stocks (that’s because, simply put, their returns are conjectural, not “fixed”). But investors have a sense for these things. “Historically S&P stocks have returned 10%, and I’ll only buy them if I think they’re going to keep doing so.” So in theory, the capital stock investor determines earnings per share, earnings growth rate and dividend payout ratio and inputs them into a valuation model to determine at the price from which S&P stocks will return 10% (although I’m not sure the process is nearly that methodical in actuality). “And if I think stocks should return more; I won’t buy on the NASDAQ unless I think I’m going to get 13%.”

From there it’s onward and upward. “If I can get 10% from S&P stocks, I need 15% to accept the illiquidity and uncertainty associated with real estate. And 25% if I’m going to invest in buyouts. And 30% to induce me to go for venture capital, with its low success rate.”

That’s the way it’s supposed to work, and in fact I think it does (although the requirements aren’t the same at all times). The result is a capital market line of the sort that has become familiar to many of us, as shown below.



The process described above results in the formation of the risk continuum or “capital market line.” The process establishes the general relationship of return relative to risk, as well as the quantum of incremental return—or the “risk premium”—that will be expected for the bearing of incremental risk. In a rational world, the result will be as follows:

- Investments that seem riskier will be priced so that they appear to offer higher returns.
- The increase in return per unit of incremental risk will be reasonable and appropriate.
- The increase in expected return generally will appear to be proportional to the incremental risk (that is, bearing a unit of risk at one point on the continuum will appear likely to bring a similar amount of incremental return as bearing a similar unit of risk elsewhere on the continuum).
- Thus there won’t be particular points on the continuum where risk-bearing is rewarded much more or much less than at others (that is, investments whose promised risk-adjusted return is obviously superior to the rest).

In a rational world, any violations of these provisions would be corrected. As a result, the prices of mispriced assets are bid up or pushed down to move such that the prices are fair. As a result:

- the violations would be corrected,
- all investments would offer risk-adjusted returns that are fair relative to each other, and
- investors could increase their returns only by increasing the amount of risk they bear.

If investors always behaved that way, their actions would cause the market to be marked by “efficient markets” where no investment offers a higher risk-adjusted return than any other. Of course, markets don’t always operate that way; they’re supposed to—things certainly aren’t always priced right—but the general suggestion of efficiency is too logical to be disregarded. Efficiency is another essential topic, but I won’t go into it further here. (See 2 of *The Most Important Thing*, as well as the latter half of the book from January 2014.)



The key thing to note is that fluctuations in attitudes toward risk are exceptions to the principles described here. Sometimes investors become

risk-averse, and sometimes they relax their risk aversion and become too tolerant.

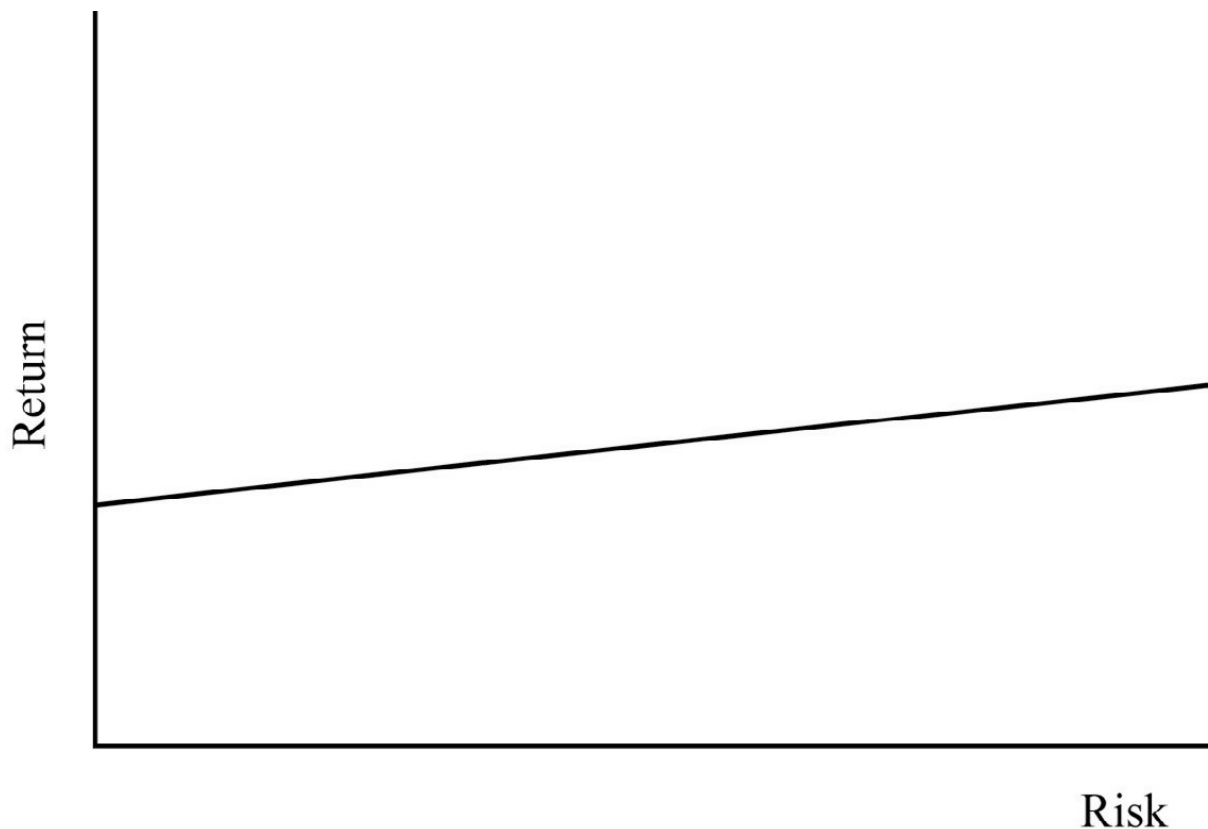
When positive events occur as described in the last chapter and optimism and greed rise, investors tend to become less risk-averse than they should be. What are the effects (follow the list of the functions investors perform on [page 106–107](#))?

- Since they feel better about the environment and are more optimistic about likely outcomes, they reduce the amount of caution they use in the investing process.
- Since they no longer consider investing to be risky, they don't need for painstaking analysis.
- They tend to make assumptions that are more generous, and they show less skepticism with credulousness.
- They're willing to make do with a reduced margin of safety.
- Viewing risk as less worrisome, they no longer demand risk compensation as cushy as in the past.
- They behave less as sticklers, since they're more attracted to high returns on risky investments and less leery of the risk they involve.

It is for these reasons that, as you'll see in the next chapter, financings are completed in the most buoyant economies and financial markets. Good times cause people to become more optimistic, jettison caution, and settle for skimpy risk premiums on risky investments. Since they are less pessimistic and less alarmed, they tend to lose interest in the safer end of the risk/return continuum. This combination of elements makes the prices of risky assets rise relative to safer assets. Thus, it comes as a surprise that more unwise investments are made in good times than in bad. This happens even though the higher prices on risky investments mean the prospective risk premiums offered for making those riskier investments are skimpier than they were in more risk-conscious times.

The reduced insistence on adequate risk premiums causes the slope of the capital market line to flatten. Going back to high school geometry, we know that the slope of a line on a graph is the distance traveled per unit change along the horizontal axis. The slope of the capital market line reflects the amount of incremental potential return that is offered for incremental risk borne. Thus it is a direct indication of the degree of risk aversion present in the market.

In times of obliviousness toward risk—or high risk tolerance—the demand in terms of risk premiums causes the slope of the line to flatten, and the amount of risk compensation to shrink.



The lower slope of the capital market line means, by definition, there's less of a return increment per unit increase in risk. In the payoff for risk-bearing is sub-par.

In my opinion, all of the above follows logically from direct observation. The process is as follows:

- positive events lead to increased optimism,
- increased optimism makes people more risk-tolerant,
- an increase in risk tolerance causes lower risk premiums to be demanded,
- a reduction in demanded risk premiums equates to lower demanded returns on risky assets,
- a reduction of demanded returns on risky assets causes their prices to rise, and
- higher prices make assets even riskier (but also attract buying of "momentum investors" who chase rising stocks).

It follows from the above that risk is high when investors feel risk is high. And risk compensation is at a minimum just when risk is at a maximum (meaning risk compensation is most needed). So much for the rational investor!

For me, the bottom line of all of this is that the greatest investment risk is the belief that there is no risk. Widespread risk tolerance or a high degree of investor comfort with risk—is the greatest subsequent market declines. But because most investors are following the progression described just above, this is rarely perceived at the time of perceiving it—and turning cautious—is most important.



Since the other side of the coin should be obvious, I won't belabor it. I will take a minute to talk about what happens when the downward turn in the cycle in attitudes toward risk makes investors more risk-averse.

One of the most important characteristics of psychological cycles is extremeness. Cycles swing not only in directions and degrees that make sense, but also in wacky ways and to excess. For example, investors come from time to time and say, "Let's disregard risk. We'll all get rich. The ardor and excitement cause them to bid prices to levels that are so high that they have to accept tales that clearly are so unrealistic—that after the fact, the whole thing is laughable if the damage done weren't so great.

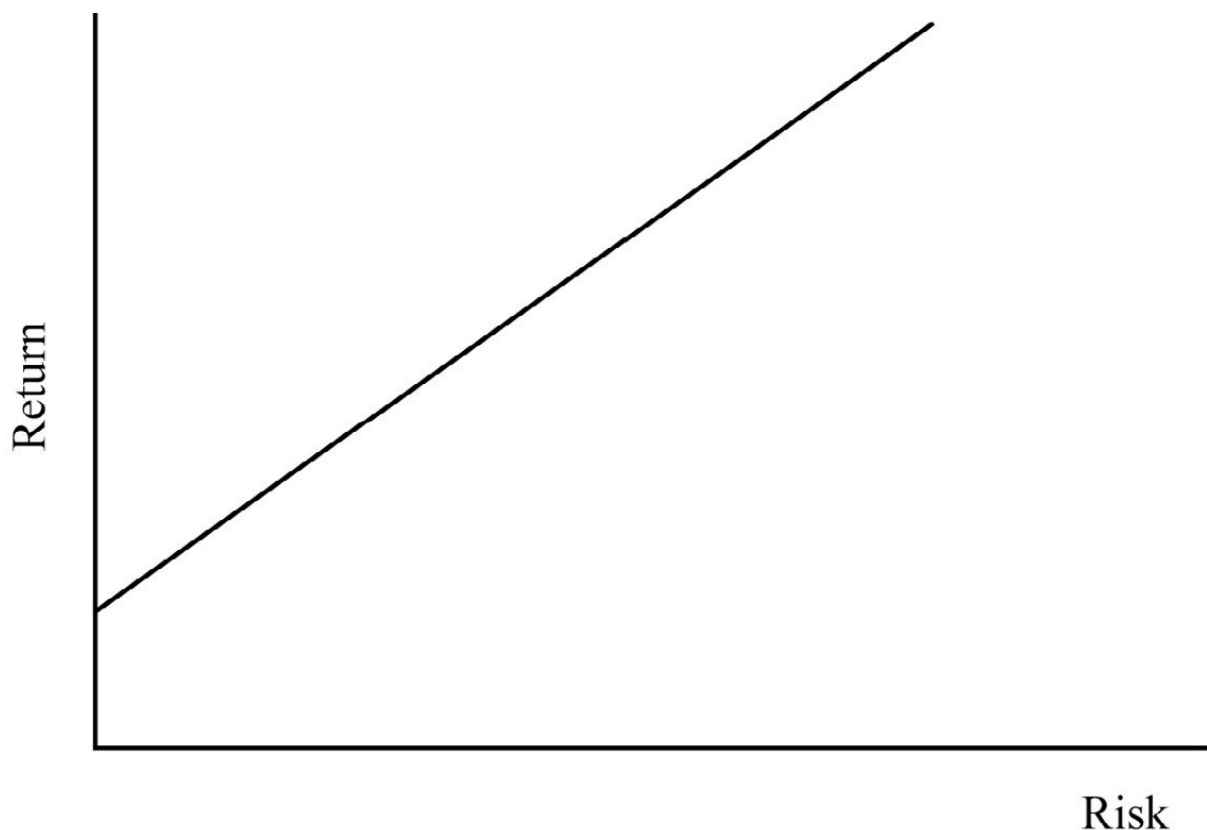
And after they have committed these sins of excess—and lost a lot of money in the resultant downswing toward greater prudence—they chastise themselves for their excesses of greed and credulousness. They wonder how they could have behaved so foolishly. They confess that they never really understood the exotic and exciting investment activities in which they had engaged. And they vow never to do it again.

Just as the inadequacy of their risk aversion allowed them to push up prices and buy at the top—egged on by the vision of easy money in the market—they couldn't discern any risk—now they push prices down and sell at the bottom. Their unpleasant recent experience convinces them—contrary to what they had thought when everything was going well—that investing is a field in which they shouldn't engage. And, as a consequence, their risk aversion goes all the way from inadequate to excessive.

- Given their recent painful experience and the negativity they've developed about what lies ahead, they amp up their caution.
- Since they now associate investing with loss rather than profit, their investment process comes to emphasize the avoidance of further loss and the prospecting for opportunity.
- They ensure that their assumptions are conservative enough to cover all potential for disappointment, and they apply extreme skepticism.
- They find it impossible to identify—even to imagine—investments that offer an adequate margin of safety.

- Since they see risk everywhere, they consider even the current swollen risk premiums insufficient.
- They become worrywarts. Just as risk tolerance had positioned them as buyers of overpriced assets at the highs, now their screaming risk aversion makes them sellers—certainly not buyers—at the bottom.

That's the point. Under these circumstances, the perception of risk is exaggerated and the slope of the capital market line becomes excessive.



In precisely the reverse of my description of the flat capital market, the higher slope of this one means the expected return increment per unit of risk is unusually generous. This is a risk-averse market, and the exaggerated payoff for risk-bearing. Thus the reward for bearing risk is greatest at just the moment when—no, rather, just because they absolutely refuse to bear it.

As risk attitudes swing from high to low, so do opportunities for profit and loss. When everything's going well and asset prices are soaring, investors tend to view the future as rosy, risk as their friend, and profit as just around the corner. Everyone feels the same, meaning little risk aversion is incorporated in prices, and thus they're precarious. Investors become risk-tolerant just when they should increase their risk aversion.

And when events are down, so are investors. They think of the place to lose money, risk as something to be avoided at all depressingly likely. As I described at the end of the last excess of caution that prevails, (a) no one will accept possible incorporate any optimism at all and (b) they likewise cannot count possibility that an assumption could be “too *bad* to be true.”

Just as risk tolerance is unlimited at the top, it is non-existent. This negativity causes prices to fall to levels from which losses are highly unlikely and gains could be enormous. But the sting of the prior decline tends to increase risk aversion and send investors to the sidelines (and thus risk) are at their lowest.



I'd like to provide a real-life example of changing attitudes, of some quantification, taken from “The Happy Medium” (July 2004):

The prevalence of risk-tolerance (or risk-obliviousness) in the late 1990s was clear. I personally heard a prominent brokerage house strategist say, “Stocks are overpriced, but not enough to keep them from being a buy.” And we all heard the man on the street say, “I'll hold up so much in my 401(k), it wouldn't bother me if it fell.” (Where was that guy two or three years later?)

No, those risk-tolerant attitudes will not persist forever. Eventually something will intrude, exposing securities' imperfections and too-high prices. Prices will decline. Investors will like them less at \$100 than they did at \$100. Fear of losing the remaining \$60 will overtake the urge to make back the lost \$40. Risk aversion eventually will reassert itself (and usually go to excess).

How about some quantification of this cycle? In mid-1998, just before the collapse of Long-Term Capital Management brought investors other than techies to their senses, only \$12.5 billion of defaulted bonds yielded more than 20% (one possible threshold for the label “distressed debt”). Because investors weren't very worried about risk, they demanded ultra-high returns from relatively few non-defaulted bonds; the word “blithe” might best describe their attitude.

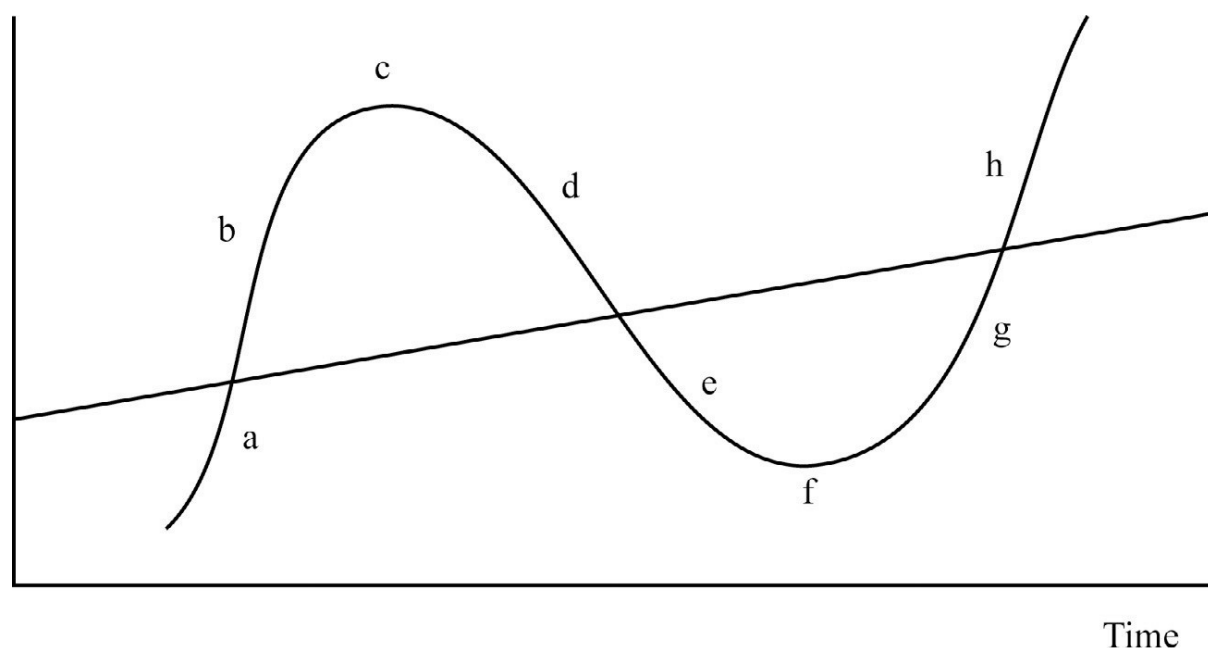
But Long-Term's demise awakened investors to the existence of risk, and a year later, the amount of bonds yielding more than 20% had more than tripled to \$38.7 billion. By mid-2002, when the corporate scandals held the debt market in a grip of terror, yielders had grown to \$105.6 billion, eight and a half times just four years earlier. Risk aversion had come a long way from inadequate and, as later events showed, had become excessive. E

March 31, 2004, this figure had fallen 85%, back to just \$16.2 billion; risk aversion had subsided (and possibly had become inadequate again). I'm sure that fundamentals didn't fluctuate anywhere near the degree reflected in prices, yields and thus the distressed debt to usual, reality was greatly exaggerated by swings in psychology.

When investors in general are too risk-tolerant, security prices embody more risk than they do return. When investors are too risk-averse, prices can offer more return than risk.

The title of the memo cited above—"The Happy Medium"—was by my mother's wisdom and her constant reminders that we should avoid extremes in behavior. Rather we should tend toward the middle things—toward a reasonable balance between too much and too little.

But my experience as an investor convinces me that the happy medium is rarely seen. If you reflect on what I said about the graph of a cyclical phenomenon back on [page 25](#), you may be impressed to find that in stages "a," "g," cyclical phenomena tend to return from extremes and move toward the more reasonable mean. What a rational thing that is!



But then, as I pointed out, it's usually the case that those "cyclical extremes" continue past the fair midpoint in stages "b," "e" and "h," toward the opposite extreme.

A statistician who looks at the above graphic will tell you that *on average* the phenomenon charted is at the central value or astride the fair midpoint. But the rest of us see it as almost always in motion: back toward those midpoints. In fact, it spends about as much time

highs and lows as it does at the happy medium. Most investors' attitudes toward risk do the very same.

What's the greatest source of investment risk? Does it come from economic developments? Corporate events that fall short of forecasts? Companies whose products become uncompetitive? Earnings declines? creditworthiness? No, it comes when asset prices attain excessively high levels as a result of some new, intoxicating investment rationale that can't be justified on the basis of fundamentals, and that causes unreasonable valuations to be assigned. And when are these prices reached? When risk aversion and caution evaporate and risk tolerance and optimism take over. This condition is the investor's greatest enemy.

What Happens When Risk Aversion Is Insufficient

The Global Financial Crisis of 2007–08 represented the greatest financial downswing of my lifetime, and consequently it presents the best opportunity to observe, reflect and learn. The scene was set for its occurrence by a series of developments. Here's a partial list:

- Government policies supported an expansion of home ownership—which by definition meant the inclusion of people who historically couldn't afford to buy homes—at a time when home prices were soaring;
- The Fed pushed interest rates down, causing the demand for mortgage yielding instruments such as structured/levered mortgage securities to increase;
- There was a rising trend among banks to make mortgage loans, package them and sell them onward (as opposed to retaining them);
- Decisions to lend, structure, assign credit ratings and invest were based on the basis of unquestioning extrapolation of low historic default rates;
- The above four points resulted in an increased eagerness to extend mortgage loans, with an accompanying decline in lending standards;
- Novel and untested mortgage backed securities were developed that promised high returns with low risk, something that has great appeal in non-skeptical times;
- Protective laws and regulations were relaxed, such as the Glass-Steagall Act (which prohibited the creation of financial conglomerates), the uptick rule (which prevented traders who had bet against a stock from forcing them down through non-stop short selling), and the rules that limited banks' leverage, permitting it to nearly triple;

- Finally, the median articles stating that risk had been eliminated by a combination of:
 - the adroit Fed, which could be counted on to inject stimulus whenever economic sluggishness developed,
 - confidence that the excess liquidity flowing to China for its exports and to oil producers would never fail to be recycled back into the markets, buoying asset prices, and
 - the new Wall Street innovations, which “sliced and diced” risk so finely, spread it so widely and placed it with so many holders that no one could bear it.

The existence of all the above elements indicated the presence of a high level of risk tolerance. In fact, they couldn't have arisen if risk tolerance hadn't been part of the psyches of investors, lenders, borrowers and regulators. The existence of such risk tolerance like that seen in the years immediately preceding the crisis should be very worrisome, as it implies an absence of worry and healthy skepticism.

It is inescapable that these developments—and the risk tolerance or obliviousness that was behind them—ultimately would lead to unsound financial behavior, particularly via the issuance of financial instruments that were unsound and likely to fail. The ability to borrow large amounts of money at low interest rates caused asset buyers to consider the period of low rates as a permanent feature. But it wasn't marked by the availability of sound, bargain-priced assets. Rather, the ready availability of leverage made it easy to invest heavily in assets whose prices had risen a great deal, and in innovative, untested, synthetic, levered investment products, many of which would go on to fail.

Perhaps most importantly among the contributing factors, the period was marked by risky behavior on the part of financial institutions. When a period is characterized by benign macro events, hyper-financial activity, rapid technological innovation, there is a tendency for providers of capital to compete for a share in a process I call “the race to the bottom” (I'll make a reference to a memo of that name). The mood in the years 2005–07 was captured by Citigroup CEO Charles Prince in June 2007, virtually on the eve of the Financial Crisis, in a statement that became emblematic of the period: “The music stops, in terms of liquidity, things will be complicated. But as long as the music's playing, you've got to get up and dance. We're not dancing.”

In other words, banks had to do—and were doing—things that depended for their success on the continuation of abnormally favorable conditions. Those conditions that would turn into problems if those conditions normalized. But no one could decline to participate for fear of losing market share. The instruments were untested and potentially defective, but no one was willing to

share. That's the kind of crowd behavior that typifies . . . exacerbates cycles.

In theory a bank CEO could have declined to join in this for the realities of the times, anyone who sat out the dance, lost failed to rake in the "easy money" that his competitors were reaping forced out of his job by activist investors. Thus banks bid aggressive opportunity to provide capital as if the music would never stop. cognizance of cycles makes it clear that eventually it will. This tolerance and risk obliviousness plays an essential part in the up-p precedes—and sets the scene for—every dramatic down-phase.

As the period 2005–07 was rolling along, it presented a observe events that made manifest market participants' attitudes and to reach helpful conclusions. I believe the following exce Race to the Bottom," a memo I wrote on the subject in few months before the first indication that bad times were coming—p an excellent example. It demonstrates the potential value of infer from isolated and perhaps anecdotal experiences:

While the last few years have given me many opportunities to marvel at excesses in the capital markets, in this case the one t battle cry—"that calls for a memo"—hit the newspapers in Englan during my last stay. As the *Financial Times* reported on Novem 2006:

Abbey, the UK's second-largest home loans provider, has raised the standard amount it will lend homebuyer to five times either their single or joint salaries, eclipsing the traditional borrowing levels of around three and a half times salary. It followed last week's decision by Bank of Irel Mortgages and Bristol and West to increase standard salary multiples from four to 4.5 times.

In other words, there had been a traditional rule of thumb s borrowers can safely handle mortgages with a face amount equal t three-plus times their salaries. But now they can have five times—roughly 50% more. What inference should be drawn? There are a least four possibilities:

- the old standard was too conservative, and the new one's right;
- conditions have changed, such that the new standard is as conse for today as the old one was for its times;

- it's reasonable for mortgage lenders to accept higher default rates and thus lower net returns, because their cost of capital has risen.
- the rush to place money has caused a supplier of capital to lower standards.

Now, I am no expert on the U.K. mortgage market, and it is not my intention in this memo to comment on general capital markets or not any one sector. Further, it's certainly true that today's low interest rates mean a given salary can support a bigger mortgage than that's likely to hold true so long as (1) borrowers keep their jobs and (2) their mortgages carry fixed rates). But if you think Abbey's rush for taking this step might be a logical one like that, the question is "why now?"

Logical reasons and sober decision making might be involved. But so might competition to put out money and the usual late-stage belief that "it's different this time." Lenders and investors invariably depart from time-honored disciplines when cycles move to extremes. It's out of a belief that current conditions are different from those that prevailed in the past, when those disciplines were appropriate. And just as invariably, they're shown that cycles repeat and nothing changes.

What did we see in the U.S. mortgage market as home prices rose and interest rates declined? First, low teaser rates. Then higher loan-to-value ratios. Then 100% financing. Then low-amortization loans. Then no-amortization loans. Then loans requiring no documentation of employment or credit history. All these things made it possible for more buyers to stretch for more expensive homes, but at the same time they made mortgages riskier for lenders. And these developments took place when home prices were sky-high and interest rates were at multi-generation lows. In the end, buyers took out the biggest mortgage possible given their incomes and prevailing interest rates. Such mortgages would land them in the houses of their dreams . . . and leave them there for as long as conditions deteriorate, which they invariably do.

Do you remember the game Bid-a-Note from the TV show *Name That Tune*? Contestant x said, "I can name that tune in six notes." Then contestant y said, "I can name that tune in five notes." Then contestant x said, "I can name that tune in four notes." The contestant who eventually got the chance to guess the name of the tune was the one who was willing to accept the riskiest proposition—to take the basis of the least information.

So the Bank of Ireland entered the competition to lend money on home purchases and said, "I'll lend four and a half times the borrower's salary." And Abbey said, "I'll lend five times." The so-called winner in this auction is the one who'll put out the money with the least safety. Whether that's really winning or losing will become clear when the cycle turns, as it did in the U.S. last year. Certainly there's a race to the bottom going on. . . . It's the institution that'll make loans with the slightest margin for error. . . .

Any way you slice it, standards for mortgage loans have dropped in recent years, and risk has increased. Logic-based? Perhaps. Cycle-induced (and exacerbated)? I'd say so. The *FT* quoted John-Paul Crutchley, a banking analyst at Merrill Lynch, as saying that Abbey are [*sic*] lending a multiple of five times salary, that's perfectly sensible—or it could be tremendously risky." Certainly mortgage lending was made riskier. We'll see in a few years whether that was intelligent risk taking or excessive competitive ardor. . . .

Today's financial market conditions are easily summed up: There's a global glut of liquidity, minimal interest in traditional investments, little apparent concern about risk, and skimpy prospective returns everywhere. Thus, as the price for accessing returns that are potentially adequate (but lower than those promised in the past), investors are readily accepting significant risk in the form of heightened leverage, untested derivatives and weak deal structures. The current cycle isn't unusual in its form, only its extent. There's a little mystery about the ultimate outcome, in my opinion, but at any point in the cycle it's the optimists who look best.

As is often the case, I could have made this a shorter memorandum simply invoking my two favorite quotations, both of which have a place here.

The first is from John Kenneth Galbraith, who passed away last year. I was fortunate to be able to spend a few hours with Mr. Galbraith a year and a half earlier and to have the benefit of his wisdom firsthand. This quote, once again, is from his invaluable book *A Short History of Financial Euphoria*. It seems particularly apt under the current circumstances:

Contributing to . . . euphoria are two further factors noted in our time or in past times. The first is the extreme brevity of the financial memory. In consequence, financial disaster is quickly forgotten. In further consequence, when the same or closely similar circumstances occur again, sometimes

in only a few years, they are hailed by a new, often youthful, and always supremely self-confident generation as a brilliantly innovative discovery in the financial and larger economic world. There can be few fields of human endeavor in which history counts for so little as in the world of finance. Past experience, to the extent that it is part of memory at all, is dismissed as the primitive refuge of those who do not have the insight to appreciate the incredible wonders of the present.

The second is Warren Buffett's bedrock reminder of the need to adjust our financial actions based on the investor behavior playing out around us. Fewer words, but probably even more useful:

The less prudence with which others conduct their affairs, the greater the prudence with which we should conduct our own affairs.

This memo can be summed up simply: there's a race to the bottom going on, reflecting a widespread reduction in the level of prudence on the part of investors and capital providers. No one can point out that those who participate will be punished, or that their long-run performance won't exceed that of the naysayers. But that is the usual pattern.

If you refuse to fall into line in carefree markets like today's, it's likely that, for a while, you'll (a) lag in terms of return and (b) look like an old fogey. But neither of those is much of a problem. It means keeping your head (and capital) when others eventually lose theirs. In my experience, times of laxness have always been followed by corrections in which penalties are imposed. It may not happen this time, but I'll take that risk. In the meantime, Oaktree and its people will continue to apply the standards that have served us well over the last twenty years.

Warren Buffett puts it well in the above quote; it's one I think it aptly sums up this phenomenon, as well as the contrarian behavior that is required as a result. When others fail to worry about risk and don't apply caution, as Buffett says, we must turn more cautious. But it can be said that when other investors are panicked and depressed and can't imagine conditions under which risk would be worth taking, we should be more aggressive.

What Happens When Risk Aversion Is Excessive

The symmetry of financial cycles, described at length in chapter 1, was guaranteed that the risk-tolerant environment of 2005–07—and the boom in financial instrument issuance that it abetted—would be followed by a sharp correction. And, of course, such a correction materialized.

As I said earlier, the Global Financial Crisis of 2007–08 provides a hope will turn out to be a once-in-a-lifetime opportunity to witness an irrational swing of the pendulum of emotion to total negativity, and the cycle in attitudes toward risk in the direction of excessive risk aversion.

The actions that had been abetted by excessive risk tolerance in the years as just described—were exposed as having been foolhardy, and as they brought great pain and loss:

- It became clear that the effort to expand home ownership had led to purchases of homes by many people who couldn't afford them. Thousands lost the money they had put into home equity, along with moving costs and any improvements they made.
- Sub-prime mortgages that had been issued without proof of income and employment were shown to have been unwise.
- Weak mortgage lending practices—along with extensive mortgage fraud—that those practices failed to catch—turned out, not surprisingly, led to the issuance of many mortgages on which borrowers were unable to make the required payments.
- The extrapolation of low historic default rates on mortgages had ignored the possibility that weakened mortgage lending practices would result in defaults at unprecedented rates.
- Since actual defaults exceeded the historic rates that had been the basis for security structuring decisions, debt ratings and loss projections, structured and levered securities that had been built from sub-prime mortgages also defaulted in stunning numbers, proving their high ratings to have been erroneous.
- Levered mortgage backed securities (and financial derivatives, most of which contain high levels of built-in leverage) generally switched from being return-enhancing tools to weapons of financial mass destruction, as levered funds and securities breached loan covenants, and issuers ultimately proved unable to service their debt.
- Of course, the new financial products demonstrated—as usual—that financial innovations promising high returns with low risk rarely keep that promise.

- As for the relaxed regulations, the financial conglomerates permitted the revocation of the Glass-Steagall Act had extensive problems; the uptick rule allowed the stocks of financial institutions to go down relentlessly; and several banks proved unable to survive high levels of leverage that had been allowed.

Because of everything that had gone before, the consequences included massive mortgage defaults and home repossessions; downgradings and failures on the part of mortgage backed securities; collapsing home prices; the inability to sell existing homes; collapsing stock and corporate bond markets and the disappearance of liquidity; a total drying up of credit availability; and failures, bailouts and bankruptcies at a number of major financial institutions.

Of course, it's improbable events that brought on the credit crunch. Lots of bad things happened that had been considered unlikely (if not impossible), and they happened at the same time, to investors who'd taken on significant leverage. ("The Limits to Negativism," October 2008)

What was the cumulative psychological effect of all of the above on investors and other financial-system participants? In short, it scared them to death. When total fear replaces a high degree of confidence, excessive risk aversion takes the place of unrealistic risk tolerance. And that's what happened in late 2008 following the bankruptcy of Lehman Brothers. Investors came forward in droves, exhibiting a high degree of urgency. They stayed on the sidelines. Asset prices collapsed. And market liquidity went to zero.

All these things resulted from the replacement of high risk tolerance with high risk aversion. And they all contributed to still more fear, still more risk aversion, further negative events, and widespread fear of more of the same. The same people who had bought untried instruments on the basis of optimistic assumptions and promises that were too good to be true now were convinced that the entire financial system might melt down.

To provide an example of the swing toward excessive risk aversion and its impact, I'll share the story of something that happened a few weeks after the Lehman bankruptcy. It's the event that inspired me to write my low-key memo quoted above, "The Limits to Negativism."

Participating to a minor degree in the general trends that were sweeping the industry, Oaktree formed its first levered funds in the years just preceding the crisis. We used less leverage than others—for example, four times equity in our first European senior loan fund versus the more conventional seven or eight times. We tried to be conservative about what assets we bought, but even so, we nevertheless brought us to the brink of a meltdown. Prior to

or “leveraged” loans—even those with credit problems—had rarely traded prices below 96 cents on the dollar. Thus we felt we were well insulated from the possibility of margin calls (demands from lenders for additional capital) that, under our borrowing agreement, could come only if the average market price of the loans in the portfolio got down to 88.

But in the aftermath of the Lehman bankruptcy, loan prices fell to unprecedented levels, pressured by, among other things, banks’ portfolios abandoned by levered holders who received margin calls on their own and failed to meet them. Thus 88—and a margin call and melt-down—became a real possibility for us. We were able to get time to talk to our lender, and we set about raising additional equity from the fund’s investors, with which to reduce the fund’s leverage from 4-to-1 to 2-to-1. We asked the investors to put up more additional equity; they understood that this represented an opportunity to retain loans at discounted prices rather than give up on them; to enjoy the loans’ high implied yields; and to benefit from the fund’s low-cost leverage. Thus most of them came forward with the equity we requested. At the newly reduced level of leverage, the fund was protected from a margin call unless the average price of our loans fell to an unimaginable 65.

But with the total absence of buyers and the continuation of the sell-off and hedge-fund-withdrawal-related selling, the loan market continued to spiral downward, as the notion of “the right price” gave way to widespread concern that no price could be counted on to hold. Thus the average price of our loan portfolio neared 70. It fell to me to get the leverage ratio down from 2 to 1 to 1-to-1, in which case we could completely eliminate the contractual covenant that introduced the risk of a margin call.

Now I was offering the fund’s investors a chance to put up additional loans at yields to maturity that were well into double digits, and with potential returns on the overall fund in the 20s (before fees and potential defaults). Of course, if a pre-existing investor failed to put up his share of the additional equity and allowed someone else to do so, that would be tantamount to selling off part of his interest in the fund at those yields.

And yet, the combination of non-stop price declines, portfolio liquidation, and a total absence of buyers made it challenging for some fund managers to take the step of again adding capital. Some were fatigued from having to deal with the issues popping up everywhere in their portfolios. Some viewed the situation as a chance not as rescuing their investments, but as possibly “throwing good money after bad.” Some didn’t have liquid funds on hand. And some just didn’t have the willingness to defend additional investment to their bosses. At the bottoms, it can be extremely hard to take actions that require great fortitude and staunchness. And that led to the event I’m going to describe.

I went to a pension fund that was an investor in the fund for an additional equity investment. The yields I laid out were attractive, admitted, but they were worried about the possibility of loan defaults. The conversation went like this:

Pension fund: What about the potential that defaults will render the investment unsuccessful?

HM: Well, our average default rate over the last 26 years in high yield bonds—which are junior in the capital structure to loans like the ones we have—has been about 1% a year (and bear in mind that there are recoveries in the case of default, meaning our credit losses have been less than a percentage a year). Thus defaults at our historic rate would do little to diminish the promised return in the 20s.

Pension fund: But what if it's worse than that?

HM: The worst five-year period we've ever had showed defaults averaging 3% per year; obviously not a problem relative to the yields we're earning about.

Pension fund: But what if it's worse than that?

HM: The average default rate in the high yield bond universe—without assuming any ability to avoid defaults through skillful credit selection—has been 4.2% per year. Resulting credit losses of 2–3% clearly wouldn't do much to jeopardize the results on this investment.

Pension fund: But what if it's worse than that?

HM: The worst five years in history for the universe averaged 7.5% defaults, not a problem.

Pension fund: But what if it's worse than that?

HM: The worst one-year default rate in high yield bond history was 10%. That still leaves plenty of return here.

Pension fund: But what if it's worse than that?

HM: One and a half times the worst year in history would be 15% defaults. That would still make a little money given the portfolio yield in the 20s. Even with such a minimal return to be the result, defaults of that order would have to happen every year—not just once.

Pension fund: But what if it's worse than that?

At this point I asked, "Do you have any equities?" and I told them I did—and really believed the doomsday scenarios toward which they had been pushing me—they'd better leave the room immediately and sell them all.

My point is that, in a negative environment, excessive risk aversion can cause people to subject investments to unreasonable scrutiny and make negative assumptions (just as they may have performed little or no due diligence and applied rosy assumptions when they made investments in the preceding heady times). During panics, people spend 100% of their time making

there can be no losses. . . . at just the time that they should be worried about missing out on great opportunities.

In times of extreme negativism, exaggerated risk aversion is likely to drive prices to already be as low as they can go; further losses to investors are avoided, and thus the risk of loss to investors is minimal. As I've indicated earlier, the most common thing in the world is the belief that there's no risk. By the same token, the safest (and most rewarding) time to buy usually comes when everyone is so convinced there's no hope.

If I could ask only one question regarding each investment I have under consideration, it would be simple: How much optimism is factored into the price? A high level of optimism is likely to mean the favorable possibilities have been priced in; the price is high relative to intrinsic value, and there's little margin for error in case of disappointment. But if optimism is low or absent, it's likely that the price is low; expectations of future negative surprises are unlikely; and the slightest turn for the better will result in appreciation. The pension fund meeting described above was important for the simple reason that it indicated that all optimism had been wrung out of investors' thinking.

After that meeting, I practically ran to my office to write "The Limits of Negativism" at roughly the point in that cycle when one could have bought the most debt at the lowest prices. In it I shared the

Lots of bad things happened that had been considered unlikely (if not impossible), and they happened at the same time, to investors who'd been taken on significant leverage. So the easy explanation is that the people who were hurt in the credit crisis hadn't been skeptical—or pessimistic—enough.

But that triggered an epiphany: *Skepticism and pessimism aren't synonymous. Skepticism calls for pessimism when optimism is excessive. But it also calls for optimism when pessimism is excessive.*

I'll write some more on the subject, but it's really as simple as that. Contrarianism—doing the opposite of what others do, or "leaning against the wind"—is essential for investment success. But as the credit crisis reached a peak last week, people succumbed to the wave rather than resisting. I found very few who were optimistic; most were pessimistic to some degree. Some became genuinely depressed—even a few great investors I know. Increasingly negative tales of the coming meltdown were exchanged via email. *No one applied skepticism, or said "that horror story's unlikely to be true."* Pessimism fed on itself. People's only concern was bullet-proofing their portfolios to get through the coming collapse, or raising cash to meet redemptions. The one thing they weren't doing last week

was making aggressive bids for securities. So prices fell and fell several points at a time—the old expression is “gapped down.”

Hopefully this contemporary account will give you a sense for what excessive, unreasonable risk aversion feels like, and also a sense for what you should be doing under circumstances like those.

Postscript: a few of the fund's investors (including the one I met that day) declined to post the additional equity. Feeling that I should have been able to keep the fund afloat, I put it up in their place. The result was a levered portfolio of depressed senior loans at a time of high risk aversion made that one of the best investments I've ever made. . . . The unwillingness of others to participate in that market had rendered it absurdly cheap.



This chapter regarding the cycle in attitudes toward risk has grown to be one of the longest in this book. There's a good reason for that: it's one of the most important cycles. In *The Most Important Thing*, I talk about the importance of knowing where we stand in the various cycles. Understanding how investors are thinking about and dealing with risk is perhaps the most important thing to strive for. In short, excessive risk tolerance contributes to the creation of danger, and the swing to risk aversion depresses markets, creating some of the greatest buying opportunities.

The rational investor is diligent, skeptical and appropriately risks at all times, but also on the lookout for opportunities for potential return that more than compensates for risk. That's the ideal. But in good times, when most people say, “Risk? What risk? I don't see much that could go wrong,” you look how well things have been going. And anyway, risk is my friend. The more risk I take, the more money I'm likely to make.” Then, in a moment, they switch to something simpler: “I don't care if I ever make money in the market; I just don't want to lose any more. Get me out!”

It's essential to note that since rational, unemotional investors are so much in the minority, the totality of investors rarely achieve equilibrium regarding their attitudes toward risk, or any of the other aspects of the cycle, or emotion as to which the cycle oscillates or the pendulum swings. It's not to maintain a healthy balance between the risk aversion that urges them to be cautious and the risk tolerance that urges them on: usually the other is in the pronounced ascendancy. The same is very much true with regard to greed and fear, skepticism and credulousness, the willingness to sometimes see only positives and sometimes only negatives, and many other

things. The cycles in psychology amply demonstrate that investors spend little of their time at the happy medium.

The fluctuation—or inconstancy—in attitudes toward risk is both the cause of some cycles and the cause or exacerbator of others. And it goes on, since it seems to be hard-wired into most people's psyches to be more optimistic and risk-tolerant when things are going well, and then more worried and risk-averse when things turn downward. That means they're willing to buy when they should be most cautious, and most reluctant to sell when they should be most aggressive. Superior investors recognize this and strive to behave as contrarians.

IX

THE CREDIT CYCLE



Superior investing doesn't come from buying high-quality assets, but from buying when the deal is good, the price is low, the potential return is substantial, and the risk is limited. These conditions are much more the case when the credit markets are in the less-euphoric, more-stringent part of their cycle. The slammed-shut phase of the credit cycle probably does more to make bargains available than any other single factor.

Now our foundation is complete. We've covered the economic cycles that provide the fundamental backdrop for investment activity, also the psychological and attitudinal swings that occur in response to changes in fundamentals (and that tend to exaggerate them). Now to take up some specific types of financial cycles. You will note that fluctuations in all the above strongly affect the cycles covered in the following chapters.

As we've already discussed, some activities—like home buying—are highly responsive to movements in the economic cycle, and others—like purchasing food—are not. Some cycles have a profound effect on other aspects of the economy and on other cycles, and some do not. In this chapter, the credit cycle is in each case the former: it is responsive to economic developments and highly influential. Lastly, it is extremely volatile. Thus its movements are powerful and extreme, and greatly affect activity in many other areas. And all these things are exacerbated by the swings of psychology described in chapters VII and VIII.

Please note that the subject here is sometimes referred to as the capital market cycle rather than the credit cycle. I don't find the distinction important. Strictly speaking, "capital" refers to all the money used by a business, while "credit" refers to that portion of a company's capital made up of debt rather than equity. In practice, these two cycle names can be used interchangeably, although I do find many fewer references to the capital market cycle. I'm going to stick to "the credit cycle" when

talking about debt markets, and I may lapse into “the capital market” when I’m talking about the general availability of financing. But either way, and most importantly, the considerations that apply to one apply equally to the other.

The following passage from my memo “Open and Shut” (December 2000) calls on some of what I wrote earlier and places the credit cycle within the range of cycles:

Consider this: the ups and downs of economies are usually based on fluctuations in corporate profits, and fluctuations in profits for the stock and fall of securities markets. However, in recessions and recoveries, economic growth usually deviates from its trendline rate by only a few percentage points. Why, then, do corporate profits increase and decrease so much more? The answer lies in things like financial leverage and operating leverage, which magnify the impact on profits of rising and falling revenues.

And if profits fluctuate this way—more than GDP, but still relatively moderately—why is it that the securities markets soar and collapse so dramatically? I attribute this to fluctuations in the availability of capital, and, in particular, to the profound influence of psychology on the availability of capital.

In short, whereas economies fluctuate a little and profits a lot, the credit window opens wide and then slams shut. . . . Thus, the credit cycle is the most volatile of all cycles, and has the greatest impact. Thus it deserves a great deal of attention.

And here’s how I put it, more succinctly, in “You Can’t Predict. You Can Only Prepare.” (November 2001):

The longer I’m involved in investing, the more impressed I am by the power of the credit cycle. It takes only a small fluctuation in the economy to produce a large fluctuation in the availability of capital, with great impact on asset prices and back on the economy itself.

Changes in the availability of capital or credit constitute one of the most fundamental influences on economies, companies and markets. Even though the credit cycle is less well-known to the man on the street than most other cycles discussed in this book, I consider it to be of the greatest importance and profound influence.

As suggested in the citation just above, the credit cycle can be understood through the metaphor of a window. In short, sometimes it’s open and sometimes it’s closed. And, in fact, people in the financial

frequent reference to just that: “the credit window,” as in “the place to borrow money.” When the window is open, financing is plentiful and obtained, and when it’s closed, financing is scarce and hard to come by. It’s essential to always bear in mind that the window can go from being open to slammed shut in just an instant. There’s a lot more to this cycle—including the reasons for these cyclical movements and their impact—but that’s the bottom line.



Why does this cycle have the importance I ascribe to it? First, capital is an essential ingredient in the productive process. Thus the ability of companies (and economies) to grow usually depends on the availability of incremental capital. If the capital markets are closed, it can stunt growth.

Second, capital must be available in order for maturing debt to be refinanced. Companies (as well as most other economic units, such as governments and consumers) generally don’t pay off their debts. Most of the time they merely roll them over. But if a company is unable to do so at the time that its existing debt comes due, it may default and go into bankruptcy. Where we stand in the credit cycle—whether credit is readily available or difficult to obtain—is the greatest determinant of whether debt can be refinanced at a given time.

Many corporate assets are long-term in nature (like buildings, machinery, vehicles and goodwill). Yet corporations often raise the money to buy those things by issuing short-term debt. They do this because the cost of borrowing is generally lowest on short maturities. This arrangement—“borrowing short to invest long”—works out well most of the time, but when the credit market is open and fully functioning, meaning debt can be rolled over with ease when it comes due. But the mismatch between long-term assets that can’t be easily liquidated and shorter-term liabilities can easily bring on a crisis if the credit cycle turns negative so that maturing debt can’t be refinanced. This classic mismatch, when combined with stringency in financing markets, is often the cause of the most spectacular financial meltdowns.

When the Global Financial Crisis began to build in 2007 and the credit markets froze up, the U.S. Department of the Treasury took the unprecedented step of guaranteeing all commercial paper. If it weren’t for these debt instruments—which have maturities of 270 days or less—might not have been capable of being rolled over, and thus could have caused defaults among even the strongest of companies. In fact, the defaults well have been concentrated among top companies, since they issue so much commercial paper in billions precisely for the reason that their strong

creditworthiness gives them easy access to the commercial paper market. (The importance of the market being open—and of the ability to maturing paper—underlines the essential difference between positive worth and liquidity. Even a wealthy company can get into trouble have cash on hand and can't obtain enough to meet its debt maturities and other calls on cash as they arise.)

Third, financial institutions represent a special, exaggerated case of on the credit markets. Financial institutions are in the business of money, and they need access to financing to keep that business also are often the site of the greatest short/long mismatches and potential meltdowns. Consider, for example, the bank that takes deposits that can be withdrawn any day and uses them to make mortgage loans that won't be repaid for thirty years. What happens if all the depositors demand their money back on the same bad day (a "run on the bank")? If there's no credit market (and no government bailout), that bank may fail.

Fourth and finally, the credit market gives off signals that have a psychological impact. A closed credit market causes fear to spread of proportion to businesses' negative realities. Difficult conditions can lead the capital market to close . . . and closed capital markets have a negative impact on business conditions (as well as on market participants and businesses). This kind of "vicious circle" is part of most financial

The Credit Cycle in Operation

By now you should have an understanding of the nature and the credit cycle. The next topic to tackle is why credit cycles occur: what causes credit to be more available at some times and less available at other times?

The credit window doesn't have a mind of its own, opening and closing on its own accord. Rather, it follows events elsewhere. In "You Can Prepare." (November 2001), I put the expand-and-contract cycle in context and explained it in some detail:

The process is simple:

- The economy moves into a period of prosperity.
- Providers of capital thrive, increasing their capital base.
- Because bad news is scarce, the risks entailed in lending and investing seem to have shrunk.
- Risk averseness disappears.

- Financial institutions move to expand their businesses—that is, to provide more capital.
- They compete for market share by lowering demanded returns (e.g., cutting interest rates), lowering credit standards, providing more capital for a given transaction, and easing covenants.

At the extreme, providers of capital finance borrowers and provide loans that aren't worthy of being financed. As *The Economist* said earlier this year, “the worst loans are made at the best of times.” This leads to capital destruction—that is, to the investment of capital in projects where the cost of capital exceeds the return on capital, and eventually to cases where there is no return on capital.

When this point is reached, the up-leg is reversed.

- Losses cause lenders to become discouraged and shy away.
- Risk averseness rises, and along with it, interest rates, credit restrictions and covenant requirements.
- Less capital is made available—and at the trough of the cycle, only to the most qualified of borrowers.
- Companies become starved for capital. Borrowers are unable to roll over their debts, leading to defaults and bankruptcies.
- This process contributes to and reinforces the economic contraction.

Of course, at the extreme the process is ready to be reversed. Because the competition to make loans or investments is low, high returns can be demanded along with high creditworthiness. Contrarians who commit capital at this point have a shot at high returns, and those tempting potential returns begin to draw in capital. In this way, a recovery begins to be fueled.

Sometimes people are eager to put money to work, and that makes the credit window open wide. But when circumstances cause them to change their minds, financing can become unavailable. As with so many other things in this book, it's essential that the reader gain a clear understanding of the workings of the cycles, and especially of the way in which each cycle leads to the next. Thus the thorough, step-by-step explanation provided here is essential and must be absorbed.

But in “You Can't Predict. You Can Prepare.” I went on to reduce the process to just a few words. They truly do constitute the bottom of the credit cycle, and they make clear the never-ending chain-reaction nature of the cycle:

Prosperity brings expanded lending, which leads to unwise lending which produces large losses, which makes lenders stop lending, which ends prosperity, and on and on.

A market is an auction house where the item being offered goes to the person who'll pay the most for it. The financial markets are no exception. The opportunity to make an investment or provide a loan goes to the participant who'll pay the most for that opportunity. The bidding takes the price to higher dollar levels and higher valuation parameters (such as price/earnings ratios). In the credit market, a high price or high valuation parameter translates directly into a low yield on the debt instrument in question, and the chance to provide capital goes to the lender offering the lowest yield.

My memo "The Race to the Bottom," published in February 2000, was primarily about the eagerness of providers of capital to expand their business in good times, and its effect:

It helps to think of money as a commodity. Everyone's money is pretty much the same. Yet institutions seeking to add to loan volume and private equity funds and hedge funds seeking to increase their fees, all want to move more of it. So if you want to place more money—*that is*, get people to go to you instead of your competitors—financing—you have to make your money cheaper. As with the commodities market, low price is the most dependable route to increased market share.

One way to lower the price for your money is by reducing the interest rate you charge on loans. A slightly more subtle way is to agree to a higher price for the thing you're buying, such as a higher p/e ratio for a common stock or a higher price when you're buying a company. Any way you slice it, you're settling for a lower prospective return. But there are other ways to cheapen your money, and they're the primary subject of this memo.

With debt, structure is important. Thus easier terms for the borrower can bring added risk for the lender. For example, lenders would like protective covenants that limit the extent to which a borrower can engage in certain actions that increase the risk that is present. They may limit the debt the borrower can assume, limit the cash it can pay out in dividends, require that it maintain a certain minimum net worth. But, in a heated credit market, the "best buyer" of debt—the most eager

might be willing to accept a structure with fewer covenants, and risk.

So in an auction, the chance to make a loan or buy a the provider of capital who's willing to accept a combination and the riskiest structure. When risk aversion is present and the lending opportunities is reasonable relative to the supply of opportunities, the bidding is usually prudent. But when risk tolerance over and lenders compete avidly for opportunities, the bidding become overheated. The resulting opportunity to lend is likely to be high a price: a yield that's too low and/or risk that's excessive. Thus an overheated auction in the credit market—as elsewhere—is likely to produce a “winner” who's really a loser. This is the process I call the race to the bottom.

On the other hand, there are times when buyers show up for auctions in small numbers, and the few who do attend are interested in buying at giveaway prices. The bidding stalls, and the result is low prices, low yields, and loan structures that afford excellent protection. Unlike the overheated climate that spawns the race to the bottom, in an ice-cold market which no one's eager to lend can create real winners.

The degree of openness of the credit window depends almost entirely on whether providers of capital are eager or reticent, and it has a significant impact on economies, companies, investors, and the prospective return and riskiness of the investment opportunities that result.

In the short term, the effect of a generous capital market condition is to make more money available to more companies for more reasons, at lower rates of interest and with fewer covenants. This leads to higher levels of acquisitions, buyouts and corporate expansion (not to mention rapid recapitalizations of buyout companies and thus high short-term rates of return). In the short run, this contributes to a high level of general financial activity.

Another effect is to forestall financial stringency at weak companies. When lenders are strict and covenants are tight, operating problems can lead quickly to both technical defaults (violations of covenants) and “money defaults” (non-payment of interest or principal). But looser conditions can permit default to be forestalled if covenants are lax or absent; if borrowers have the option to convert cash-pay bonds into payment-in-kind bonds (through a recent innovation, “toggle bonds”); or if they can raise new money to postpone the day of reckoning.

Eventually, many of the forestalled defaults will demonstrate their inevitability, with the companies falling from more highly leveraged heights. And certainly the capital markets' willingness to finance

than-deserving companies will lead ultimately to a higher level of corporate distress. Thus, everything else being equal, the bigger the boom—and the greater the excesses of the capital markets in the upward direction—the greater the bust. Timing and extent are not predictable, but the occurrence of cycles is the closest thing to inevitable. (“The Race to the Bottom,” February 2007)

The Impact of the Credit Cycle

One of the main points in this book is the extent to which events in the credit cycle have influence in other fields and on other kinds of cycles. Now this is clearer than in the credit cycle.

In “Genius Isn’t Enough” on the subject of Long-Term Capital Management (October 1998), I wrote “Look around the market, there’s a crisis; you’ll probably find a lender.” Over-permissive providers of capital frequently aid and abet financial bubbles. There have been numerous recent examples where loose capital markets contributed to booms that were followed by famous collapses: real estate in 1989–92; emerging markets in 1994–98; Long-Term Capital in 1998; the movie exhibition industry in 1999–2000; venture capital funds and telecommunications companies in 2000–01. In each case, lenders and investors provided too much cheap money and the result was over-expansion and dramatic losses. In the movie *Fields of Dreams* Kevin Costner was told, “If you build it, they will come.” In the financial world, if you offer cheap money, they will borrow and build—often without discipline, and with very negative consequences.

The capital cycle contributed tremendously to the tech bubble. Money from venture capital funds caused far too many companies to be created, often with little in terms of business justification or prospects. Wild demand for IPOs caused their hot stocks to rise meteorically, enabling venture funds to report triple-digit returns and attract still more capital requiring speedy deployment. The greed of the capital markets let telecom companies sign on for huge projects that were only partially financed, secure in the knowledge that more financing would be available later, at higher p/e’s and interest rates as the projects were further along. This ease caused more optical fiber capacity to be built than was needed at the time. A lot of which is sitting idle. Much of the investment that went into

may never be recovered. Once again, easy money has led to capital destruction.

In making investments, it has become my habit to worry about the economic future—which I'm sure I can't know much about—than I do about the supply/demand picture relating to capital. Being positioned to make investments in an uncrowded arena conveys vast advantages. Participating in a field that everyone's throwing money at is a formula for disaster. ("You Can't Predict. You Can't Predict.")

To restate, here's further proof from "The Happy Medium" (July 2001) that a lengthy and important description like that above with regard to the process can be summed up in fewer words:

From time to time, providers of capital simply turn the spigot off—as in so many things, to excess. There are times when a borrower can get any amount of capital for any purpose, and times when even the most deserving borrowers can't access reasonable amounts for worthwhile projects. The behavior of the capital markets is a good indicator of where we stand in terms of psychology and a great contributor to the supply of investment bargains.

I went on in that memo to discuss the way in which the credit cycle contributes to the creation of excess:

Looking for the cause of a market extreme usually requires watching the videotape of the credit cycle a few months or years. Most recent bull markets are abetted by an upsurge in the willingness to supply capital, usually imprudently. Likewise, most collapses are preceded by a wholesale refusal to finance certain companies, industries, or the entire gamut of would-be borrowers.

My goal in what follows is to describe the effect of the credit cycle. I do so I'll return again to the Global Financial Crisis (GFC), since that was the best teaching moment.

In the late 1960s, in my early years as an equity analyst, I learned a lot about the economic cycle and the way corporate profits rise and fall with it. While I had a lot more to learn, I already knew a lot about the psychology and risk attitudes (and about their importance). But I had no appreciation for the role or operation of the credit cycle. In order to fully understand most phenomena in the investment world unless you've lived through them. Now I've come to the conclusion that the credit cycle

really big deal. In fact, when asked about the causes of the Global Crisis of 2007–08, I put it right at the top of the list.

The very choice of the name “Global Financial Crisis” for the 2007–08 experience reflects the fact that it was essentially a *financial* phenomenon—caused almost entirely by events within the financial system—and not one with primarily economic or other origins. Here are the attitudinal and behavioral factors in the financial world that led to reaching crisis:

- The existential cause was the too-liberal attitudes toward financial markets described on [pages 119–120](#).
- Those carefree attitudes were inflamed by strong demand for yielding investments that resulted from the Fed’s lowering of general interest rates.
- Those two factors led to, among other things, an excessive optimism on the part of investors to accept innovative financial products and to swallow whole the favorable extrapolation of history and the optimistic assumptions on which those products were based.
- The predominance of mortgage backed securities among those innovations gave rise to a rapidly increasing need for mortgages which to fashion the new securities.
- That demand facilitated the selling onward of mortgages, which allowed mortgage lenders to be careless in choosing the home buyers to whom they would lend. Since mortgage originators wouldn’t be retaining the mortgages they created, they didn’t have to worry about their soundness. In an extreme example of this trend, a category of “sub-prime” mortgages was created for borrowers who couldn’t satisfy traditional lending standards in terms of employment, income, or who chose to pay higher interest rates rather than these things. The fact that weak borrowers like these could borrow large sums was indicative of irrational credit market conditions.
- Relaxed credit diligence on the part of mortgage lenders, and the availability to home buyers of generous sub-prime financing, made home ownership possible for more Americans than ever, including many who wouldn’t have been able to afford it under the tighter traditional mortgage standards.
- Seduced by potential profits from rating vast numbers of securities backed by sub-prime mortgages (and enabled by their own naiveté and perhaps their greed), the credit-rating agencies competed for business by offering inflated ratings, in a race to the bottom of the scale.
- Home affordability increased substantially, fueled by the fact that interest rates are low on short maturities, as originators hit upon

initial monthly payments as a way to maximize mortgage issuance. This contributed to the widespread use of floating-rate mortgages entailing low starting monthly payments based on low initial “teaser” rates of interest. Obviously, these non-fixed rates posed a potential risk to borrowers who could barely afford their payments before they increased. But borrowers were assured that, thanks to the generous capital market conditions, they would always be able to refinance into yet another mortgage, again with a sub-market teaser rate.

- Investment banks were eager to turn the raw material of plentiful prime mortgages into tranching mortgage backed securities with the highest average credit rating, in order to maximize their salability. Their ardor for this activity at just the time that “financial engineering” came into favor gave rise to ratings for tranches that turned out to be totally divorced from how they actually would perform under stress.
- The investment banks that created and sold these securities often were willing to retain the equity layer at the bottom of the structure in order to facilitate a high volume of issuance or simply to hold high-yielding assets (i.e., even they were oblivious to the nature of their product). And other banks took advantage of the high levels of permissible leverage to create assets at very favorable yield spreads by using low-cost borrowings to buy the risky, high-yielding equity and junior debt tranches of structured mortgage securities.

As you can see from the above, virtually all the conditions on which the GFC was built were endogenous to the financial system and the credit market. The developments that constituted the foundation for the Crisis weren’t caused by a general economic boom or a widespread surge in corporate profits. The key events didn’t take place in the general business cycle or the greater world beyond that. Rather, the GFC was a large phenomenon that resulted entirely from the behavior of financial institutions. The main forces that created this cycle were the easy availability of capital; the lack of experience and prudence sufficient to temper the unbridled enthusiasm that pervaded the process; imaginative financial engineering; the separation of lending decisions from loan retention; and irresponsibility and downright greed.

It must be noted, however, that this chain reaction was abetted by government officials who were eager to expand the American dream of home ownership and naively thought it would be great if everyone was enabled to own a home. In a speech in October 2002, President George W. Bush repeated what had been told by one of his friends: “You don’t have to have a lousy first-time home buyer. If you put your mind to it, the first-time home buyer, even the low-income home buyer can have just as nice a house as anybody else.”

wonder if the people who heard that statement at the time found it as it seems today.

After the GFC hit, Congressman Barney Frank, formerly one of the strongest proponents of broadened home ownership, said: “Home ownership is a good thing. But we have made a great mistake in this society: we allowed people in the society who should not be allowed to borrow to buy their home. And we have pushed people into home ownership who should not be there.” (As you can see, political rhetoric is cyclical, too.)

In other words, the events that provided the basis for the GFC were all about money. The pursuit of money took a powerful upswing, and economic realities that reflect on and constrain the attainment of goals were often ignored. And the caution and risk aversion that usually bear the willingness of market participants to provide money were largely ignored. Thus the capital cycle rose to an irrational extreme, the consequences of which are generally foreseeable.

When the cycle rises to an extreme, it invariably can't stay there for long. Sometimes it corrects under its own weight, and sometimes this happens because of events outside the cycle. In this case it was more the laying of the foundation for the Crisis, the unraveling of which was essentially all financial in nature, although the first step came from the real world.”

- Most influentially, in 2006, sub-prime mortgage borrowers began to default in large numbers. Some of the borrowers, who had received loans without proving their ability to make payments, turned out to be unable. Some loans, secured through fraud, went bad when fictitious borrowers disappeared. Other loans, for the full home purchase price—which had allowed home buyers to gamble on continued home appreciation without risking any money of their own—were also in trouble when the market stopped rising.
- Regardless of the reason, the historic basis that allowed sub-prime mortgage backed securities to achieve high leverage and high returns—insistence that there wouldn't be a nationwide wave of mortgage defaults—failed to hold. As it turned out, lending decisions had been made unwisely, with undue reliance on that history. Importantly, lenders and investors had ignored the chance that such reliance would lead to lending behavior so careless that it in itself would render the system irrelevant.
- Large numbers of mortgage defaults led to downgradings, covenant breaches and payment defaults on mortgage backed securities.
- The downgradings, breaches and defaults caused the prices of mortgage backed securities to collapse, and the resultant loss of confidence in the system.

- the market liquidity for these instruments to dry up.
- With terrified buyers taking to the sidelines—and terrified increasingly eager to sell (or forced to sell by margin calls)—was a dramatic downward spiral in the prices of mortgage backed securities.
 - These negative developments collided head-on with new regulations, designed to increase transparency, which required banks' assets "marked to market." But with prices in free-fall and liquidity existent, it was hard to have faith in any price chosen. When marked down their assets to be appropriately conservative, the losses shocked investors, contributing to further panic, which caused prices to decline further, and so on.
 - In many cases the very viability of banks came into question. Many to be absorbed by other banks (with support from the government) bailed out by the government.
 - Each bank failure, acquisition (at pennies on the dollar) or brought losses to investors and further sapped confidence. In the interlocking relationships among banks caused grave concern regarding the remaining ones' ability to rely on amounts due others. "Counterparty risk" became the newest source of
 - Banks reported massive losses. The rising prices quoted for swaps—derivatives used to bet against banks' creditworthiness—increased odds of insolvency. Shareholders dumped bank stocks in response, forcing down their prices. Short sellers sold unremittingly adding to the downward pressure, rendering their pessimistic predictions self-fulfilling, and further extending the vicious circle.
 - Ultimately, Lehman Brothers was denied absorption or bailout, leading to its bankruptcy. That collapse, when added to the many other unsettling events taking place simultaneously, led to nothing short of
 - Whereas the markets had reacted adversely to the problems in mid-2007 but overlooked the potential for contagion to other late 2008 everyone threw in the towel on everything. The prices of assets other than Treasuries and gold collapsed.
 - Funds that had invested using borrowed capital—"leverage" or "margin"—saw asset values marked down precipitously and received demands from lenders to post additional capital. When they appealed to the banks for more time, the banks generally couldn't or wouldn't provide it. The result was distressed selling of portfolios *en masse*, increased the downward pressure on prices.
 - The capital markets slammed shut in this environment, meaning financing became virtually impossible in all quarters of the markets, even those totally unrelated to homes and mortgages.

- Given the sum of the above, all economic units pulled back, buy, investor expand. The result was an economic retrenchment been labeled “the Great Recession.”

Taken to its extreme in the last fifteen weeks of 2008, the credit cycle seemed universal and unstoppable. Few people were able to imagine any force capable of arresting it or—as described in the book—any hypothetical scenario that was too dire to come true. A total collapse of the financial system was considered a real possibility.

The bottom line is that the willingness of potential providers of credit to make it available on any given day fluctuates violently, with a profound impact on the economy and the markets. There’s no doubt that the recent credit crisis was as bad as it was because the credit markets froze up and capital became unavailable other than from governments. (“Open and Shut”)

I truly believe a system meltdown—with ramifications like the Great Depression—could have occurred. Former Treasury secretary Timothy Geithner’s book *Stress Test* bears this out. Fortunately, however, the U.S. government took steps that turned the tide. These included the guaranteeing of commercial paper, mentioned earlier, as well as the creation of market funds. The bank bailouts showed that help was available, and the September 2008 bankruptcy of Lehman Brothers suggested that the government was differentiating between the banks that were worth saving and those that weren’t. Whereas panicky market participants were convinced that Morgan Stanley was next in line for collapse after Lehman—and Goldman Sachs would follow that—the downward spiral was arrested when Japan’s Mitsubishi UFJ went through with a promised \$9 billion bailout in Morgan Stanley.

Importantly, events in the credit markets eventually demonstrated that credit cycles can’t go in one direction forever, even given widespread catastrophic events. Debt prices were generally in free-fall from the time of the Lehman bankruptcy on September 15 through the end of 2008. By the end of 2009, however, the ingredients for a solid market recovery were in place.

- The over-levered funds that had received margin calls either raised additional capital, sold assets to de-lever as required, or liquidated.
- Funds and investment managers that received notices from investors desiring to withdraw at year-end either put up “gates” to postpone withdrawals or completed the asset sales needed to meet them.

- The prices of debt securities reached a point where they implied so high that selling was unpalatable and buying became attractive.
- And, ultimately, market participants demonstrated that when negative psychology is universal and “things can’t get any worse,” they When all optimism has been driven out, and panicked risk aversion everywhere, it becomes possible to reach a point where prices can’t go any lower. And when prices eventually stop going down, people feel relief, and so the potential for a price recovery begins.

The quoted prices for debt continued downward in the first quarter as composure, confidence and a “base” of buying power had run out. But the ability of investors to buy in large scale dried up began, because of the factors listed just above. And when buying materialized in the second quarter—perhaps because distressed debt came to the realization that they had shrunk unreasonably from the task of “catching a falling knife”—the dearth of supply for a powerful move to the upside.

The Global Financial Crisis shows the credit cycle at the greatest since the Great Depression. Debt markets historically had been marked by general conservatism, meaning excesses on the upside were limited. Bubbles took place in the equity market. Certainly it was the Crash of 1929.

But the creation of the high yield bond market in the late 1970s, liberalization of debt investing, and the generally positive economic environment of the subsequent three decades provided those who ventured with a favorable overall experience. This combination led to a move toward acceptance of low-rated and non-traditional debt instruments.

There were periods of weakness in debt in 1990–91 (related to bankruptcies among the highly levered buyouts of the 1980s) (stemming from excessive borrowing to fund overbuilding in the telecom industry, which led to prominent downgrades that coincided with several high-profile corporate accounting scandals). But the effects of these were limited because of the isolated nature of their causes. It was not until that the financial markets witnessed the first widespread, debt-induced downturn with ramifications for the entire economy. Thus the GFC provided the ultimate example of the credit cycle’s full effect.



As I described it in “Open and Shut,” the capital market operation, and its message is easy to perceive. An uptight, cautious market usually stems from, leads to or connotes things like these:

- fear of losing money
- heightened risk aversion and skepticism
- unwillingness to lend and invest regardless of merit
- shortages of capital everywhere
- economic contraction and difficulty refinancing debt
- defaults, bankruptcies and restructurings
- low asset prices, high potential returns, low risk and excessive premiums

Taken together, these things are indicative of a great time to invest. Of course, however, because of the role played by fear and risk aversion in their creation, most people shy away from investing while they are in force. That makes it difficult for most people to invest in a capital cycle that is negative, just as it is potentially lucrative.

On the other hand, a generous capital market is usually associated with the following:

- fear of missing out on profitable opportunities
- reduced risk aversion and skepticism (and, accordingly, reduced diligence)
- too much money chasing too few deals
- willingness to buy securities in increased quantity
- willingness to buy securities of reduced quality
- high asset prices, low prospective returns, high risk and skimpy premiums

It's clear from this list of elements that excessive generosity in capital markets stems from a shortage of prudence and thus serves to give investors one of the clearest red flags. The wide-open capital market arises when the news is good, asset prices are rising, optimism is riding high, and all things seem possible. But it invariably brings about the issuance of unsound and overpriced securities, and the incurring of debt levels that ultimately will result in ruin.

The point about the quality of new issue securities in a wide capital market deserves particular attention. A decrease in risk aversion and skepticism—and increased focus on making sure opportunities aren't missed rather than on avoiding losses—makes investors open to a greater quantity of issuance. The same factors that make investors willing to buy issues of lower quality.

When the credit cycle is in its expansion phase, the statistics on new issuance make clear that investors are buying new issues in greater amounts. But the acceptance of securities of lower quality

bit more subtle. While there are credit ratings and covenants to look at, it can take effort and inference to understand the significance of these things. In feeding frenzies caused by excess available funds, recognizing and resisting this trend seem to be beyond the ability of the majority of market participants. This is one of the reasons why the aftermath of an overly generous capital market includes losses, economic contraction, and a subsequent unwillingness to lend.

The bottom line of all of the above is that generous credit conditions usually are associated with elevated asset prices and subsequent losses, while credit crunches produce bargain-basement prices and great profit opportunities. (“Open and Shut”)



The ultimate purpose of this book isn't to help you understand cycles that they've taken place, like the Global Financial Crisis as described at length. Rather it is to enable you to sense where we stand in cycles in real time, and thus to take the appropriate action.

The key to dealing with the credit cycle lies in recognizing that it reaches its apex when things have been going well for a while, news has been good, risk aversion is low, and investors are eager. That makes it easy to raise money and causes buyers and investors to compete for the opportunity to provide it. The result is cheap financing, low credit standards, weak deals, and the unwise extension of credit. Borrowers hold the cards when the credit window is wide open—not lenders or investors. The implications of all of this should be obvious: proceed with caution.

The exact opposite becomes true at the other extreme of the cycle. At the nadir is reached when developments are unpleasant, risk aversion is heightened, and investors are depressed. Under such circumstances, no one wants to provide capital, the credit market freezes up, and people go begging. This puts the cards into the hands of providers of capital rather than the borrowers.

Because borrowing is difficult and capital is generally unavailable to those who possess it and are willing to part with it, they can apply rigorous standards, insist on strong loan structures and protective covenants, and demand high prospective returns. It's things like these that provide the margin of safety required for superior investing. When these boxes can be ticked, investors should swing into an aggressive mode.

Superior investing doesn't come from buying high-quality assets, but from buying when the deal is good, the price is low, the potential return is substantial, and the risk is limited. These conditions are much more likely to be found when the credit markets are in the less-euphoric, more-stringent part

cycle. The slammed-shut phase of the credit cycle probably does more to make bargains available than any other single factor.

X

THE DISTRESSED DEBT CYCLE



Few lenders and bond buyers are imprudent enough to advance money that won't be repaid if conditions remain as they are. In sober times, they insist on a margin of safety sufficient to ensure interest and principal will be paid even if conditions for the company deteriorate.

But when the credit market heats up—when the race to the bottom causes avid lenders to finance less-deserving borrowers and weaker debt structures—bonds are issued that lack that margin of safety and won't be able to be serviced if things get a little worse. This is the unwise extension of credit. This process, as with Oaktree, “stacks the logs in the fireplace” for the next bonfire.

I was fortunate to partner with Bruce Karsh three decades ago, and in 1988 we formed our first fund for distressed debt investing, which I believe was one of the very first from a mainstream financial institution. It took us into a highly specialized investment niche.

Rather than companies that are doing well or have bright futures, our distressed debt investments are generally in companies that are doing so poorly that they've defaulted on their outstanding debt or are considered highly likely to do so: they're either in bankruptcy or viewed as such. To be clear, our typical company isn't challenged operationally, just overloaded with debt; thus our mantra is “good company, bad balance sheet.”

Normally investors buy debt securities or make loans because they expect to be paid interest periodically and have their principal repaid when the security matures. With distressed debt, however, the consensus is that these things won't occur: instead it is expected that the debt won't “stay current” or be serviced.” So if interest and principal aren't expected to be paid, what is a distressed debt investor's motivation?

The answer is that debtholders who aren't paid as scheduled have a “creditor claim” against the debtor. In short—and to over-simplify—the company goes through bankruptcy, the old owners are wiped out and

creditors become the new owners. Each creditor receives his share of the value of the company—depending on the amount and seniority of claims he holds—in some combination of cash, new debt and ownership of the company going forward.

A distressed debt investor tries to figure out (a) what the bankruptcy is worth (or will be worth) at the time it emerges from bankruptcy, (b) that value will be divided among the company's creditors and other claimants, and (c) how long this process will take. With correct answers to these questions, he can determine what the annual return will be on a company's debt if purchased at a given price.

Getting our start in distressed debt investing in 1988 was extremely advantageous, as there were few competitors and the field was little known and little understood—two conditions that can help make for superior returns if possible in any field. As a result, our funds have been able to achieve an average return over the 29 years since. But, as with many things, the success is of only limited significance. The funds we formed at less-than-average prices generally have yielded good returns, but the results on funds formed at right times have been superlative.

In other words, the opportunities for top returns in distressed debt do go. Given the subject matter of this book, what I'll review here is what makes the opportunities rise and fall. Unsurprisingly, the answer lies in the fluctuations in the distressed debt cycle. What, in turn, causes those fluctuations?



The opportunities to profit in distressed debt are highly cyclical and determined by developments in other cycles. Thus they are illustrative of the workings of cycles and ripe for discussion here.

In the beginning—in 1988, 1989 and early 1990—our funds benefitted from the often-ignored benefits of distressed debt investing, and our returns were good. But in the latter half of 1990, the market for sub-invested debt collapsed in the first of the three major crises Bruce and I have lived through together. In addition to creating the low purchase price for our 1990 funds above-average gainers, this episode was highly educative; it gave us our first glimpse of the process through which superior opportunities arise in distressed debt.

The first of the two essential ingredients in their creation consists in the unwise extension of credit.” Given the discussion in the last chapter, you should have an understanding of what I have in mind and how I'll explain through the example of high yield bonds:

- At the start, appropriately risk-averse investors apply stringent credit standards to the issuance of high yield bonds.
- The same healthy economic environment that facilitates bond issuance makes it easy for companies to service their existing debt (defaults are scarce).
- Thus high yield bonds—with their generous interest coupons and damage from defaults—provide solid realized returns.
- Those returns convince investors that high yield bond investing is attracting increased capital to the market.
- Increased capital for investment translates into increased demand for bonds. Since Wall Street never allows demand to go unmet, there is increased bond issuance.
- The same condition that allows larger amounts of bonds to be issued by strong investor demand—invariably also allows bonds of lesser creditworthiness to be issued.

Few lenders and bond buyers are imprudent enough to advance loans that won't be repaid if conditions remain as they are. And in sober moments, they insist on a margin of safety sufficient to ensure, as I said, that interest and principal will be paid even if conditions for the debtor deteriorate.

But when the credit market heats up—when the race to the bottom begins—avid lenders to finance less-deserving borrowers and accept weaker structures—bonds are issued that lack that margin of safety and won't be serviced if things get a little worse. This is the unwelcome side of credit. This process, as noted earlier, “stacks the logs in the fire for the next bonfire.”

But that's only the first half the process. Even after the fuel for a fire has been assembled, there won't be a conflagration until the second half arrives: an igniter. It usually comes in the form of a sharp decline in corporate profits to decline. This is often accompanied by a sharp drop in the slamming shut of the credit window—such that existing debt can't be refinanced and goes into default instead. And often conditions are exacerbated by exogenous events that sap confidence and damage the economy and financial markets. In 1990 these consisted of:

- the Gulf War, which was set off by the Iraqi invasion of Kuwait
- the bankruptcy of many of the prominent, highly levered buyout firms of the 1980s; and
- the imprisonment of Michael Milken (the principal investment banker behind high yield bonds) and the collapse of Drexel Burnham Lambert, his employer and the investment bank most closely associated with high yield bonds). With Drexel and Milken gone from the scene,

exchanges that had helped weakened companies avoid default were to effect.

When the igniter arrives, bonds that shouldn't have been issued—and perhaps even some that should have—begin to fail.

- A slower economy makes it harder for companies to
- With the credit market closed, refinancing can't be accomplished, meaning defaults rise.
- Rising defaults damage investor psychology.
- Investors who were risk-tolerant when things were going well now become risk-averse.
- Advancing capital to financially distressed companies—which seemed like a good idea just a short time earlier—now goes out of
- Potential debt buyers back away, refusing to “catch a falling
- saying they'll wait until the uncertainty has been resolved.
- Capital that is mobile flees from the market. Buyers become
- sellers predominate.
- Selling of bonds increases; bond prices cascade downward; funds that receive withdrawals become forced sellers; and eventually bonds available for sale at any price.

These are the conditions that give rise to the ability to effect large purchases of distressed debt, and thus to opportunities for elevated returns. Of course, the cycle doesn't go in just one direction. Eventually the economy begins to recover and the credit market reopens. These developments cause the default rate on high yield bonds to recede. The combination of an improving economy and a declining default rate causes selling to abate. Thus the downward pressure on bond prices eases, and buying begins. Prices rise rather than fall, and balance sheet restructuring to restore companies to viability, remove stumbling blocks and unlock value. When gains on bond purchases made at the bottom begin to be noticed, additional capital is attracted to the market. The combination of rising prices and increased capital causes the demand for bonds to grow. And we've gone full cycle: we're back to the beginning.

A while back I put together a concise synopsis of the way the issuance underlies the rise and fall of the distressed debt cycle. Here

- Risk-averse investors limit quantities issued and demand high quality
- High-quality issuance leads to low default rates.
- Low default rates cause investors to become complacent and risk-tolerant.

- Risk tolerance opens investors to increased issuance and lower quality.
- Lower-quality issuance eventually is tested by economic difficulty gives rise to increased defaults.
- Increased defaults have a chilling effect, making investors once more.
- And so it resumes.

This is a cycle I've seen in action repeatedly. My 29 years of experience with distressed debt tells me its themes definitely do rhyme. And the description of the cycle gives me an excellent opportunity to make my point that each event in a cycle causes the one that follows. In fact, the description specifically to achieve that goal. Take a look at the progression just described: you'll see that the final words of each line are the same as the first words of the line that follows. This is a true chain reaction, and we expect to continue in the future.



As you can see, the rise and fall of opportunities in the market for distressed debt stems from the interaction of other cycles: in the economy, investor psychology, risk attitudes and the credit market.

- The economic cycle influences investor psychology, company profitability and the incidence of default.
- The cycle in psychology contributes to fluctuations in credit market conditions and the desire of investors to lend, buy and sell.
- The cycle in attitudes toward risk facilitates the issuance of capital at the top and denies capital for refinancing at the bottom.
- The credit cycle has a profound effect on the availability of refinancing and the degree to which would-be debt issuers are subjected to tighter credit standards.

Hopefully it's clear that multiple underlying cycles have effects on the distressed debt market that are far from discrete and isolated. As we saw earlier, each of these cycles rises and falls; each causes the other to fall; and each is affected by the rise and fall of others. But this is a dramatic cycle in distressed debt opportunities, and one that needs a more complete explanation.

XI

THE REAL ESTATE CYCLE

Much of investing is subject to gross generalizations and sweeping statements—usually stressing the positives, because of humans’ tendency toward greed and wishful thinking—and for some reason this seems particularly true in real estate. Over the course of my career I’ve heard investment in real estate rationalized by easily digested statements like “they’re not making any more” (in connection with land), “you can always live in it” (in connection with houses), and “it’s a hedge against inflation” (in connection with properties of all types). What people eventually learn is that regardless of the merit behind these statements, they won’t protect an investment that was made at too high a price.

The cycle in real estate has a lot in common with other cycles, one that controls the provision of capital or credit.

- Positive events and increased profitability lead to greater enthusiasm and optimism.
- Improved psychology encourages increased activity. That includes doing more of something; doing it on the basis of rosier assumptions; higher prices to do it; and/or lowering the standards that if one is to do it. All of these things tend to entail the greater risk.
- The combination of positive psychology and the increase in activity causes asset prices to rise, which encourages still more activity, further price increases, and greater risk-bearing.
- Inevitably this virtuous circle takes on the appearance of being unstoppable, and this appearance causes asset prices and the level of activity to go too far to be sustained.

But when the news eventually turns less positive and the environment becomes less hospitable, the levels of psychology, activity and risk-bearing prove to have been excessive, and the same goes for asset prices. The

resulting price correction causes psychology to become less positive, causes disinvestment, which puts further downward pressure on prices forth.

These are all elements that most financial cycles have in common includes the cycle in real estate. But the real estate cycle incorporates an ingredient that the others generally don't share: the long lead times required for real estate development to take place.

In the credit market, for instance, good news and rosy psychology will lead to increased lending as soon as investment bankers can line up borrowers and print prospectuses. Thus lenders' increased ardor translates almost instantaneously into increased demand for securities, lower demand yields, lower lending standards, and increased levels of lending and issuance.

But in the market for physical real estate—the world of “so-called bricks and mortar”—there can be significant delays. Before a new building can come to market, adding to the supply of space (and thus putting pressure on prices for space if demand does not increase apace), feasibility studies have to be performed; a site has to be found and the building has to be designed; environmental impact studies performed; permission to build has to be secured from authorities, and sometimes zoning modifications; financing has to be obtained; and construction has to be completed. This process can run to several years in the case of a major project it can exceed a decade. But it can change very significantly in the interim.

I'll use a description of the cycle in real estate development from 1929 (January 2013) to illustrate. As I said there, “it's usually clear, simply and regularly recurring”:

- Bad times cause both the level of building activity to be low and the availability of capital for building to be constrained.
- In a while the times become less bad, and eventually even good.
- Better economic times cause the demand for premises to rise.
- With few buildings having been started during the soft period and now coming on stream, this additional demand for space causes the supply/demand picture to tighten and thus real estate sale prices to rise.
- This improves the economics of real estate ownership, reawakening developers' eagerness to build.
- The better times and improved economics also make providers of capital more optimistic. Their improved state of mind causes financing to become more readily available.

- Cheaper, easier financing raises the projected returns on potential projects, adding to their attractiveness and increasing developers' desire to pursue them.
- Higher projected returns, more-optimistic developers and more-generous providers of capital combine for a ramp-up in starts.
- The first completed projects encounter strong pent-up demand. They lease up or sell out quickly, giving their developers good returns.
- Those good returns—plus each day's increasingly positive headlines—cause still more buildings to be planned, financed and green-lighted.
- Cranes fill the sky (and additional cranes are ordered from the factory, but that's a different cycle).
- It takes years for the buildings started later to reach completion. In the interim, the first ones to open eat into the unmet demand.
- The period between the start of planning and the opening of a building is often long enough for the economy to transition from boom to bust. Projects started in good times often open in bad times, meaning their space adds to vacancies, putting downward pressure on rents and sale prices. Unfilled space hangs over the market.
- Bad times cause the level of building activity to be low and the availability of capital for building to be constrained.

Note that as in the case of many of the other cycles discussed, one step leads to the next. In particular, the step at the bottom of the one that kicks off the next iteration. This is a good example of how cycles are self-perpetuating.



In lending, since the time lags inherent in the process are brief and business conditions in force at the time that the lender will to lend a loan is conceived generally are still in force when the loan is funded, conditions change materially in the relatively brief interim, the lender is able to pull his commitment under a “material adverse change” clause in the contract. Thus there's relatively little risk in general lending from the gap between idea and action.

But given the many years that can pass between a building's conception and its opening, conditions can change enormously, as just described, adds an element to real estate development that makes it potentially riskier. Developers hope this risk will be offset by the fact that they can utilize

extensive external financing (and thus they risk relatively little of their money and are able to lever up the return on it to a great

When I moved to Los Angeles in 1980, steel skeletons stood along the “Wilshire corridor” in Westwood, where gleaming condominium buildings had been imagined. The developers who had initiated these projects in the 1970s boom were caught out. The positive conditions that had supported initiation of those projects had turned negative, as the economy slowed and as supply created by quicker builders sopped up what was left.

Some of those rusting skeletons remained uncompleted for years. The developer who had dreamed of a high return on a project’s \$100 million cost instead lost his \$5 million or \$10 million of equity. A good part of the construction loans they had extended for those projects illustrates the downside of the real estate cycle and the effect of the time lags.

But investors who bought those halted projects (often from lenders that had repossessed them) and completed them often benefitted from:

- the ability to buy them for less than the developers had invested in planning, entitling and construction of the framework,
- the reduced cost to complete them at lower prices for labor and materials in a non-boom environment,
- the shorter period remaining between the onset of their involvement and the completion of the building, and
- the possibility that—just as they had been approved in good times and run into bad times—stalled projects purchased in bad times might be sold onto the market in good times.

The long lead time in real estate development gave rise to this potential, and my team participated in it. It illustrates the impact of cycles on real estate. Initiating projects in boom times can be a source of loss, while completing them in weak times can be very profitable. It all depends on what you do when you do it. Or as they say in golf, “Every putt makes a difference.”



There’s another aspect of cycles that can be seen clearly in the real estate arena—although it affects cycles in many other areas as well—and the fact that people’s decisions often fail to take into account what others are doing. Here’s an example:

When prosperity is rolling along and wealth and good feelings are increasing, there generally will be increases in the demand for homes—resulting in increases in the price of homes—and increases in the availability of financing to home buyers. Often this causes a housing shortage

demand for homes increases relative to supply, which adjusts home prices combine with eager lenders' financing of homebuilders to encourage the construction of new homes to meet the demand.

A homebuilder might conclude there is unmet demand for 10 town. Out of an abundance of caution—and because of the limit and his access to funds—he decides to build just 20 new homes good.

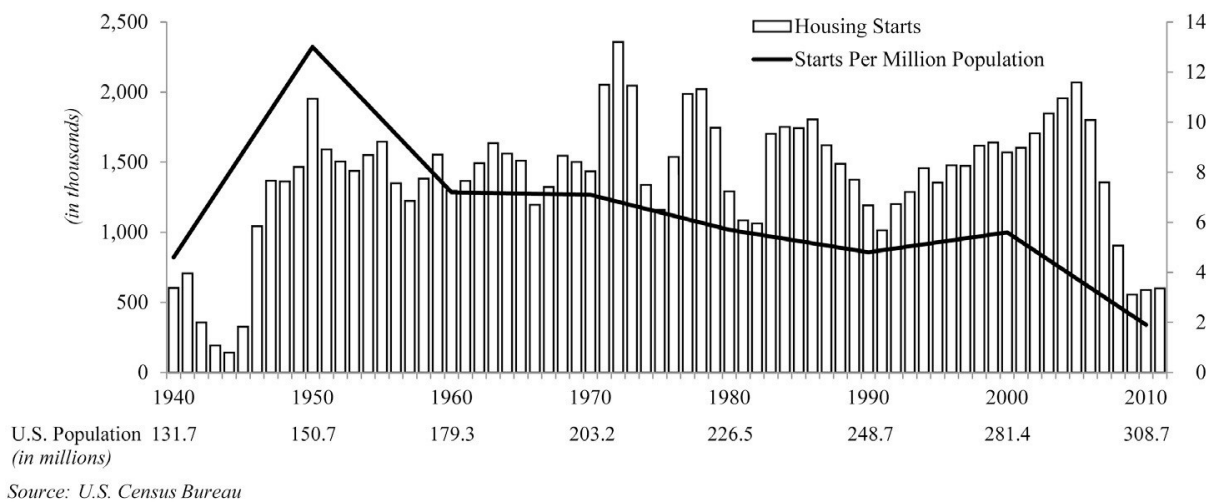
But what if ten homebuilders all make the same decision? In the homes will be built. First, more homes will be built than there's And second, by the time those homes come on the market, the have cooled; people may not feel as prosperous; and thus demand might be sharply lower. In that case, the 200 new homes may dearth of demand, meaning they'll go unsold or sell for on which the developers had based their decision to build.

Now conditions have reversed. The economy is slow. Access to financing dries up, making it hard for would-be home buyers to obtain there's a sizable inventory of unsold homes. Clearly the builders to stop building. So they all do so at the same time next time the economy improves, there may not be enough homes increase in demand. And so forth.

This is a straightforward description of one aspect of a cycle it isn't hypothetical. At the 2012 Oaktree Conference, my partner showed one of the most compelling graphs I've ever seen:

This graph presents the record of annual housing starts in the 1940 through 2010. The reason it made such an impression on me whereas it showed that 2010 housing starts were at the lowest wartime 1945 (and just equal to 1940's slightly less-depressed level observation tells only part of the story. It fails to take into account since the '40s in the U.S. population, the source from which long-run increases in demand for housing arise.

U.S. Housing Production



Thus, while the number of starts in 2010 was the same as in 1940, the ratio of starts to population—arguably a far more meaningful figure—was *half* the highly depressed 1940 level. The explanation is that practically one had resumed building homes following the sub-prime mortgage housing bust, and Global Financial Crisis of 2007–08. The key inference to draw from this observation was that the supply of new homes in the immediately following clearly would be insufficient to meet a demand for homes.

Of course, “conventional wisdom” at the time held that there could be a pickup in demand for homes. Instead, most people believed the American dream of home ownership was over; demand for homes would remain depressed forever; and thus the overhang of unsold homes would be absorbed only very slowly. They cited the trend among young people of renting rather than buying, and as usual they extrapolated it rather than questioned its durability. As in so many of the examples in this book, for most people, psychology-driven extrapolation took the place of an understanding and belief in cyclicality.

It was clear to me and my Oaktree colleagues, from the graph and our knowledge of the data behind it, that because the greatest economic downturn in almost eighty years had halted additions to the housing supply, the economy could recover strongly if there was any material increase in demand. Rejecting conventional wisdom, we were convinced that housing demand would prove cyclical as usual, and thus would pick up sometime in the intermediate-term future. This conclusion—supported by other data and our analysis—contributed to our decision to invest heavily in non-performing home mortgages and non-performing bank loans secured by land for residential construction, and to purchase North America’s largest private

homebuilding company. These investments worked out quite well. (It's interesting in this context to note what the *Wall Street Journal* said in its 12, 2017 article headlined "Generation of Renters Now Buying": "In time home buyers] have accounted for 42% of buyers this year, up from 31% in 2015 and 31% at the lowest point during the recent housing boom. So much for extrapolating widespread abandonment of home ownership.

This is an example of a case where awareness of the nature of our position in the current one permitted a profitable inference. In a case where a cycle standing at an extreme—this time housing starts at a trough—sent a signal calling for a highly profitable action.

The proper course seems clear now, when we look back on the event. The reasons behind successful decisions invariably are obvious in retrospect. But in this case, dispassionate analysis of cyclical data permitted a conclusion to be reached in real time . . . which is where



While I'm on the subject of cycles in real estate, I want to mention one more time but not the last—on the way people occasionally conclude that the financial phenomenon has ceased being cyclical, and to that end, one of the greatest cases. When things go well, people tend to think that times will roll on forever. In fact, not much time has to have elapsed since the last correction before cyclical history is overlooked in its entirety. It is appropriate to again cite John Kenneth Galbraith's observation regarding attitudes toward history:

There can be few fields of human endeavor in which history counts for so little as in the world of finance. Past experience, to the extent that it is part of memory at all, is dismissed as the property of those who do not have the insight to appreciate the incredible volatility of the present.

Much of investing is subject to gross generalizations and sweeping statements—usually stressing the positives, because of humans' tendency toward greed and wishful thinking—and for some reason this seems particularly true in real estate. Over the course of my career I've seen investment in real estate rationalized by easily digested statements like "they're not making any more" (in connection with land), "you can live in it" (in connection with houses), and "it's a hedge against inflation" (in connection with properties of all types). What people eventually learn, regardless of the merit behind these statements, they won't protect an investment that was made at too high a price.

In the late 1990s and early 2000s, as I mentioned earlier, a elected officials decided that as a step toward benefitting society, increasing access to the American dream, it would be swell to have more owned homes. As a result, government-sponsored mortgage lenders sent the message that home financing should be made more easily available, and they complied. The combination of this message, the resulting increase in mortgage availability, and the sharp decline in interest rates that was in place at the time had a powerful stimulative effect on potential home ownership.

The flow of money for mortgages was strongly encouraged by an estate platitude: “mortgage lending is safe.” This was based on conviction that a nationwide wave of mortgage defaults couldn’t happen. The combination of strong economic growth, generally moderate economic fluctuations, prudent mortgage lending practices had prevented the occurrence of such a wave in the period since the Great Depression . . . long enough for such episode to be forgotten, as Galbraith says. But that didn’t mean lenders were incapable of lapsing into lending practices so generous and so lax that a powerful recession could bring on such a wave of defaults.

In the early years of the 21st century, the combination of strong economic growth and plentiful mortgage money—abetted by optimistic media coverage—caused home prices to rise strongly. Thus we started to hear yet another sweeping generalization concerning real estate: “home prices always go up” (see the section that follows).

I hope by now you’ve caught on to the fact that the mortgage market is a question matter only so much, and certainly they can’t be strong enough to always carry the day. Human emotion inevitably causes the price of even worthwhile assets—to be transported to levels that are extreme and unsustainable: either vertiginous highs or overly pessimistic lows.

In short, conscientious belief in the inevitability of cycles like I’m talking about means that a number of words and phrases must be excluded from an intelligent investor’s vocabulary. These include “never,” “always,” “forever,” “can’t,” “won’t,” “will” and “has to.”



In the years leading up to the sub-prime mortgage crisis of 2007–08, the Financial Crisis of 2007–08, a great deal of bullish behavior (which was shown to have been reckless) was underlaid by the belief that home prices depended on to appreciate steadily and not prove cyclical. As part of contributing to—the bullish trend, some researchers rendered supportive statements and optimistic projections:

- According to a *New York Times Magazine* article from March 2007 titled “This Very, Very Old House,” a vice president of the Federal Reserve Bank of New York said:

- Reserve Bank of New York had concluded that “the sharp rise in prices is in line with economic conditions . . . not a reality.” It even quoted him as saying, “We sometimes wonder if home prices haven’t increased much more, given the tremendous increase in the size of mortgage the average family can afford.”
- The article also mentioned “like-minded experts [from Columbia University and the Wharton School] who focus on what they call ‘superstar cities,’ places so desirable that they not only are for a correction but they also can sustain ‘ever-increasing’ prices compared with less-sought-after cities.” (Of course, willingness to employ terms like “ever-increasing” should serve as an absolute for the alert investor.)

But there was a lot to question regarding the reasonableness of conclusions:

- the brevity of the data on home prices,
- the fact that statements made about the trend in the price of the home sold in a given year don’t necessarily say anything about the performance of a given home or of all homes standing adjustment for physical changes in the average house over time, mix of homes sold that year relative to all homes), and
- likewise, there is no adjustment for the fact that neighborhoods whole cities go in and out of favor over time, affecting the value homes. For this reason, statements about homes in a given neighborhood wouldn’t necessarily be applicable to homes in

Thus I found it very interesting when “This Very, Very Old House” was published in 2006. It reported on a study that solved many of methodological problems by following the price of a single house in Amsterdam: one that Pieter Frantz built in 1625. It had stood since virtually unaltered, and had changed hands six times, but its location Herengracht canal neighborhood had remained Amsterdam’s most desirable throughout. Thus the article provided insight not as to whether average home had risen, but how the price of a particular home had performed.

In contrast to the long-term optimists on housing quoted earlier, Robert Shiller said of the Frantz house and its neighbors, “Local Herengracht data is very instructive, because you can see 50-year growth, then it turns around. That’s more realistic than the super-optimist argument.” The author of the study, Piet Eichholtz, was described as “skeptical of those who claim that property values can continue to

infinitum.” He considered the economic arguments for it being “different time” but said, “you can see similarly rosy assessments made over and over, which are then quashed by circumstances.” Amen.

Here’s more from the *Times* article:

“There is a myth which says that real-estate values go up steadily over time, . . .” Eichholtz said, “. . . but the data e . . . that myth.”

That is to say, where everyone from your wise old uncle to the broker who sold you your house holds it as gospel that real estate is one of the best long-term investments, this longest of long-term indices suggests that, on the contrary, it sort of stinks. Eichholtz and 1973 (the period of Eichholtz’s original study), real property values on the Herengracht—adjusted for inflation—went up a mere 0.2 percent per year, worse than the stingiest bank savings account. As Shiller wrote in his analysis of the Herengracht index, prices did roughly double, but took nearly 350 years to do so. . . . It’s only in recent years, Shiller says, that huge increases in real-estate prices have become the norm and that people have come to expect them.

. . . if this description of the past few years [in which prices went up amazingly”] typifies the brave new world we live in, putting it into the perspective of time—rise, fall, rise, fall—leads us back to what may be the oldest history lesson of all: it tends to repeat itself. [emphasis added]

The long-term data on home prices is valuable, but the most important lesson is that, in times of rising asset prices, people turn bullish and commentators provide authoritative support. That’s only natural. Rationalization for price appreciation that has taken place (and prediction of still more to come) invariably occurs at highs, not lows. For real estate, to commentators who issue sober statements in bullish times, or who argue against negativity when markets are down.



The real estate industry is subject to cyclical ups and downs. But real estate cycles can be amplified by special factors:

- the time lags between conception and readiness for sale,
- the extremely high financial leverage that is typical, and
- the fact that the supply is generally too inflexible to be able to meet demand fluctuations. (That is, a manufacturer can eliminate a

lay off workers or reduce production if demand for his
But a landlord, hotelier or real estate developer has a much
reducing the premises he has on offer if demand falls short.)

The cycle in real estate illustrates and exemplifies the ways in which cyclical factors lead to and cause each other, as well as the tendency to go to extremes. It's not for nothing that they often say cynics are right in tougher times, when optimistic generalizations can no longer be summed up in a word—namely, "only the third owner makes money." Not the developer who conceived and initiated the project. And not the banker who loaned money for its construction and then repossessed the project from the developer in the down-cycle. But rather the investor who bought the property from the developer amid distress and then rode the up-cycle.

Of course this is an exaggeration, like all generalizations. But it serves as a reminder of the relevance of cyclicality to the real estate market, especially of the way cyclicality can function in the less-good times.

XII

PUTTING IT ALL TOGETHER—THE MARKET CYCLE



The first time an inexperienced investor lives through an upward market cycle, the beginnings of the progression may seem logical, as the positives compound in a bull market or bubble. The flood of much good news and good feeling can end in losses can come as a surprise. It is inescapable that it will seem so to the uninitiated course, because if the progressions weren't permitted to go to extremes on the basis of errors in judgment, markets wouldn't rise from bull market tops to collapse from (or bear market bottoms to rise from).

Our job as investors is simple: to deal with the prices of where they stand today and making judgments regarding how they will change in the future. Prices are affected primarily by developments in two areas: fundamentals and psychology.

- Fundamentals, which I've been calling "events," can largely be broken down into earnings, cash flow, and the outlook for the two. They are many things, including trends in the economy, profitability and the availability of capital.
- And psychology—how investors feel about fundamentals and value them—is likewise affected by many things, particularly investors' levels of optimism and attitude toward risk.

There are cycles in the elements mentioned above, and several repeat each cycle. The themes behind the cycles' behavior—and the ways they interact and combine—have a repeating, understandable pattern, which we have discussed at length. They all come together—and they combine with idiosyncratic and random influences—to cause the behavior of the market.

It's my goal in this chapter to give you a feel for the cyclical ups and downs of the market. Not for the fact that it rises and falls, so in the past, or what these movements were in reaction to. But for the forces—and specifically the non-fundamental, non-economic forces—that cause it to go up and down, often in manic fashion.

If the market were a disciplined calculator of value based exclusively on company fundamentals, the price of a security wouldn't fluctuate more than the issuer's current earnings and the outlook for earnings. In fact, the price generally should fluctuate less than earnings, since quarterly changes in earnings often even out in the long run and, therefore, necessarily reflect actual changes in the company's long-term potential.

And yet security prices generally fluctuate much more than earnings. The reasons, of course, are largely psychological, emotional and non-fundamental. Thus price changes exaggerate and overstate fundamental changes. Here is a shorthand version as to why:

- Events in the economy and in corporate profits turn increasingly positive.
- Positive events feed investor psychology. Emotion, so-called “animal spirits” and investors' tolerance for risk all rise with those positive events (or sometimes despite negative ones).
- Rising psychology causes investors to be less demanding in terms of protection and prospective return.
- The combination of positive events, strengthening psychology and lowering of investors' return requirements causes asset prices to rise.
- Eventually, however, the process goes into reverse. Events fail to live up to expectations, perhaps because the environment that produced them becomes less hospitable, or perhaps because expectations were unrealistically high.
- Investors eventually prove that psychology can't remain positive forever. Cooler heads conclude that prices have reached levels that are unjustified, or psychology may soften for any of a million possible reasons (or for no apparent reason).
- Prices fall when events are less positive or come to be viewed negatively. Sometimes it happens simply because prices have reached levels that are unsustainable, or because of negative developments in the environment.
- Having turned downward, asset prices continue to decline until they reach a level so low that the stage is set for their recovery.

It's important to understand the way fundamentals and psychology interact as described above. But it's essential that I repeat something about

process: while the description above is of one that is orderly and the process is nowhere as neat as this description may make it sequence in which these things occur is subject to change, as is direction of causality.

- Sometimes events cause psychology to strengthen, and sometimes improving psychology has a positive impact on events (for bolstering the economy and corporate profits).
- And while it's obvious that improved investor psychology makes prices rise, it's just as obvious that rising prices make investors feel wealthier, smarter and more optimistic.

So, in other words, these relationships can work in both directions even do so simultaneously. And each can cause the other. The speed at which things play out is highly variable from cycle to cycle and over the given cycle. And lastly, cycles don't necessarily progress smoothly; rather, they can be marked by dips, recoveries and feints along the way.

It's for reasons like these that investing can't be described as science. It can't be depended on to work the same every time. I keep coming back to Mark Twain's observation that "history doesn't repeat itself, but it does rhyme." The reasons and results are never the same as in the usually reminiscent of developments we've seen before.

Regardless of the imprecision of the process, it's clear that past events, expected future events combine with psychology to determine asset prices. Events and psychology also influence the availability of credit, and availability of credit greatly affects asset prices, just as it feeds back to influence events and psychology.

In sum, these things all come together to create the market cycle. We think about it every day, most prominently in connection with the ups and downs of the stock market, but also with regard to markets for things like oil and currencies. This is where many cycles intersect, and it's the subject of this chapter.



Financial theory portrays investors as "economic men": objective, rational optimizers. Thus it suggests that the market, which they collectively form, is an impartial author and investor (and Warren Buffett's teacher) Ben Graham's "weighing machine": a disciplined assessor of the value of assets.

In stark contrast, however, the truth is that financial facts and figures are only a starting point for market behavior; investor rationality is not the rule; and the market spends little of its time calmly digesting data and setting prices free of emotionality.

Investment fundamentals are rather straightforward. Past events have already taken place and been recorded, and many people have the quantitative skills required to analyze them. Current performance is captured in financial statements, which sometimes present an accurate picture and sometimes require skillful adjustment. And future events are unknown to everyone (although some investors are better able than others to foresee them). Fundamentals aren't the most variable part of investing or the part that most intrigues me. And anyway, I can't write a book telling you more than others about future events. Doing a superior job of that requires the elements of foresight, intuition and "second-level thinking" that I can't be reduced to paper or taught.

The part of investing that fascinates me—that I find myself thinking about most, and where my Oaktree colleagues and I have made some of our most significant contributions to our clients' welfare—regards the ways in which investors deviate from the assumption of rationality, and the contribution of those deviations to the oscillation of cycles.

A large number of elements play parts in this aspect of investing. Some are making, interfering with the process of arriving at purely economic decisions. They may come under the headings of human nature, psychology or sociology—but the distinction isn't straightforward or important for our purposes here—and they absolutely are capable of dominating investor behavior and market movements. Some but not all vary cyclically, and they all can affect or be affected by cycles. Here are the most important influences:

- the way investors fluctuate rather than hold firmly to rationality and the resulting irrational decisions;
- the tendency of investors to hold distorted views of what's really going on, engaging in selective perception and skewed interpretation;
- quirks like confirmation bias, which makes people accept what confirms their thesis and reject that which doesn't, and the tendency toward non-linear utility, which causes most people to value gains more highly than a dollar made (or a dollar of potential profit lost);
- the gullibility that makes investors swallow tall tales of profit in good times, and the excessive skepticism that makes them reject all possibility of gains in bad times;
- the fluctuating nature of investors' risk tolerance and risk aversion, and thus of their demands for compensatory risk premiums;
- the herd behavior that results from pressure to fall into line with others are doing, and as a result the difficulty of holding independent positions;
- the extreme discomfort that comes from watching others make money doing something you've rejected;

- thus the tendency of investors who have resisted an asset bubble ultimately succumb to the pressure, throw in the towel and buy though—no, *because*—the asset that is the subject of the appreciated substantially);
- the corresponding tendency to give up on investments that are and unsuccessful, no matter how intellectually sound, and
- finally, the fact that investing is all about money, which introduces powerful elements such as greed for more, envy of the money making, and fear of loss.

Bulls and Bears

Investors have been characterized for at least a hundred years as (optimists who think stocks will rise and behave aggressively as “bulls” (pessimists who think they’ll fall and who thus behave deferentially as “bears”). Consequently, people apply the label of “bull market” to a market that has risen, is rising or will rise (it’s quite imprecise) and “bear market” to the opposite.

About 45 years ago—in the early 1970s—I received one of the greatest gifts I was ever given, when an older and wiser investor introduced me to the “three stages of a bull market”:

- the first stage, when only a few unusually perceptive people believe that things will get better,
- the second stage, when most investors realize that improvement is actually taking place, and
- the third stage, when everyone concludes that things will get better forever.

The arrival of this simple truth opened my eyes to the non-rational psychological extremes and the impact of those extremes on market behavior. Like many of the great quotations and adages, it captures disproportionate wisdom in a few simple words. It’s all about the changeable patterns they follow over the course of a cycle, and how they lead to error.

In the first stage, because the possibility of improvement is in the mind of most investors and thus unappreciated, security prices incorporate little or no optimism. Often the first stage occurs after prices have been pounded in a crash, and the same downtrend that decimated prices also has wiped out the market psychology, turning the members of the crowd against the market and making them swear off investing forever.

In the last stage, on the other hand, events have gone well for so long that they have been reflected so powerfully in asset prices, further lifting the market—that investors extrapolate improvement to infinity and prices to reflect their optimism. Trees generally don't grow to the sky, but in this stage investors act as if they will . . . and pay up for what they perceive. Few things are as costly as paying for potential that has been overrated.

It follows from the above that someone who invests in the first stage when almost no one can see a reason for optimism—buys assets at prices from which substantial appreciation is possible. But someone who enters in the third stage invariably pays a high price for the market's excessive enthusiasm and loses money as a result.

The description of the three stages of the bull market offers a lot of food for thought with great economy. But not long after learning about the three stages, I wrote across something even better and briefer—essentially the same message in just 14 words: “What the wise man does in the beginning, the fool does in the end.”

I consider this the number-one piece of investment wisdom and an incredible distillation of the import of market cycles. Again, the early investor who by definition has to be that rare person who sees the future before others and has the inner strength to buy without validation from the crowd garners undiscovered potential at a bargain price. But every investor eventually is overdone and bid up too far, so that the buyer in the third stage pays for potential that is overrated. He ends up with capital punishment for overappreciation.

“What the wise man does in the beginning, the fool does in the end.” That's what you get 80% of what you have to know about market cycles and their timing. Warren Buffett has said much the same thing even more concisely: “First come the innovator, then the imitator, then the idiot.”

Of course, cycles work in both directions, and the depths of the 2008 Financial Crisis gave me an opportunity to invert the old saying and describe the three stages of a bear market in “The Tide Goes Out” (March 2009):

- the first stage, when just a few thoughtful investors recognize that, despite the prevailing bullishness, things won't always be rosy,
- the second stage, when most investors recognize that things are deteriorating, and
- the third stage, when everyone's convinced things can only get worse.

I've mentioned capitulation before. It's a fascinating phenomenon, and there's a dependable cycle to it, too. In the first stage of either a bull or bear market, most investors refrain (by definition) from joining in on the

only a tiny minority does. This may be because they lack the special insight that underlies that action; the ability to act before the case becomes obvious and others have flocked to it (after which it's no longer unappreciated); or the spine needed to take a stand reflected in market prices); or the spine needed to take a stand against the herd and behave as a non-conforming contrarian.

Having missed the opportunity to be early, bold and right, investors continue to resist as the movement takes hold and gathers steam. When the market has resulted in market movement, they still may not join in. With discipline, they refuse to buy into the market, asset class or industry that has been lifted by bullish buyers, or to sell once selling has caused prices to fall below intrinsic value. It's not for them to join late.

But most investors do capitulate eventually. They simply run out of resolve needed to hold out. Once the asset has doubled or tripled on the way up—or halved on the way down—many people feel they've been wrong, and are so envious of those who've profited from the market that they've stepped the decline, that they lose the will to resist further. My favorite quote on this subject is from Charles Kindleberger: "There is nothing more important to one's well-being and judgment as to see a friend get rich" (*Manias, Panics, and Crashes: A History of Financial Crises*, 1989). Market participants are pained by the money that others have made and they've missed out on. They're afraid the trend (and the pain) will continue further. They conclude that joining the herd will stop the pain, so they surrender. Eventually, they sell the asset well into its rise or sell after it has fallen a great deal.

In other words, after failing to do the right thing in stage one, investors compound the error by taking that action in stage three, when it has become obvious that the wrong thing to do. That's capitulation. It's a highly predictable investor behavior during cycles, and a great example of psychological error at its worst.

Of course, when the last resister has given up and bought well into the decline—or sold well into the decline—there's no one left to fall in line. The last buyer means the end of the bull market, and vice versa. The last seller makes the top or bottom and sets the scene for a cyclical change in direction. He is the "fool in the end."

The following account from history shows that even the most brilliant among us can fall prey to capitulation:

Sir Isaac Newton, who was the Master of the Mint at the time of the "South Sea Bubble," joined many other wealthy Englishmen in investing in the stock [of the South Sea Company]. It rose from £100 in January of 1720 to £1,050 in June. Early in this rise, he realized the speculative nature of the boom and sold his shares at a loss.

£7,000 worth of stock. When asked about the direction of the market, he is reported to have replied “I can calculate the motion of heavenly bodies, but not the madness of the people.”

By September 1720, the bubble was punctured and the stock price fell below £200, off 80% from its high three months earlier. It turned out, however, that despite having seen through the bubble, Isaac, like so many investors over the years, couldn't stand the pressure of seeing those around him make vast profits. He bought back the stock at its high and ended up losing £20,000. Not even one of the world's smartest men was immune to this tangible law of gravity! (bubble.com, January 2000)

Bubbles and Crashes

We've always had rising and falling markets, and we always continue to a significant extent, they're called bull markets and Even further and they're called booms, manias and crazes; busts panics. The most popular terms today for describing extreme bull and markets are “bubble” and “crash.”

These latter terms have been around for a long time. The “South Sea Bubble” cited above, a mania for investing in the company that would pay off the national debt by exploiting a monopoly to America, caught England by storm in 1720. And the market collapse that kicked off the Great Depression is called the Great Crash of 1929. But the “tech bubble,” “Internet bubble” and “dot-com bubble” of 1995 and the housing and mortgage bubbles that ended in 2007, bringing significant crashes in markets around the world—that brought the “bubble” into everyday use.

As a result of the foregoing, there's a tendency these days—the part of the media—to call any big market rise a bubble. In fall 2017, the S&P 500 index of U.S. stocks has roughly quadrupled (including dividends) from its low in March 2009, and the yield on yield bonds has fallen to a piddling 5.8%. Thus I'm often asked whether we're in a new bubble of one kind or another, perhaps implying an imminent crash. That's why I want to spend some time on my conviction that every big rise is a bubble. For me, the term “bubble” has specific psychological connotations that should be understood and looked

I've lived through bubbles much older than those in tech stocks and housing mentioned on the preceding page. One of the best examples is the 1960s mania for the “Nifty Fifty” stocks—the shares of the highest-quality

fastest-growing companies in America. As far as I'm concerned, there's a common thread that runs through bubbles and was exemplified by the Nifty Fifty: conviction that, as far as the subject asset is concerned, "there's nothing as a price too high." And of course it follows that if you pay, you're sure to make money.

There's only one form of intelligent investing, and that's figuring out something's worth and buying it for that price or less. You can't have intelligent investing in the absence of quantification of value and on an attractive purchase price. Any investment movement that's built around a concept other than the relationship between price and value is irrational.

The idea of "growth stocks" began to be popularized in the early 1960s based on the goal of participating in the rapidly growing profits of companies benefitting from advances in technology, marketing and management techniques. It gathered steam, and by 1968, when I had a summer investment research department of First National City Bank (the predecessor of Citibank), the Nifty Fifty stocks—the fastest growing and best—had appreciated so much that the bank trust departments that did most of the investing in those days generally lost interest in all other stocks.

Everyone wanted a piece of Xerox, IBM, Kodak, Polaroid, Motorola, Hewlett-Packard, Texas Instruments, Coca-Cola and Avon. These companies were considered to be so great that nothing bad could ever happen to them. And it was accepted dictum that it absolutely didn't matter what you paid. If it was a little too high, no matter: the companies' fast growth would soon grow into it.

The result was predictable. Whenever people are willing to invest regardless of price, they're obviously doing so based on emotion and popularity rather than cold-blooded analysis. So the Nifty Fifty stocks that had been selling at 80–90 times earnings in 1968, in the vanguard of the bull market, came down to earth when ardor cooled. Thus many sold at 10 times earnings in the much weaker stock market of 1973, meaning that "America's best companies" had lost 80–90% of their money. And that several of the "flawless" companies mentioned have since gone bankrupt or experienced serious brushes with distress.

So much for "no price too high." No asset or company is so valuable that it can't become overpriced. Certainly that notion must have been banished forever.

But lest you think this lesson truly was learned, let's fast-forward to the 1990s. Now it was technology stocks that were commanding wide attention. Just as corporate innovation had sparked the growth stock gains in telecommunications (cell phones and transmission via optical fiber) and media (including the limitless demand for "content" to fill the new

entertainment channels), and information technology (especially the Internet) were firing investors' imaginations.

"The Internet will change the world," was the battle cry, followed by "for an e-commerce stock, there's no price too high." When the Nifty Fifty stocks had sold at inflated multiples of their companies' earnings, it wasn't a problem with Internet stocks: these companies had no earnings. Not only was the investing purely conceptual, but so were many of the companies. So instead of p/e ratios, the stocks sold at multiples of (earnings per any) or "eyeballs": the number of consumers visiting their websites.

Just as with the Nifty Fifty, there was a grain of truth underlying the investment fad; one is usually required in order for a bubble to burst. Investors slipped the moorings of reason and discipline when they thought that price didn't matter. They were right; the Internet certainly did change the world, which is unrecognizable today from what it was twenty years ago. But the companies behind the vast majority of the Internet stocks that existed in 2000 no longer exist. The Nifty Fifty losses of 80–90% are enviable; the companies' investors lost 100%.

The bottom line is clear: I think "price doesn't matter" is a component—and a hallmark—of a bubble. Likewise, in bubbles, investors often conclude that you can make money by borrowing money to invest in a mania. No matter what the interest rate is on your loan, the investment will appreciate at a rate above that. Clearly this is another example of a suspension of analytical disbelief.

"No price too high" is the ultimate ingredient in a bubble, and a foolproof sign of a market gone too far. There is no safe way to exit a bubble, only danger. It should be noted, however, that the end of a bubble is often synonymous with "going down tomorrow." Many fads roll on well beyond the time when they first reach bubble territory. Several prominent investors threw in the towel in early 2000 because their resistance to the tech bubble had proved so painful. Some saw clients withdraw a large part of their assets; some became dispirited and quit the business, and others gave up and sold into the bubble . . . just in time to see it collapse,



The following progression serves to sum up the ups and downs of the market cycle. It shows how cycles in economics, profits, psychology, and media behavior combine to move market prices well beyond intrinsic value, and how one development contributes to the next.

- The economy is growing, and the economic reports are positive.
- Corporate earnings are rising and beating expectations.
- The media carry only good news.

- Securities markets strengthen.
- Investors grow increasingly confident and optimistic.
- Risk is perceived as being scarce and benign.
- Investors think of risk-bearing as a sure route to profit.
- Greed motivates behavior.
- Demand for investment opportunities exceeds supply.
- Asset prices rise beyond intrinsic value.
- Capital markets are wide open, making it easy to raise money over debt.
- Defaults are few.
- Skepticism is low and faith is high, meaning risky deals can be
- No one can imagine things going wrong. No favorable development seems improbable.
- Everyone assumes things will get better forever.
- Investors ignore the possibility of loss and worry only about missing opportunities,
- No one can think of a reason to sell, and no one is
- Buyers outnumber sellers.
- Investors would be happy to buy if the market dips.
- Prices reach new highs.
- Media celebrate this exciting event.
- Investors become euphoric and carefree.
- Security holders marvel at their own intelligence; perhaps more.
- Those who've remained on the sidelines feel remorse; thus they capitulate and buy.
- Prospective returns are low (or negative).
- Risk is high.
- Investors should forget about missing opportunity and worry only about losing money.
- This is the time for caution!

The most important thing to note is that maximum psychology, maximum availability of credit, maximum price, minimum potential return and maximum risk all are reached at the same time, and usually these coincide with the last paroxysm of buying.

Likewise, the following progression outlines what happens in a market downswing.

- The economy is slowing; reports are negative.
- Corporate earnings are flat or declining, and falling short of
- Media report only bad news.

- Securities markets weaken.
- Investors become worried and depressed.
- Risk is seen as being everywhere.
- Investors see risk-bearing as nothing but a way to lose money.
- Fear dominates investor psychology.
- Demand for securities falls short of supply.
- Asset prices fall below intrinsic value.
- Capital markets slam shut, making it hard to issue securities debt.
- Defaults soar.
- Skepticism is high and faith is low, meaning only safe deals can or maybe none at all.
- No one considers improvement possible. No outcome seems too to happen.
- Everyone assumes things will get worse forever.
- Investors ignore the possibility of missing opportunity and worry about losing money.
- No one can think of a reason to buy.
- Sellers outnumber buyers.
- “Don’t try to catch a falling knife” takes the place of
- Prices reach new lows.
- The media fixate on this depressing trend.
- Investors become depressed and panicked.
- Security holders feel dumb and disillusioned. They realize they really understand the reasons behind the investments they made.
- Those who abstained from buying (or who sold) feel validated and celebrated for their brilliance.
- Those who held give up and sell at depressed prices, adding the downward spiral.
- Implied prospective returns are sky-high.
- Risk is low.
- Investors should forget about the risk of losing money and worry about missing opportunity.
- This is the time to be aggressive!

In the reverse of the “top” that results from the upswing of cycle, now we see that the nadir of psychology, a total inability to credit, minimum price, maximum potential return and minimum risk all coincide at the bottom, when the last optimist throws in the towel.

The progressions outlined here are simplistic. In fact, they can seem cartoonish depictions of the road to failure. But they are not imagined exaggerated. It is absolutely logical that each event will bring on

in both directions . . . until an illogical extreme is reached and the market collapses.

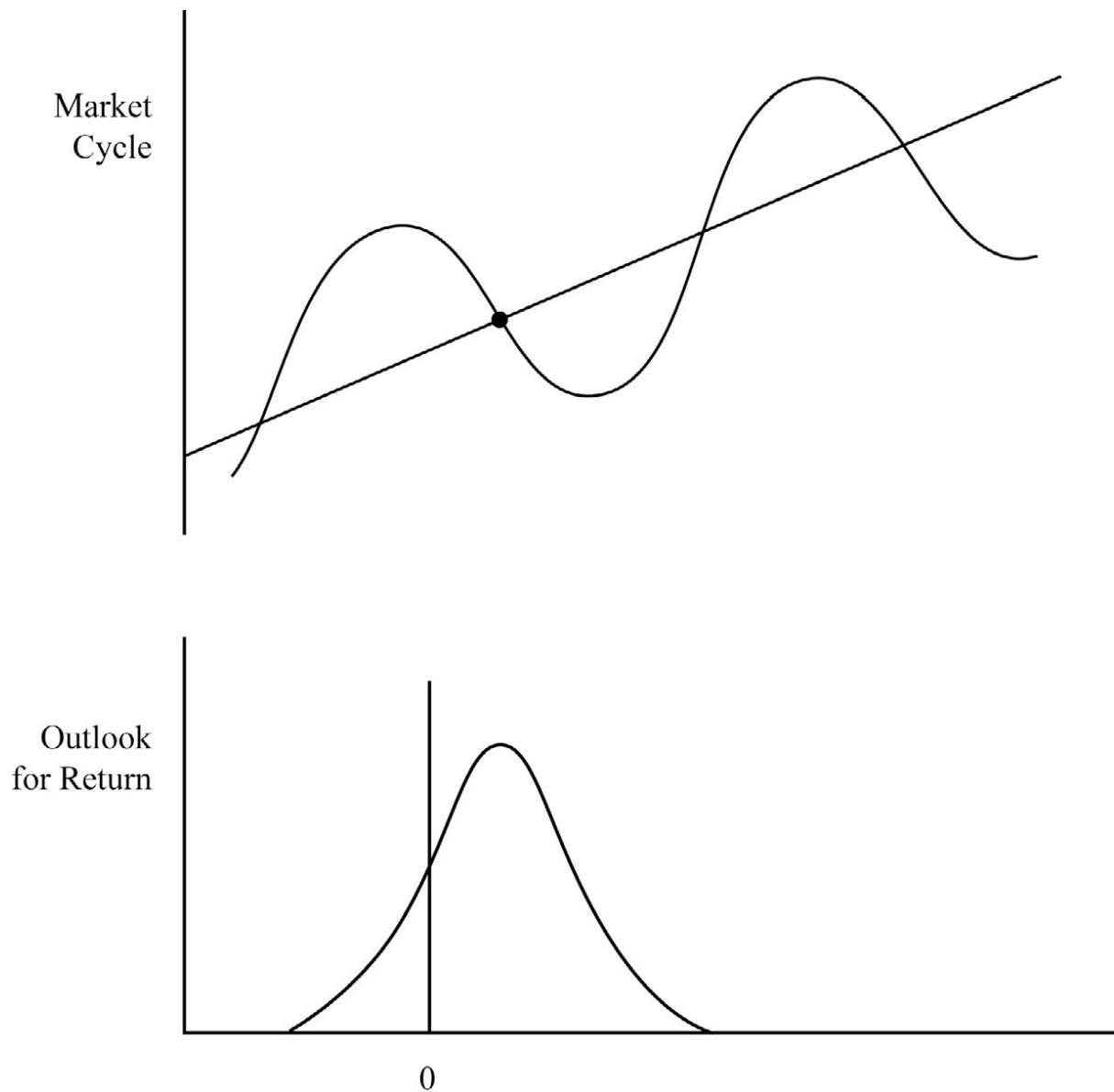
The events don't always occur in the same order, and not all of them are necessarily present in every market cycle. But these behaviors are real, and they certainly are elements that rhyme in the markets from decade to decade.

The first time an inexperienced investor lives through an upward market cycle, the beginnings of the progression may seem logical, as the position compounds in a bull market or bubble. The fact that so much good feeling can end in losses can come as a surprise. It is not surprising that it will seem so to the uninitiated, of course, because if the progression were permitted to go on to extremes on the basis of errors in judgment, it wouldn't reach bull market tops to collapse from (or bear market bottoms to recover from).

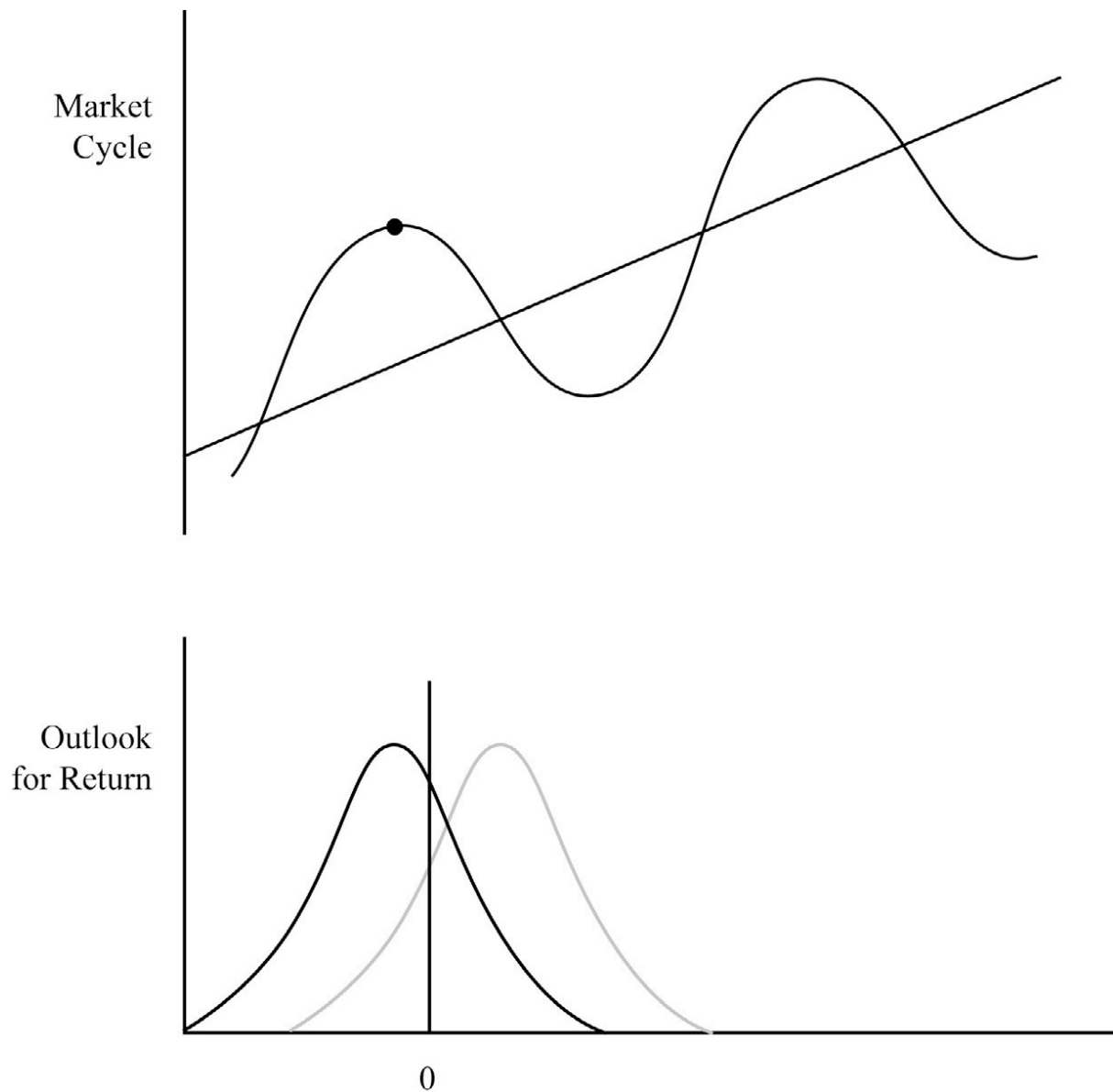


On [pages 19–21](#), I discussed the relationship between where we stand in the market cycle and what that implies for prospective returns. Now, to close this chapter, I want to further illustrate that connection.

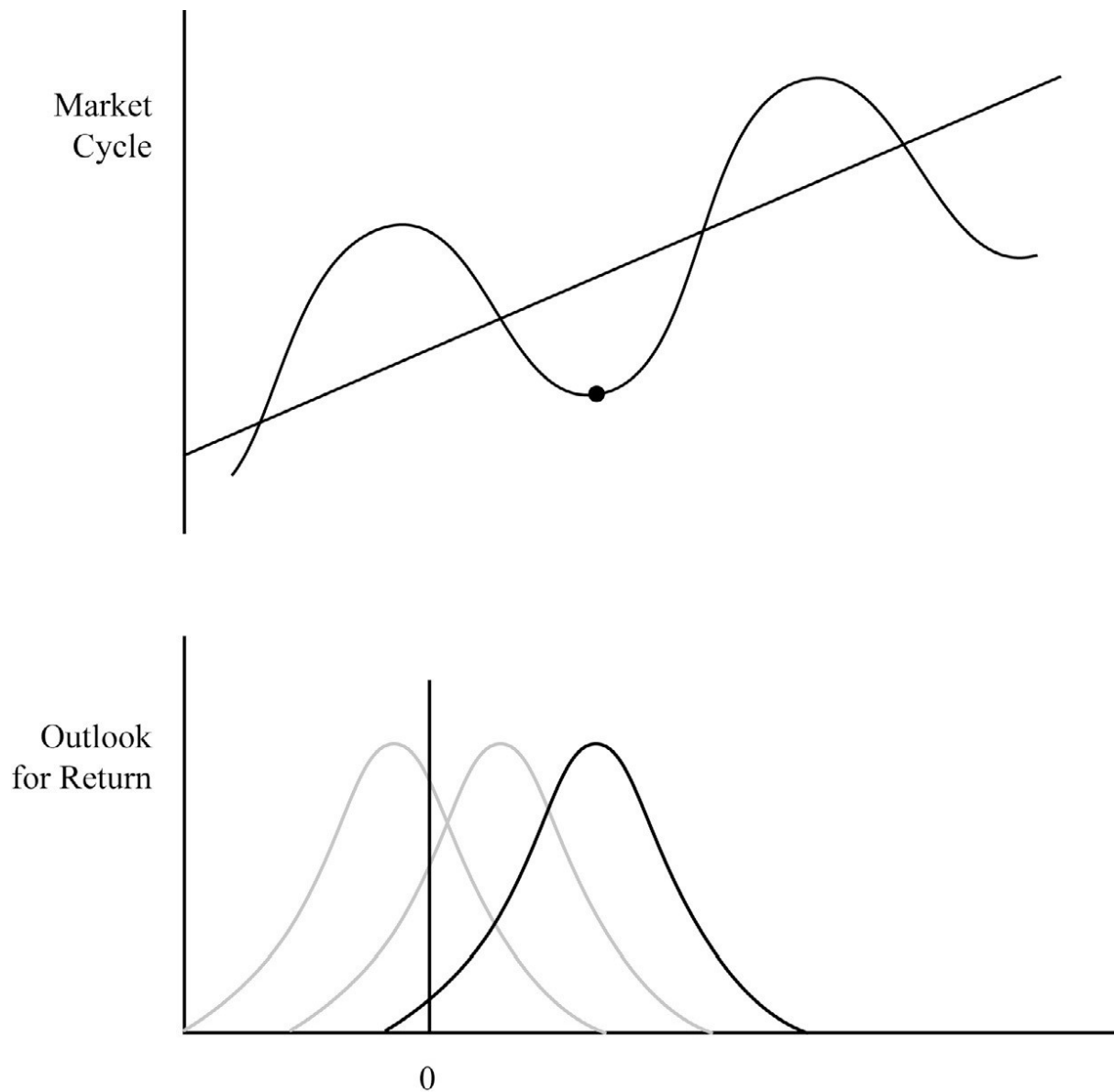
In the last few weeks—while on the verge of submitting my final draft of this book—I hit upon a way to show the relationship I have in mind. I will assume first that the market cycle is at its midpoint. That usual state of affairs—economic growth is on-trend, profits are normal, valuation metrics are reasonable in the context of history, asset prices are in line with intrinsic value, and emotions aren't extreme. Given all the above, the outlook for future returns, too, is “normal,” meaning the probability distribution governing future returns looks like the [one on page 19](#).



But what happens if the market is, instead, at a cyclical peak where what's going on in terms of fundamentals, that means valuations are stretched, prices far exceed intrinsic value, and a high level of optimism is baked in. At such a point, the outlook for returns is sub-par or even negative, as shown by the new distribution below.



And how about at a cyclical low? Now, thanks to depressed investment psychology, valuation metrics are at historically low levels that suggest the presence of bargains, and thus asset prices are well below intrinsic value. Now the distribution regarding future returns is shifted well to the right, implying unusually high potential for profit.



This conceptual depiction indicates the relationship between cycle level and potential for return. It's far from scientific, but everything I it's right.

XIII

HOW TO COPE WITH MARKET CYCLES

What's the key in all of this? To know where the pendulum of psychology and the cycle in valuation stand in their swings. To know when to buy—and perhaps to sell—when too-positive psychology and the willingness to assign too-high valuations cause prices to soar to peak levels. And to buy when downcast psychology and the desertion of valuation standards on the downside cause panicky investors to catch bargains by selling, despite the low prices that result.

The investor's goal is to position capital so as to benefit from market developments. He wants to have more invested when the market rises than when it falls, and to own more of the things that rise more or fall more than the others. The objective is clear. The question is how to achieve it. The first step is to decide how you will deal with the future. If you don't believe in economic and market forecasting, and in taking action on the basis of such forecasts, you must demand. Thus they invest more aggressively when forecasts call for favorable events, and vice versa.

As I've made clear, I don't believe in forecasting. Very few people know enough about what the future holds for it to add to their return record. The record of most forecasters—in terms of both predicting events better than others and having better investment performance than others as a result—is quite lackluster. A few people become famous in each period because of their spectacular successes, but usually their next correct forecast doesn't come for many years.

In the absence of the ability to see the future, how can you position your portfolios for what lies ahead? I think much of the answer lies in a better understanding of where the market stands in its cycle and what that means for its future movements. As I wrote in *The Most Important Thing*, “we can never know where we're going, but we'd better have a good idea where we are.”

To do that requires an understanding of the basic nature of market movements in general: what gives rise to their movements, what causes them to progress

toward peaks and troughs, and what causes them to retreat from extremes? We've touched on the key elements that concern us:

- the tendency of basic themes to repeat and of history to
- the tendency of things to rise and fall, especially those determined by human nature,
- the way each development in a cycle has implications for
- the way the various cycles interact and influence each other,
- the role of psychology in pushing cyclical phenomena beyond levels,
- thus the tendency of cycles to go to extremes,
- their tendency to move from extremes back toward a midpoint,
- the regularity with which that movement continues past the midpoint, toward the opposite extreme.

These are the generalities that affect cycles of all kinds. In addition, we have to bear in mind the specific elements that influence the market:

- the economic and profit cycles that shape the investment environment
- the tendency of psychology to overreact to developments in the environment,
- the way risk is considered non-existent and benign at some times and then enormous, inescapable and lethal at others, and
- the way market prices reflect only positives and overstate them at one point, and then reflect only negatives and ignore all the positives at another.

These are the fundamentals—the basics—regarding cycles in general. We must understand the specific ways they work in the markets—that we must perceive, attend to, and obey. We must use the insights we garner to assess where the market is positioned, what that implies for its future movements, and what we should do as a result.



Equipped with a deep understanding of the above, we turn to figuring out where we stand in the cycle.

What's the key in all of this? To know where the pendulum of the market and the cycle in valuation stand in their swings. To refuse to buy—perhaps to sell—when too-positive psychology and the willingness to assign too-high valuations cause prices to soar to peak levels. And to buy when downcast psychology and the desertion of valuation standards on the downside cause panicky investors to create bargains by selling despite

prices that prevail. As Sir John Templeton put it, “To buy when others despondently selling and sell when others are greedily buying requires the greatest fortitude and pays the greatest reward.”

The upward movement of prices from fair value to excess usually results from the presence of some combination of important elements:

- generally good news,
- complacency regarding events,
- uniformly upbeat treatment by the media,
- the unquestioning acceptance of optimistic accounts,
- a decline in skepticism,
- a dearth of risk aversion,
- a wide-open credit market, and
- a positive general mood.

Conversely, the collapse of prices from fair value to bargain levels is usually marked by some or all of the following:

- generally bad news,
- rising alarm regarding events,
- highly negative media accounts,
- the wholesale acceptance of scare stories,
- a strong rise in skepticism,
- a significant increase in risk aversion,
- a credit market that has slammed shut, and
- a mood of general depression.

The question is how we can tell where the market stands in the cycle. Importantly, the elements that contribute to the market's rise manifest themselves via valuation metrics—p/e ratios on stocks, yields on bonds, capitalization ratios on real estate, and cash flow multiples on buyouts. All of these things are elevated relative to historic norms. All of these things are low prospective returns. The reverse is true when a market collapses: asset prices to bargain valuations. These things can be observed and quantified.

In addition, our understanding of cycle positioning can be greatly enhanced by an awareness of how investors are behaving. To respond to market movements, one realization is more important than all others: the risk in investing doesn't come primarily from the economy, the companies, the securities, the stock certificates or the exchange buildings. It comes from the behavior of the market participants. So do most opportunities for exceptional returns.

When investors act in a prudent manner, reflect risk aversion, skepticism and restrain their positive emotions, security prices tend to be reasonable relative to underlying value, making the market a safe place. On the other hand, when investors become euphoric, their overenthusiastic buying lifts prices to dangerous levels. And when they are despondent, their panicked selling takes prices to bargain lows.

Warren Buffett tells us, “The less prudence with which others conduct affairs, the greater the prudence with which we should conduct affairs.” When others are euphoric, we should be terrified. And when we are terrified, we should turn aggressive.

It’s not what you buy that determines your results, it’s what you pay. And what you pay—the security’s price and its relationship to intrinsic value—is determined by investor psychology and the resulting behavior. The key to being able to behave in a way that’s appropriate given the market lies significantly in assessing psychology and the behavior of other investors. We have to know whether the market is red-hot and thus overpriced, or is thus a bargain.

In “It Is What It Is” (March 2006)—and also in *The Most Important*—I included what I called a guide to market assessment. I can’t reason not to include it again here, or of a better substitute. These points are non-scientific, non-quantitative and non-quantifiable, and even somewhat jocular. But they should give you a sense for the watchout for:

Economy:	Vibrant	Sluggish
Outlook:	Positive	Negative
Lenders:	Eager	Reticent
Capital markets:	Loose	Tight
Capital:	Plentiful	Scarce
Terms:	Easy	Restrictive
Interest rates:	Low	High
Yield Spreads:	Narrow	Wide
Investors:	Optimistic	Pessimistic
	Sanguine	Distressed

	Eager to buy	Uninterested in buying
Asset owners:	Happy to hold	Rushing for the exits
Sellers:	Few	Many
Markets:	Crowded	Starved for attention
Funds:	Hard to gain entry	Open to anyone
	New ones daily	Only the best can raise money
	General Partners hold the cards	Limited Partners have bargaining power
Recent performance:	Strong	Weak
Asset prices:	High	Low
Prospective returns:	Low	High
Risk:	High	Low
Popular qualities:	Aggressiveness	Caution and discipline
	Broad reach	Selectivity
The right qualities:	Caution and discipline	Aggressiveness
	Selectivity	Broad reach
Available mistakes:	Buying too much	Buying too little
	Paying up	Walking away
	Taking too much risk	Taking too little risk

As I said in introducing the guide, “for each pair, check off the one you think is most descriptive of the current market. And if you find that your checkmarks are in the left-hand column, hold on to your wallet. These sorts of markers can tell us where we stand in the cycle and what that’s likely to imply for the future. Thus they help in what I call the temperature of the market.” Here’s how I described the process

Important Thing:

If we are alert and perceptive, we can gauge the behavior of the world around us and from that judge what we should do.

The essential ingredient here is *inference*, one of my favorite words. Everyone sees what happens each day, as reported in the media. But how many people make an effort to understand what the everyday events say about the psyches of market participants, the investment climate, and thus what should be done in response?

Simply put, we must strive to understand the implications of what is going on around us. When others are recklessly confident and buying aggressively, we should be highly cautious; when others are frightened into inaction or panicked selling, we should become aggressive.

So look around, and ask yourself: Are investors optimistic or pessimistic? Do the media talking heads say the markets should be piled into or avoided? Are novel investment schemes readily accepted or dismissed out of hand? Are securities offerings and fund openings being treated as opportunities to get rich or possible pitfalls? Is the credit cycle rendered capital readily available or impossible to obtain? Are price/earnings ratios high or low in the context of history? Are yield spreads tight or generous?

All of these factors are important, and yet none of them entail forecasting. We can make excellent investment decisions on the basis of present observations, with no need to make guesses about the future.

The key is to take note of things like these and let them tell you what to do. While the markets don't cry out for action along these lines every day, they do at the extremes, when their pronouncements are highly important.

At any time, lots of things are happening in the world, the economic and investment environment. No one can study, parse, understand and incorporate all of them into investment decisions. And no one need try. At any time, different events occur in each cycle, and in a different sequence, with different results.

My point here is that not all the details are important. Rather, (a) figure out which are the important ones, (b) make inferences about what is going on from the important ones (and then perhaps consider as many less-important ones as we can), and (c) conclude from those inferences what are the one or two things that most characterize the investment environment and what action they call for. In other words, being attuned to cyclical developments and their significance.

In particular, however, you can't have the extremes of market cycle without a departure of valuation metrics from their norms. Valuation is a result—and thus symptomatic or indicative—of investor psychology.

The psychological and emotional elements I've listed have their primary impact by convincing investors that past valuation standards have become irrelevant and can be departed from. When investors are flying high and making money, they find it easy to come up with convenient reasons why assets should be untethered from the constraints of valuation norms. The explanation usually begins with, "it's different this time." Watch for an ominous sign of the willing suspension of disbelief. Likewise, when prices collapse in a crash, it's usually because of an assumption that the things that supported value in the past can be trusted to work again.

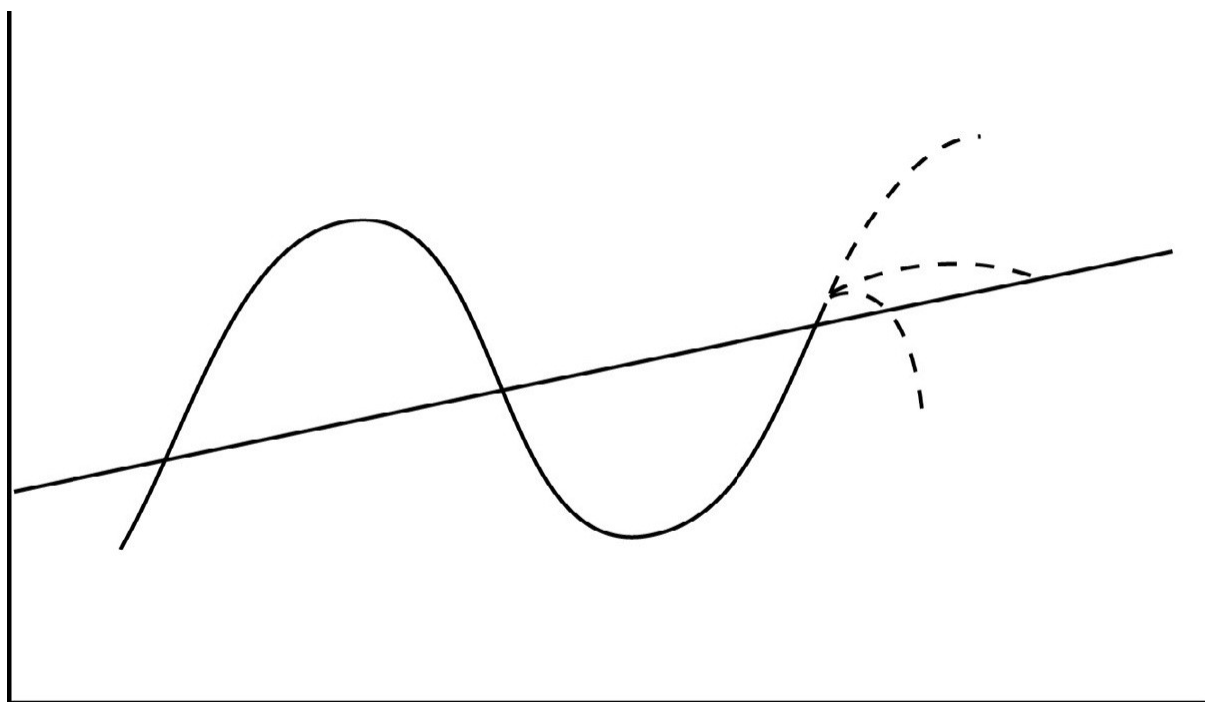
So the key to understanding where we stand in the cycle depends on the forms of assessment:

- The first is totally quantitative: gauging valuations. This is an appropriate starting point, for if valuations aren't out of line with the market cycle is unlikely to be highly extended in the future.
- And the second is essentially qualitative: awareness of what's going on around us, and in particular of investor behavior. Importantly, it's possible to be disciplined even in observing these largely non-quantitative phenomena.

The key questions can be boiled down to two: how are things priced? and how are investors around us behaving? Assessing these two elements—consistently and in a disciplined manner—can be very helpful. The answers will give us a sense for where we stand in the cycle.

In closing on this subject, I want to repeat something I've been saying: even the best of temperature-taking can't tell us what will happen next or the tendencies.

Since market cycles vary from one to the next in terms of the pace and duration of their fluctuations, they're not regular enough for us to be sure what'll happen next on the basis of what has happened. From a given point in the cycle, the market is capable of moving in any direction: up, flat or down.



But that doesn't mean all three are equally likely. Where we influences the tendencies or probabilities, even if it does not determine developments with certainty. All other things being equal, when the market is high in its cycle, a downward correction is more likely than continued growth, and vice versa. It doesn't have to work out that way, of course, but it's a safer bet. Assessing our cycle position doesn't tell us what will happen, just what's more and less likely. But that's a lot.



The best way to teach the recognition of cyclical excesses is through examples of how it can be done at the extremes, where it matters most. Over the following pages I'm going to review the formation of two market bubbles and the crash that followed the second of them. Even though the bubbles were different, but each of these three market events illustrates the importance of assessing the market's temperature.

First, let's review the meteoric rise of stocks in the late 1990s, and especially the formation of the Internet bubble. What should an alert investor have noted during that time?

- In the decade of the 1990s, the U.S. economy enjoyed peacetime expansion in its history.
- In December 1996, when the S&P 500 index of equities stood at a record high, Fed Chairman Alan Greenspan asked, "How do we know when irrational exuberance has unduly escalated asset values?" But he was never

from again on this subject, even as the S&P more than doubled of 1527 in 2000.

- In 1994, Prof. Jeremy Siegel of the Wharton School published *Stocks for the Long Run*, in which he pointed out that there had been a long period of time in which stocks had failed to outperform bonds, cash and inflation.
- Whereas researchers at the University of Chicago had earlier estimated that the normal return on U.S. equities was in the vicinity of 10% in the 1990s the average return on the S&P was nearly 20%.
- The better stocks performed, the more capital investors allocated to them. This was especially true of technology stocks, which were the clear market leaders.
- More tech stocks were added to equity indices like the S&P 500, meaning index and quasi-index investors had to buy more of them—which made them go up—which attracted still more capital to them. This was a classic “virtuous circle,” and no one could imagine it ending.
- The absence of earnings at most “new-economy” companies eliminated any requirement that their stocks should sell at reasonable price/earnings ratios.
- In the latter stages of the bubble, the prices of some dot-com stocks increased several hundred percent on the day of their initial offerings. To be willing to buy the new stocks at their inflated prices, buyers who thought about it would have to have believed either (a) the companies’ founders had been happy to sell the stocks at a fraction of their true value or (b) the founders knew less about their stocks’ value than the buyers did. Both of these conclusions were tenuous.
- In order to share in this miracle—and to avoid the pain of others profited—investors took part in IPOs of companies that had no earnings (and in some cases no revenues) and about whose business models they knew little or nothing.
- The year 1999 saw the publication of *Dow 36,000*, a book whose authors, James Glassman and Kevin Hassett, argued that because interest rates were so low in risk (see [Siegel](#) on [page 217](#)), they needn’t pay a risk premium as they historically had. That meant their prices should immediately rise to levels from which they would offer appropriately reduced prospective returns. Thus the Dow Jones Industrial Average deserved to more than triple from just over 10,000 at the time.
- The p/e ratio on the S&P 500—fueled by rampaging growth in technology stocks—reached a high of 33 during the tech-stock boom, fully double the post-war norm and the highest valuation in history up to that point in time.

So what ingredients actually were present?

- good economic news,
- favorable articles and books,
- carefree, risk-oblivious behavior on the part of investors,
- unusually strong investment returns,
- extremely high valuations relative to history,
- widespread willingness to pay prices that could not be justified analytically, and
- belief in a perpetual motion machine that would never stop.

This last point is particularly important. Every bubble starts with a grain of truth, as I noted earlier. But the import and profit potential of the bubble is overestimated in bubbles; disbelief is suspended; and it is widely believed that the gains can roll on forever.

Certainly an objective temperature-taker could have known the pieces were in place for a substantial bubble that was likely to fall to earth. I described phenomena as I described [them in “bubble.com”](#) (January 2000).

- Webvan Group, which started in business in 1999, had sales of \$100 million and a \$350,000 profit in the September quarter. The company currently values it at \$7.3 billion.
- On December 9, VA Linux went public at 30 and soared to \$239, for a market value of \$9.5 billion, half that of Microsoft. At that date, the company's 1999 sales were \$17.7 million and it had lost \$10 million (versus Apple's profit of \$600 million in the most recent 12 months).
- Because the price/earnings ratios of Internet companies are so high—usually negative—one may be forced to look to the price/earnings ratio in order to speak about valuation. Red Hat, for example, sells at about 1,000 times its annualized revenues in the August 1999 quarter.
- Among non-Internet tech companies, Yahoo! is worth \$119 billion, more than General Motors and Ford together. At the current stock price of \$432, its p/e ratio on 1999 estimated earnings is just over 1,000.

Under these unusual circumstances, the *Wall Street Journal* wrote on December 10, 1999 that “stock valuations take on an unusually large importance in gauging a business's performance.” In other words—the absence of other signs—people must look to the share price for the indication of how the company is doing. Isn't that backwards? In the old days, investors figured out how the business was doing and then set the share price to reflect it.

In this valuation parameter vacuum, a “lottery ticket mentality” seems to govern the purchase decision. The model for investing in the tech and dot-com companies isn’t the likelihood of a 20% annual return based on projected earnings and p/e ratios, but a 1,000% gain based on a concept. The pitch may be, “We’re looking for first-round financing for a company valued at \$30 million that we think can IPO in two years at \$2 billion.” Or maybe it’s “We’ll be priced at \$20. It may end the day at \$100 and be a winner in 12 months.”

Would you play? Could you stand the risk of saying no and being wrong? The pressure to buy can be immense.

There have always been ideas, stocks and IPOs that produced great profits. Yet the pressure to participate wasn’t as great as it is now because in the past the winners made millions, not billions, and it took years, not months. The upside in the deals that’ve worked so far has been 100-to-1 (give or take a zero). With that kind of potential, (a) the upside becomes irresistible and (b) it doesn’t take a very high probability of success to justify the investment. I have said in the past that while the market is usually driven by fear and greed, in this case the biggest motivator is the fear of missing out. Never was there as much pressure as today. This only intensifies the pressure to join in and come out on that limb of risk.

The bottom line is that in an extreme bubble like this one, an investor doesn’t have to make fine distinctions. All you have to do is identify nutty behavior when you see it. To me—as an observer looking in detached fashion from the sidelines, rather than someone with skin in the tech-investing game—the events of the tech/Internet craze had the feel of a Hans Christian Andersen tale. Those who participated in it so early rolled on that no one would step forward to say the emperor had no clothes. Developments like those just described were signs of the mass hysteria that is a part of every bubble.

Those who were enriched by the performance of equities in the 1990s, especially by the tech and Internet stocks—took it to foretell prosperity and wealth creation, and they responded to the returns they enjoyed by ratcheting up their expectations for future returns. (As Warren Buffett quotes the ancient Greek statesman and orator Demosthenes, “What each man wishes, that he also believes to be true.”) But Buffett concluded that euphoria-driven buying in the 1990s had overpaid for the performance and borrowed from future performance, with highly negative implications. It just took a couple of years for them to be proved right.

The eventual result in 2000–02 was (a) the first three-year decline in the stock market since 1929–31, with a 49% drop in the S&P 500 from its peak to bottom (ignoring income), (b) massive declines in tech stocks, and (c) a 100% drop in many Internet and e-commerce stocks.



Usually one would expect that the painful deflating of such a bubble had been driven by excesses of optimism and credulousness—would have had a serious instructional impact, meaning it would take a decade for another such bubble to arise. But that was not to be the case. More important, I'm going to repeat once more what John Kenneth Galbraith said about the learning of financial lessons:

Contributing to the . . . euphoria are two further factors that have not occurred in time or in past times. The first is the extreme brevity of the human memory. In consequence, financial disaster is quickly forgotten. As a further consequence, when the same or closely similar circumstances occur again, sometimes in only a few years, they are hailed by a new, often youthful, and always supremely self-confident generation as a brilliantly innovative discovery in the financial and larger economic world.

The events that only a few years later began to lead to the subprime mortgage boom—adequately described earlier in this book—provided a good example of the way cyclical excesses are built on error:

- As I wrote on [page 119](#), some U.S. leaders decided that only economic growth could come from increased home ownership.
- The federal agencies charged with issuing mortgages got the message and ramped up the availability of mortgages.
- Declining interest rates made mortgages—and thus home ownership—increasingly affordable.
- Increased home affordability and increased mortgage availability allowed home buying to increase and the demand for homes to grow.
- This increased demand caused the prices of homes to increase substantially. The canard that “home prices only go up” became an accepted truth, feeding back to produce still more demand for homes.
- The further platitude that “there can never be a nationwide wave of mortgage defaults” caused mortgage-backed securities to be prime candidates for investment, particularly by financial institutions.
- Wall Street came forward with a model for securitizing prosaic, home mortgages into tranches of collateralized debt obligations—the next

high-return, low-risk thing.

- The construction and selling of CDOs and other mortgage securities (MBS) added greatly to bank profits.
- With relaxed regulations allowing banks to employ materially greater leverage, large amounts of capital were available for investment equity required for MBS creation.
- The meteoric growth of MBS packaging created rapidly increasing demand for the essential raw material in the process: newly originated mortgages.
- In order to expand the volume of mortgages they issued, lenders found novel ways to increase their appeal to borrowers: interest-only mortgages that minimized monthly payments by eliminating the traditional requirement that the principal balance be paid down; adjustable-rate mortgages that allowed borrowers to benefit from the ultra-low interest rates at the short end of the yield curve; and, most importantly, “no-doc prime” mortgages (sometimes called “liar loans”) that didn’t require applicants to document income and employment.
- With sub-prime mortgages being packaged into securities and sold off to investors, as opposed to being retained as in the past, lenders’ focus shifted from borrowers’ creditworthiness to loan volume. With lenders being paid fees simply for making loans—and able to sell them off immediately and thus retain no risk of default—there was no incentive for them to worry about the creditworthiness of their borrowers. Clearly, this gave them perverse incentives. (Incentives like these—which allow participants to engage in pro-risk behavior without having to bear the consequences—were described in the Global Financial Crisis as creating “moral hazard,” a term that came into widespread use. Even if it’s heard less often these days, the concept survives, and it remains dangerous.)
- The key to the purported success of sub-prime mortgage backed securities lay in “financial engineering” performed by “quants”—Ph.D.’s, many of them in their first jobs. They modeled risk based on a flawed assumption that mortgage defaults would remain uncorrelated and benign as in the past.
- The creation of large amounts of sub-prime MBS meant there was a lot of business for the rating agencies whose imprimatur was essential for these profitable assignments to go to the agency that provided the highest rating. This led to ratings shopping and created powerful incentives, this time supporting widespread ratings inflation.
- These trends caused the rating agencies to issue many thousands of triple-A ratings to mortgage backed securities (as compared to the few U.S. companies enjoying triple-A ratings at the time). Rating

personnel clearly lacked the ability to fully understand the they were assessing.

- Banks and other financial institutions bought huge amounts of abetted by “risk-management” techniques such as Value at Risk relied heavily on the short, benign history and overly general and thus vastly understated the risk buyers were assuming.
- All of the above occurred in and was facilitated by risk-oblivious attitudes characterized by risk-is-gone rhetoric (see [page 120](#)).

So what was at the root of the formation of this bubble? A “Now It’s All Bad?” (September 2007):

. . . a standard combination that proved perfectly

- underlying greed,
- good returns in the up-leg of the cycle,
- euphoria and complacency,
- a free-and-easy credit market,
- Wall Street’s inventiveness and salesmanship, and
- investors’ naiveté.

The resulting developments that fed the upswing are clear:

- a huge decrease in risk aversion, and thus the disappearance of skepticism,
- the acceptance of sweeping positive generalizations regarding home mortgages,
- excessive faith in new tools like financial engineering and risk management, and
- widespread blindness to the impact of improper incentives on participants in the process.

In addition to short memories, psychological excesses and logical the bubble that arose in mortgage backed securities was abetted additional factors:

- Because this new bubble arose in mortgage-land—a part of the markets completely separate from that which had been visited by tech and Internet bubble—the fixed income investors and financial institutions it appealed to were ones who hadn’t been affected by the other, and hadn’t learned from it.
- The terrible recent performance of equities had so discouraged investors—and interest rates brought low by the dovish Fed

diminished the yields available on fixed income investments—that investors gave up on obtaining strong returns from stocks. That rendered them highly susceptible to the promise of a return without risk: mortgage backed securities.

This is a good time for a valuable aside. Falling for the su asset that will provide return without risk, or what I call the “silver bullet” (dating back to *The Lone Ranger*, a 1950s TV Western about a gunshots never missed)—is one of investors’ greatest recurring failings. It exemplifies “whatever man wishes,” as Demosthenes said, but it makes no sense. If there were a silver bullet:

- Why would the purveyor offer it to you rather than buying it?
- And wouldn’t everyone else buy it and drive up its price to where it’s no longer a sure thing?

I’ve seen dozens of silver bullets touted over the course of my career. Not one has proved out. No investment strategy or tactic can deliver a high return without risk, especially to buyers lacking investing skill. Outstanding investment results can only come from exceptional skill (or perhaps in isolated moments from good luck).

As I said earlier, bubbles generally spring from a grain of truth taken too far, and their eventual puncturing brings the greatest pain to the investment world. Belief in a silver bullet underlies many bubbles. The fact that investors are willing to swallow a promise of return without risk is an infallible indicator that skepticism is in short supply, psychology is overheated, and the subject asset is overestimated and thus probably overpriced. Watch out for it, and resist if you can. By definition, bubbles come to an end, and few people do.

I would sum up the key conclusion this way: The sub-prime mortgage bubble arose from broad acceptance that the phenomena would work that had never before been seen in action. The analysis of investment vehicles that entail (a) the application of skepticism and conservative assumptions and (b) examination over a long history that includes some trying times. These were clearly missing.

In fact, most bubbles, if not all, are characterized by the unacceptance of things that have never held true in the past; of valuations that are dramatically out of line with historic norms; and/or of techniques and tools that haven’t been tested.

The sub-prime mortgage bubble demonstrated an extremely important principle in action that I haven't touched upon before now. The investment environment—and the performance of investment techniques and instruments—is not immutable. Rather, as I've said many times with regard to market cycles, these things are affected by the involvement of people.

There is no such thing as a market that is separate from—the people who make it up. The behavior of the people in the market changes the market. When their attitudes and behavior change, the market will change.

In the case at hand:

- market forces changed the participants' motivations, as just
- these changed motivations changed their behavior, and
- their changed behavior clearly determined the results.

An essential progression was at work, as is clear to see:

- History was taken as saying there couldn't be a nationwide wave of mortgage defaults.
- Acceptance of this benign history caused vast sums to be invested in mortgage backed securities.
- It permitted the securities to be structured aggressively.
- It also caused rating agencies to extrapolate the benign experience and grant high ratings.
- Most importantly, the strong demand for mortgage backed securities created a need for the raw material—mortgage loans—and led to an eagerness to issue them that caused lending standards to deteriorate.
- These developments, taken together, virtually guaranteed that there *would be* a nationwide wave of mortgage defaults.

As I mentioned earlier, because of the impact of careless assumptions—the import of which went largely unnoticed and unremarked at the time—the default experience on mortgages issued from the late 1990s through the early 2000s turned out to be much worse than it historically had been; worse than what security structurers and rating agencies had thought possible; and worse than what buyers' models had said it would be. Ignoring this possibility was a significant contribution to the creation of the sub-prime mortgage bubble. The occurrence caused the resulting crisis.

The key is to understand that the behavior of investors can alter the market by changing the results that investors can expect the market to produce. This reflects George Soros's theory of reflexivity:

In situations that have thinking participants, the participants' distorted views can influence the situation to which they relate because false views lead to inappropriate actions. ("Soros: General Theory of Reflexivity," *Financial Times*, October 26, 2009)

People trying to understand how things work in the economic and financial worlds should take this lesson very much to heart.



All it takes for the perpetual motion machine to grind to a halt is one or two assumptions and the operation of some general rules:

- Interest rates can go up as well as down.
- Platitudes can fail to hold.
- Improper incentives can lead to destructive behavior.
- Attempts to quantify risk in advance—particularly as to novel products for which there is no history—will often be unavailing.
- The “worst case” can indeed be exceeded on the downside.

The error in all these things is always clear in hindsight. But the practices with regard to mortgages and mortgage backed securities—became so prominent and important—were taking place in an obscure corner of the financial world. Thus they were invisible to chief investment officers, portfolio strategists, equity investors, alternative investment managers, traditional bond buyers, and seemingly even mortgage investors.

We at Oaktree were fortunate in the years in which the mortgage crisis arose—leading up to the Global Financial Crisis—to understand that the credit cycle was becoming extended in the upward direction, and thus that markets were increasingly precarious. This led us to sell assets; replace liquidating distressed debt funds with smaller ones; increase our level of consciousness and conservatism; and raise a stand-by fund several times the largest ever in order to take advantage of the distressed debt opportunities that might materialize.

What was the basis on which we did this? In retrospect it was easy, although it never seems as easy in real time. All you really had to do was make the following general observations:

- the Fed had reduced the base rate of interest to very low levels to ward off the depressing effects of the tech bubble's bursting, as concern over Y2K;
- because of the low yields available on Treasuries and high grade debt, as well as the disenchantment with equities that had resulted

three-year decline in 2000–02, investors were eager to put money in alternative instruments;

- investors had shrugged off the pain of the collapse of the 2000 and the telecom meltdown and corporate scandals of 2001–
- thus little risk aversion was present (especially in areas other than equities, which remained out of favor), rendering investors generally eager for investments in exotic, structured and synthetic instruments;
- as a result of all the above, the markets were wide open to low-quality debt, poorly structured instruments and untested alternatives.

These were our observations, and it was the last that most called out attention to the negative trends that were afoot. It felt like there was no one on which either Bruce Karsh or I didn't visit each other's office to complain about a newly issued security, saying, "It shouldn't be possible to have a piece of junk like this. The fact that it is means there's something wrong with the market." Those risky deals told us fear, skepticism and risk aversion were insufficient, and greed, gullibility and risk tolerance were in the ascendant. The implications of this combination are never good.

All the points noted here were obvious and not subject to debate. What mattered is whether you made these observations and drew the appropriate conclusions. You didn't have to fully understand what was wrong with prime mortgages or deconstruct mortgage backed securities and highly structured collateralized debt obligations. *We certainly didn't.*

And, by the way, in those years when the mortgage bubble was bursting, stocks weren't doing well or selling at lofty multiples, and the market wasn't booming (and thus necessarily heading for a recession). But if you made the observations just listed, you likely would have concluded, as I did, that it was time to reduce the quantum of risk in your portfolio. It took all it took.

And here are the results of the bursting of the mortgage bubble and the associated contagion, as shown in the performance of some of the investment indices in 2008, the year it all came apart. Clearly it was enormously important to have reduced risk.

Standard & Poor's 500 Stocks	(37.0)%
Dow Jones Industrial Average	(31.9)
NASDAQ Composite	(40.0)
MSCI Europe, Australasia, Far East Stocks	(45.1)
Citigroup High Yield Market Index	(25.9)
Merrill Lynch Global High Yield European Issuer Index (in Euro)	(32.6)
Credit Suisse Leveraged Loan Index	(28.8)
Credit Suisse Western European Loan Index (in Euro)	(30.2)

Finally in discussing how to detect and respond to market cycles, I want to return once more to the widespread panic that followed the bankruptcy filing by Lehman Brothers in September 2008.

Although the sub-prime mortgage crisis originated in a small corner of the financial and investment world, the impact was soon felt widely, by the financial institutions that had underestimated the risk in mortgage-backed securities and thus invested too heavily in them. As a threat to these essential institutions, the impact metastasized to the bond markets in all countries—and then to economies all around in the form of the Global Financial Crisis.

Thus, as I described earlier, money market funds and commercial paper were to be guaranteed by the U.S. government. A number of prominent financial institutions failed or had to be bailed out/rescued/absorbed. No one knew how far the carnage would spread. The equity and debt markets collapsed. Now the generalizing was on the negative side: “the financial system could totally melt down” in a vicious circle without end.

Since the generalizations were on the downside, the error-making machine went into reverse. No greed, only fear. No optimism, only pessimism. No tolerance, only risk aversion. No ability to see positives, only negative willingness to interpret things positively, only negatively. No ability to imagine good outcomes, only bad. Thus we reached the day on which the discussion mentioned back on [pages 131–132](#), in which the pension head was unable or unwilling to accept that any assumption regarding possible defaults could be conservative enough.

What was the essential observation? Here’s what I wrote in “The Negativism” (October 2008):

Contrarianism—doing the opposite of what others do, or “leaning against the wind”—is essential for investment success. But as the credit crisis reached a peak last week, people succumbed to the wave rather than resisting. I found very few who were optimistic; most were pessimistic to some degree. Some became genuinely depressed—even a few great investors I know. Increasingly negative tales of companies coming meltdown were exchanged via email. No one applied skepticism, or said “that horror story’s unlikely to be true.” People fed on itself. People’s only concern was bullet-proofing their portfolios to get through the coming collapse, or raising cash to meet redemptions. The one thing they weren’t doing last week was

making aggressive bids for securities. So prices fell and fell, several points at a time—the old expression is “gapped down.”

The key—as usual—was to become skeptical of what “everyone” was saying and doing. One might have said, “Sure, the negative may turn out to be true, but certainly it’s priced into the market. There’s little to be gained from betting on it. On the other hand, if it turns out not to be true, the appreciation from today’s depressed prices will be enormous. I buy!” The negative story may have looked compelling, but it’s the positive story—which few believed—that held, and still holds, the greater potential for profit.

At this market cycle extreme, all the news truly was negative. It was certainly not imaginary. The only questions I received were “How far will it go?” and “What will be the effects?” Given that asset prices reflected such abject pessimism regarding these things—I’d say near-suicidal thinking—the key to profiting lay in recognizing that even in the face of bad news and a very poor outlook, pessimism can be overdone, and thus prices can become too cheap.

It was the excessiveness of the prevailing pessimism that led me to write “The Limits to Negativism” at the credit market’s low ebb in 2008. In it I pointed out, as mentioned in the chapter on attitudes, that the superior investor’s essential skepticism “calls for pessimism when optimism is excessive. But it also calls for optimism when pessimism is excessive.” That variety of skepticism was totally lacking in the darkest days, of course.

Shortly after Lehman’s bankruptcy filing on September 15, 2008, Bruce Karsh and I reached the conclusion that (a) no one could know how a financial institution meltdown would go, but (b) negativity was certainly rampant and very possibly excessive, and assets looked terribly cheap. Thinking strategically, we decided that if the financial world ended, no one could rule out—it wouldn’t matter whether we’d bought or not—the world didn’t end and we hadn’t bought, we would have failed our job.

So we bought debt aggressively. Oaktree invested more than a billion dollars a week over the fifteen weeks from September 15 through December 31 of the year. Some days we thought we were going too fast, and some days we went slow; that probably meant we had it about right. The world didn’t end; the vicious cycle of financial institution implosion stopped with Lehman Brothers; the capital markets reopened; the financial institutions came back to life; debt was again able to be refinanced; bankruptcies turned out to be few relative to history; and the assets we bought appreciated substantially. In short, paying heed to the cycle was rewarded.

While we're reviewing the climate in late 2008, this is an appropriate a discussion of investor behavior on the way to, and at, market bottom.

First, what is a bottom? It's the point when the lowest prices of a security are reached. Thus a bottom can be viewed as the day the price falls, or the last day on which sellers predominate relative to buyers. Whatever reason, it's the last day prices go down, and thus the security reaches their nadir. (Of course these definitions are highly exaggerated. The expression "a bottom"—like "a top"—describes a period of time, not a single day. Thus the phrase "the last day" is mostly a figure of speech.) After prices rise, since there are no holders left to capitulate and sell, or no buyers now want to buy more strongly than the sellers want to sell.

The question I want to touch on here is, "When should one buy?" I made reference in earlier chapters to "falling knives," which is an important concept. When a market is cascading downward, investors often can be heard to say, "We're not going to try to catch a falling knife." In other words, "The trend is downward and there's no way to know when it will stop, so why should we buy before we're sure the bottom has been reached?"

What I think they're really saying is, "We're scared—in particular about buying before the decline has stopped, and thus of looking foolish by going to wait until the bottom has been reached, the dust has settled, and uncertainty has been resolved." But hopefully by now I've made it abundantly clear that when the dust has settled and investors' nerves have steadied, the best bargains will be gone.

At Oaktree, we strongly reject the idea of waiting for the bottom before buying.

- First, there's absolutely no way to know when the bottom has been reached. There's no neon sign that lights up. The bottom can only be recognized only after it has been passed, since it is defined as the point before the recovery begins. By definition, this can be identified only after the fact.
- And second, it's usually during market slides that you can find the best quantities of the thing you want, from sellers who are throwing in the towel and while the non-knife-catchers are hugging the sidelines. Once the slide has culminated in a bottom, by definition there are no sellers left to sell, and during the ensuing rally it's buyers who predominate. Thus the selling dries up and would-be buyers face growing competition.

We began to buy distressed debt immediately after Lehman filed for bankruptcy protection in mid-September 2008 as described on page 235, we continued through year-end, as prices went lower and lower. By quarter of 2009, other investors had collected themselves, caught on values that were available, and gathered some capital for investment the motivated sellers done selling and buying having begun, them to buy in size without pushing up prices.

Like so many other things in the investment world that might be tried on the basis of certitude and precision, waiting for the bottom to great example of folly. So if targeting the bottom is wrong, buy? The answer's simple: when price is below intrinsic value. What if price continues downward? Buy more, as now it's probably an even greater bargain. All you need for ultimate success in this regard is intrinsic value, (b) the emotional fortitude to persevere, and (c) even have your estimate of value proved correct.



Here's how the prominent investment indices did the following year. returns that were available in 2009 show the importance of having a cycle at a negative extreme and having bought in (or through) the accompanying chaos.

Standard & Poor's 500 Stocks	26.5%
Dow Jones Industrial Average	22.7
NASDAQ Composite	45.4
MSCI Europe, Australasia, Far East Stocks (in US\$)	27.8
Citigroup High Yield Market Index	55.2
BofA Merrill Lynch Global High Yield European Issuance Index	43.0
Credit Suisse Leveraged Loan Index	44.9
Credit Suisse Western European Loan Index (in Euro)	47.2

It's time for another aside: If you look at the last two tables—showing big losses in 2008 and big gains in 2009—it's easy to conclude the two years together were something of a non-event. For example, put \$100 into the Credit Suisse Leveraged Loan Index on the first of 2008, you would have lost 29% over the course of the year and left at the end. But then you would have gained 45% in 2009 with \$103 at the conclusion of the two-year period, for a net two-year results in the asset classes listed above ranged from moderate losses to moderate net gains.

It matters enormously, however, what you did in between. You would have enabled you to recoup most or all of your losses.

with results as described above. But if you lost your nerve and
trough—or if, having bought with borrowed money, you received
call you couldn't meet and saw your positions sold out from under you—
experienced the decline but not the recovery, and your net result in
event” two-year period was disastrous.

For this reason, it's important to note that exiting the market
—and thus failing to participate in a cyclical rebound—is truly
sin in investing. Experiencing a mark-to-market loss in the down
of a cycle isn't fatal in and of itself, as long as you hold through
upward part as well. It's converting that downward fluctuation into a
permanent loss by selling out at the bottom that's really terrible.

Thus understanding cycles and having the emotional and financial
wherewithal needed to live through them is an essential ingredient
investment success.



Before I declare victory in the Global Financial Crisis and
state firmly that the success my colleagues and I enjoyed in
this cycle wasn't inevitable. That's because—reflecting Elroy Dimson's
theme—the good outcome we got wasn't the only outcome that could
materialized. I'm convinced that if Hank Paulson, Tim Geithner and
Bernanke hadn't acted when they did, or if they had acted differently,
their actions hadn't been as successful as they were, a financial
replay of the Great Depression absolutely could have occurred. In that
our actions wouldn't have been cause for celebration.

I fear that people may look back at the decline of 2008 and
that followed and conclude that declines can always be depended
recouped promptly and easily, and thus there's nothing to worry
down-cycles. But I think those are the wrong lessons from the
the outcome that actually occurred was so much better than some of
“alternative histories” (as Nassim Nicholas Taleb calls them) that could have
occurred instead. And if those incorrect lessons are the ones that
as I believe they may have been, then they're likely to bring on
increases the amplitude of another dramatic boom/bust cycle someday,
one with more serious and long-lasting ramifications for investors and
of society.

But things did break for us in the recovery from the Crisis, and
“long” investors. Certainly we positioned our clients' portfolios correctly
the future that materialized, and a lot of it was because of the
way in which psychological and market cycles operate. Given the
predict the future, that's the best anyone can do.



Bubbles and crashes have a pattern all their own: a logic—or essence of which rhymes from one instance to the next. The three reviewed here show the oscillation of the cycle at its most extreme, hopefully they provide an indication of how cycles can be recognized and dealt with.

I want to make a few last, essential points.

- First, every element in the progressions I've described was clear the time, providing investors could keep their emotions and distorted perception from getting in the way.
- Second, drawing the essential inferences—and thus taking the appropriate actions—didn't require any forecasting at all. The descriptions of the actual progressions are compelling, with no guesswork needed regarding the future. The events and resulting cyclical behavior dictated profitable behavior.
- Finally, however, while I say the events were self-evident and implications obvious, I want to state definitively that nothing the time. Even the best and least emotional among us are subject to the same inputs and stimuli as everyone else. We were never sure, but we did the right thing nevertheless. And while the errors leading up to the Global Financial Crisis were easily recognizable, the timing of correcting was absolutely beyond predicting. The best investors can act in light of what they see in the environment. But they remind me of what John Maynard Keynes is reputed to have said: "The market can remain irrational longer than you can remain solvent."



To close on the subject of dealing with cyclical events, I'll give you one more example, from 1991. Leveraged buyouts had boomed in the 1980s, thanks to the ability of buyers of companies to access debt capital, often ranging up to 95% of the total purchase price. Many of the subject companies being saddled with debt they wouldn't be able to service in the recession that followed; to large numbers of companies bankruptcies; and thus to high yield bonds' first crisis. These developments came to a head just after we formed our Funds II and IIB for Europe. Here's how I assessed the environment in a letter to their clients, March 23, 1991:

In general, the market prices of distressed company debt declined during 1990. Some of this was driven by fundamentals, as the value of all assets weakened along with the economy, and some was due to “technical conditions,” meaning the supply of this sort of debt ballooned and buyers were discouraged and withdrew.

The worsening economic and psychological climate gives us the opportunity to pick and choose among a large number of potential investments at bargain prices. The environment is dreary, and what one buys is soon quoted lower, and there is no ebullience.

These are exactly the conditions under which we want to be working. When buyers are enjoying themselves, and when they are exchanging “high fives” because everything they buy goes higher the next day and makes them feel smart, the “pain index” is too high. Buyers are emboldened.

Today’s conditions tell me we are more likely to be getting bargains than if we were having fun. There are few competing buyers to drive up the prices of the things we want to buy. Each purchase price is more likely to turn out to have been a “low” than a “high.” In short, this is a good time to be putting money to work in an investment area such as ours.

We cannot presume to begin investing on the day the economy hits market bottom. Our greatest hope is that the low point will come sometime during the period when we are actively investing and that we will buy on the way to, during, and after that point.

This is a good example of taking the temperature of the market at the time. It was not just because it was me who took it, and it turned out to be right (the distressed debt funds we were investing at that time had some of the highest returns we’ve ever achieved). Rather, it was because it detected and pushed back against the depressing emotional influence that we were keeping others from buying. It shows we knew the “dreary” and mark-to-market losses that were driving buyers away were more likely to have favorable rather than unfavorable implications for subsequent returns, and that declining prices are good for buyers, not bad.

Understanding what things really mean—rather than how they make investors feel—is the first step toward doing the things that are right at the right times.



Having allowed this chapter to reach great length, I’m going to end with a general discussion of how to think about positioning a portfolio as the market moves through its cycle.

I think it's helpful to take an organized approach to what I risks." What I'm talking about here is the fact that investors have with two possible sources of error. The first is obvious: the risk money. The second is a bit more subtle: the risk of miss. Investors can eliminate either one, but doing so will expose them entirely the other. So most people balance the two.

What should an investor's normal stance be regarding the two balanced, or favoring one or the other? The answer depends on goals, circumstances, personality and ability to withstand risk (and on the same things with regard to one's clients, if any).

And separate and apart from his normal posture, should the investor the balance from time to time? And if so, how? I think investors appropriately adjust their stance if they (a) feel they have the required and (b) are willing to expend effort and bear the risk of being wrong. They should do this based on where the market is in its cycle. In a market that is high in its cycle, they should emphasize limiting the loss of money, and when the market is low in its cycle, they should be reducing the risk of missing opportunity.

How? Try to travel into the future and look back. In 2023, do you think you're more likely to say, "Back in 2018, I wish I'd been more aggressive?" "Back in 2018, I wish I'd been more defensive"? And is there anything about which you'd be likely to say, "In 2018, I missed the chance to buy xyz"? What you think you might say a few years down the road will help you figure out what you should do today.

The above decisions relate directly to the choice between aggressiveness and defensiveness. When an investor wants to reduce his chance of losing money, he should invest more defensively. More worried about missing opportunity? In that case, increased aggressiveness is called for. Varying one's stance should be done in response to where the market is in its cycle and, again, this can be approached in terms of how the market is behaving and how other investors are behaving—the two elements in assessing the market mentioned earlier.

When most investors are behaving aggressively, that's a good signal that the market is a risky place, since little risk aversion is being applied. If investors' aggressiveness is likely to have resulted directly in elevated prices. In both ways, as I [mentioned on page 211](#), the aggressiveness makes the market risky for us.

A good way to think about this decision is to consider which attitude is right for the current market environment. In late 2008/early 2009, a conservative investor needed only two things to make a lot of money: money to invest and a market that was about to invest in. If he had those two things, he made a lot of money that followed. In retrospect, what he didn't need was caution, conservatism.

risk control, discipline, selectivity and patience; the more of those things he had, the less money he made.

Does that mean “money and nerve” is always a surefire formula for investment success? Absolutely not. If an investor had money and nerve in early 2007, he bore the full brunt of the Global Financial Crisis. If he had risk control, discipline, selectivity and patience, he needed caution, conservatism, risk control, discipline, selectivity and patience. Moreover, even in late 2008/early 2009, intelligent investors couldn’t completely abandon caution and discipline, because there was no way for them to know that the recovery from the GFC would be as smooth as the aftermath so relatively painless for investors. At Oaktree, we focused a lot, but we emphasized the senior debt of high-quality companies, not the junior debt and weaker issuers in which, as it turned out, investors made even more money.

Among the many factors that make investing interesting is the fact that there’s no tactic or approach that always works. The only way to be correctly positioned as the cycle moves is to make well-reasoned bets and adjust the attributes employed. But it’s not easy.

One way people tend to respond to the challenges these days is to ask, “What inning are we in?” Ever since the financial meltdown in 2008, I’ve encountered this question on a regular basis. What people really mean is, “Where do we stand in the cycle?” In the fourth quarter of 2008, I wondered, “How much of the pain has been felt already, and how much lies ahead?” More recently they’ve been inquiring primarily about the end of the cycle: how much longer will it continue on the upswing—making it difficult to borrow—and when will credit availability begin to tighten?

I consider these questions in the light of my sense for how far the cycle has gone, and I answer in the form the questioners want: second inning (game just getting started), fifth inning (game half over), or eighth inning (game near end). But recently I’ve become more conscious of the limitation of this approach: unlike a normal baseball game, we have no way to know when a particular cycle will go on. There is no regulation length. A baseball game will last nine innings, but an economic or market cycle could last nine, twelve or fourteen. These things are not knowable.

None of these approaches offers a foolproof technique for deciding how to position portfolios. There’ll always be just ways to think systematically about something that isn’t subject to easy answers. But hopefully they suggest a route that’s superior to deciding on the basis of emotion, guesswork or simply following the herd.

How you deal with cycles is one of the most important things in investing. Cycles will happen to you. What you do in response is key.

XIV

CYCLE POSITIONING



Successfully positioning a portfolio for the market movements t ahead is dependent on what you do (turning aggressive or defensive and when you do it (based on a superior understanding of the market imply for future market moves).

I once knew a man who was congenitally optimistic and aggressive perhaps because he'd had the good sense to be born rich and had a charmed life. He never exhibited self-doubt or seemed to question the accuracy of his forecasts or the likelihood that his stratagems would work. He was aggressive as long as I knew him, in what turned out to be the right time for aggressiveness. That experience inspired me to coin a phrase to describe the forces at work:

There are three ingredients for success—aggressiveness, timing and skill—and if you have enough aggressiveness at the right time, you don't need that much skill.



In February 2017, I found myself working on the final chapters while on vacation in India. One day I visited one of the world's oldest cities, Jaipur's Amer Fort—and attempted to capture a small fraction of its history in photos. While reviewing my photos afterwards, I had the good luck to stumble on some I had taken a few months earlier in another part of China.

In the course of that visit, a Beijing-based client had asked me some provocative questions, and in answering them I made some scribbles, as I often do, on a whiteboard. Answers came to me that day that hadn't occurred to my mind before. Recognizing that, I took photos of the board before leaving. (What an innovation: twenty years ago I would not have had a camera at a presentation.) Three months later, while reviewing my photographic handiwork at the Amer Fort, I came across those photos.

China and thought back to those ideas for the first time. They'll provide the grist for this chapter.

Now I found myself in bed in India, after a night's sleep cut by a 10½-hour time difference from New York. For some reason, a connection dawned between the above saying about success, my photos from China, and the question of how to deal with cycles. In short, I had figured out how to parse the main components of investment skill.

(Note that I say "a connection dawned" rather than "I made a connection." I consciously choose the passive form here, because I often do not realize that ideas "come to me," rather than feeling I develop them through an effortful, intentional process. That's how many of my insights arise, often aided by the reduction of a thought to a graphic representation, as I did when I made in China. That's just how my mind works.)

Back when I first arrived at the explanation of success stated above, the word "timing" I had meant "lucky timing." After all, what could be better than being aggressive at a fortuitous time? But lying in bed in India, I realized that good timing doesn't have to be exclusively the result of luck. Good timing in investing can come from diligently assessing where we are in a cycle and then doing the right thing as a result. The study of portfolio theory about how to position your portfolio for the possible outcomes that can result. That, in one sentence, is what this book is about.



I'd like to return to that simple sentence and apply a little more detail regarding the formula for investment success. I conclude that it can be considered in terms of six main components, or rather three pairings:

- **Cycle positioning**—the process of deciding on the risk posture of your portfolio in response to your judgments regarding the principal cycle.
- **Asset selection**—the process of deciding which markets, market sectors, and specific securities or assets to overweight and underweight.

Positioning and selection are the two main tools in portfolio management. It may be an over-simplification, but I think everything investors do can be put under one or the other of these headings.

- **Aggressiveness**—the assumption of increased risk: risking more of your capital; holding lower-quality assets; making investments that are more reliant on favorable macro outcomes; and/or employing financial leverage or high-beta (market-sensitive) assets and strategies.
- **Defensiveness**—the reduction of risk: investing less capital and more in cash instead; emphasizing safer assets; buying things that are less risky than can be.

relatively well even in the absence of prosperity; and/or shunning leverage and beta

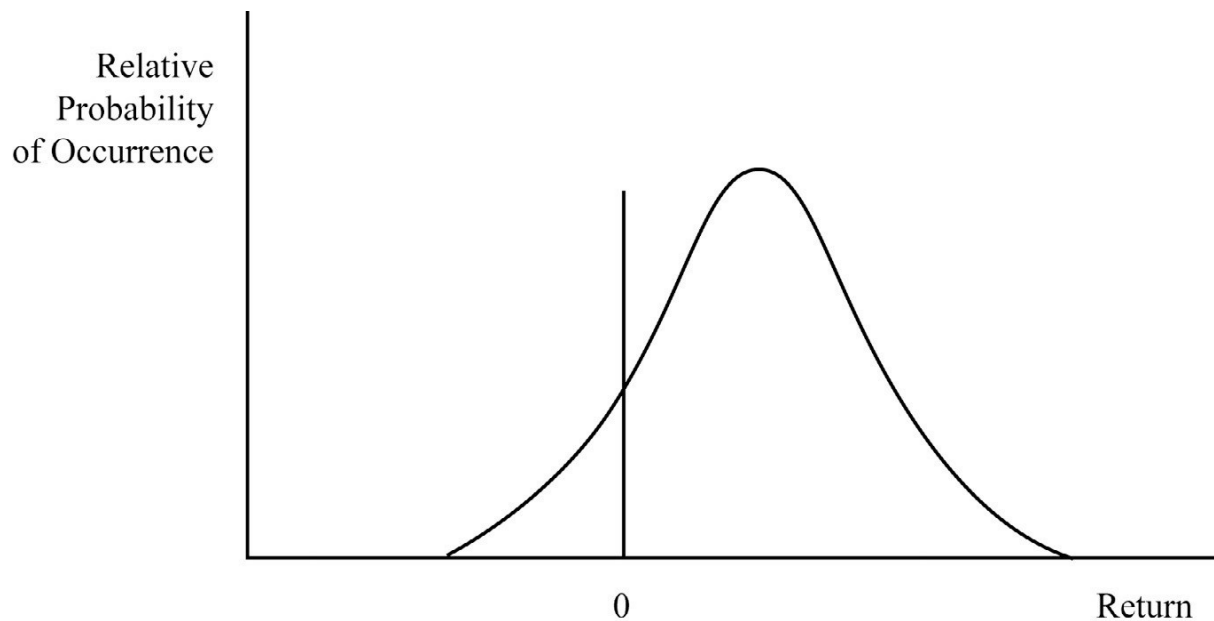
The choice between aggressiveness and defensiveness is the primary dimension in which investors position portfolios in response to where they think they stand in the cycles and what that implies for future market developments.

- **Skill**—the ability to make these decisions correctly on balance (certainly not in every case) through a repeatable intellectual process on the basis of reasonable assumptions regarding the future. This has come to be known by its academic name: “alpha”
- **Luck**—what happens on the many occasions when skill and reasonable assumptions prove to be of no avail—that is, when randomness has more effect on events than do rational processes, whether resulting in “breaks” or “tough luck”

Skill and luck are the prime elements that determine the success of portfolio management decisions. Without skill on an investor's part, success shouldn't be expected to produce success. In fact, there's some negative skill, and for people who are saddled with it, flipping a coin or abstaining from decisions would lead to better results. And luck is the wild card; it can make good decisions fail and bad ones succeed, but only in the short run. In the long run, it's reasonable to expect skill to prevail.

Part of my Indian awakening—stemming from my ruminations in China—concerned the dichotomy between selection and positioning, and the way skill influences the outcome of those two pursuits.

A market will do what it will do. Some of the outcome will be determined by economic events and corporate profitability; some will be determined by investor psychology and the resulting behavior; and some will be determined by randomness or luck. We may have some ideas about what the future will bring in terms of market performance, but they may be based on sound or defective reasoning, and they may prove to be right or wrong. But the market's future performance—whatever it will be, whether knowable or not—as the starting point for our discussion here. Let's express the expectation for a market's behavior as a probability distribution:



This is the starting point or baseline—the canvas, if you will, for an investor's actions. The question is whether he has the skill required to improve on the market's performance through active decision making, or whether he should give up on doing so and instead invest passively for market performance.

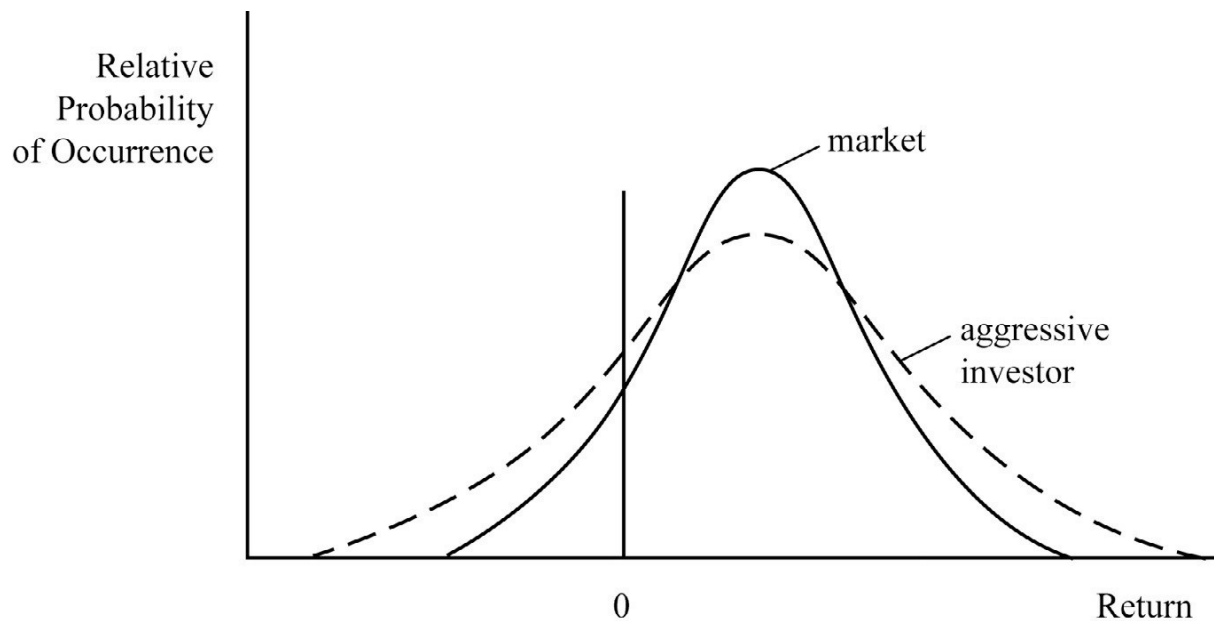
I mentioned above the two main ways in which an investor can add to his returns: cycle positioning and asset selection. I'll start by going into greater depth regarding the first.

As I also said, I believe cycle positioning primarily consists of moving between aggressiveness and defensiveness: increasing and decreasing exposure to market movements.

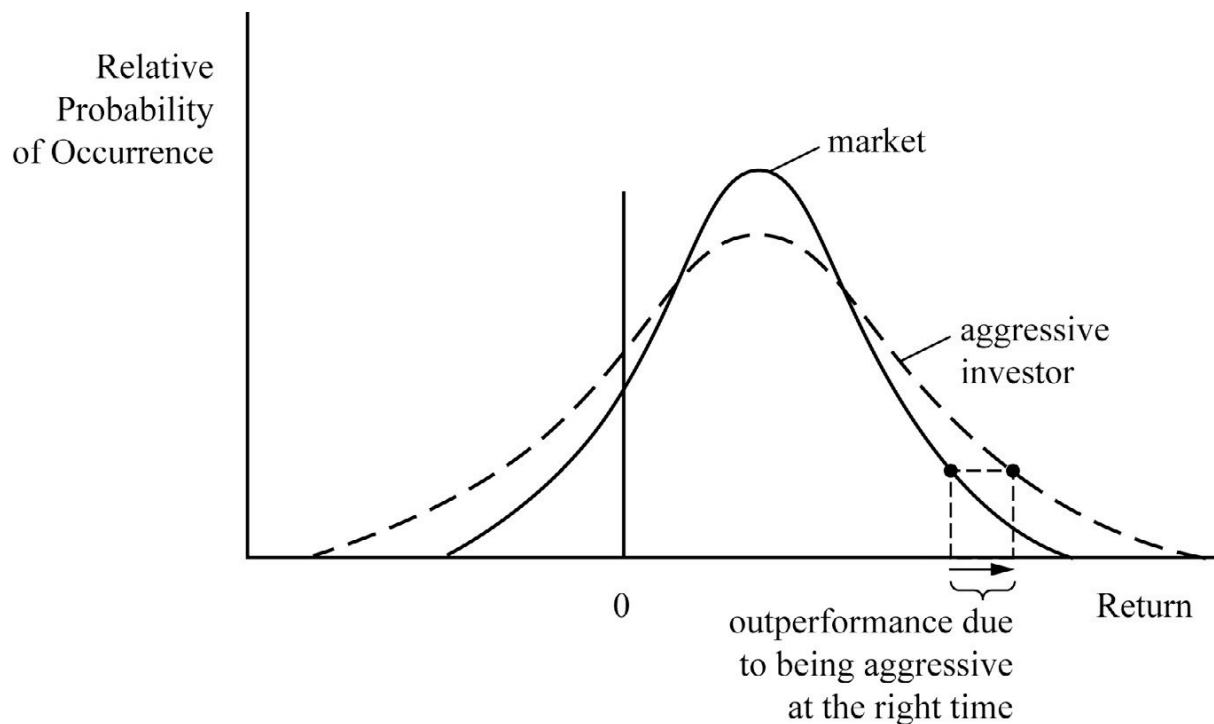
Let's say you conclude you're in a propitious environment:

- the economic and profit cycle are on the rise and/or likely to exceed people's expectations,
- investor psychology and attitudes toward risk are depressed (or sober) rather than feverish, and thus
- asset prices are moderate to low relative to intrinsic value.

In such a case, aggressiveness is called for. So you increase your commitments and add to your portfolio's risk posture and "beta" (sensitivity). The dotted line in the graphic below shows the outcome for performance. You have increased your potential for gains if the market goes well *and* for losses if it does poorly.



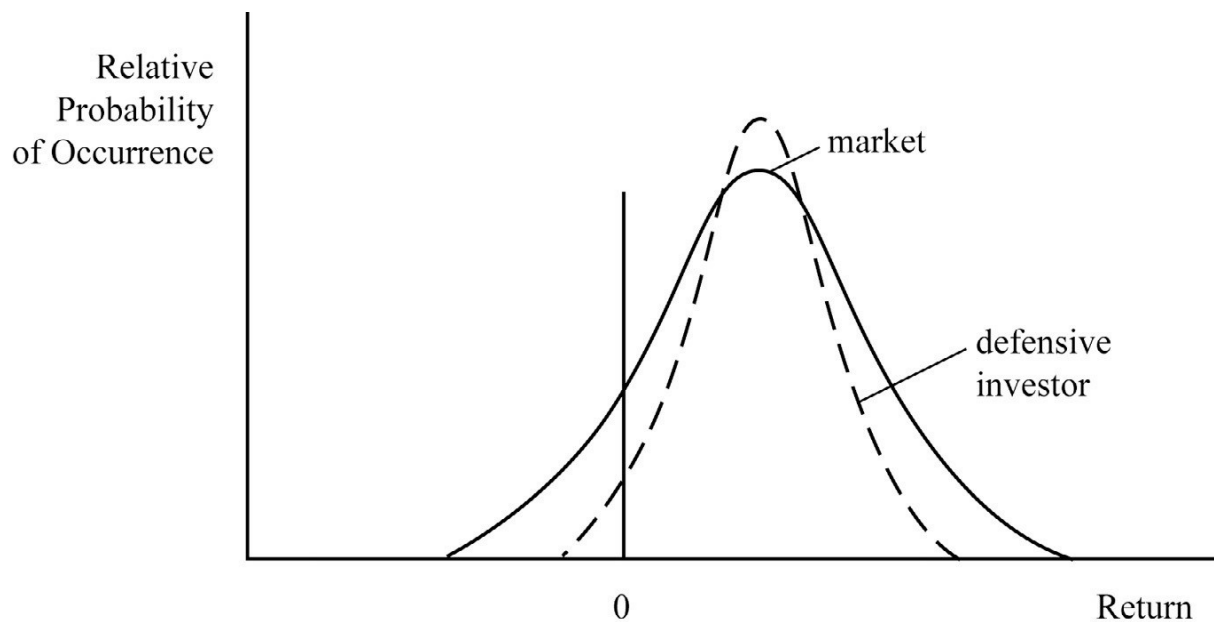
If your judgments are validated by a market rise, your aggressively positioned portfolio, with its enhanced market sensitivity, will rise even more, making you an outperformer, as shown in the following graph.



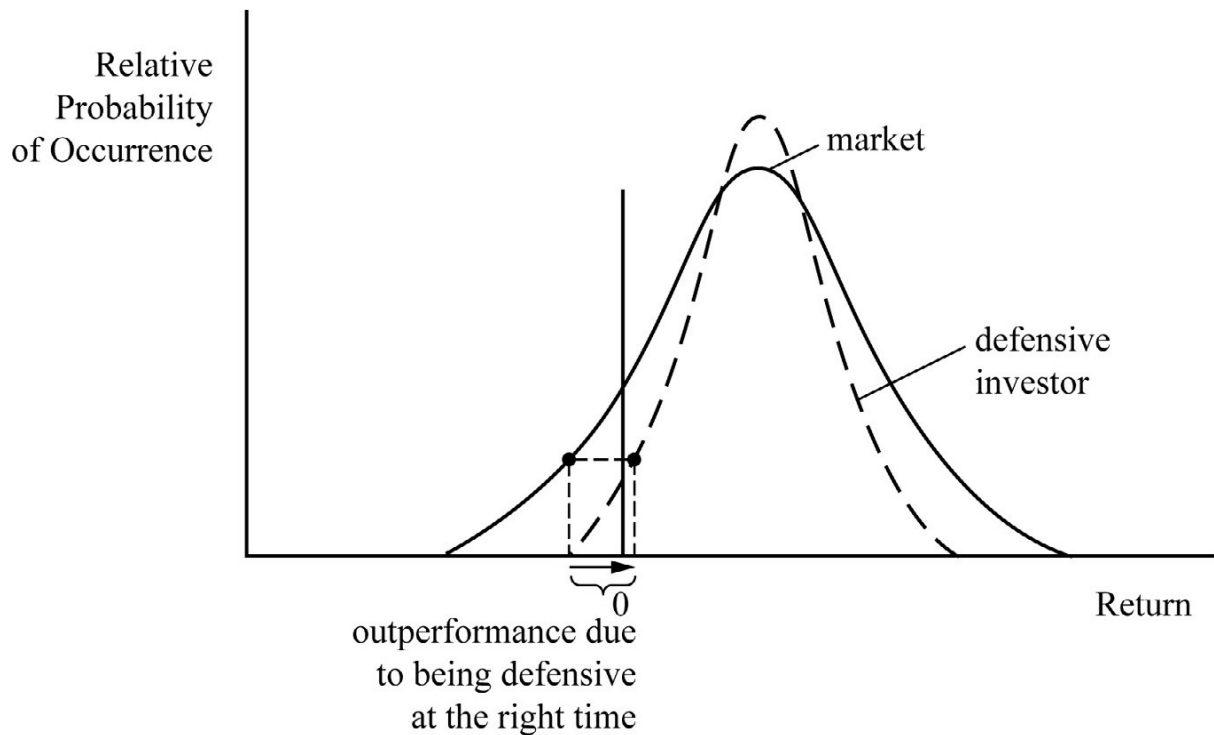
The recipe for success here consists of (a) thoughtful analysis of market stands in its cycle, (b) a resulting increase in aggressive positioning being proved right. These things can be summed up as “skill” or “cycle positioning.” Of course, “c”—being proved right—isn’t fully a matter within anyone’s control, in particular because of the degree

subject to randomness. So being proved right won't happen every to skillful investors who reason things out well.

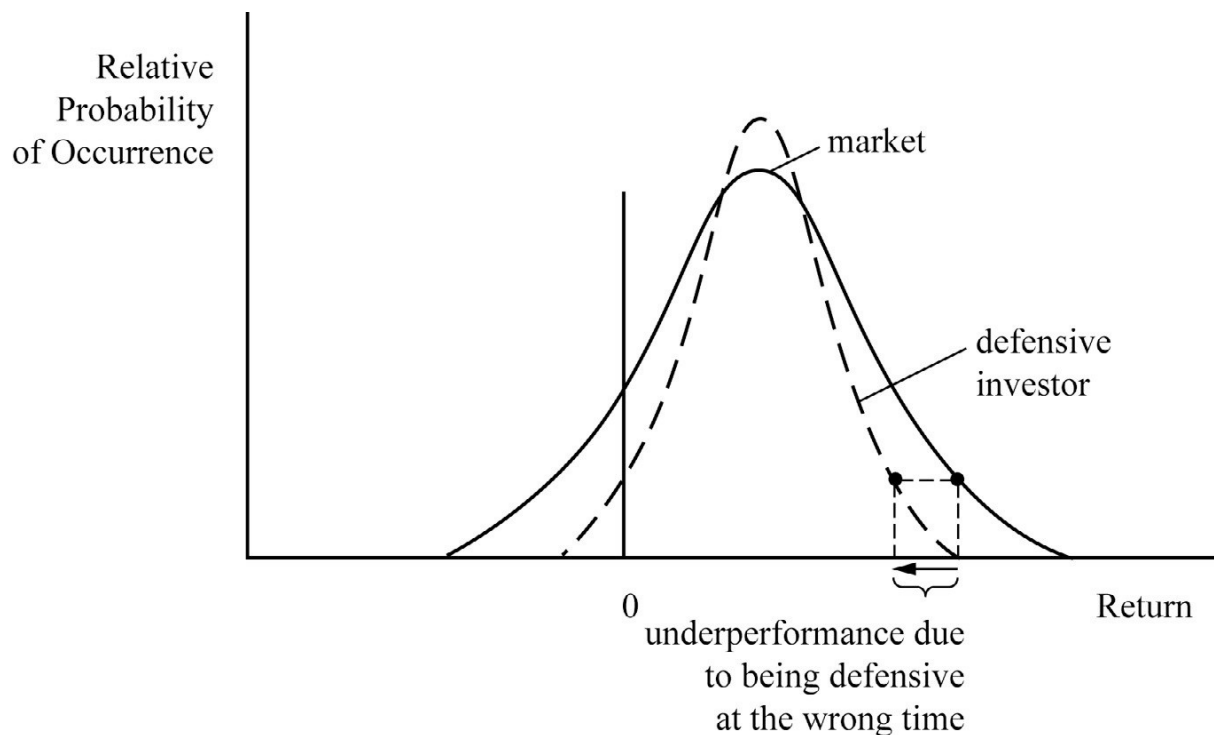
On the other hand, your analysis might tell you the cycle positioning is poor—the economy is tiring, psychology is excessively optimistic, and thus so are asset prices—meaning you should tend toward defensiveness. In this case, you should take some capital off the table and otherwise cut with a portfolio described by the graphic at the top of the



Now you've cut your beta and prepared for bad times. If you're in a weak market, the market's performance will be drawn from the left tail of the probability distribution, and your defensive positioning will make you an outperformer in that market, losing less, again as shown by the graph below. Your portfolio, being defensive, is less exposed to market movements and thus right for a weak market:



Of course not everyone has a superior understanding of cyclical efforts at positioning are successful. Suppose an investor who lacks regarding positioning decides to turn defensive and cuts his market as shown. If the market surprises on the upside, he turns his investments underperform:



In the first chapter I introduced the subject of “tendencies.” the market should be considered via a probability distribution, and distribution—if constructed accurately—will give you a sense for its likely tendency. The market’s movement through the cycle repositions the distribution and thus influences its likely future tendency, as I showed [pages 204–206](#).

When the market is low in its cycle, gains are more likely than losses are less likely. The reverse is true when the market is high. Positioning moves, based on where you believe the market stands, amount to trying to better prepare your portfolio for the events that will happen. While you can always be unlucky regarding the relationship between what logically should happen and what actually does happen, good positioning decisions can increase the chance that the market’s tendency—and your chance for outperformance—will be on your side.

In 1977, New York City experienced a wave of “lovers’ lane” murders perpetrated by a serial killer labeled “the Son of Sam.” In 2014, the obituary of Timothy Dowd, the detective who caught him, I saw where it quoted him as saying it was his job “to prepare for my view of the future as indeterminate and subject to considerable randomness. I think that’s a great way to think about it. While it’s like I’m advocating being passive and leaving things to chance, the fact is that superior investors have favorably skewed distributions of outcomes, not batting averages of 1.000. They still need favorable outcomes: the right tickets to be drawn from the bowl.

One of the best ways to enjoy the skewed distribution of outcomes is to get the market’s tendency on your side. The outcome will never be under your control, but if you invest when the tendency is biased toward favorable, you’ll have the wind at your back. If the tendency is biased toward unfavorable, the reverse will be true. Analysis of cycles can give you a better-than-average understanding of the market’s likely tendency and thus enable you to improve your chances by properly positioning your portfolio for what lies ahead.



What follows is completely unrelated to the subject of cycles, but if you’re interested in, you needn’t read it. But I want to complete the list of the actions investors can take to improve performance by adding another component of portfolio management: asset selection.

Asset selection consists of identifying markets, market sectors, and individual assets that will do better or worse than the rest, and over- and underweighting them in portfolios. The higher an asset’s price is relative to its intrinsic value, the less well it should be expected to do (all

equal), and vice versa. The key prerequisite for superior performance regard is above-average insight into the asset's intrinsic value future changes in that value, and the relationship between its intrinsic and its current market price.

All investors who follow a given asset have (or should have) opinions regarding its intrinsic value. The market price of the asset reflects consensus of those opinions, meaning investors collectively have set the price. That's where buyers and sellers agree to transact. The buyer because they think it's a smart investment at the current price, and sell because they think it's fully priced or overpriced there. What do you think about the accuracy of those views?

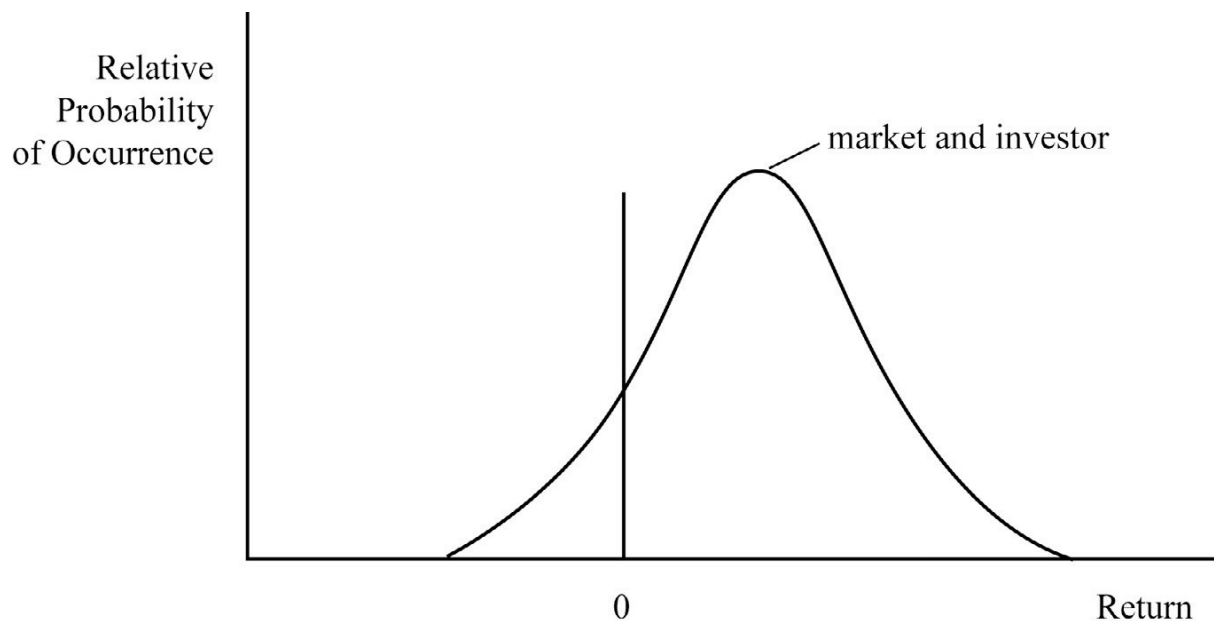
- **Theoretical**—The efficient market hypothesis states that all available information is incorporated into prices “efficiently,” so that the assets are fair and investors can't “beat the market” by choosing them.
- **Logical**—What we're talking about is the ability to make those judgments better than the average investor and thus achieve above-average performance. Yet the one thing we know for sure is that on average, all investors are average. Thus logic tells us they can't make above-average judgments.
- **Empirical**—Performance studies show that very few investors are consistently more right than others about those judgments. Most investors do worse than the markets, especially after the subtraction of transaction costs, management fees and expenses. That's the reason for the rising popularity of passive index investing.

That's not to say no one beats the market. Lots of people do, but usually no more than would be the case under an assumption of randomness. A few do it more consistently than randomness would predict, and some of those become famous. The essential ingredient—superior insight into intrinsic value—is what gives them that ability. I call it “second-level thinking”: the ability to think differently from the consensus and better.

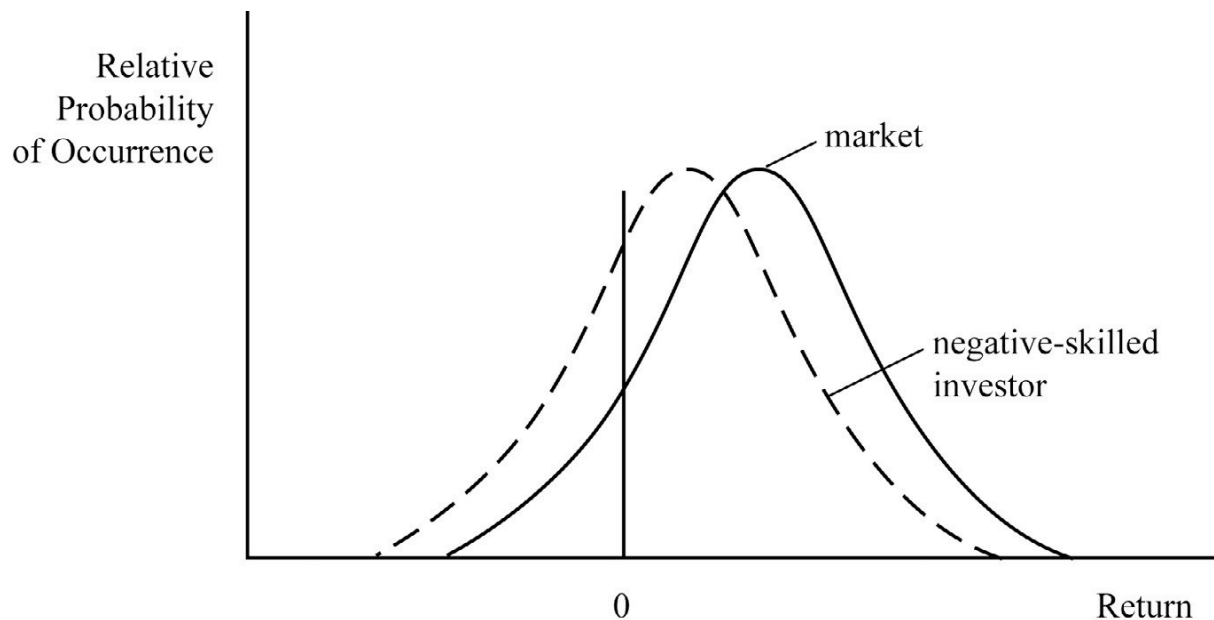
I'm not going to go any further regarding intrinsic value, the relationship of price to value, or second-level thinking, since they're all covered in *The Most Important Thing*. But the bottom line is that the superior investor, who's capable of second-level thinking—is able to pick more outperforming assets than underperformers, and thus to invest more in the former and less in the latter. The recipe for superior asset selection is that simple.

And what is the hallmark of that superiority? Asymmetrical results. The investor who has no skill at selection has the same ratio of winners to losers as the market. Thus he does well when the market does well,

poorly when it does poorly:

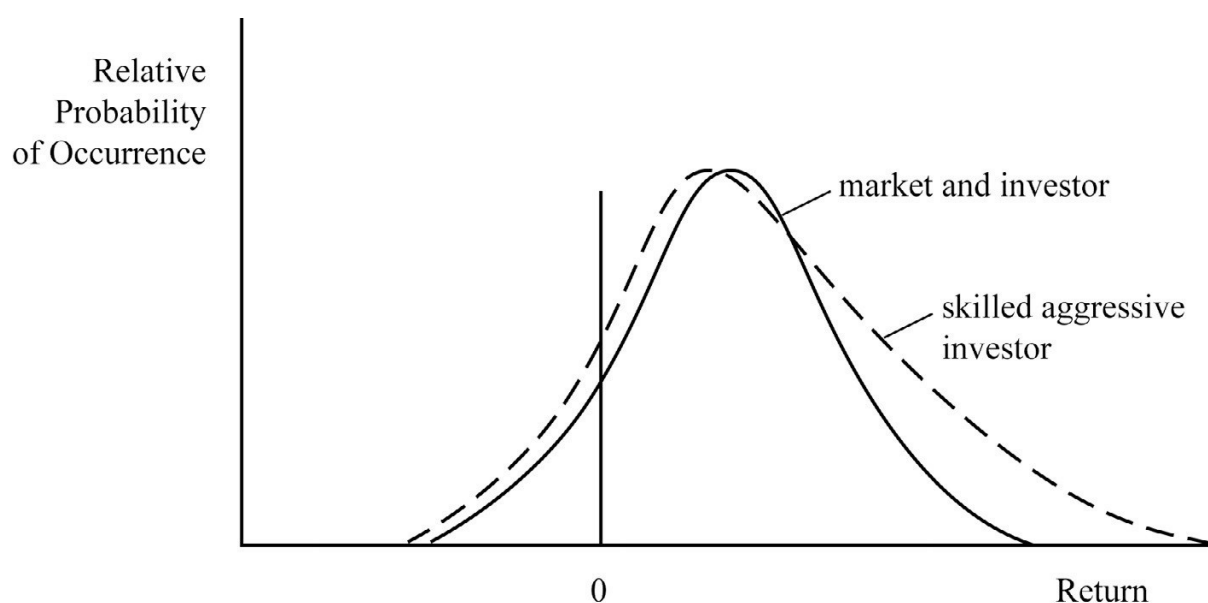


An investor with negative skill at selection picks proportionally more losers than winners and thus does worse than the market, both when it goes up and when it goes down, as shown below. In other words, his distribution is shifted to the left of the market's:

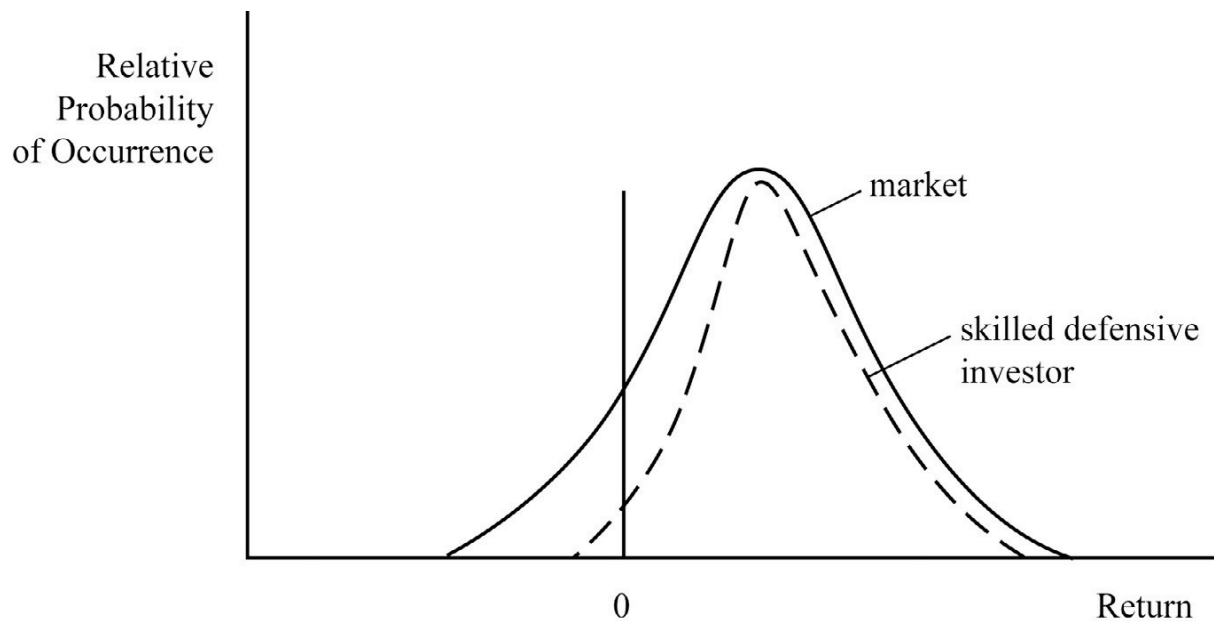


But the skillful selector has a better ratio of winners to losers than the market; he is able to invest more in his winners than his losers. Winners are more successful than his losers are unsuccessful.

- The holdings of a habitually aggressive investor who is capable of superior selection will go up more than the market when the market goes up and may go down more than the market when it goes down. The margin of superiority on the upside will exceed his degree of inferiority on the downside, as it comes from his ability to select assets with more upside potential without entailing commensurate downside risk. The result he'll do better than the market when it goes up, but not when the market goes down as his aggressiveness would suggest. This is an example of the asymmetry that marks the superior investor:

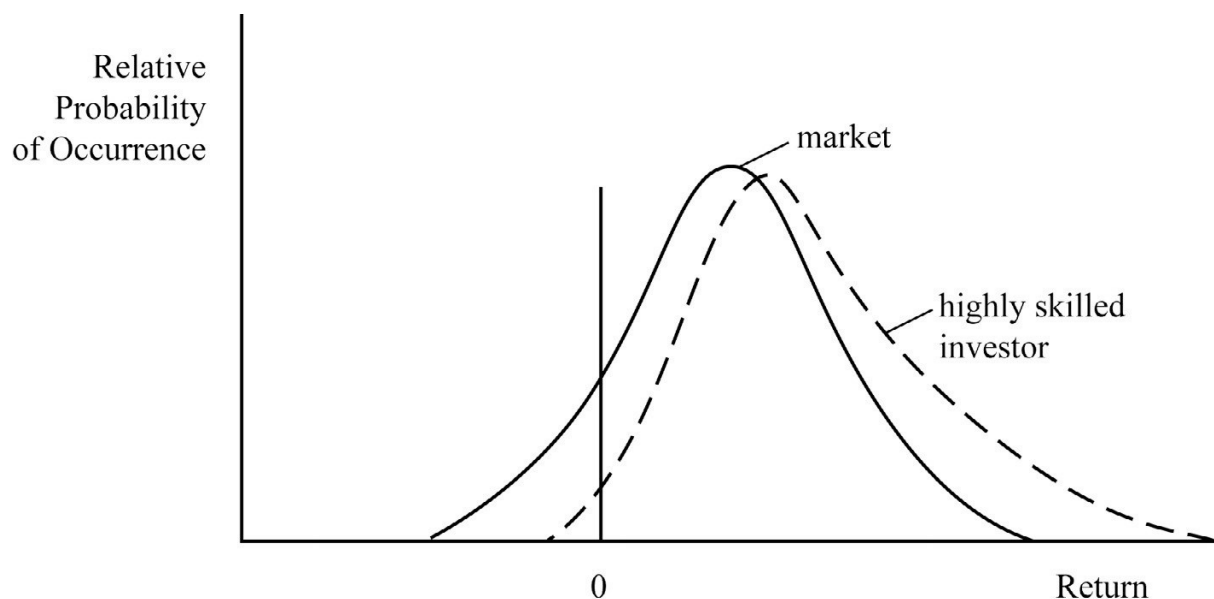


- Likewise, the habitually defensive investor with superior selection will do better than the market when it goes down, but his selection will keep him from underperforming in rising markets to the full extent that his defensiveness alone would suggest. His skill in selection enables him to find defensive assets that have upside participation that is disproportionate to their downside risk. It is an asymmetrical distribution as well:



Both of the investors with superior selection skill—the aggressive one and the defensive one—exhibit asymmetry relative to the market. That is, both have performance distributions that are biased favorably. Both have upside potential that is disproportional to their downside risk (although in different ways). That's how alpha in asset selection manifests itself.

Finally, the investor who is neither habitually aggressive nor habitually defensive—but who possesses skill at both cycle positioning and asset selection—correctly adjusts market exposure at the right time *and* exhibits asymmetrical performance that comes from a better-than-average ratio of winners to losers. This is the best of all worlds:



Almost anyone can make money when the market rises and loses money when it falls, and almost anyone can have the same ratio of winners to losers.

as the market overall. It takes superior skill to improve in produce the asymmetry that marks the superior investor.

Please note that in this discussion I have separated skill at cycle from skill at asset selection. This bifurcation is somewhat artificial. I describe the two elements that influence performance, but many great investors have both, and most of the rest have neither. Investors who are capable of both have a better sense for the market's likely tendency and put together portfolios that are better suited for the market environment. The likely lies ahead in terms of the ratio of winners to losers. That makes them great . . . and rare.



My epiphany in India taught me that successfully positioning a portfolio in the market movements that lie ahead is dependent on what you do (aggressive or defensive) and when you do it (based on a superior understanding of what cycles imply for future market moves). It's the help in this book to help in those regards.

XV

LIMITS ON COPING

In my opinion it's entirely reasonable to try to improve investment results by altering positions on the basis of an understanding of the market cycle. But it's essential that you understand the limitations, as well as the skills that are required and how difficult it is.

I undertook to write this book to give myself an opportunity to write down what I know about cycles, and because I enjoy writing, but mainly for the reasons above—to help the reader deal with the market's ups and downs.

In the preceding pages, I have covered many of the considerations on the process of understanding cycles, as well as the vagaries of the market. I restrain the confidence anyone should have regarding his ability to achieve his goal here is to repeat these considerations and sum up.

Investing, as I've said, consists of positioning capital so as to take advantage of future events. I also said we never know what the future holds, and thus we don't know where we're going. But we should do all we can to know where we're going, since the current position of the cycle has powerful implications for the future. We should cope with its possible future.

Where we stand in cycles has a profound influence on the market's tendencies: on what is likely to happen, and perhaps even when. As discussed in chapter I and illustrated in the last chapter, our position shifts the probability distribution governing the future.

Many things can happen. We know we face uncertainty and risk. All we can know about the future—at best—is what the probabilities are. Knowing the probabilities can help us be more right than others on average. It's essential to remember that knowing the probabilities is far from knowing precisely what's going to happen.

We generally have no choice but to be content with knowing the probabilities. But the sample in terms of each outcome (e.g., each year's growth or each stock's gain next year) will often be limited to one experience—meaning many things can happen but only one will. There

won't be enough observations to allow us to assume that the one the probabilities say is most likely. . . and most likely thing will happen any time soon.

For example, let's take the correction of a euphoria-driven bubble. Theoretically, it doesn't have to happen ever. But the realities that (a) it will happen eventually and (b) the more time that passes happening—and the longer the cycle continues to move upward—the likely (and usually the more imminent) the expected correction becomes.

Of course, the more time that passes before this logical event the further that the cycle swings in the upward direction—the will conclude that the rules of cycles have been somehow suspended the called-for correction will *never* happen. This can lead to the kind buying we saw in 2000, and eventually to an extremely painful

We have to safeguard our portfolios (and our investment management businesses) against the danger stemming from the fact that the thing most likely to happen—which our understanding of cycles can tell not happen until long after it first becomes likely. And we have ourselves emotionally so as to be able to live through the potential lag between reaching a well-reasoned conclusion and having it correct.



How about a review of history? In the mid-1990s, the galloping rocketing tech sector presented an opportunity for conservative investors conclude that stocks were highly overvalued. The reasoning might have been solid—based on an effective interpretation of the relevant data—for caution may have been strong. However, it would have taken the market to prove those investors right, and as one of the most important investment adages says, “Being too far ahead of your time is indisputable from being wrong.” The fact that their conclusions were well-founded would have done the investors only limited good; before the correction of the bubble finally set in, they might have lost much of the capital under their management.

However, the keen intellect that had led those investors to their conclusions, combined with sufficient conviction, should have allowed them to stick to their guns. Hopefully they would have remained cautious rather than capitulate and buy in higher. If so, they would have been a few years later and recovered their reputations and their assets. But the interim they lived through certainly would have been painful.

Fast-forward to the current decade and the chance for more. Cautious investors have been provided another chance to conclude that stocks are overheated, reduce holdings, and thus miss out on more

gains. Clients might have bailed again, and cautious firms' assets could have shrunk (in a rising market).

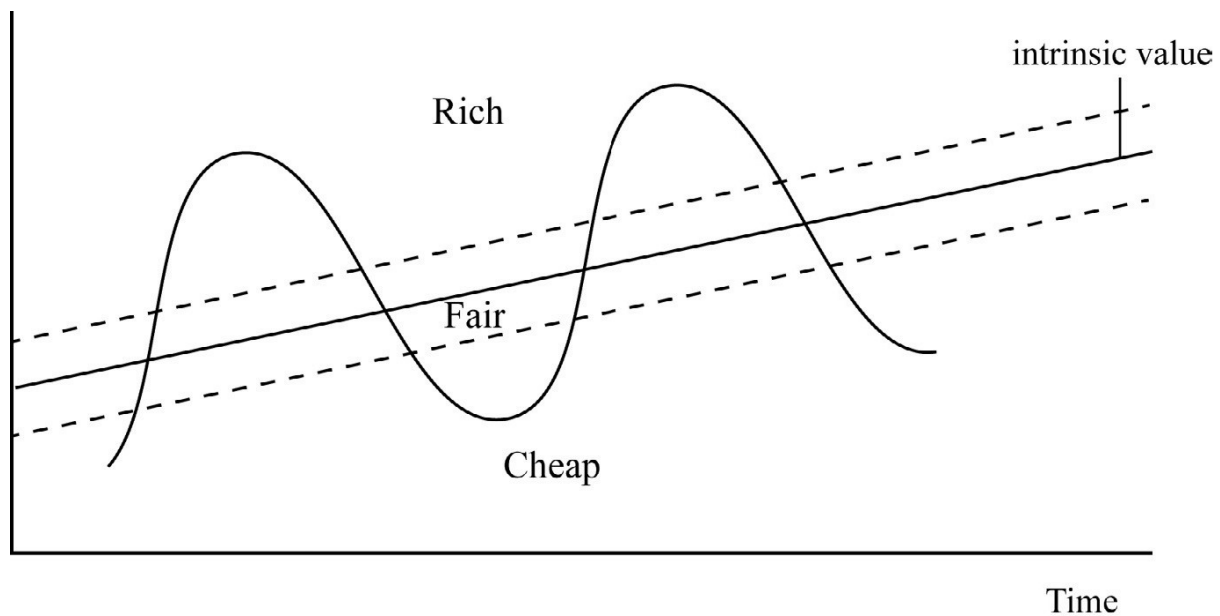
Is caution appropriate again? Will it be proved appropriate by events? Will a correction come soon enough for cautious investors the benefits of being right? Will they be viewed as perma-bears are validated every once in a while by downturns? Or as brilliant who are right in principle but so far thwarted by the undependability and effect in the investment world? These questions are largely unanswered. But the most important thing is for the reader to note this key positioning for cycles isn't easy.



In my opinion it's entirely reasonable to try to improve long-term results by altering positions on the basis of an understanding of the market cycle. But it's essential that you also understand the limitations, as well as the skills that are required and how difficult it is.

Importantly, I want to call attention to the obvious fact that the everyday ups and downs of the market—the clear examples in chapter XII all concerned “once-in-a-lifetime” cyclical extremes (these days seem to happen about once a decade). First, the extremes of boom and crash—and, in particular, the process through which they arise—most clearly illustrate the cycle in action and how to respond to it. When dealing with pronounced extremes that we should expect the likelihood of success.

Here's how I think about the impact of market movements on the investment environment we work in. It's admittedly a simplistic vision, but it suggests a world that is discernable and much more regular than the one we see. But it has worked for me for decades as a general framework. Heck out of trying to understand the world as a series of irregular, jagged, and zags:



Between the extremes of “rich” and “cheap”—when the cycle is in middle ground of “fair”—the state of the relationship between price is, by definition, nowhere as clear-cut as at the extremes. As

- It’s hard to make frequent distinctions and hard to do so correctly.
- Thus distinctions in the middle ground aren’t as potentially profitable as at the extremes, and those distinctions can’t be expected to work dependably.

Detecting and exploiting the extremes is really the best we can hope for. And I believe it can be done dependably—if you’re analytical, experienced (or well-versed in history) and unemotional. That means, however, that you shouldn’t expect to reach profitable conclusions daily, monthly or even yearly.

We can’t create great opportunities to time the market through our understanding of cycles. Rather, the market will decide when to take them. Remember, when there’s nothing clever to do, the mistake lies in trying to be clever.

The reasonableness of the effort at cycle timing depends simply on what you expect of it. If you frequently try to discern where we are in the sense of “what’s going to happen tomorrow?” or “what’s in store next month?” you’re unlikely to find success. I describe such efforts as “trying to be cute.” No one can make fine distinctions like those often made by market timers, or consistently right enough to add materially to investment results. No one knows when the market will make developments that efforts at cycle timing label “probable” will materialize.

On the other hand, positioning portfolios for the major cycles has been a big contributor to Oaktree's success. My colleagues and I became cautious in 1990–93, 2002 and 2008, and we turned cautious and pulled back in 1994–95 and 2005–06, and to some extent in the last few years. We tried to use the cycle to our advantage and to add value for our clients, and we got the positioning largely right on those occasions. Also, there were no major opportunities to do so that we missed.

So thus far our major cycle calls have all proved correct. The world makes it sound like this is a battle that's consistently winnable. But personally, "all" consists of four or five times in 48 years. By making calls only at the greatest cyclical extremes, I've maximized my chances of being right. No one—and certainly not I—can succeed regularly, other than at extremes.

So please note—as I always try to remind people—that I don't want to give the impression that readers should expect to be disappointed when they don't. As I said in "On the Couch" (2016):

I want to make it abundantly clear that when I call for caution in 2006–07, or active buying in late 2008, or renewed caution in early 2016, or a somewhat more aggressive stance here in early 2016, I do so with considerable uncertainty. My conclusions are the result of my reasoning, applied with the benefit of my experience (and collaboration with my Oaktree colleagues), but I never consider them 100% likely to be correct, or even 80%. I think they're right, of course, but I always make my recommendations with trepidation.

I read the same newspapers as everyone else. I see the same economic data. I'm buffeted by the same market movements. The same factors appeal to my emotions. Maybe I'm a little more confident in my reasoning, and certainly I have more experience than most. But the key is that—for whatever reason—I'm able to stand up to my emotions and follow my conclusions. None of my conclusions can be documented or proved. If they could be, most intelligent people would reach the same conclusions, with the same degree of confidence. I tell you this only to communicate my feeling that I should fear he's not up to the task just because he's unsure of his conclusions. These aren't things about which certainty is attainable.

Peter Bernstein contributed some useful wisdom on this subject. I'll conclude this chapter with his thoughts:

After 28 years at this post, and 22 years before this in money management, I can sum up whatever wisdom I have accumulated in one way: The trick is not to be the hottest stock-picker, the vintner, the forecaster, or the developer of the neatest model; such victories are transient. The trick is to survive! Performing that trick requires a strong stomach for being wrong because we are all going to be wrong more often than we expect. The future is not ours to know. Experience helps to know that being wrong is inevitable and normal, not something terrible tragedy, not some awful failing in reasoning, not even bad luck in most instances. Being wrong comes with the franchise of this activity whose outcome depends on an unknown future. . . .
“Being Wrong and Still Making Money,” Seeking Alpha, March 2017, emphasis added)

XVI

THE CYCLE IN SUCCESS

The important lesson is that—especially in an interconnected, informed world—everything that produces unusual profitability will attract incremental capital until it becomes overcrowded and institutionalized, at which point its prospective risk-adjusted return will move toward the mean (or worse).

And, correspondingly, things that perform poorly for a while eventually will become so cheap—due to their relative depreciation and the lack of investor interest—that they’ll be primed to outperform. Cycles like these hold the key to success in investing in trees that everyone is assuming will grow to the sky.

Hopefully you’re now equipped for the task of recognizing, responding to cycles. That can contribute substantially to your investment success. But as Peter Bernstein said, even the best investors won’t be successful all the time. Understanding that is an important part of to live with the effort. Success, like the other things mentioned in comes and goes.

In fact, over the course of my career, I’ve detected a cycle to a large degree, the ebb and flow of success—like the other cycles described—stems from the role played by human nature. And, once each development in the cycle leads to the next. I’ve long held the I mentioned back on [page 34](#)—and it’s been strongly reinforced over years of involvement with distressed debt and distressed companies—that “success carries within itself the seeds of failure, and failure success.”

Peter Kaufman, Charlie Munger’s biographer and the CEO of Glenair, exceptional producer of aerospace components, describes the workings of dialectical materialism as follows: “As any system grows toward its maximum or peak efficiency, it will develop the very internal contradictions and weaknesses that bring about its eventual decay and demise” (his essay

#49: “The Perpetual See-Saw,” 2010). This captures the process that guarantees that success will prove cyclical.

The Role of Human Nature

Another way I put it is that “success isn’t good for most people. Success can change people, and usually not for the better. People think they’re smart. That’s fine as far as it goes, but it has negative ramifications. Success also tends to make people richer, which can lead to a reduction in their level of motivation.

In investing there’s a complex relationship between humility and confidence. Since the greatest bargains are usually found among things that are undiscovered or disrespected, to be successful an investor needs enough confidence in his judgment to adopt what David Swensen, the successful head of Yale’s high-performing endowment, describes as “uncomfortably idiosyncratic portfolios, which frequently appear downright imprudent in the eyes of conventional wisdom” (*Pioneering Portfolio Management*, 2000). By definition, pronounced bargain prices are most likely to be found among things that conventional wisdom dismisses, that most investors find uncomfortable, and whose merits are hard to compare. Investing in them requires considerable inner strength.

When one of those positions initially fails to rise as the investor expects or perhaps goes in the opposite direction—the investor has to have enough confidence to hold on to his position or even add to it. He mustn’t decline as a sure “sell” signal; in other words, it can’t be that the market knows more than he does.

But, on the other hand, the investor also has to know his limitations and not assume he’s infallible. He has to understand that no one knows the macro future holds. While he’s likely to have opinions regarding the future course of economies, markets and interest rates, he should acknowledge that they’re not necessarily correct. And, counter to what he mustn’t always assume that he’s right and the market’s wrong, he mustn’t hold or add without limitation and without rechecking his facts and reasoning. That’s hubris.

As successes accrue, it’s common for people to conclude that they’ve got investing mastered. And after making a lot of money in a strongly rising market, their faith increases in their own opinion and instinct. Their investing comes to reflect less self-doubt, meaning they think less about the possibility of being wrong and worry less about the possibility of loss. This can cause them to no longer insist on the full margin of safety.

that gave rise to their earlier successes. This is the reason for and most important investment adages: “don’t confuse brains with a market.”

The plain truth is, there’s little of value to be learned from success who are successful run the risk of overlooking the fact that they were or that they had help from others. In investing, success teaches making money is easy, and that they don’t have to worry about risk—particularly dangerous lessons.

They may conclude that the small opportunity that gave them their big winner is infinitely scalable, which most are not. And many people—including investors who became famous for having had a single success—conclude that they can branch out into any number of other fields with the same intelligence that produced that first epic success must be broadly applicable.

Factors like these make investment success hard to replicate, meaning success can prove cyclical rather than serial. In fact, rather than imply that another success is coming, one success may in and of itself make a second one unlikely. As quote Henry Kaufman, formerly Salomon Brothers’ chief economist: “There are two kinds of people who lose a lot of money: those who know nothing and those who know everything” (“Archimedes on Wall Street,” *Forbes*, October 19, 1998).

It’s not for nothing that there are famous jinxes, such as the curse of the cover of *Sports Illustrated* or *Forbes* magazine. A cover appearance is the result of a singular accomplishment that may have resulted from a lucky break, a unique, non-replicable opportunity, or the bearing of a heavy burden. Or the good outcomes that land people on a magazine cover—including successful investors lauded by *Forbes*—may cause them to become confident and cocksure, and less disciplined and hardworking. In short, there is no formula for success.

The Role of Popularity

One of the principal ways in which success carries the seeds of failure is through increased popularity. I wrote just before that the most often found among those things that are hard to comprehend, uncomfortable and easily dismissed by the crowd. Short-term investment performance is largely a popularity contest, and most bargains are made for the simple reason that they haven’t yet been taken up by the herd. Popular. On the contrary, assets that have performed well are usually those that have gained in popularity because of their obvious merit and then become high-priced.

Let's think about investment strategies. It's essential to grasp that nothing will work forever: no approach, rule or process can outperform all. First, most securities and approaches are right for certain environments parts of the cycle, and wrong for others. And second, past success itself renders future success less likely.

In the 1960s, when the idea of investing in stocks was first gaining popularity among Americans, the emphasis was on industry leaders called "blue-chip" securities. Small-capitalization stocks were largely overlooked at first, but eventually they were noticed and bought. This led them to do better than large-caps. When people noticed small-cap stocks' superior returns in this "catch-up" phase, their buying produced even higher returns for them. . . . until it caused them to be fully priced. At that point, interest rotated to large-caps, which then regained the lead.

Likewise, growth and tech stocks did far better than more-mundane value stocks in the late 1990s. This divergence reached a maximum in 1999, with growth stocks outperforming value stocks that year by almost 50 percentage points. But growth stocks' dramatic outperformance rendered them overpriced, and when the stock market corrected in 2000–02, they lost more than value stocks, which previously had languished cheap.

In other words, "outperformance" is just another word for one thing appreciating relative to another. And, clearly, that can't go on forever. Regardless of how great its merits may be, "a" is unlikely to remain more valuable than "b." That means if "a" keeps appreciating relative to "b," there has to be a point at which it will become overvalued relative to "b," just when the last person gives up on "b" because it's been performing poorly and jumps to "a," it will be time for "b" (now compellingly attractive relative to "a") to outperform.

Strong forces create a tendency for strategies, investors or investment management firms that succeed for a while to cease doing so. I suspect that most ideas aren't infinitely scalable. One essential truth about investing is that, generally speaking, good results will bring more money. "Hot" money managers and strategies, and if allowed to grow unchecked, more money will bring bad performance.

In the mid-2000s, convertible arbitrage rapidly gained in popularity. Investors without a view on the prospects for a given stock were able to buy bonds convertible into it, as long as they were able to short the underlying shares in an appropriate "hedge ratio" (see my memo "A Case Study in Convertible Arbitrage," 2005). Convertible arbitrageurs reported outstanding risk-adjusted returns in all kinds of market environments. . . . until so much money was chasing the strategy that no one could find positively attractively priced as those of the past.

The important lesson is that—especially in an interconnected, in world—everything that produces unusual profitability will attract incremental capital until it becomes overcrowded and fully institutionalized, at point its prospective risk-adjusted return will move toward the mean (or worse).

And, correspondingly, things that perform poorly for a while eventually will become so cheap—due to their relative depreciation and the investor interest—that they’ll be primed to outperform. Cycles are the key to success in investing, not trends that everyone is assured to the sky.

It’s all a matter of ebb and flow. In investing, things work until they don’t. Or as Ajit Jain of Berkshire Hathaway told me about investing the “it’s easy until it isn’t.”

- Cheap small-caps outperform until they reach the point where they’re no longer cheap.
- Trend-following or momentum investing—staying with the winners—works for a while. But eventually rotation and buying the top over as the winning strategy.
- “Buying the dips” lets investors take advantage of momentary weakness up until the time when a major problem surfaces (or the market no longer recovers), causing price declines to be followed by price declines, not quick rebounds.
- Risky assets outperform—coming from valuations where they were excessively penalized for their riskiness—until they’re priced more like safe assets. Then they underperform until they once again offer adequate risk premiums.

The bottom line is clear: nothing works forever. But it’s essential to recognize that when everyone becomes convinced that something will keep working forever, that’s the very time when it’ll become certain not to. “In investing, everything that’s important is counter-intuitive, and everything that’s obvious to everyone is wrong.”

Perhaps the greatest example of the influence of popularity (in the market) is seen in 1979, and few investors who were around at the time have forgotten it. On August 13 of that year, after almost a decade of stellar performance on the part of stocks, *Business Week* magazine published a story entitled “The Death of Equities.” Its conclusion—that stocks were overvalued for—was based on the opposite of everything recommended in this book.

The article cited a litany of reasons why the poor performance would continue unabated:

- Sevenmillion people had given up on investing in stocks.
- Many other forms of investment had done better.
- Pension funds were turning to “hard assets” like gold.
- Inflation had sapped corporations’ ability to increase profits.

It went on:

Even institutions that have so far remained in the financial market are pouring money into short-term investments and such “alternate equity” investments as mortgage-backed paper, foreign securities, venture capital, leases, guaranteed insurance contracts, indexed stock options, and futures.

And here was its conclusion:

Today, the old attitude of buying solid stocks as a one’s life savings and retirement has simply disappeared. Says a young U.S. executive: “Have you been to an American stockholders’ meeting lately? They’re all old fogies. The stock market is where the action’s at.”

In short, what “The Death of Equities” said was that stocks had become unpopular that they would never do well again. It takes a highly first-level thinker to conclude that poor past performance has led to unpopularity today, which implies poor performance tomorrow. Rather, an insightful second-level thinker says poor past performance has led to unpopularity today, *which implies low prices today*, which in turn leads to *good* performance tomorrow.

“The Death of Equities” was published just a few years before—and in essence laid out the entire case for—the 1982 kickoff of the great bull market in history. At the time it was published, the S&P 500 index was at 100. In March of 2000 it reached 1527. That’s a price gain of 527% or 13.7% per year for almost 21 years (and those figures ignore dividends, which took the total gain to more than 28 times and the annualized total return to 17.6%). The lesson is simple: investors should be leery of unpopularity. Rather, it’s unpopularity that is the buyer’s friend.

The Role of Companies

Companies, too, are subject to ups and downs in terms of success on a series of cause-and-effect events. One of the most pronounced processes I've witnessed took place at Xerox.

The office-copying giant—the first to make it possible to avoid the process of photostatting, which required that documents be sent out to a lab to be duplicated—was one of the first companies I visited as an office-equipment analyst in the late 1960s. At that time, Xerox had a monopoly on “dry” copying and appeared to be in full command of its destiny. My senior analyst and I used to meet with Xerox’s “analysts” and for each copier model in the product line he would show me the company’s projections in terms of the number of machines it had in the field in the coming year and the annual rental revenue.

Because of Xerox’s dominant market position, it was largely guaranteed that those projections come true. It could charge monopolistic prices that enabled it to fine-tune the size of its rental fleet and that gave it very high margins. It also insisted on a rental-only model, refusing to sell machines and lose control of them. What a perpetual motion machine!

But Xerox management may have ignored the possibility that those margins would prove unsustainable. In 1975, Xerox resolved an antitrust complaint over control of the copier market by entering into a licensing agreement requiring it to make its powerful portfolio of patents available to competitors. Competitors began to produce and sell their own copiers. They were undercut Xerox’s prices and displace part of its rental population. This reduced Xerox’s U.S. market share in copiers from nearly 100% to the teens, and it ate significantly into Xerox’s profits. As the incumbents had trouble responding to the price competition, as this would have hurt their existing business. The competitors did so in an early example of what is now called disruption.

In 1968, thanks to its monopolistic position, strong growth and profitability, Xerox was a leader of the Nifty Fifty I described in my book. Companies that were considered so strong and so fast growing that “nothing bad could happen” and “no stock price was too high.” But trees rarely grow to the sky, and success is rarely unending.

Because Xerox’s behavior had invited competition that it was unwilling to respond to—and for other reasons as well—by the early years of the 21st century it was experiencing serious difficulty.

Companies, like people, have the potential to respond to success with behavior that dooms that very success. Thus companies can:

- get complacent and become “fat and happy,”
- become bureaucratic and slow-moving,
- fail to take action to defend their positions,

- cease to be innovative and non-conforming, and join the crowd mediocrity, and/or
- conclude that they can do pretty much anything, and thus venture areas beyond their competence.

In these ways and more, success truly carries the seeds of failure. Good news—as I said earlier—is that failure also carries the seeds

- Once under attack, companies can regain their motivation and sense of purpose.
- They can shed bureaucratic fat and get serious about meeting competition and making money.
- And in the ultimate form of success, they can go through slim down, and shed losing business lines, unprofitable locations, onerous contracts and burdensome debt. (Of course, however, the owners of companies that go into bankruptcy usually lose their positions.)

Thus Jahan Janjigian wrote about Xerox in 2002:

[In 2000,] a new management team implemented a number of restructuring initiatives aimed at returning the company to profitability. These included aggressive cost cuts and the elimination of 13,600 positions. Xerox also sold off its Chinese and Indian operations, as well as 50% of Fuji Xerox to Fuji Photo Film. Furthermore, Xerox allowed GE Capital to take over the financial receivables, for which it already has received \$2.7 billion. And Xerox has exited the struggling small-office and home-office business.

In April, Xerox agreed to pay \$10 million to settle the year investigation. It later restated all of its past financial results required under the agreement. Additionally, the company was successful in working with creditors in renegotiating its debt obligations to more manageable terms. Perhaps most important, the company's products are now more competitive in terms of price and quality.

Due to the success of these efforts, Xerox returned to profitability sooner than expected. . . . Given significant operational improvements, we think Xerox makes for a compelling buy at current levels. (“Xerox Back from the Brink,” *Forbes Growth Investor*, October 2002)

Because companies don't last as long as economies and markets do, the long-term cycle in company success may not take as long either. Over their lifetimes, companies' gains can lead to losses, and losses can lay the groundwork for gains. There is a cycle in business success.

The Role of Timing

Among the factors that can contribute greatly to an individual's or a company's success is timing. Among other things, it helps to do things at an above-average time. That's what put Xerox on the map in the 1960s, and it helped me out as well.

In August 1978, shortly after shifting from running Citibank's equity research department to managing bond portfolios, I got the phone call that changed my life. "There's a guy named Milken in California, and he's involved in something called high yield bonds. A client wants a portfolio of those bonds. Can you figure out what that means?"

It was just in 1977 or 1978 that high yield bond investing had its institutional inception. That's when Michael Milken achieved his first major success in convincing investors that it's okay for below-investment grade companies to issue bonds—and for institutions to buy them—if the interest rate was high enough to compensate for the risk. The high yield universe consisted of more than \$3 billion of bonds at the time I first got involved. The major investing organizations had a rule against buying bonds rated below investment grade, which were commonly called "junk bonds." And Moody's categorically rejected B-rated bonds, saying they "fail to possess the characteristics of a desirable investment."

How could these unpopular bonds *not* have been underrated bargains? How could early participation *not* have been a boon?

And then, a decade later, Bruce Karsh brought his legal skills and strategic insight to my team, complementing Sheldon Stone's expertise in finance. So we organized one of the first distressed debt funds from a major financial institution. What could be more risky and thus more unseemly than investing in the debt of companies in bankruptcy or considered destined for liquidation? The idea could people be more averse? In other words, where else could money be made so safely?

In both these instances and more, I've been lucky to come across these asset classes when they were still undiscovered, uncrowded and thus bargain-priced. There's little that can make investing as easy as having a market largely to oneself. It beats the heck out of trying to wring a few cents out of a market that everybody has discovered, figured out, taken to and

into. The latter is far from a formula for success. The latecomer crowded field isn't "the wise man in the beginning"; rather, he's more likely to be "the fool in the end." Those of us who've been lucky early rather than late know—or certainly should know—that our success wasn't all our doing. The times have to cooperate.

And that brings me to a great observation from Henry Phipps, celebrated partner of Andrew Carnegie and Henry Clay Frick, two century's greatest businessmen. In 1899, Phipps wrote as follows:

Good times like the present, make bad times; a law sure as the swing of the pendulum. We have experience to know these elementary truths. Have we the sense to put them into practice? (George F. R. Henry Clay Frick: *The Man*, 2002)

As with people—whose successes may constitute isolated instances rather than indicators of great things to come as they might be wont to believe—times may not remain conducive to continued success. Good times can encourage investment decisions that depend on the perpetuation of good in order to succeed. But those good times can lead instead to bad times that test the decisions in ways they can't withstand.

Not only are good times followed sooner or later by bad times, but with so many other examples of cycles—good times often *produce* bad times. Good times can lead to unwise debt issuance (as we saw in the 1920s and the distressed debt cycle) or to overbuilding (as we saw in the 1980s and the real estate cycle).

Likewise as Ruchir Sharma, chief global strategist at Morgan Stanley Investment Management, wrote in his book *The Rise and Fall of Nations* regarding the effect of fresh new reformers, "Reform leads to growth in good times, and good times encourage an arrogance and complacency that leads to a new crisis."

Phipps pointed out that history shows these trends clearly. The question is whether we are astute and unemotional enough to recognize that good times don't necessarily lead to further good times, and thus that success can be cyclical.

Let's return to Charlie Munger's quote from Demosthenes: "For that which a man wishes, that he will believe." In other words, wishful thinking takes hold. This can cause investors to believe that good times will be followed by more good times. But that ignores the cyclical nature of success, and especially of investment.

XVII

THE FUTURE OF CYCLES

The tendency of people to go to excess will never end. *A* since those excesses eventually have to correct, neither will the occurrence of cycles. Economies and markets have never moved in a straight line in the past, and neither will they do so in the future. That means investors with the ability to understand cycles will find opportunities for profit.

Thus far I've covered a lot of the past and a bit of the future. To conclude, I want to turn to the future.

Over the course of my career, I've witnessed numerous occasions when pundits said the occurrence of one type of cycle or another had come to an end. Whether because of economic vitality, financial innovation, astute corporate management, or the supposed omniscience of central bank heads of Treasury, they observed that the fluctuations of either the economic cycle or the cycle in profits would be seen no more.

I spent some time on this subject in "Will It Be Different This Time?" (November 1996). First I described a newspaper article that had appeared a few days earlier:

It recounted the case currently being made for this remaining a continuous, recession-free economic expansion. As its lead paragraph says:

From boardrooms to living rooms and from government offices to trading floors, a new consensus is emerging. The big, bad business cycle has been tamed.

The current expansion, at 67 months, has already far exceeded the post-war average. Nevertheless, 51 of the 53 "top economists" surveyed by Blue Chip newsletter (my favorite experts and the source of my July 1996 memo, "The Value of Predictions II") predict a global

next year of 1.5% or more. And the University of Michigan survey finds that among consumers, more expect five more good years than expect bad times to emerge.

The Chairman of Sears states “There is no natural law that we have to have a recession.” According to Amoco’s Chairman, “I don’t see any reason to believe [the recovery] can’t go on until the end of the century.” Sara Lee’s CEO says “I don’t know what could happen to make a cyclical downturn.” (“The Business Cycle is Tamed, Most Say, Alarming Others,” *Wall Street Journal*, November 15, 1996)

Clearly these statements, made in 1996, did not truly mark the end of business cycles. Rather, there was a modest recession in 2001 and then, years later, the Great Recession of 2008–09, the most powerful cyclical downturn ever experienced by most people alive today.

I went on in “Will It Be Different This Time?” to quote a number of assertions from other respected leaders:

“There will be no interruption of our present prosperity.”

“I cannot help but raise a dissenting voice to the statements that our prosperity in this country must necessarily diminish and recede in the future.”

“We are only at the beginning of a period that will go down in history as the golden age.”

“The fundamental business of the country . . . is to maintain a prosperous basis.”

In assessing the reasonableness of these statements, it’s important to note that they were made, respectively, by the president of the Pierce-Fenner & Smith Car Company, the president of the New York Stock Exchange, the president of the Bush Terminal Company, and the president of the United States. The positions of the first and third should serve as a giveaway that these statements came in the distant past, even without your knowing that the president in question was Herbert Hoover. The timing of these statements, in 1928 and 1929—on the doorstep of the Great Depression that plagued the world for more than a decade—was less than auspicious. So much, however, for prosperity without slowdowns, and for the end of cyclicality.

But then, in the 2000s, the expectation of “permanent prosperity” reemerged. While not specifically arguing that there would be no more business cycles, a lot of investors, bankers and members of the media embraced the belief that risk was gone—essentially the same thing.

In his autobiography, *Stress Test*, former Treasury secretary Tim Geithner describes the climate when he arrived at the Fed in 2003:

Economists were starting to debate whether America's long stretch of stability constituted a new normal, a Great Moderation, a quasi-permanent era of resilience to shocks. There was growing confidence that derivatives and other financial innovations designed to hedge and distribute risk—along with better monetary policy to respond to downturns and better technology to smooth out inventory cycles—had made devastating crises a thing of the past.

The fact that this purportedly permanent prosperity was given a name like “the Great Moderation” suggests it had entered the popular consciousness. And thus it met my requirement for the riskiest possible environment, which there's widespread belief that there's no risk. On [page 120](#), I cite the media accounts to that effect that enumerated the forces that had caused risk's elimination: Fed omniscience, unremitting demand for securities from cash-flow-rich nations, and the latest inventions from Wall Street.

Not only were all these “cycles are over” proclamations wrong, but very much worth noting that each of them coincided with—and certainly contributed to—the upward march to a cyclical high. And that followed those highs would be unusually painful: the Great Depression of 1929–39, a three-year stock market decline in 2000–02 (the first since 1929) and the Global Financial Crisis of 2007–08.

In “Will It Be Different This Time?” I went on from recalling Pollyanna statements just cited to drawing what I think are the essential conclusions on this subject:

Of course, what these observations signaled wasn't that cycles wouldn't repeat, but rather that the onlookers had grown too confident. Cycles in economies, companies and markets will continue to occur at least as long as people are involved in making decisions—which I believe means forever.

There is a right time to argue that things will be different—that's when the market is on its backside and everyone else is in a panic at giveaway prices. It's dangerous when the market's at record levels to reach for a positive rationalization that has never held true in the past. But it's been done before, and it'll be done again.

“It's different this time” are four of the most dangerous words in the business world—especially when applied, as is often the case, to some market that has reached what in prior times would have been called an extreme.

When people say “it's different” in this case, what they often mean is that the rules and processes that produced cycles in the past have been

But the cyclical behavior of the financial past did not result from operation of physical or scientific rules. In science, cause and effect is a dependable and repeatable relationship, so that it's possible to say with confidence, "if a, then b." But while there are some principles that operate in the world of finance and business, the resulting truth is very different from that in science.

The reason for this—as I've harped on repeatedly—is the involvement of people. People's decisions have great influence on economic, business and market cycles. In fact, economies, business and markets consist of transactions between people. And people don't make their decisions scientifically.

Some take history, facts and data into account, and some approach their decisions as "economic men." But even the most unemotional and stoic among them are subject to human influences and the loss of objectivity.

Richard Feynman, the noted physicist, wrote, "Imagine how much harder physics would be if electrons had feelings!" That is, if electrons didn't always do what science expects, the rules of physics would work only some of the time.

The point is that people *do* have feelings, and as such they are not bound by inviolable laws. They'll always bring emotions and foibles to their trading and investing decisions. As a result, they'll become euphoric at the right time and despondent at the wrong time—exaggerating the upside when things are going well and the downside risk when things are going poorly—and thus they'll take trends to cyclical extremes.

A few paragraphs from *The Most Important Thing* provide a good foundation for a recap of the main points regarding cycles' persistence.

The basic reason for the cyclicity in our world is the involvement of humans. Mechanical things can go in a straight line. Time moves ahead continuously. So can a machine when it's adequately powered. But processes in fields like history and economics involve people, and when people are involved, the results are variable and cyclical. The main reason for this, I think, is that people are emotional and inconsistent, not steady and clinical.

Objective factors do play a part in cycles, of course—factors like quantitative relationships, world events, environmental changes, technological developments and corporate decisions. But it's the application of psychology to these things that causes investors to overreact or underreact, and thus determines the amplitude of the cyclical fluctuations.

When people feel good about the way things are going and are optimistic about the future, their behavior is strongly impacted. The

spend more and save less. They borrow to increase their enjoyment of their profit potential, even though doing so makes their financial position more precarious (of course, concepts like precariousness are forgotten in optimistic times). And they become willing to pay for current value or a piece of the future.

Investors tend to look at the processes that are afoot, attribute more dependability to them, rely on that dependability, and extrapolate the processes. What they overlook is the role of emotions: greed on the up and fear on the downswing.

Emotions operate on cycles two ways: they magnify the forces that create extremes that eventually require correction, and they cause market participants to overlook the cyclicity of cyclical things at just those moments when recognition of excesses is most essential and most potentially profitable: stage three of the bull and bear markets [described on page 192](#) and [193–194](#).

The following passage from *The Most Important Thing* may serve to wrap up regarding the outlook for cycles' recurrence:

Cycles will never stop occurring. If there were such a thing as a completely efficient market, and if people really made decisions in a calculating and unemotional manner, perhaps cycles (or at least extremes) would be banished. But that'll never be the case.

Economies will wax and wane as consumers spend more or less, responding emotionally to economic factors or exogenous events, geopolitical or naturally occurring. Companies will anticipate a rosy future during the up cycle and thus over-expand facilities and inventories; these will become burdensome when the economy turns down. Providers of capital will be too generous when the economy is doing well, abetting over-expansion with cheap money, and then they'll pull the reins too tight when things cease to look as good. Investors will overvalue companies when they're doing well and undervalue them when things get difficult. . . .

Ignoring cycles and extrapolating trends is one of the most dangerous things an investor can do. People often act as if companies that are doing well will do well forever, and investments that are outperforming will outperform forever, and vice versa. Instead, it's the opposite that is more likely to be true.

A deep understanding of emotions and the overdoing they lead to is a key bottom line of this book. Cyclical deviations from the trendline are produced largely by overdoing and its eventual correction. This is

true in the securities markets, which are nothing but a collection of people making decisions (often in a herd-like manner) that they hope will be profitable. But it is no less true of economies and companies; they are not like independent, well-oiled machines, but they, too, are nothing but people making decisions, with all that implies.

The first time rookie investors see this phenomenon occur, it is understandable that they might accept that something that has never happened before—the cessation of cycles—could happen. But the second time or the third time, those investors, now experienced, should realize it's never going to happen, and turn that realization into their advantage.

The next time you're approached with a deal predicated on cycles having ceased to occur, remember that invariably it's a lie.
(*The Most Important Thing*)

I was 22 years old and a newcomer to investing in 1968, with my initial exposure to the Nifty Fifty. People far more experienced than I held forth regarding the companies' greatness, their boundless potential for growth, the fact that nothing bad could ever happen to them, and the absence of any limits on their stock prices. I swallowed those stories whole. I don't remember pushing back on the illogicality of their extreme valuations. Thus I was lucky to learn my first lessons regarding cyclicity, value, and risk at an early age and with relatively little bet on the erroneous conclusions.

I was a little less naive when I organized Citibank's response to the oil embargo of 1973, when the price of oil went from \$20 a barrel to over \$40. Energy analysts saw few impediments to continued increases. And the wonders of computers led to the formation of far more disc-drive capacity than in the 1980s than would turn out to be needed.

But later, with a few decades of experience under my belt, I was able to recognize the excesses of the tech/Internet/e-commerce bubble of the late 1990s, and of the unquestioning capital market behavior that led to the Global Financial Crisis of 2007–08. This learning process regarding the excesses in the markets—and their contribution to cycles—is an indispensable part of every investor's education.



The tendency of people to go to excess will never end. And the excesses eventually have to correct, neither will the occurrence of bubbles. Economies and markets have never moved in a straight line in history, and neither will they do so in the future. And that means investors who understand cycles will find opportunities for profit.

XVIII

THE ESSENCE OF CYCLES

I'm going to conclude by pulling together some of the book's paragraphs that I think hold the keys to understanding cycles, their genesis, and how they should be dealt with. I'll alter them only as necessary to allow them to stand alone here, out of context. This won't be a summary of the book, but rather a recap of some observations. (And for those who wish, reading just the bolded sentences will provide a good synopsis of the recap.)

H.M.

Investment success is like the choosing of a lottery winner, determined by one ticket (the outcome) being pulled from a bowlful (the full range of possible outcomes). In each case, one outcome is from among the many possibilities.

Superior investors are people who have a better sense for what ticket is in the bowl, and thus for whether it's worth participating in the lottery. In other words, while superior investors—like everyone else—don't know exactly what the future holds, they do have an above-average understanding of future tendencies. [\(page 14\)](#)



The odds change as our position in the cycles changes. If we take a defensive investment stance as these things change, we're being passive regarding cycles; in other words, we're ignoring the chance to tilt the odds in our favor. But if we apply some insight regarding cycles, we can increase our aggressiveness and place them on more aggressive investments when the odds are in our favor, and we can take money off the table and increase our defensiveness when the odds are against [us](#). [\(page 21\)](#)



In my view, the greatest way to optimize the positioning of a portfolio at a given point in time is through deciding what balance it should strike between aggressiveness and defensiveness. And I believe the

aggressiveness/defensiveness balance should be adjusted over time in response to changes in the state of the investment environment number of elements stand in their cycles.

The key word is “calibrate.” The amount you have invested, your allocation of capital among the various possibilities, and the riskiness of your own all should be calibrated along a continuum that runs from aggressive to defensive. When we’re getting value cheap, we should be aggressive; when we’re getting value expensive, we should pull back. Calibrating one’s portfolio position is what this book is mostly about. (page 12)



In the world investors inhabit, cycles rise and fall, and pendulums swing back and forth. Cycles and pendulum swings come in many forms and relationships, but the underlying reasons for them—and the patterns they produce—have a lot in common, and they tend to be consistent over time. Or as Mark Twain is reputed to have said (there’s no evidence he actually said it), “History doesn’t repeat itself, but it does rhyme.”

Whether Twain said it or not, that sentence sums up a lot of what this book is about. Cycles vary in terms of reasons and details, and the ups and downs (and the reasons for them) will occur forever, producing changes in the investment environment—and thus in the returns. That’s called for.

The fact is that the performance of these things is heavily influenced in the short run by, among other things, the involvement of people, which is far from steady. Rather they fluctuate from time to time, often because of things we can lump under the broad heading of “psychology.” Thus people’s behavior varies, and it certainly varies as the environment varies, and in the absence of changes in the environment, too. (pages 24–25)



The cycle oscillates around the midpoint. The midpoint of a cycle is thought of as the secular trend, norm, mean, average or “happy medium,” generally as being in some sense as “right and proper.” The extremes of the cycle, on the other hand, are thought of as aberrations or excesses returned from, and generally they are. While the thing that’s cycling tends to spend much of the time above or below it, eventual movement back in the direction of the midpoint is usually the rule. The movement from a high or a low extreme back toward the midpoint is often described as “regression toward the mean,” a powerful and very reasonable tendency in human life. But it can also be seen that the cyclical pattern generally

of movement from the reasonable midpoint toward a potentially impr extreme as it does going from an extreme back toward the midp

The rational midpoint generally exerts a kind of magnetic pull, br thing that's cycling back from an extreme in the direction of "nor usually doesn't stay at normal for long, as the influences respo swing toward the midpoint invariably continue in force and thus cause swing back from an extreme to proceed through the midpoint and further, toward the opposite extreme.

It's important to recognize and accept the dependability of th details vary, but the underlying dynamics are usually similar. (pag



The themes that provide warning signals in every boom/bust are ones: that excessive optimism is a dangerous thing; that risk aversion essential ingredient for the market to be safe; and that overly markets ultimately lead to unwise financing, and thus to danger participants. In short, the details are unimportant and can be irrelevant. the themes are essential, and they absolutely do tend to recur. Under that tendency—and being able to spot the recurrences—is one of the important elements in dealing with cycles. (page 36)



Cycles have more potential to wreak havoc the further they progress midpoint—i.e., the greater the aberrations or excesses. If the swin one extreme goes further, the swingback is likely to be more viole more damage is likely to be done, as actions encouraged by the operation at an extreme prove unsuitable for life elsewhere in the

In other words, the potential for havoc increases as the moveme from the midpoint increases: as economies and companies do “too well” stock prices go “too high.” Advances are followed by mere corrections bull markets by bear markets. But booms and bubbles are follo more harmful busts, crashes and panics. (pages 28–29)



Most people think of cycles as series of events that follow usual sequence: upswings are followed by downswings, and then even by new upswings. But to have a full understanding of cycles, enough. The events in the life of a cycle shouldn't be view being followed by the next, but—much more importantly—as each *causing* the next. (page 30)



The things I call cycles do not stem completely—or sometimes at the operation of mechanical, scientific or physical processes. They would be much more dependable and predictable if they did—but much less potentially profitable. (This is because the greatest profits come from seeing things before others do, and if cycles were totally dependable and predictable there would be no such thing as superiority in seeing them.) Some cycles follow an underlying principle (and sometimes not), but much variation is attributable to the role of humans in creating cycles. The involvement of human psychology in the process enables their emotion- and psychology-induced tendencies to influence cyclical phenomena. Chance or randomness also plays a big role in some cycles, and human behavior contributes to their existence, too. Humans are a big part of the reason these cycles exist, but also—along with randomness—for their inconsistency and thus their undependability.

The effort to explain life through the recognition of patterns—and to come up with winning formulas—is complicated, in large part, because we live in a world that is beset by randomness and in which people's behavior is the same from one instance to the next, even when they intend to do otherwise. The realization that past events were largely affected by these things—and that future events aren't fully predictable—is unpleasant, as it makes life more subject to anticipation, rule-making and rendering safe. Thus people seek explanations that would make events understandable . . . beyond that which is appropriate. This is as true in investing as in other aspects of life. (pages 41–42)



Why is the pendulum of psychology important? In essence, the too-swinging upward and downward swings of the cycles I'm covering in this book result from—and represent—psychological excesses in action.

In business, financial and market cycles, most excesses are on the upside—the inevitable reactions to the downside, which also tend to overshoot—the result of exaggerated swings of the pendulum of psychology. Thus, understanding and being alert to excessive swings is an entry-level requirement for avoiding harm from cyclical extremes, and hopefully for profiting from them.

The norms in terms of growth and appreciation are in so many ways "healthy." And if the participants built their behavior around those norms—instead of occasionally building up hopes for more and thus setting the stage for eventual moves toward less—the world would be a more temperate, and less-error-prone place. But that's not the nature of life. (pages 85–86)



It all seems so obvious: investors rarely maintain objective, rational, and stable positions. First they exhibit high levels of optimism, greed, tolerance and credulousness, and their resulting behavior causes assets to rise, potential returns to fall, and risk to increase. But then, —perhaps the arrival of a tipping point—they switch to pessimism, aversion and skepticism, and this causes asset prices to fall, prospective returns to rise, and risk to decrease. Notably, each group of phenomena to happen in unison, and the swing from one to the other often goes beyond what reason might call for.

That's one of the crazy things: in the real world, things go between “pretty good” and “not so hot.” But in the world of investor perception often swings from “flawless” to “hopeless.” The pendulum careens from one extreme to the other, spending almost no time at “the medium” and rather little in the range of reasonableness. First there's and then there's capitulation. ([page 96](#))



At the greatest extremes of the pendulum's swing, a process appears as a virtuous circle or a vicious circle. When even predominantly positive psychology is rosy, negative developments to be overlooked, everything is interpreted favorably, and things are often thought to be incapable of taking a turn for the worse. When things have been going badly for months or years and psychology is highly negative, it's the potential for improvement that can be

The superior investor—who resists external influences, remains emotionally balanced and acts rationally—perceives both positive and negative events, weighs events objectively, and analyzes them dispassionately. But the truth is that sometimes euphoria and optimism lead most investors to view things more positively than is warranted, and sometimes depression and pessimism make them see only bad and interpret events with a negative cast. Refusing to do so is one of the mistakes of investing.

Usually, when either set of polar extremes is in the ascendancy, the swing is readily observable, and thus the implications for investors should be clear to objective observers. But of course, the swing of the market from one extreme or the other occurs for the simple reason that market participants are moving in the same direction in a herd. ([pages 98–99](#))



My view that risk is the main moving piece in investing makes it clear that at any given point in time, the way investors collectively are viewing

and behaving with regard to it is of overwhelming importance in the investment environment in which we find ourselves. And the state of the environment is key in determining how we should behave with respect to it at that point. Assessing where attitudes toward risk stand in their cycle is perhaps the most important topic covered in [this](#) book. ([page 103](#))



Good times cause people to become more optimistic, jettison their caution, and settle for skimpy risk premiums on risky investments. Further, they are less pessimistic and less alarmed, they tend to lose interest in the lower end of the risk/return continuum. This combination of elements makes the prices of risky assets rise relative to safer assets. Thus it shouldn't come as a surprise that more unwise investments are made in good times than in bad. People are more inclined to make risky investments in good times even though the prices are higher, prices often mean the prospective risk premiums offered are skimpier than they were in more risk-conscious times. And when negative events occur, the lack of adequate risk premiums and margin for error makes the investments to have been unwise.

It follows from the above that risk is high when investors feel risk. And risk compensation is at a minimum just when risk is at its peak (meaning risk compensation is most needed). So much for the rational investor!

For me, the bottom line of all of this is that the greatest investment risk is the belief that there is no risk. Widespread risk tolerance or a high degree of investor comfort with risk—is the greatest risk. When subsequent market declines. But this is rarely perceived at the time of the decline, perceiving it—and turning cautious—is [most important](#). ([pages 111–112](#))



Just as the inadequacy of investors' risk aversion allows them to get carried up and buy at the top—egged on by the vision of easy money and the hope that they can't discern any risk—in less positive times they push prices down and sell at the bottom. Their unpleasant experience convinces them the contrary to what they had thought when everything was going well—that investing is a risky field in which they shouldn't engage. And, as a consequence, their risk aversion goes all the way from inadequate to excessive.

They become worrywarts. Just as risk tolerance had positioned them to become buyers of overpriced assets at the highs, now their screaming risk aversion makes them sellers—certainly not buyers—at [the bottom](#). ([pages 114–115](#))



During panics, people spend 100% of their time making sure there are no losses. . . . at just the time that they should be worrying about and looking out on great opportunities.

In times of extreme negativism, exaggerated risk aversion is likely to set in. Prices to already be as low as they can go; further losses to be avoided and thus the risk of loss to be minimal. Thus, the safest time to invest comes when everyone is convinced there's no hope. (page 132)



As risk attitudes swing from high to low, so do opportunities for profit. When everything's going well and asset prices are soaring, investors tend to view the future as rosy, risk as their friend, and profit as easily achieved. Everyone feels the same, meaning little risk aversion is incorporated into their decisions. Prices, and thus they're precarious. Investors become risk-tolerant just when they should increase their risk aversion.

And when events are down, so are investors. They think of the future in the place to lose money, risk as something to be avoided at all costs, and the future as depressingly likely. Under the excess of caution that prevails, they cannot accept possibilities that incorporate any optimism at all and (b) they cannot countenance the possibility that an assumption could be "too true."

Just as risk tolerance is unlimited at the top, it is non-existent at the bottom. This negativity causes prices to fall to levels from which losses are high and gains unlikely and gains could be enormous. But the sting of prior declines increases risk aversion and sends investors to the sidelines just as prices begin to rise (thus risk) are at their lowest. (page 116)



Understanding how investors are thinking about and dealing with risk is perhaps the most important thing to strive for. In short, excessive risk tolerance contributes to the creation of danger, and the swing to risk aversion depresses markets, creating some of the greatest buying opportunities.

The fluctuation—or inconstancy—in attitudes toward risk is both the result of some cycles and the cause or exacerbator of others. And it is on, since it seems to be hard-wired into most people's psyches to be more optimistic and risk-tolerant when things are going well, and then more worried and risk-averse when things turn downward. That means they're willing to buy when they should be most cautious, and most reluctant

when they should be most aggressive. Superior investors recognize this and strive to behave as contrarians. ([pages 134–135](#))



Changes in the availability of capital or credit constitute one of the fundamental influences on economies, companies and markets. Even though the credit cycle is less well-known to the man on the street than most other cycles discussed in this book, I consider it to be of great importance and profound influence.

When the credit window is open, financing is plentiful and easily obtained and when it's closed, financing is scarce and hard to get. Finally, to always bear in mind that the window can go from wide open to shut in just an instant. There's a lot more to fully understand the cycle, including the reasons for these cyclical movements and their impact—but that's the bottom [line. \(page 138\)](#)



Prosperity brings expanded lending, which leads to unwise lending, which produces large losses, which makes lenders stop lending, which ends the prosperity, and on and [on. \(page 143\)](#)



Looking for the cause of a market extreme usually requires a videotape of the credit cycle a few months or years. Most market extremes are abetted by an upsurge in the willingness to provide capital imprudently. Likewise, most collapses are preceded by a wholesale loss of confidence in finance—certain companies, industries, or the entire gamut of would-be borrowers. ([page 147](#))



The key to dealing with the credit cycle lies in recognizing that it reaches its apex when things have been going well for a while, news has been good, risk aversion is low, and investors are eager. That makes it easy for them to raise money and causes buyers and investors to compete for capital to provide it. The result is cheap financing, low credit standards, weak oversight, and the unwise extension of credit. Borrowers hold the cards when the credit window is wide open—not lenders or investors. The implications of this should be obvious: proceed with caution.

The exact opposite becomes true at the other extreme of the credit cycle. The nadir is reached when developments are unpleasant, risk aversion is heightened, and investors are depressed. Under such circumstances, no one wants to provide capital, the credit market freezes up, and prop-

go begging. This puts the cards into the hands of providers of capital rather than borrowers.

Because borrowing is difficult and capital is generally unavailable to those who possess it and are willing to part with it, lenders can apply rigorous standards, insist on strong loan structures and protective covenants, and demand high prospective returns. It's things like these that provide the margin of safety required for superior investing. When these boxes can be ticked, investors should swing into an aggressive mode.

Superior investing doesn't come from buying high-quality assets, but from buying when the deal is good, the price is low, the potential return is substantial, and the risk is limited. These conditions are much more likely to occur when the credit markets are in the less-euphoric, more-stringent part of the cycle. The slammed-shut phase of the credit cycle probably does more to make bargains available than any other single factor. [\(pages 159–160\)](#)



The merits of the asset in question matter only so much, but they can't be strong enough to always carry the day. Human emotion causes the prices of assets—even worthwhile assets—to be transported to levels that are extreme and unsustainable: either vertiginous highs or overwhelming pessimistic lows.

In short, conscientious belief in the inevitability of cycles like I'm describing means that a number of words and phrases must be excluded from an intelligent investor's vocabulary. These include "never," "always," "forever," "can't," "won't," "will" and "has to." [\(pages 179–180\)](#)



About 45 years ago—in the early 1970s—I received one of the greatest pieces of wisdom ever given, when an older and wiser investor introduced me to the stages of a bull market:

- the first stage, when only a few unusually perceptive people believe things will get better,
- the second stage, when most investors realize that improvement is actually taking place, and
- the third stage, when everyone concludes things will get better forever.

The arrival of this simple truth opened my eyes to the non-rational psychological extremes and the impact of those extremes on markets. Like many of the great quotations and adages, it captures disproportionate wisdom in a few simple words. It's all about the changeability of

pattern they follow over the course of a cycle, and how they error.

In the first stage, because the possibility of improvement is in most investors and thus unappreciated, security prices incorporate little or optimism. Often the first stage occurs after prices have been pounded in a crash, and the same downtrend that decimated prices also has wiped out psychology, turning the members of the crowd against the market and making them to swear off investing forever.

In the last stage, on the other hand, events have gone well for so long that they have been reflected so powerfully in asset prices, further lifting the market—that investors extrapolate improvement to infinity and prices to reflect their optimism. Trees generally don't grow to the sky, but in this stage investors act as if they will . . . and pay up to the limit they perceive. Few things are as costly as paying for potential that has been overrated.

It follows from the above that someone who invests in the first stage when almost no one can see a reason for optimism—buys asset prices from which substantial appreciation is possible. But someone who invests in the third stage invariably pays a high price for the market's excessive enthusiasm and loses money as a result. (pages 191–193)



The most important thing to note is that maximum psychology, maximum availability of credit, maximum price, minimum potential return and maximum risk all are reached at the same time, and usually these coincide with the last paroxysm of buying. (page 201)



In the reverse of the “top” that results from the upswing of the market, we see that the nadir of psychology, a total inability to access the minimum price, maximum potential return and minimum risk all coincide at the bottom, when the last optimist throws in the towel. (page 202)

Since the generalizations at the lows of the financial crisis of 2008, on the downside, the error-making machine went into reverse. No optimism. No risk tolerance, only risk aversion. No ability to see positives, only negatives. No willingness to interpret positively, only negatively. No ability to imagine good outcomes, only pessimism. (page 233)



What is a bottom? It's the point when the lowest prices of the market are reached. Thus a bottom can be viewed as the day the last

The truth is that financial facts and figures are only a starting point for understanding market behavior; investor rationality is the exception, not the rule; the market spends little of its time calmly weighing financial data and prices free of emotionality. [\(page 189\)](#)



The investor's goal is to position capital so as to benefit from market developments. He wants to have more invested when the market rises than when it falls, and to own more of the things that rise more or fall more than the others. The objective is clear. The question is how to achieve it. In the absence of the ability to see the future, how can we construct portfolios for what lies ahead? I think much of the answer lies in a better understanding of where the market stands in its cycle and what that means for its future movements. As I wrote in *The Most Important Thing*, "we never know where we're going, but we'd better have a good idea where we are." [\(pages 207–208\)](#)



What's the key in all of this? To know where the pendulum of market valuation stands in its cycle and to refuse to buy—perhaps to sell—when too-optimistic psychology and the willingness to assign too-high valuations cause prices to soar to peak levels. And to buy when pessimistic psychology and the desertion of valuation standards on the downside cause panicky investors to create bargains by selling despondently at prices that prevail. As Sir John Templeton put it, "To buy when others are despondently selling and sell when others are greedily buying requires the greatest fortitude and pays the greatest reward." [\(pages 209–210\)](#)



The essential ingredient here is *inference*, one of my favorite words. The investor sees what happens each day, as reported in the media. But how can he make an effort to understand what those everyday events say about the psychology of market participants, the investment climate, and thus what should be done in response?

Simply put, we must strive to understand the implications of what is going on around us. When others are recklessly confident and buying aggressively, we should be highly cautious; when others are frightened into inaction or panicked selling, we should become aggressive.

Psychological and emotional elements have their primary impact by convincing investors that past valuation standards have become irrelevant and can be departed from. When investors are flying high and making money, they find it easy to come up with convenient reasons why assets should

untethered from the constraints of valuation norms. The explanation usually begins with, “it’s different this time.” Watch out for this ominous willing suspension of disbelief. Likewise, when asset prices collapse in a crash, it’s usually because of an assumption that none of the things supported value in the past can be trusted to work in the future (page 215)



“It’s different this time” are four of the most dangerous words in the world—especially when applied, as is often the case, to something that has reached what in prior times would have been called an extreme.

When people say “it’s different” in this case, what they mean is that the rules and processes that produced cycles in the past have been suspended. The cyclical behavior of the financial past did not result from the physical or scientific rules. In science, cause and effect enjoy a dependable and repeatable relationship, so that it’s possible to say with confidence, “then a, then b.” But while there are some principles that operate in the world of finance and business, the resulting truth is very different from that.

The reason for this—as I’ve harped on repeatedly—is the influence of people. People’s decisions have great influence on economic, business, and market cycles. In fact, economies, business, and markets consist of transactions between people. And people don’t make their decisions scientifically.

People have feelings, and as such they aren’t bound by inviolable rules. They’ll always bring emotions and foibles to their economic and business decisions. As a result, they’ll become euphoric at the wrong time and despondent at the wrong time—exaggerating the upside potential when things are going well and the downside risk when things are going poorly—and they’ll take trends to cyclical extremes. (pages 288–290)



Cycle positioning is the process of deciding on the risk posture of a portfolio in response to your judgments regarding the principal cycle. Asset selection is the process of deciding which markets, market segments, or specific securities or assets to overweight and underweight. These are the main tools in portfolio management. It may be an over-simplification to think everything investors do falls under one or the other of these headings (page 248)



Cycle positioning primarily consists of choosing between aggressiveness and defensiveness: increasing and decreasing exposure to market movements.

The recipe for success here consists of (a) thoughtful analysis of market stands in its cycle, (b) a resulting increase in aggressiveness, and (c) being proved right. These things can be summed “skill” or “alpha” at cycle positioning. Of course, “c”—being proved right—isn’t a matter fully within anyone’s control, in particular because of the degree to which it is subject to randomness. So being proved right doesn’t happen every time, even to skillful investors who reason things out
([page 252](#))



When the market is low in its cycle, gains are more likely than usual and losses are less likely. The reverse is true when the market is high. Positioning moves, based on where you believe the market stands, don’t amount to trying to better prepare your portfolio for the events of the future. While you can always be unlucky regarding the relationship between what logically should happen and what actually does happen, good positioning decisions can increase the chance that the market’s tendency—and your chance for outperformance—will be on your side. ([pages 254–255](#))



In my opinion it’s entirely reasonable to try to improve long-term results by altering positions on the basis of an understanding of the market cycle. But it’s essential that you also understand the limitations, as well as the skills that are required and how difficult it is.

Importantly, I want to call attention to the obvious fact that the everyday ups and downs of the market—the clear examples of which this book all concern “once-in-a-lifetime” cyclical extremes (which seem to happen about once a decade). First, the extremes of bubbles and crashes—and, in particular, the process through which they arise—most clearly illustrate the cycle in action and how to respond to it. And second, dealing with pronounced extremes that we should expect the high likelihood of success.

Between the extremes of “rich” and “cheap”—when the cycle is in the middle ground of “fair”—the state of the relationship between price and value is, by definition, nowhere as clear-cut as at the extremes. As

- It’s hard to make frequent distinctions and hard to do so correctly.
- Thus distinctions in the middle ground aren’t as potentially profitable as at the extremes, and those distinctions can’t be expected to work dependably.

Detecting and exploiting the extremes is really the best we can hope for. And I believe it can be done dependably—if you're analytical, experienced (or well-versed in history) and unemotional. That means, however, that you shouldn't expect to reach profitable conclusions daily, monthly or even yearly.

The reasonableness of the effort at cycle timing depends simply on what you expect of it. If you frequently try to discern where we are, the sense of "what's going to happen tomorrow?" or "what's in the next month?" you're unlikely to find success. I describe such efforts as "trying to be cute." No one can make fine distinctions like those often made, or consistently right enough to add materially to investment results. No one knows when the market will develop trends that efforts at cycle timing label "probable" will materialize.

As Peter Bernstein said, "The future is not ours to know. But it is ours to know that being wrong is inevitable and normal, not some terrible disaster, not some awful failing in reasoning, not even bad luck in most instances. Being wrong comes with the franchise of an activity whose results depend on an unknown future. . . ." (pages 265–269)



The tendency of people to go to excess will never end. And excesses eventually have to correct, neither will the occurrence of recessions. Economies and markets have never moved in a straight line in the past, and neither will they do so in the future. And that means investors who understand cycles will find opportunities for profit. (page 293)

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About the Author



HOWARD ARKES is cochairman and cofounder of Oaktree Capital Management, a leading investment firm responsible for over \$120 billion in assets. His previous book on investing, *The Most Important Thing: Uncommon Sense for the Thoughtful Investor*, was a critically acclaimed bestseller. He lives in New York City.

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